INCH-POUND

MIL-STD-1560B
w/CHANGE 1
18 May 2012
SUPERSEDING
MIL-STD-1560B
20 July 2007

DEPARTMENT OF DEFENSE
INTERFACE STANDARD

INSERT ARRANGEMENTS FOR MIL-DTL-38999, MIL-DTL-27599 AND SAE-AS29600 SERIES A ELECTRICAL CIRCULAR CONNECTORS

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Comments, suggestions, or questions on this document should be addressed to Commander, DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990 or emailed to CircularConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

SUMMARY OF CHANGE 1 MODIFICATIONS

1. Changed “MIL-C-29600 Series A” to “SAE-AS29600” throughout the document. Added new paragraph 2.2.3, “Non-Government publications” to section 2, and included documents SAE-AS29600, SAE-AS39029/59, SAE-AS39029/60, SAE-AS39029/90 and SAE-AS39029/91 in the document listing.

2. Updated the foreword on page ii with “DLA Land and Maritime” (formerly called “Defense Supply Center Columbus”) and its associated email address.

3. Updated contents on page iv with “6.4, Changes from previous issue”, and on page v with “TABLE”, referencing tables I and II.

4. Updated section 5 with a new subparagraph to provide dimensional requirements for size 10 contacts (see page 3).

5. Combined MIL-STD-1560B subparagraphs 5.1.i and 5.1.j into 5.1.j. This editorial change is intended to provide overarching dimensional information for the connector gauge features, which ensures that the “NOTES” following apply to both the plugs and receptacles.

6. Added new insert arrangement 9-7 to Figure 1. The new arrangement, using size 22D contacts, is for a weapon system application.

7. Reinstated center contact cavity “PP” to Figure 9, insert arrangement 25-61. Reinstated “DD” cavity, which is included in revision “A” of this document, but which is missing from revision “B”. Also reinstated contact cavity “n” from previous revision “A”, in place of contact cavity “o”, which was transcribed incorrectly into revision “B”.

8. Updated data in the table for insert arrangement 25-46 with the required SAE-AS39029/59 and /60 size 8 coax contacts, and added a note stating to use identical insert arrangement 25-90 if size 8 twinax contacts in accordance with SAE-AS39029/90 and /91 are specified.

9. Updated notes for insert arrangements 17-2, 19-18, 21-75, 25-7, 25-8, 25-20 and 25-90 with the following statement: “heat shrink sleeves used with size 8 twinax contacts are rated to 175 ºC (347 ºF) maximum”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH PAGE

FOREWORD… ii

1. SCOPE 1
1.1 Scope 1

2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents 1
2.2.1 Specifications, standards, and handbooks 1
2.2.2 Other Government documents, drawings and publications 1
2.3 Order of precedence 2

3. DEFINITIONS 2
3.1 Definitions 2

4. GENERAL REQUIREMENTS 2
4.1 Recycled, recovered, or environmentally preferable materials 2

5. DETAILED REQUIREMENTS 2
5.1 Dimensions 2
5.2 Marking 4
5.3 Drawing notes 4

6. NOTES
6.1 Intended use 5
6.2 Cross-reference 5
6.3 Subject term (key word) listing 8
6.4 Changes from previous issue 8

FIGURE

1. Insert arrangement, electrical connector, shell size 9 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 9
2. Insert arrangement, electrical connector, shell size 11 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 15
3. Insert arrangement, electrical connector, shell size 13 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 22
4. Insert arrangement, electrical connector, shell size 15 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 27
5. Insert arrangement, electrical connector, shell size 17 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 36
6. Insert arrangement, electrical connector, shell size 19 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 47
7. Insert arrangement, electrical connector, shell size 21 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) 60

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

CONTENTS

8. Insert arrangement, electrical connector, shell size 23 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600)……73

FIGURE

9. Insert arrangement, electrical connector, shell size 25 (Series I, III, IV of MIL-DTL-38999, Series I of MIL-DTL-27599, and Series A of SAE-AS29600) ……92
10. Insert arrangement, electrical connector, shell size 8 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 123
11. Insert arrangement, electrical connector, shell size 10 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 128
12. Insert arrangement, electrical connector, shell size 12 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 133
13. Insert arrangement, electrical connector, shell size 14 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 139
14. Insert arrangement, electrical connector, shell size 16 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 145
15. Insert arrangement, electrical connector, shell size 18 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 155
16. Insert arrangement, electrical connector, shell size 20 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 173
17. Insert arrangement, electrical connector, shell size 22 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 186
18. Insert arrangement, electrical connector, shell size 24 (Series II of MIL-DTL-38999, and Series II of MIL-DTL-27599)…… 206

TABLE

I. Cross-reference of figures to sections…… 5
II. Cross-reference of figures to insert arrangements……6

CONCLUDING MATERIAL…… 222

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers insert arrangements for use with MIL-DTL-38999 series I, II, III and IV; MIL-DTL-27599 series I and II; and SAE-AS29600 series A circular, electrical connectors.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-38999	-	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for.
MIL-DTL-27599	-	Connectors, Electrical, Circular, Miniature, Quick Disconnect, Environment Resistant, Solder Contacts, General Specification for.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

02003	-	Contact Electrical Connector Concentric Twinax, Socket, Shielded, Size 12 (For MIL-DTL-38999 Series I, III, and IV connectors)
02004	-	Contact Electrical Connector Concentric Twinax, Pin, Shielded, Size 12 (For MIL-DTL-38999 Series I, III, and IV connectors)

(Copies of DLA Land and Maritime drawings are available for download at internet web address http://www.landandmaritime.dla.mil/programs/milspec/DocSearch.aspx).

2.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

SAE INTERNATIONAL

SAE-AS29600	-	Connectors, Electrical, Circular, Miniature, Composite, High Density, Quick Coupling, Environment Resistant, Removable Crimp Contacts, Associated Hardware, General Specification for.
SAE-AS39029/59	-	Contacts, Electrical Connector, Socket, Crimp Removable, Shielded, Size 8 (for MIL-C-38999 Series I, III, and IV Connectors).
SAE-AS39029/60	-	Contacts, Electrical Connector, Pin, Crimp Removable, Shielded, Size 8 (for MIL-C-38999 Series I, III, and IV Connectors).
SAE-AS39029/90	-	Contact, Electrical Connector, Concentric Twinax, Pin, Size 8.
SAE-AS39029/91	-	Contact, Electrical Connector, Concentric Twinax, Socket, Shielded, Size 8.

(Copies of these documents are available at http://www.sae.org or SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Definitions. The terms used in this standard are generally accepted by the electrical and electronics industries and are commonly used in electrical connector engineering practice.

4. GENERAL REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the materials meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. DETAILED REQUIREMENTS

5.1 Dimensions. Dimensions shall be in accordance with the applicable section of this standard and the following dimensional data:

a. ↓ indicates ℄ (center line) of insert arrangement.

b. All dimensions are true position (TP). Dimensions are in inches**,** unless otherwise specified. Dimensions and markings: Front face of pin insert shown (socket insert opposite).

c. Number 12 contacts - MIL-DTL-27599: pin contacts .094 ± .001 diameter; socket insert, contact entry hole .099 minimum diameter, number of places as applicable.
MIL-DTL-38999 and SAE-AS29600 series A: pin contacts .094 ± .001 diameter; socket insert, contact entry hole .103 minimum diameter, number of places as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

d. Number 16 contacts - MIL-DTL-27599: pin contacts .0625 ± .0010 diameter; socket insert, contact entry hole .067 minimum diameter, number of places as applicable.
MIL-DTL-38999 and SAE-AS29600 series A: pin contacts .0625 ± .0010 diameter; socket insert, contact entry hole .071 minimum diameter, number of places as applicable.

e. Number 20 contacts - MIL- DTL-27599: pin contacts .040 ± .001 diameter; socket insert, contact entry hole .045 minimum diameter, number of places as applicable.
MIL-DTL-38999 and SAE-AS29600 series A: pin contacts .040 ± .001 diameter; socket insert, contact entry hole .049 minimum diameter, number of places as applicable.

f. Number 22 and 22M - MIL-DTL-27599: pin contacts .0300 ± .0005 diameter; socket insert, contact entry hole .036 minimum diameter, number of places as applicable.

g. Number 22D contacts - MIL-DTL-38999 and SAE-AS29600 series A: pin contacts .0300 ± .0005 diameter; socket insert, contact entry hole .035 minimum diameter, number of places as applicable.

h. Number 8 shielded contacts - MIL-DTL-38999 and SAE-AS29600 series A: pin contacts .218 ± .001 diameter; socket insert, contact entry hole .227 minimum diameter, number of places as applicable.

i. Number 10 contacts - MIL-DTL-38999: pin contacts .125 ± .001 diameter; socket insert, contact entry hole .134 minimum diameter, number of places as applicable.

j. Receptacle and plug gauge measurements:

Receptacle gauge measurements (note: gauging for location of socket contact cavity or tip of pin contact): Gauge features for MIL-DTL-38999 series I, II, and III, SAE AS29600 series A, and MIL-DTL-27599 series I and II receptacle connectors shall be .002 smaller than H at MMC, and .006 smaller than GG at MMC.

⊕	H -.002 Ⓜ	GG -.006 Ⓜ	See notes

Plug gauge measurements (note: gauging for location of socket contact cavity or tip of pin contact): Gauge features for MIL-DTL-38999 series I, II, and III; SAE AS29600 series A and MIL-DTL-27599 series I and II plug connectors shall be .002 larger than W at MMC and .006 larger than GG at MMC.

⊕	W +.002 Ⓜ	GG +.006 Ⓜ	See notes

NOTES:

a. .004 DIA Ⓜ — Fixed contacts: For fixed contact designs, including MIL-DTL-27599 and hermetic connectors of MIL-DTL-38999, this requirement pertains to the location of the tip of the pin contacts and the location of the socket insert, contact cavities.

b. .015 DIA Ⓜ — Removeable crimp type socket contact cavities: For MIL-DTL-38999 and SAE-AS29600 series A connectors with removable crimp type contacts, this requirement pertains to the location of the socket insert, contact cavities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

c. .024 DIA Ⓜ Removeable crimp type pin contacts: For MIL-DTL-38999 and SAE-AS29600 series A connectors with crimp type removable contacts, this requirement pertains to the location of the tip of the pin contact.

d. Feature control frame datums H, GG, and W are located on connector intermateability dimension figures for key/keyway polarization in MIL-DTL-38999.

5.2 Marking. Marking shall be in accordance with MIL-DTL-38999, SAE-AS29600 series A, MIL-DTL-27599, and as shown in the applicable section of this standard. Sawtooth lines may be used in place of straight lines.

5.3 Drawing notes. The following information is applicable to all figures of this standard.

a. Dimensions are in inches, unless otherwise specified.
b. Metric equivalents are given for information only.
c. Insert arrangements using multi-axial (i.e. coax, twinax, shielded) contacts shall not be used in firewall connector applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard covers insert arrangements for use with MIL-DTL-38999 series I, II, III and IV; MIL-DTL-27599 series I and II; and SAE-AS29600 series A circular connectors. Information pertaining to shell size, arrangement number, number of contacts, size of contacts, service rating, contact location, and superseding information is contained in this document.

6.2 Cross-reference. See table I for cross-reference of previous sections and current figures. See table II for reference links to insert arrangement numbers.

TABLE I. Cross-reference of figures to sections.

Title	Current figure	Previous section	Document	Series
Insert arrangement, electrical connector, shell size 9	1	101	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 11	2	102	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 13	3	103	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 15	4	104	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 17	5	105	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 19	6	106	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 21	7	107	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 23	8	108	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 25	9	109	MIL-DTL-38999	I,III, IV
			MIL-DTL-27599	I
			SAE-AS29600	A
Insert arrangement, electrical connector, shell size 8	10	201	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 10	11	202	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 12	12	203	MIL-DTL-38999	II
			MIL-DTL-27599	II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

TABLE I. Cross reference of figures to sections - Continued.

Title	Current figure	Previous section	Document	Series
Insert arrangement, electrical connector, shell size 14	13	204	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 16	14	205	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 18	15	206	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 20	16	207	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 22	17	208	MIL-DTL-38999	II
			MIL-DTL-27599	II
Insert arrangement, electrical connector, shell size 24	18	209	MIL-DTL-38999	II
			MIL-DTL-27599	II

TABLE II. Cross-reference to insert arrangements.

Figure	Insert arrangement	Page	Figure	Insert arrangement	Page
1	9-3	9	9	25-11	111
1	9-98	10	9	25-20	113
1	9-44	11	9	25-7	116
1	9-6	12	9	25-8	119
1	9-35	13	9	25-90	120
1	9-7	14	9	25-37	121
2	11-4	15	10	8-3	122
2	11-5	16	10	8-6	123
2	11-98	17	10	8-44	124
2	11-99	18	10	8-98	125
2	11-13	19	10	8-35	126
2	11-35	20	11	10-5	127
2	11-2	21	11	10-13	128
3	13-4	22	11	10-98	129
3	13-8	23	11	10-99	130
3	13-98	24	11	10-35	131
3	13-22	25	12	12-22	132
3	13-35	26	12	12-4	134
4	15-5	27	12	12-8	135
4	15-15	28	12	12-98	136
4	15-18	29	12	12-35	137
4	15-19	30			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

TABLE II. Cross-reference to insert arrangements - Continued.

Figure	Insert arrangement	Page	Figure	Insert arrangement	Page
4	15-37	31	12	12-3	138
4	15-97	32	13	14-5	139
4	15-35	34	13	14-15	140
5	17-6	35	13	14-18	141
5	17-8	36	13	14-97	142
5	17-11	37	13	14-35	143
5	17-99	38	13	14-37	144
5	17-26	39	14	16-8	145
5	17-2	41	14	16-26	146
5	17-35	43	14	16-99	148
5	17-55	45	14	16-55	150
6	19-11	46	14	16-42	152
6	19-28	47	14	16-35	153
6	19-30	48	14	16-6/7	154
6	19-32	49	15	18-11	155
6	19-67	51	15	18-28	156
6	19-35	53	15	18-30	158
6	19-66	55	15	18-32	160
6	19-45	57	15	18-53	162
6	19-18	58	15	18-67	164
7	21-11	59	15	18-35	166
7	21-16	60	15	18-45	168
7	21-24	61	15	18-66	170
7	21-25	62	15	18-96	172
7	21-27	63	16	20-1	173
7	21-41	64	16	20-2	175
7	21-39	66	16	20-16	177
7	21-1	68	16	20-24	178
7	21-35	70	16	20-25	179
7	21-75	71	16	20-27	180
8	23-1	74	16	20-39	181
8	23-35	77	16	20-41	183
8	23-2	79	16	20-35	184
8	23-21	80	17	22-1	186
8	23-32	82	17	22-2	189
8	23-34	83	17	22-21	191
8	23-36	84	17	22-32	192
			17	22-34	193

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

TABLE II. Cross reference to insert arrangements - Continued.

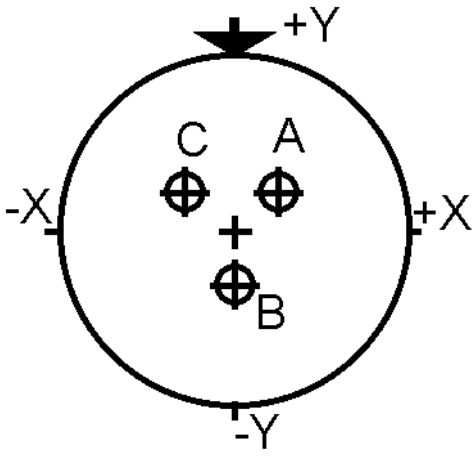
Figure	Insert arrangement	Page	Figure	Insert arrangement	Page
8	23-53	86	17	22-36	195
8	23-97	87	17	22-53	197
8	23-99	88	17	22-55	199
8	23-55	90	17	22-97	201
9	25-1	93	17	22-99	202
9	25-35	96	17	22-35	203
9	25-2	99	18	24-1	206
9	25-29	100	18	24-2	209
9	25-61	102	18	24-61	212
9	25-4	104	18	24-4	214
9	25-43	106	18	24-24	216
9	25-46	108	18	24-29	217
9	25-19	109	18	24-35	218
9	25-24	110	18	24-19	221

6.3 Subject term (key word) listing.

Contact
Insert
Shell size
Socket
Pin

6.4 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



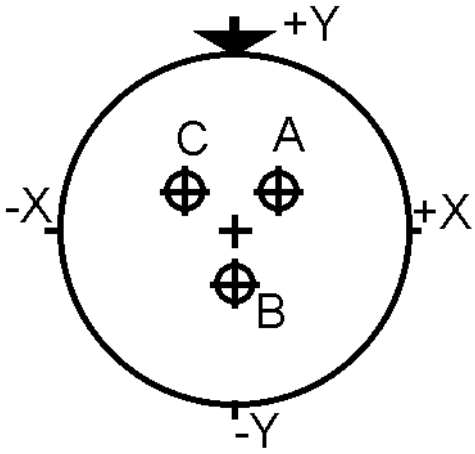
Contacts (Insert arrangement 9-3)		
Contact position identification (ID)	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.056 (1.42)	+.018 (0.46)
B	+.000 (0.00)	-.079 (2.01)
C	-.056 (1.42)	+.018 (0.46)

(Applicable to MIL-DTL-27599 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
9	-3	3	20	M	All	MS20049-3

FIGURE 1. Insert arrangement, electrical connector, shell size 9 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

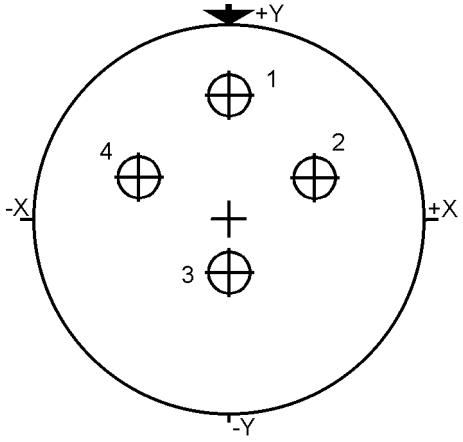


Contacts (Insert arrangement 9-98)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.038 (0.97)
B	+.000 (0.00)	-.075 (1.91)
C	-.065 (1.65)	+.038 (0.97)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
9	-98	3	20	I	All	MS20049-98

FIGURE 1. Insert arrangement, electrical connector, shell size 9 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



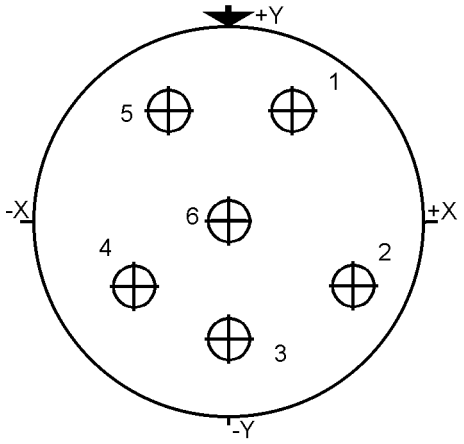
Contacts (Insert arrangement 9-44)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
1	+.000 (0.00)	+.081 (2.06)
2	+.080 (2.03)	+.021 (0.53)
3	+.000 (0.00)	-.050 (1.27)
4	-.080 (2.03)	+.021 (0.53)

(Inactive for new design for MIL-DTL-38999 and SAE-AS29600)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
9	-44	4	22	M	All	MS20049-44

FIGURE 1. Insert arrangement, electrical connector, shell size 9 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



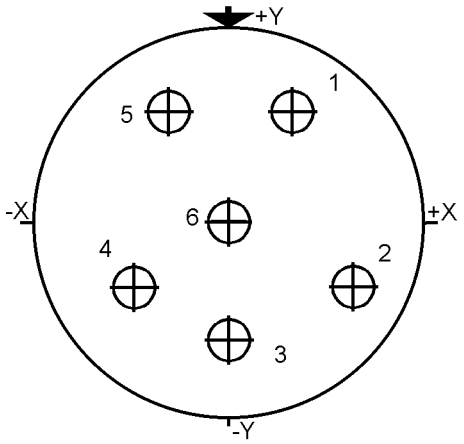
Contacts (Insert arrangement 9-6)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.078 (1.98)
2	+.078 (1.98)	-.045 (1.14)
3	+.000 (0.00)	-.090 (2.29)
4	-.078 (1.98)	-.045 (1.14)
5	-.045 (1.14)	+.078 (1.98)
6	+.000 (0.00)	+.000 (0.00)

(Inactive for new design for MIL-DTL-38999 and SAE-AS29600)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
9	-6	6	22M	M	All	MS20049-6

FIGURE 1. Insert arrangement, electrical connector, shell size 9 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



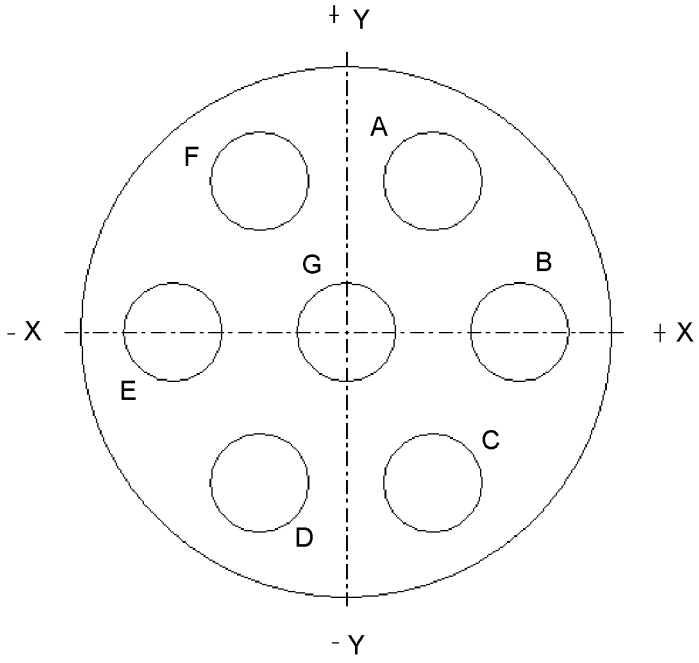
Contacts (Insert arrangement 9-35)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.078 (1.98)
2	+.078 (1.98)	-.045 (1.14)
3	+.000 (0.00)	-.090 (2.29)
4	-.078 (1.98)	-.045 (1.14)
5	-.045 (1.14)	+.078 (1.98)
6	+.000 (0.00)	+.000 (0.00)

(Applicable to MIL-DTL-38999 and SAE-AS29600)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
9	-35	6	22D	M	All	MS27525-35

FIGURE 1. Insert arrangement, electrical connector, shell size 9 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



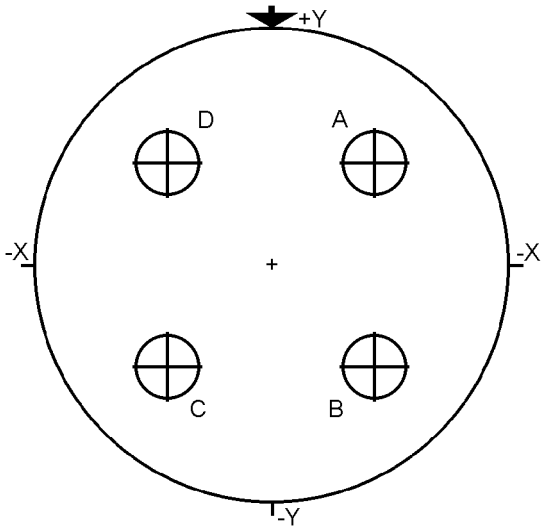
Contacts (Insert arrangement 9-7)		
Contact position ID	Contact location	
	X-axis mm (inches)	Y-axis mm (inches)
A	+1.18 (.046)	+2.06 (.081)
B	+ 2.36 (.093)	+0.00 (0.00)
C	+1.18 (.046)	- 2.06 (-.081)
D	- 1.18 (-.046)	- 2.06 (-.081)
E	- 2.36 (-.093)	- 0.00 (0.00)
F	-1.18 (- .046)	+2.06 (.081)
G	+.000 (0.00)	+.000 (0.00)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
9	- 7	7	22D	M	All

FIGURE 1. Insert arrangement, electrical connector, shell size 9 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

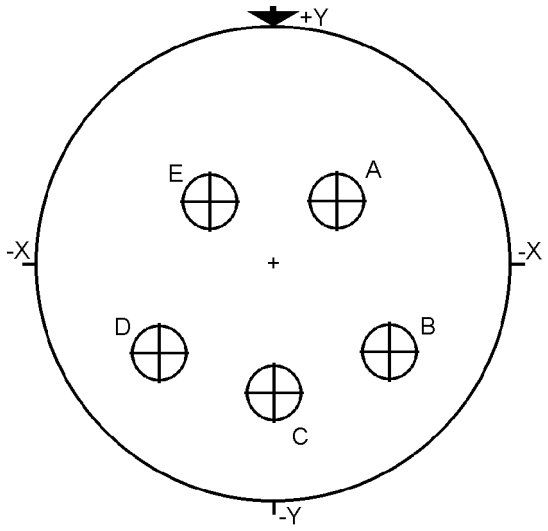


Contacts (Insert arrangement 11-4)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.065 (1.65)
B	+.065 (1.65)	-.065 (1.65)
C	-.065 (1.65)	-.065 (1.65)
D	-.065 (1.65)	+.065 (1.65)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
11	-4	4	20	I	All	MS20050-4

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

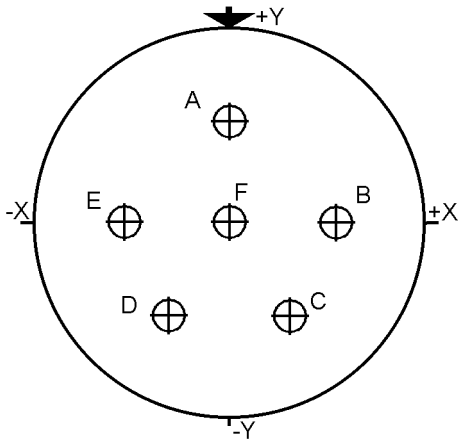


Contacts (Insert arrangement 11-5)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.056 (1.42)
B	+.113 (2.87)	-.065 (1.65)
C	+.000 (0.00)	-.130 (3.30)
D	-.113 (2.87)	-.065 (1.65)
E	-.065 (1.65)	+.056 (1.42)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
11	-5	5	20	I	All	MS20050-5

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

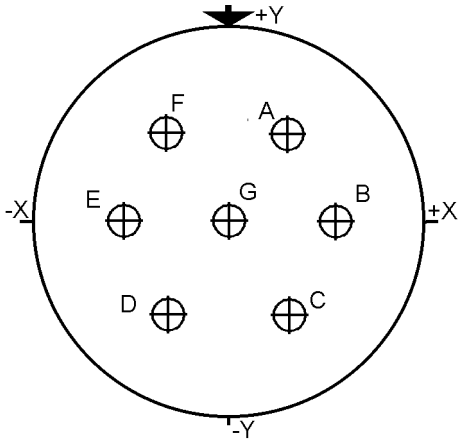


Contacts (Insert arrangement 11-98)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.130 (3.30)
B	+.130 (3.30)	+.000 (0.00)
C	+.065 (1.65)	-.113 (2.87)
D	-.065 (1.65)	-.113 (2.87)
E	-.130 (3.30)	+.000 (0.00)
F	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
11	-98	6	20	I	All	MS20050-98

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



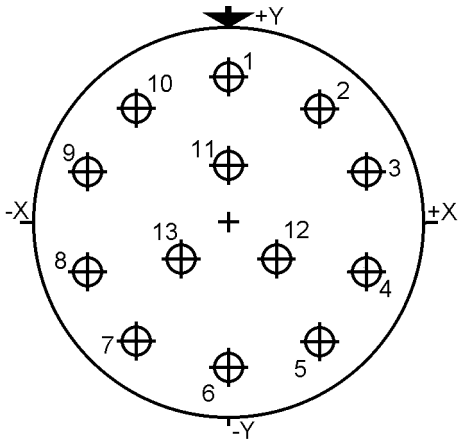
Contacts (Insert arrangement 11-99)		
Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.113 (2.87)
B	+.130 (3.30)	+.000 (0.00)
C	+.065 (1.65)	-.113 (2.87)
D	-.065 (1.65)	-.113 (2.87)
E	-.130 (3.30)	+.000 (0.00)
F	-.065 (1.65)	+.113 (2.87)
G	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
11	-99	7	20	I	All	MS20050-99

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



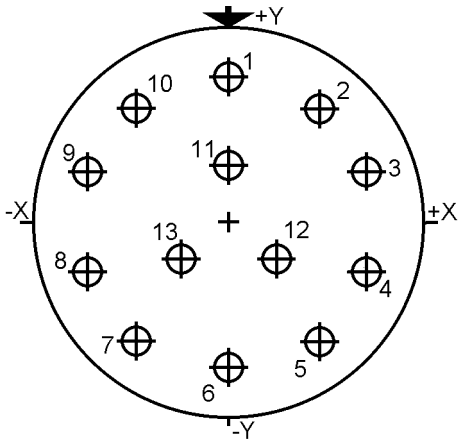
Contacts (Insert arrangement 11-13)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.000 (0.00)	+.146 (3.71)	8	-.138 (3.51)	-.045 (1.14)
2	+.085 (2.16)	+.118 (3.00)	9	-.138 (3.51)	+.045 (1.14)
3	+.138 (3.51)	+.045 (1.14)	10	-.085 (2.16)	+.118 (3.00)
4	+.138 (3.51)	-.045 (1.14)	11	+.000 (0.00)	+.056 (1.42)
5	+.085 (2.16)	-.118 (3.00)	12	+.049 (1.24)	-.035 (0.89)
6	+.000 (0.00)	-.146 (3.71)	13	-.049 (1.24)	-.035 (0.89)
7	-.085 (2.16)	-.118 (3.00)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
11	-13	13	22M	M	All	MS20050-13

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



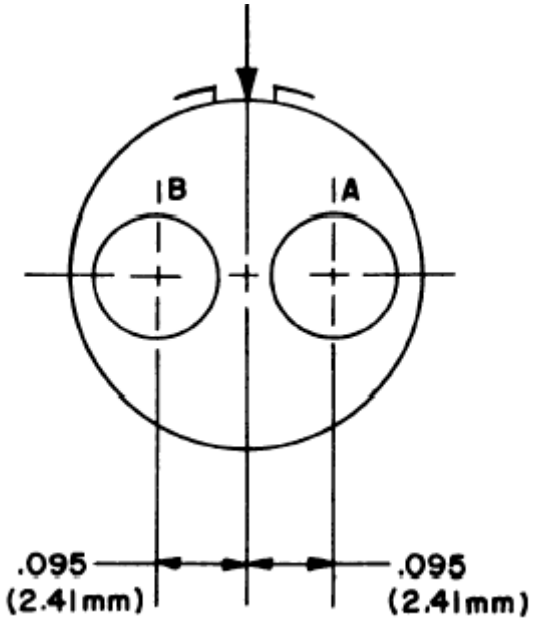
Contacts (Insert arrangement 11-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.000 (0.00)	+.146 (3.71)	8	-.138 (3.51)	-.045 (1.14)
2	+.085 (2.16)	+.118 (3.00)	9	-.138 (3.51)	+.045 (1.14)
3	+.138 (3.51)	+.045 (1.14)	10	-.085 (2.16)	+.118 (3.00)
4	+.138 (3.51)	-.045 (1.14)	11	+.000 (0.00)	+.056 (1.42)
5	+.085 (2.16)	-.118 (3.00)	12	+.049 (1.24)	-.035 (0.89)
6	+.000 (0.00)	-.146 (3.71)	13	-.049 (1.24)	-.035 (0.89)
7	-.085 (2.16)	-.118 (3.00)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
11	-35	13	22D	M	All	MS27526-35

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

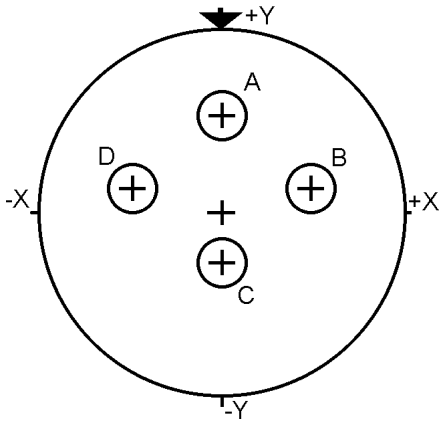


Contacts (Insert arrangement 11-2)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.095 (2.41)	+.000 (.00)
B	-.095 (2.41)	+.000 (.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
11	-2	2	16	I	All

FIGURE 2. Insert arrangement, electrical connector, shell size 11 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

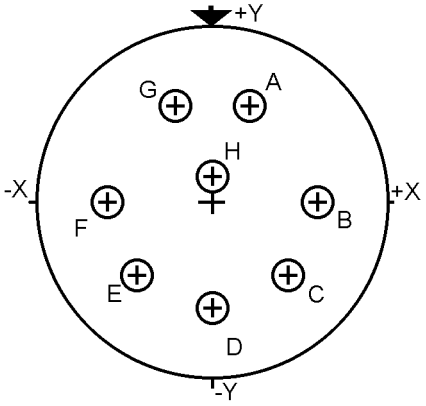


Contacts (Insert arrangement 13-4)		
Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.150 (3.81)
B	+.146 (3.71)	+.035 (0.89)
C	+.000 (0.00)	-.083 (2.11)
D	-.146 (3.71)	+.035 (0.89)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
13	-4	4	16	I	All	MS20051-4

FIGURE 3. Insert arrangement, electrical connector, shell size 13 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

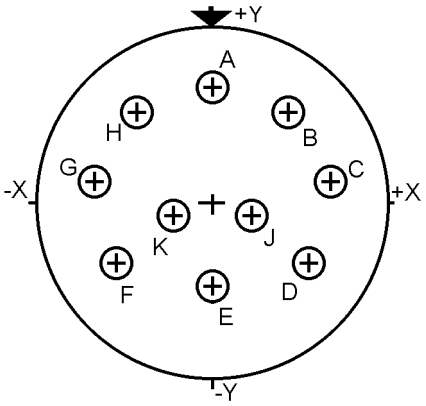


Contacts (Insert arrangement 13-8)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.157 (3.99)
B	+.170 (4.32)	+.000 (0.00)
C	+.120 (3.05)	-.120 (3.05)
D	+.000 (0.00)	-.170 (4.32)
E	-.120 (3.05)	-.120 (3.05)
F	-.170 (4.32)	+.000 (0.00)
G	-.065 (1.65)	+.157 (3.99)
H	+.000 (0.00)	+.044 (1.12)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
13	-8	8	20	I	All	MS20051-8

FIGURE 3. Insert arrangement, electrical connector, shell size 13 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 13-98)		
Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.195 (4.95)
B	+.125 (3.18)	+.150 (3.81)
C	+.193 (4.90)	+.030 (0.76)
D	+.164 (4.17)	-.105 (2.67)
E	+.000 (0.00)	-.135 (3.43)
F	-.164 (4.17)	-.105 (2.67)
G	-.193 (4.90)	+.030 (0.76)
H	-.125 (3.18)	+.150 (3.81)
J	+.065 (1.65)	-.015 (0.38)
K	-.065 (1.65)	-.015 (0.38)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
13	-98	10	20	I	All	MS20051-98

FIGURE 3. Insert arrangement, electrical connector, shell size 13 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 13-22)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.197 (5.00)	12	-.182 (4.62)	+.088 (2.24)
2	+.126 (3.20)	+.158 (4.01)	13	-.126 (3.20)	+.158 (4.01)
3	+.182 (4.62)	+.088 (2.24)	14	-.045 (1.14)	+.197 (5.00)
4	+.203 (5.16)	+.000 (0.00)	15	+.045 (1.14)	+.107 (2.72)
5	+.182 (4.62)	-.088 (2.24)	16	+.117 (2.97)	+.026 (0.66)
6	+.126 (3.20)	-.158 (4.01)	17	+.093 (2.36)	-.075 (1.91)
7	+.045 (1.14)	-.197 (5.00)	18	+.000 (0.00)	-.120 (3.05)
8	-.045 (1.14)	-.197 (5.00)	19	-.093 (2.36)	-.075 (1.91)
9	-.126 (3.20)	-.158 (4.01)	20	-.117 (2.97)	+.026 (0.66)
10	-.182 (4.62)	-.088 (2.24)	21	-.045 (1.14)	+.107 (2.72)
11	-.203 (5.16)	+.000 (0.00)	22	+.000 (0.00)	-.030 (0.76)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
13	-22	22	22M	M	All	MS20051-22

FIGURE 3. Insert arrangement, electrical connector, shell size 13 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



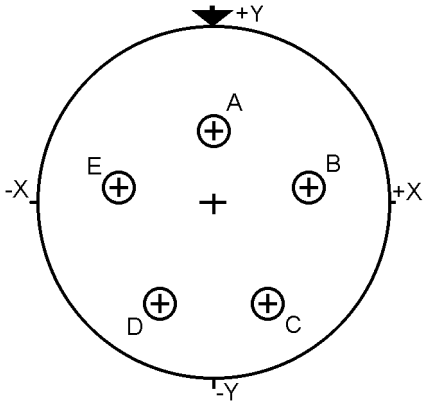
Contacts (Insert arrangement 13-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.197 (5.00)	12	-.182 (4.62)	+.088 (2.24)
2	+.126 (3.20)	+.158 (4.01)	13	-.126 (3.20)	+.158 (4.01)
3	+.182 (4.62)	+.088 (2.24)	14	-.045 (1.14)	+.197 (5.00)
4	+.203 (5.16)	+.000 (0.00)	15	+.045 (1.14)	+.107 (2.72)
5	+.182 (4.62)	-.088 (2.24)	16	+.117 (2.97)	+.026 (0.66)
6	+.126 (3.20)	-.158 (4.01)	17	+.093 (2.36)	-.075 (1.91)
7	+.045 (1.14)	-.197 (5.00)	18	+.000 (.00)	-.120 (3.05)
8	-.045 (1.14)	-.197 (5.00)	19	-.093 (2.36)	-.075 (1.91)
9	-.126 (3.20)	-.158 (4.01)	20	-.117 (2.97)	+.026 (0.66)
10	-.182 (4.62)	-.088 (2.24)	21	-.045 (1.14)	+.107 (2.72)
11	-.203 (5.16)	+.000 (0.00)	22	+.000 (.00)	-.030 (0.76)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
13	-35	22	22D	M	All	MS27527-35

FIGURE 3. Insert arrangement, electrical connector, shell size 13 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

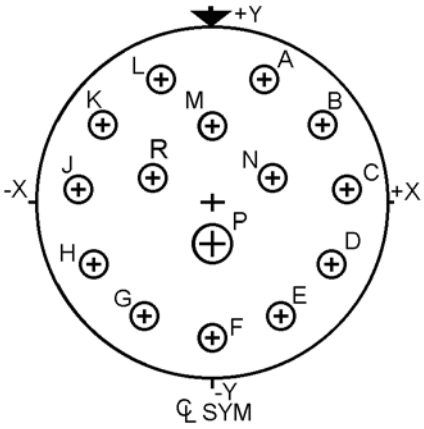


Contacts (Insert arrangement 15-5)		
Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.100 (2.54)
B	+.174 (4.42)	+.024 (0.61)
C	+.094 (2.39)	-.148 (3.76)
D	-.094 (2.39)	-.148 (3.76)
E	-.174 (4.42)	+.024 (0.61)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-5	5	16	II	All	MS20052-5

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

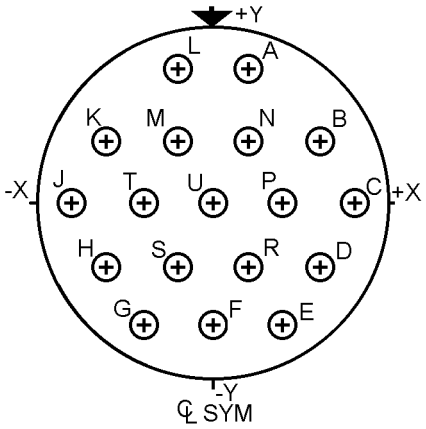


Contacts (Insert arrangement 15-15)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.225 (5.72)	J	-.244 (6.20)	+.014 (0.36)
B	+.202 (5.13)	+.140 (3.56)	K	-.202 (5.13)	+.140 (3.56)
C	+.244 (6.20)	+.014 (0.36)	L	-.100 (2.54)	+.225 (5.72)
D	+.218 (5.54)	-.113 (2.87)	M	+.000 (.00)	+.140 (3.56)
E	+.126 (3.20)	-.209 (5.31)	N	+.110 (2.79)	+.040 (1.02)
F	+.000 (.00)	-.245 (6.22)	P	+.000 (.00)	-.077 (1.96)
G	-.126 (3.20)	-.209 (5.31)	R	-.110 (2.79)	+.040 (1.02)
H	-.218 (5.54)	-.113 (2.87)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-15	1	16	I	P	MS20052-15
		14	20		All other	

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

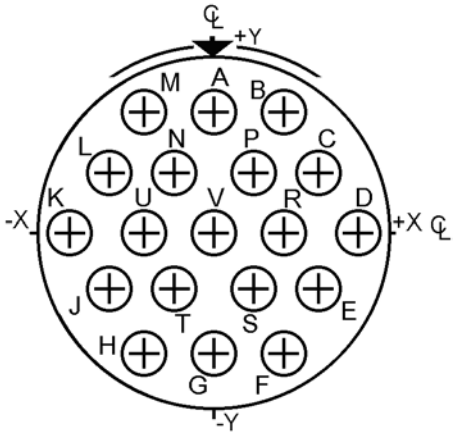


Contacts (Insert arrangement 15-18)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.252 (6.40)	K	-.195 (4.95)	+.113 (2.87)
B	+.195 (4.95)	+.113 (2.87)	L	-.065 (1.65)	+.252 (6.40)
C	+.260 (6.60)	+.000 (0.00)	M	-.065 (1.65)	+.113 (2.87)
D	+.195 (4.95)	-.113 (2.87)	N	+.065 (1.65)	+.113 (2.87)
E	+.130 (3.30)	-.225 (5.72)	P	+.130 (3.30)	+.000 (0.00)
F	+.000 (0.00)	-.225 (5.72)	R	+.065 (1.65)	-.113 (2.87)
G	-.130 (3.30)	-.225 (5.72)	S	-.065 (1.65)	-.113 (2.87)
H	-.195 (4.95)	-.113 (2.87)	T	-.130 (3.30)	+.000 (0.00)
J	-.260 (6.60)	+.000 (0.00)	U	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-18	18	20	I	All	MS20052-18

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

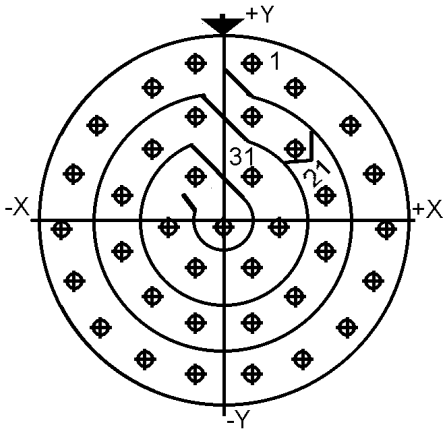


Contacts (Insert arrangement 15-19)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.225 (5.72)	L	-.195 (4.95)	+.113 (2.87)
B	+.130 (3.30)	+.225 (5.72)	M	-.130 (3.30)	+.225 (5.72)
C	+.195 (4.95)	+.113 (2.87)	N	-.065 (1.65)	+.113 (2.87)
D	+.260 (6.60)	+.000 (0.00)	P	+.065 (1.65)	+.113 (2.87)
E	+.195 (4.95)	-.113 (2.87)	R	+.130 (3.30)	+.000 (0.00)
F	+.130 (3.30)	-.225 (5.72)	S	+.065 (1.65)	-.113 (2.87)
G	+.000 (0.00)	-.225 (5.72)	T	-.065 (1.65)	-.113 (2.87)
H	-.130 (3.30)	-.225 (5.72)	U	-.130 (3.30)	+.000 (0.00)
J	-.195 (4.95)	-.113 (2.87)	V	+.000 (0.00)	+.000 (0.00)
K	-.260 (6.60)	+.000 (0.00)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-19	19	20	I	All	MS20052-19

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 15-37)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.262 (6.65)	20	+.123 (3.12)	+.119 (3.02)
2	+.123 (3.12)	+.217 (5.51)	21	+.170 (4.32)	+.040 (1.02)
3	+.211 (5.36)	+.160 (4.06)	22	+.170 (4.32)	-.050 (1.27)
4	+.254 (6.45)	+.080 (2.03)	23	+.123 (3.12)	-.127 (3.23)
5	+.266 (6.76)	-.010 (0.25)	24	+.045 (1.14)	-.172 (4.37)
6	+.247 (6.27)	-.098 (2.49)	25	-.045 (1.14)	-.172 (4.37)
7	+.200 (5.08)	-.175 (4.45)	26	-.123 (3.12)	-.127 (3.23)
8	+.130 (3.30)	-.232 (5.89)	27	-.170 (4.32)	-.050 (1.27)
9	+.045 (1.14)	-.262 (6.65)	28	-.170 (4.32)	+.040 (1.02)
10	-.045 (1.14)	-.262 (6.65)	29	-.123 (3.12)	+.119 (3.02)
11	-.130 (3.30)	-.232 (5.89)	30	-.045 (1.14)	+.172 (4.37)
12	-.200 (5.08)	-.175 (4.45)	31	+.045 (1.14)	+.074 (1.88)
13	-.247 (6.27)	-.098 (2.49)	32	+.090 (2.29)	-.004 (0.10)
14	-.266 (6.76)	-.010 (0.25)	33	+.045 (1.14)	-.082 (2.08)
15	-.254 (6.45)	+.080 (2.03)	34	-.045 (1.14)	-.082 (2.08)

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

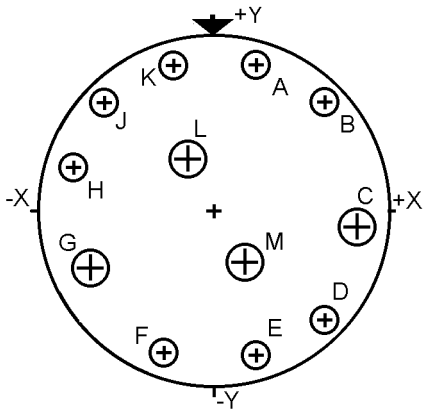
Contacts (Insert arrangement 15-37)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
16	-.211 (5.36)	+.160 (4.06)	35	-.090 (2.29)	-.004 (0.10)
17	-.123 (3.12)	+.217 (5.51)	36	-.045 (1.14)	+.074 (1.88)
18	-.045 (1.14)	+.262 (6.65)	37	+.000 (0.00)	-.004 (0.10)
19	+.045 (1.14)	+.172 (4.37)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-37	37	22M	M	All	MS20052-37

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

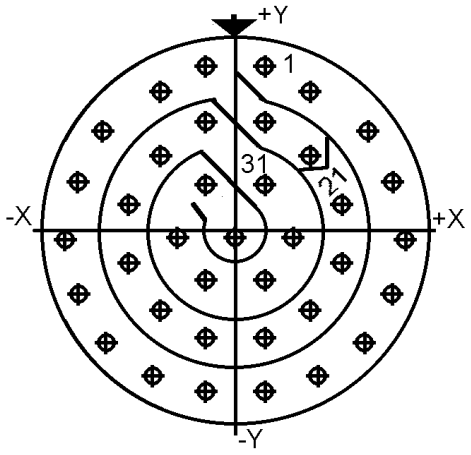


Contacts (Insert arrangement 15-97)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.234 (5.94)	G	-.207 (5.26)	-.095 (2.41)
B	+.178 (4.52)	+.178 (4.52)	H	-.234 (5.94)	+.065 (1.65)
C	+.230 (5.84)	-.023 (0.58)	J	-.178 (4.52)	+.178 (4.52)
D	+.178 (4.52)	-.178 (4.52)	K	-.065 (1.65)	+.234 (5.94)
E	+.065 (1.65)	-.234 (5.94)	L	-.047 (1.19)	+.081 (2.06)
F	-.089 (2.26)	-.235 (5.97)	M	+.047 (1.19)	-.081 (2.06)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-97	4	16	I	C, G, L, M	MS20052-97
		8	20		All others	

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 15-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.262 (6.65)	20	+.123 (3.12)	+.119 (3.02)
2	+.123 (3.12)	+.217 (5.51)	21	+.170 (4.32)	+.040 (1.02)
3	+.211 (5.36)	+.160 (4.06)	22	+.170 (4.32)	-.050 (1.27)
4	+.254 (6.45)	+.080 (2.03)	23	+.123 (3.12)	-.127 (3.23)
5	+.266 (6.76)	-.010 (.25)	24	+.045 (1.14)	-.172 (4.37)
6	+.247 (6.27)	-.098 (2.49)	25	-.045 (1.14)	-.172 (4.37)
7	+.200 (5.08)	-.175 (4.45)	26	-.123 (3.12)	-.127 (3.23)
8	+.130 (3.30)	-.232 (5.89)	27	-.170 (4.32)	-.050 (1.27)
9	+.045 (1.14)	-.262 (6.65)	28	-.170 (4.32)	+.040 (1.02)
10	-.045 (1.14)	-.262 (6.65)	29	-.123 (3.12)	+.119 (3.02)
11	-.130 (3.30)	-.232 (5.89)	30	-.045 (1.14)	+.172 (4.37)
12	-.200 (5.08)	-.175 (4.45)	31	+.045 (1.14)	+.074 (1.88)
13	-.247 (6.27)	-.098 (2.49)	32	+.090 (2.29)	-.004 (0.10)
14	-.266 (6.76)	-.010 (.25)	33	+.045 (1.14)	-.082 (2.08)
15	-.254 (6.45)	+.080 (2.03)	34	-.045 (1.14)	-.082 (2.08)

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

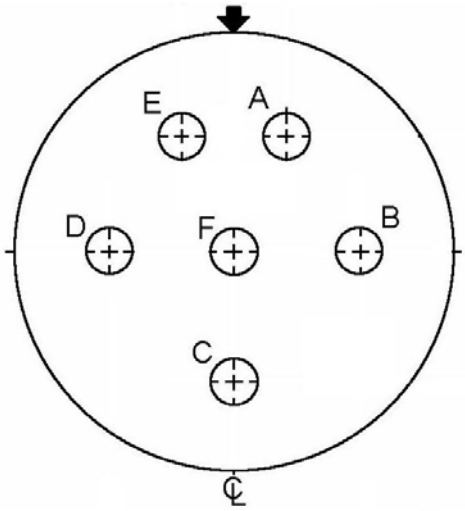
Contacts (Insert arrangement 15-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
16	-.211 (5.36)	+.160 (4.06)	35	-.090 (2.29)	-.004 (0.10)
17	-.123 (3.12)	+.217 (5.51)	36	-.045 (1.14)	+.074 (1.88)
18	-.045 (1.14)	+.262 (6.65)	37	+.000 (0.00)	-.004 (0.10)
19	+.045 (1.14)	+.172 (4.37)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
15	-35	37	22D	M	All	MS27528-35

FIGURE 4. Insert arrangement, electrical connector, shell size 15 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

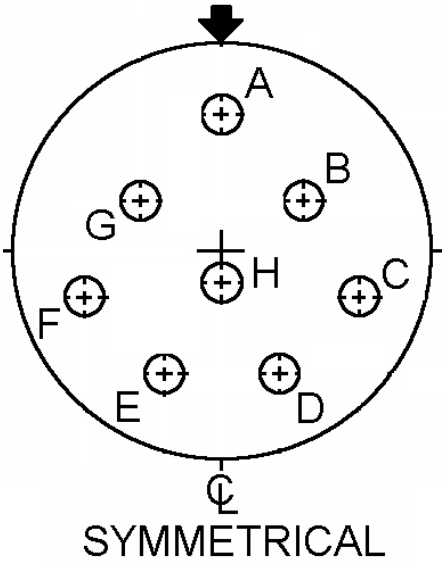


Contacts (Insert arrangement 17-6)		
Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)
A	+.121 (3.07)	+.209 (5.31)
B	+.241 (6.12)	+.000 (0.00)
C	+.000 (0.00)	-.241 (6.12)
D	-.241 (6.12)	+.000 (0.00)
E	-.121 (3.07)	+.209 (5.31)
F	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
17	-6	6	12	I	All	MS20053-6

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

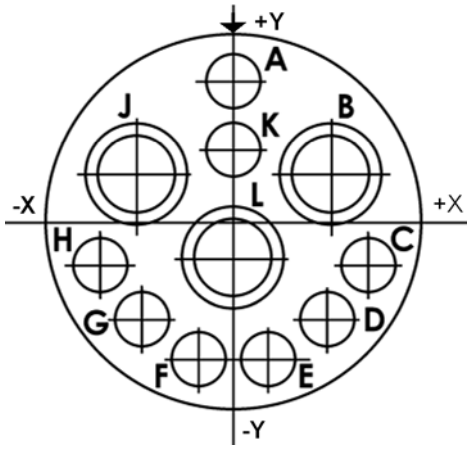


Contacts (Insert arrangement 17-8)		
Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.236 (5.99)
B	+.128 (3.25)	+.086 (2.18)
C	+.230 (5.84)	-.078 (1.98)
D	+.094 (2.39)	-.216 (5.49)
E	-.094 (2.39)	-.216 (5.49)
F	-.230 (5.84)	-.078 (1.98)
G	-.128 (3.25)	+.086 (2.18)
H	+.000 (0.00)	-.052 (1.32)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
17	-8	8	16	II	All	MS20053-8

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



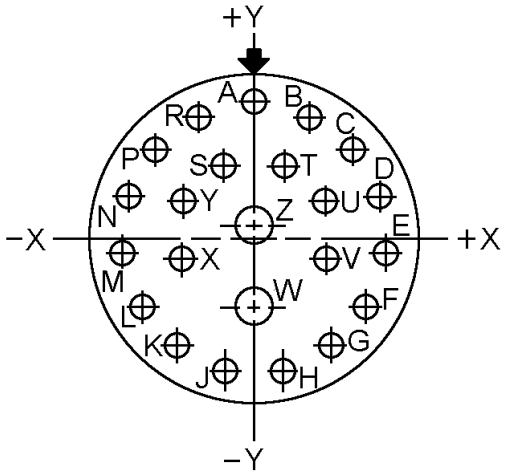
Contacts (Insert arrangement 17-11)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.290 (7.37)	G	-.190 (4.83)	-.199 (5.05)
B	+.200 (5.08)	+.100 (2.54)	H	-.275 (6.99)	-.088 (2.24)
C	+.275 (6.99)	-.088 (2.24)	J	-.200 (5.08)	+.100 (2.54)
D	+.190 (4.83)	-.199 (5.05)	K	+.000 (0.00)	+.150 (3.81)
E	+.071 (1.80)	-.282 (7.16)	L	+.000 (0.00)	-.075 (1.91)
F	-.071 (1.80)	-.282 (7.16)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact position ID	Standard contacts	
						Pin	Socket
17	-11	1	12	Coax	J	M39029/28-211	M39029/75-416
		2	12	Twinax	B, L	02004-936	02003-926
		8	20	N	A, C, D,E, F,G, H, K	M39029/58-363	M39029/56-351

(See DLA Land and Maritime drawings 02004 and 02003 for the above referenced twinax part numbers 02004-936 and 02003-926)

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

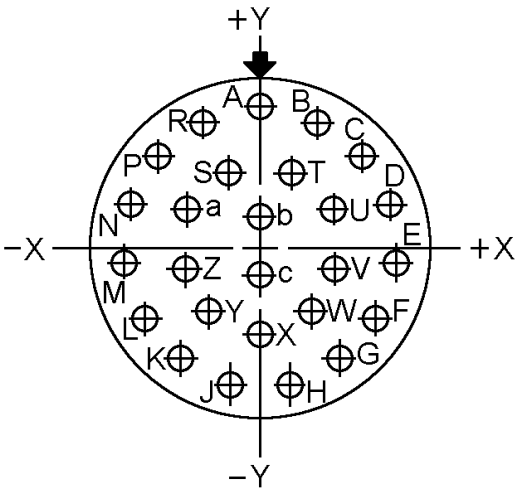


Contacts (Insert arrangement 17-99)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	.000 (0.00)	+.321 (8.15)	N	-.305 (7.75)	+.099 (2.51)
B	+.131 (3.33)	+.293 (7.44)	P	-.239 (6.07)	+.214 (5.44)
C	+.239 (6.07)	+.214 (5.44)	R	-.131 (3.33)	+.293 (7.44)
D	+.305 (7.75)	+.099 (2.51)	S	-.070 (1.78)	+.177 (4.50)
E	+.319 (8.10)	-.034 (0.86)	T	+.070 (1.78)	+.177 (4.50)
F	+.278 (7.06)	-.161 (4.09)	U	+.175 (4.45)	+.094 (2.39)
G	+.189 (4.80)	-.260 (6.60)	V	+.150 (3.81)	-.075 (1.91)
H	+.067 (1.70)	-.314 (7.98)	W	+.000 (0.00)	-.161 (4.09)
J	-.067 (1.70)	-.314 (7.98)	X	-.150 (3.81)	-.075 (1.91)
K	-.189 (4.80)	-.260 (6.60)	Y	-.175 (4.45)	+.094 (2.39)
L	-.278 (7.06)	-.161 (4.09)	Z	+.000 (0.00)	+.025 (0.64)
M	-.319 (8.10)	-.034 (0.86)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact position ID	Supersedes
17	-99	2	16	I	W, Z	MS20053-99
		21	20		All others	

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

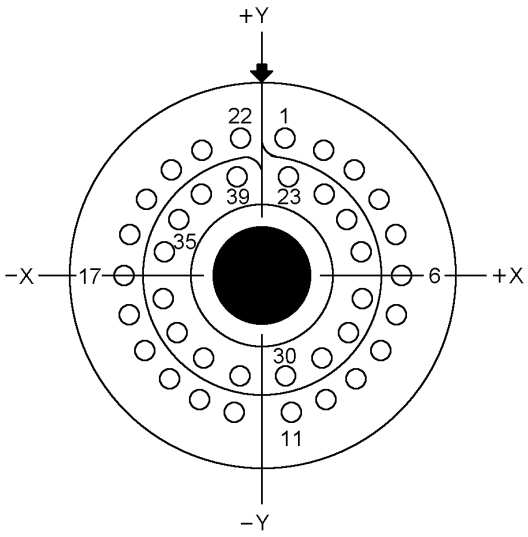


Contacts (Insert arrangement 17-26)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.321 (8.15)	P	-.239 (6.07)	+.214 (5.44)
B	+.131 (3.33)	+.293 (7.44)	R	-.131 (3.33)	+.293 (7.44)
C	+.239 (6.07)	+.214 (5.44)	S	-.070 (1.78)	+.177 (4.50)
D	+.305 (7.75)	+.099 (2.51)	T	+.070 (1.78)	+.177 (4.50)
E	+.319 (8.10)	-.034 (0.86)	U	+.175 (4.45)	+.094 (2.39)
F	+.278 (7.06)	-.161 (4.09)	V	+.178 (4.52)	-.036 (0.91)
G	+.189 (4.80)	-.260 (6.60)	W	+.119 (3.02)	-.151 (3.84)
H	+.067 (1.70)	-.314 (7.98)	X	+.000 (0.00)	-.203 (5.16)
J	-.067 (1.70)	-.314 (7.98)	Y	-.119 (3.02)	-.151 (3.84)
K	-.189 (4.80)	-.260 (6.60)	Z	-.178 (4.52)	-.036 (0.91)
L	-.278 (7.06)	-.161 (4.09)	a	-.175 (4.45)	+.094 (2.39)
M	-.319 (8.10)	-.034 (0.86)	b	+.000 (0.00)	+.065 (1.65)
N	-.305 (7.75)	+.099 (2.51)	c	+.000 (0.00)	-.065 (1.65)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact position ID	Supersedes
17	-26	26	20	I	All	MS20053-26

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 17-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.046 (1.17)	+.325 (8.26)	21	-.133 (3.38)	+.299 (7.59)
2	+.133 (3.38)	+.299 (7.59)	22	-.046 (1.17)	+.325 (8.26)
3	+.211 (5.36)	+.251 (6.38)	23	+.046 (1.17)	+.231 (5.87)
4	+.272 (6.91)	+.183 (4.65)	24	+.129 (3.28)	+.197 (5.00)
5	+.312 (7.92)	+.101 (2.57)	25	+.194 (4.93)	+.133 (3.38)
6	+.328 (8.33)	+.011 (0.28)	26	+.230 (5.84)	+.050 (1.27)
7	+.318 (8.08)	-.079 (2.01)	27	+.232 (5.89)	-.040 (1.02)
8	+.284 (7.21)	-.164 (4.17)	28	+.200 (5.08)	-.125 (3.18)
9	+.228 (5.79)	-.236 (5.99)	29	+.137 (3.48)	-.191 (4.85)
10	+.154 (3.91)	-.289 (7.34)	30	+.055 (1.40)	-.229 (5.82)
11	+.068 (1.73)	-.321 (8.15)	31	-.055 (1.40)	-.229 (5.82)
12	-.068 (1.73)	-.321 (8.15)	32	-.137 (3.48)	-.191 (4.85)
13	-.154 (3.91)	-.289 (7.34)	33	-.200 (5.08)	-.125 (3.18)
14	-.228 (5.79)	-.236 (5.99)	34	-.232 (5.89)	-.040 (1.02)
15	-.284 (7.21)	-.164 (4.17)	35	-.230 (5.84)	+.050 (1.27)
16	-.318 (8.08)	-.079 (2.01)	36	-.194 (4.93)	+.133 (3.38)

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

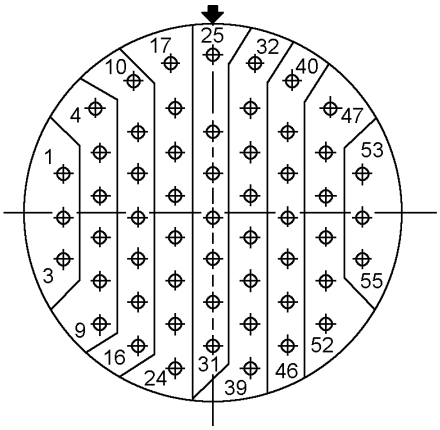
Contacts (Insert arrangement 17-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.328 (8.33)	+.011 (0.28)	37	-.129 (3.28)	+.197 (5.00)
18	-.312 (7.92)	+.101 (2.57)	38	-.046 (1.17)	+.231 (5.87)
19	-.272 (6.91)	+.183 (4.65)	39	+.000 (0.00)	-.000 (0.00)
20	-.211 (5.36)	+.251 (6.38)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact position ID	Standard contact	
						Pin	Socket
17	-2	38	22D	M	All other	M39029/58-360	M39029/56-348
		1	8 (See note)	Twinax	39	M39029/90-529	M39029/91-530

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 ºC (347 ºF) maximum.

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 17-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.312 (7.92)	+.086 (2.18)	29	.000 (0.00)	-.094 (2.39)
2	-.312 (7.92)	-.004 (0.10)	30	.000 (0.00)	-.184 (4.67)
3	-.312 (7.92)	-.094 (2.39)	31	.000 (0.00)	-.274 (6.96)
4	-.242 (6.15)	+.221 (5.61)	32	+.089 (2.26)	+.316 (8.03)
5	-.234 (5.94)	+.131 (3.33)	33	+.078 (1.98)	+.221 (5.61)
6	-.234 (5.94)	+.041 (1.04)	34	+.078 (1.98)	+.131 (3.33)
7	-.234 (5.94)	-.049 (1.24)	35	+.078 (1.98)	+.041 (1.04)
8	-.234 (5.94)	-.139 (3.53)	36	+.078 (1.98)	-.049 (1.24)
9	-.234 (5.94)	-.229 (5.82)	37	+.078 (1.98)	-.139 (3.53)
10	-.172 (4.37)	+.279 (7.09)	38	+.078 (1.98)	-.229 (5.82)
11	-.156 (3.96)	+.176 (4.47)	39	+.078 (1.98)	-.319 (8.10)
12	-.156 (3.96)	+.086 (2.18)	40	+.172 (4.37)	+.279 (7.09)
13	-.156 (3.96)	-.004 (0.10)	41	+.156 (3.96)	+.176 (4.47)
14	-.156 (3.96)	-.094 (2.39)	42	+.156 (3.96)	+.086 (2.18)
15	-.156 (3.96)	-.184 (4.67)	43	+.156 (3.96)	-.004 (0.10)
16	-.156 (3.96)	-.274 (6.96)	44	+.156 (3.96)	-.094 (2.39)
17	-.089 (2.26)	+.316 (8.03)	45	+.156 (3.96)	-.184 (4.67)
18	-.078 (1.98)	+.221 (5.61)	46	+.156 (3.96)	-.274 (6.96)

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

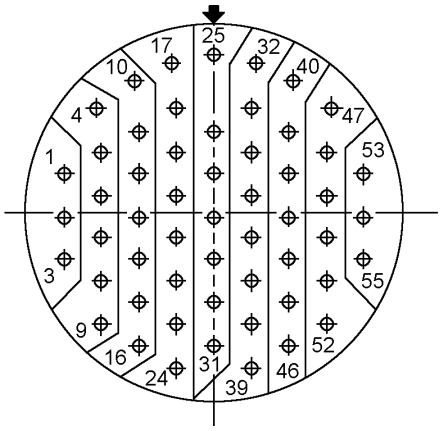
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 17-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
19	-.078 (1.98)	+.131 (3.33)	47	+.242 (6.15)	+.221 (5.61)
20	-.078 (1.98)	+.041 (1.04)	48	+.234 (5.94)	+.131 (3.33)
21	-.078 (1.98)	-.049 (1.24)	49	+.234 (5.94)	+.041 (1.04)
22	-.078 (1.98)	-.139 (3.53)	50	+.234 (5.94)	-.049 (1.24)
23	-.078 (1.98)	-.229 (5.82)	51	+.234 (5.94)	-.139 (3.53)
24	-.078 (1.98)	-.319 (8.10)	52	+.234 (5.94)	-.229 (5.82)
25	.000 (0.00)	+.329 (8.36)	53	+.312 (7.92)	+.086 (2.18)
26	.000 (0.00)	+.176 (4.47)	54	+.312 (7.92)	-.004 (0.10)
27	.000 (0.00)	+.086 (2.18)	55	+.312 (7.92)	-.094 (2.39)
28	.000 (0.00)	-.004 (0.10)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
17	-35	55	22D	M	All	MS27529-35

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 17-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.312 (7.92)	+.086 (2.18)	29	+.000 (0.00)	-.094 (2.39)
2	-.312 (7.92)	-.004 (0.10)	30	+.000 (0.00)	-.184 (4.67)
3	-.312 (7.92)	-.094 (2.39)	31	+.000 (0.00)	-.274 (6.96)
4	-.242 (6.15)	+.221 (5.61)	32	+.089 (2.26)	+.316 (8.03)
5	-.234 (5.94)	+.131 (3.33)	33	+.078 (1.98)	+.221 (5.61)
6	-.234 (5.94)	+.041 (1.04)	34	+.078 (1.98)	+.131 (3.33)
7	-.234 (5.94)	-.049 (1.24)	35	+.078 (1.98)	+.041 (1.04)
8	-.234 (5.94)	-.139 (3.53)	36	+.078 (1.98)	-.049 (1.24)
9	-.234 (5.94)	-.229 (5.82)	37	+.078 (1.98)	-.139 (3.53)
10	-.172 (4.37)	+.279 (7.09)	38	+.078 (1.98)	-.229 (5.82)
11	-.156 (3.96)	+.176 (4.47)	39	+.078 (1.98)	-.319 (8.10)
12	-.156 (3.96)	+.086 (2.18)	40	+.172 (4.37)	+.279 (7.09)
13	-.156 (3.96)	-.004 (0.10)	41	+.156 (3.96)	+.176 (4.47)
14	-.156 (3.96)	-.094 (2.39)	42	+.156 (3.96)	+.086 (2.18)
15	-.156 (3.96)	-.184 (4.67)	43	+.156 (3.96)	-.004 (0.10)
16	-.156 (3.96)	-.274 (6.96)	44	+.156 (3.96)	-.094 (2.39)
17	-.089 (2.26)	+.316 (8.03)	45	+.156 (3.96)	-.184 (4.67)
18	-.078 (1.98)	+.221 (5.61)	46	+.156 (3.96)	-.274 (6.96)

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

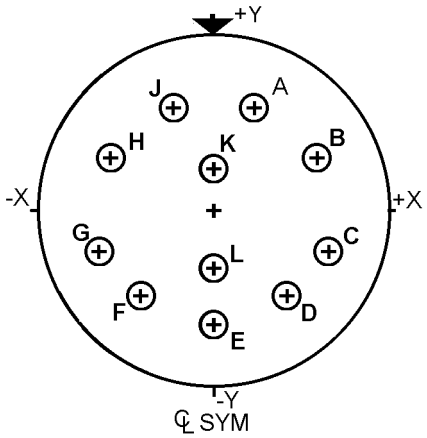
Contacts (Insert arrangement 17-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
19	-.078 (1.98)	+.131 (3.33)	47	+.242 (6.15)	+.221 (5.61)
20	-.078 (1.98)	+.041 (1.04)	48	+.234 (5.94)	+.131 (3.33)
21	-.078 (1.98)	-.049 (1.24)	49	+.234 (5.94)	+.041 (1.04)
22	-.078 (1.98)	-.139 (3.53)	50	+.234 (5.94)	-.049 (1.24)
23	-.078 (1.98)	-.229 (5.82)	51	+.234 (5.94)	-.139 (3.53)
24	-.078 (1.98)	-.319 (8.10)	52	+.234 (5.94)	-.229 (5.82)
25	+.000 (0.00)	+.329 (8.36)	53	+.312 (7.92)	+.086 (2.18)
26	+.000 (0.00)	+.176 (4.47)	54	+.312 (7.92)	-.004 (0.10)
27	+.000 (0.00)	+.086 (2.18)	55	+.312 (7.92)	-.094 (2.39)
28	+.000 (0.00)	-.004 (0.10)			

NOTE: Inactive for new design.

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
17	-55	55	22M	M	All	MS20053-55

FIGURE 5. Insert arrangement, electrical connector, shell size 17 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

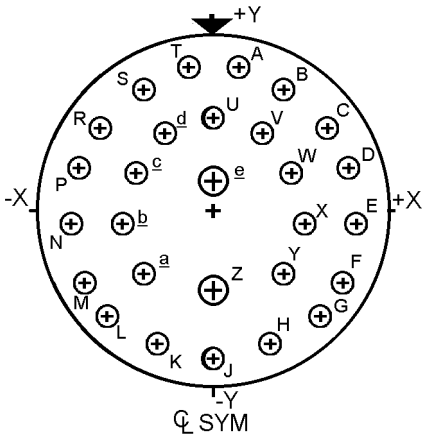


Contacts (Insert arrangement 19-11)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.105 (2.67)	+.260 (6.60)	G	-.275 (6.99)	-.053 (1.35)
B	+.250 (6.35)	+.132 (3.35)	H	-.250 (6.35)	+.132 (3.35)
C	+.275 (6.99)	-.053 (1.35)	J	-.105 (2.67)	+.260 (6.60)
D	+.179 (4.55)	-.215 (5.46)	K	+.000 (0.00)	+.105 (2.67)
E	+.000 (0.00)	-.281 (7.14)	L	+.000 (0.00)	-.092 (2.34)
F	-.179 (4.55)	-.215 (5.46)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-11	11	16	II	All	MS20054-11

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

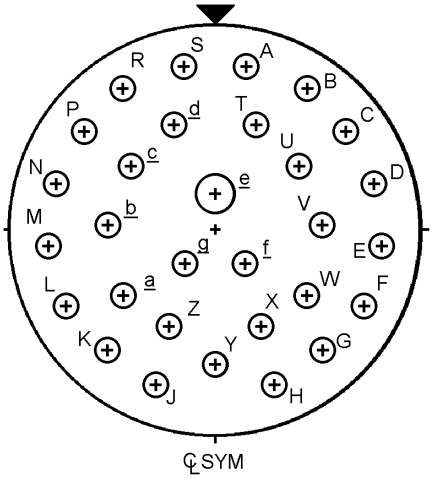


Contacts (Insert arrangement 19-28)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.066 (1.68)	+.353 (8.97)	R	-.286 (7.26)	+.217 (5.51)
B	+.189 (4.80)	+.305 (7.75)	S	-.189 (4.80)	+.305 (7.75)
C	+.286 (7.26)	+.217 (5.51)	T	-.066 (1.68)	+.353 (8.97)
D	+.345 (8.76)	+.098 (2.49)	U	+.000 (0.00)	+.230 (5.84)
E	+.357 (9.07)	-.033 (0.84)	V	+.124 (3.15)	+.193 (4.90)
F	+.321 (8.15)	-.160 (4.06)	W	+.209 (5.31)	+.095 (2.41)
G	+.242 (6.15)	-.265 (6.73)	X	+.228 (5.79)	-.033 (0.84)
H	+.130 (3.30)	-.335 (8.51)	Y	+.174 (4.42)	-.151 (3.84)
J	+.000 (0.00)	-.359 (9.12)	Z	+.000 (0.00)	-.191 (4.85)
K	-.130 (3.30)	-.335 (8.51)	a	-.174 (4.42)	-.151 (3.84)
L	-.242 (6.15)	-.265 (6.73)	b	-.228 (5.79)	-.033 (0.84)
M	-.321 (8.15)	-.160 (4.06)	c	-.209 (5.31)	+.095 (2.41)
N	-.357 (9.07)	-.033 (0.84)	d	-.124 (3.15)	+.193 (4.90)
P	-.345 (8.76)	+.098 (2.49)	e	+.000 (0.00)	+.062 (1.57)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-28	2	16	I	e, Z	MS20054-28
		26	20		All others	

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

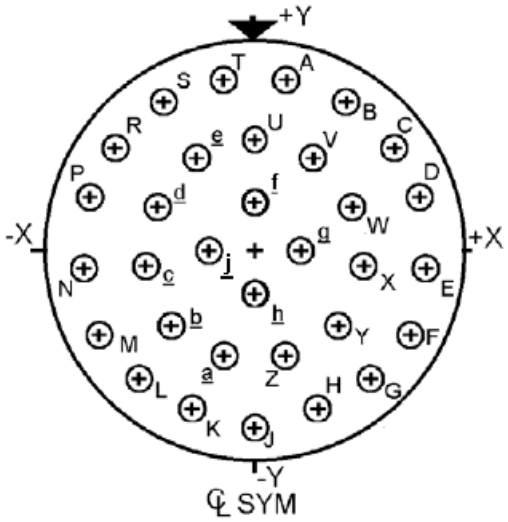


Contacts (Insert arrangement 19-30)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.346 (8.79)	S	-.065 (1.65)	+.346 (8.79)
B	+.186 (4.72)	+.299 (7.59)	T	+.084 (2.13)	+.217 (5.51)
C	+.282 (7.16)	+.210 (5.33)	U	+.181 (4.60)	+.129 (3.28)
D	+.340 (8.64)	+.093 (2.36)	V	+.228 (5.79)	+.008 (0.20)
E	+.351 (8.92)	-.033 (0.84)	W	+.193 (4.90)	-.117 (2.97)
F	+.315 (8.00)	-.158 (4.01)	X	+.096 (2.44)	-.203 (5.16)
G	+.236 (5.99)	-.261 (6.63)	Y	+.000 (0.00)	-.290 (7.37)
H	+.124 (3.15)	-.330 (8.38)	Z	-.096 (2.44)	-.203 (5.16)
J	-.124 (3.15)	-.330 (8.38)	a	-.193 (4.90)	-.117 (2.97)
K	-.236 (5.99)	-.261 (6.63)	b	-.228 (5.79)	+.008 (0.20)
L	-.315 (8.00)	-.158 (4.01)	c	-.181 (4.60)	+.129 (3.28)
M	-.351 (8.92)	-.033 (0.84)	d	-.084 (2.13)	+.217 (5.51)
N	-.340 (8.64)	+.093 (2.36)	e	+.000 (0.00)	+.072 (1.83)
P	-.282 (7.16)	+.210 (5.33)	f	+.069 (1.75)	-.076 (1.93)
R	-.186 (4.72)	+.299 (7.59)	g	-.069 (1.75)	-.076 (1.93)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-30	1	16	I	e	MS20054-30
		29	20		All others	

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

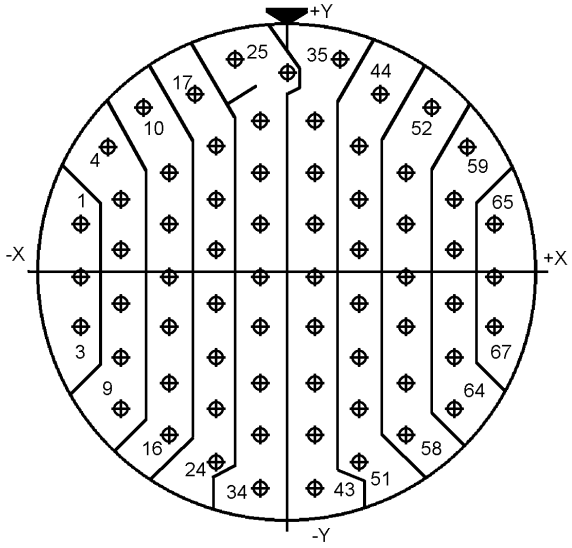


Contacts (Insert arrangement 19-32)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.066 (1.68)	+.353 (8.97)	T	-.066 (1.68)	+.353 (8.97)
B	+.189 (4.80)	+.305 (7.75)	U	+.000 (0.00)	+.230 (5.84)
C	+.286 (7.26)	+.217 (5.51)	V	+.124 (3.15)	+.193 (4.90)
D	+.345 (8.76)	+.098 (2.49)	W	+.209 (5.31)	+.095 (2.41)
E	+.357 (9.07)	-.033 (0.84)	X	+.228 (5.79)	-.033 (0.84)
F	+.321 (8.15)	-.160 (4.06)	Y	+.174 (4.42)	-.151 (3.84)
G	+.242 (6.15)	-.265 (6.73)	Z	+.065 (1.65)	-.221 (5.61)
H	+.130 (3.30)	-.335 (8.51)	a	-.065 (1.65)	-.221 (5.61)
J	-.000 (0.00)	-.359 (9.12)	b	-.174 (4.42)	-.151 (3.84)
K	-.130 (3.30)	-.335 (8.51)	c	-.228 (5.79)	-.033 (0.84)
L	-.242 (6.15)	-.265 (6.73)	d	-.209 (5.31)	+.095 (2.41)
M	-.321 (8.15)	-.160 (4.06)	e	-.124 (3.15)	+.193 (4.90)
N	-.357 (9.07)	-.033 (0.84)	f	+.000 (0.00)	+.096 (2.44)
P	-.345 (8.76)	+.098 (2.49)	g	+.096 (2.44)	+.000 (0.00)
R	-.286 (7.26)	+.217 (5.51)	h	+.000 (0.00)	-.096 (2.44)
S	-.189 (4.80)	+.305 (7.75)	j	-.096 (2.44)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-32	32	20	I	All	MS20054-32

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 19-67)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.081 (2.06)	35	+.088 (2.24)	+.360 (9.14)
2	-.357 (9.07)	-.009 (0.23)	36	+.045 (1.14)	+.261 (6.63)
3	-.357 (9.07)	-.099 (2.51)	37	+.045 (1.14)	+.171 (4.34)
4	-.304 (7.72)	+.213 (5.41)	38	+.045 (1.14)	+.081 (2.06)
5	-.279 (7.09)	+.126 (3.20)	39	+.045 (1.14)	-.009 (0.23)
6	-.279 (7.09)	+.036 (0.91)	40	+.045 (1.14)	-.099 (2.51)
7	-.279 (7.09)	-.054 (1.37)	41	+.045 (1.14)	-.189 (4.80)
8	-.279 (7.09)	-.144 (3.66)	42	+.045 (1.14)	-.279 (7.09)
9	-.279 (7.09)	-.234 (5.94)	43	+.045 (1.14)	-.369 (9.37)
10	-.244 (6.20)	+.280 (7.11)	44	+.156 (3.96)	+.301 (7.65)
11	-.201 (5.11)	+.171 (4.34)	45	+.123 (3.12)	+.216 (5.49)
12	-.201 (5.11)	+.081 (2.06)	46	+.123 (3.12)	+.126 (3.20)
13	-.201 (5.11)	-.009 (0.23)	47	+.123 (3.12)	+.036 (0.91)
14	-.201 (5.11)	-.099 (2.51)	48	+.123 (3.12)	-.054 (1.37)
15	-.201 (5.11)	-.189 (4.80)	49	+.123 (3.12)	-.144 (3.66)
16	-.201 (5.11)	-.279 (7.09)	50	+.123 (3.12)	-.234 (5.94)
17	-.156 (3.96)	+.301 (7.65)	51	+.123 (3.12)	-.324 (8.23)

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

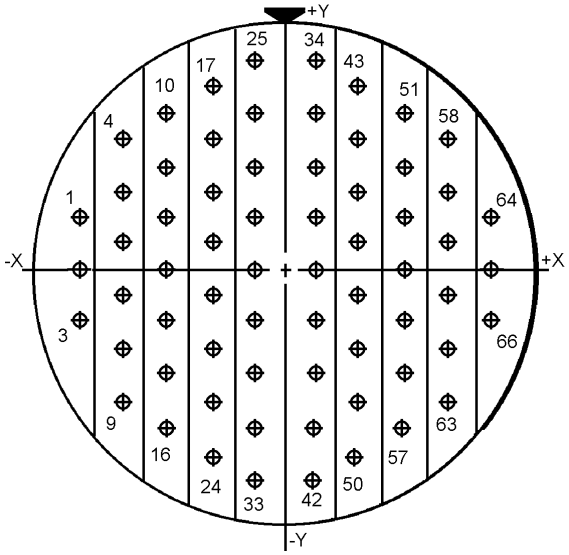
Contacts (Insert arrangement 19-67)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.123 (3.12)	+.216 (5.49)	52	+.244 (6.20)	+.280 (7.11)
19	-.123 (3.12)	+.126 (3.20)	53	+.201 (5.11)	+.171 (4.34)
20	-.123 (3.12)	+.036 (0.91)	54	+.201 (5.11)	+.081 (2.06)
21	-.123 (3.12)	-.054 (1.37)	55	+.201 (5.11)	-.009 (.23)
22	-.123 (3.12)	-.144 (3.66)	56	+.201 (5.11)	-.099 (2.51)
23	-.123 (3.12)	-.234 (5.94)	57	+.201 (5.11)	-.189 (4.80)
24	-.123 (3.12)	-.324 (8.23)	58	+.201 (5.11)	-.279 (7.09)
25	-.088 (2.24)	+.360 (9.14)	59	+.304 (7.72)	+.213 (5.41)
26	-.000 (0.00)	+.339 (8.61)	60	+.279 (7.09)	+.126 (3.20)
27	-.045 (1.14)	+.261 (6.63)	61	+.279 (7.09)	+.036 (0.91)
28	-.045 (1.14)	+.171 (4.34)	62	+.279 (7.09)	-.054 (1.37)
29	-.045 (1.14)	+.081 (2.06)	63	+.279 (7.09)	-.144 (3.66)
30	-.045 (1.14)	-.009 (0.23)	64	+.279 (7.09)	-.234 (5.94)
31	-.045 (1.14)	-.099 (2.51)	65	+.357 (9.07)	+.081 (2.06)
32	-.045 (1.14)	-.189 (4.80)	66	+.357 (9.07)	-.009 (0.23)
33	-.045 (1.14)	-.279 (7.09)	67	+.357 (9.07)	-.099 (2.51)
34	-.045 (1.14)	-.369 (9.37)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-67	67	22M	M	All	MS20054-67

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 19-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.090 (2.29)	34	+.045 (1.14)	+.360 (9.14)
2	-.357 (9.07)	+.000 (0.00)	35	+.045 (1.14)	+.270 (6.86)
3	-.357 (9.07)	-.090 (2.29)	36	+.045 (1.14)	+.180 (4.57)
4	-.279 (7.09)	+.225 (5.72)	37	+.045 (1.14)	+.090 (2.29)
5	-.279 (7.09)	+.135 (3.43)	38	+.045 (1.14)	+.000 (0.00)
6	-.279 (7.09)	+.045 (1.14)	39	+.045 (1.14)	-.090 (2.29)
7	-.279 (7.09)	-.045 (1.14)	40	+.045 (1.14)	-.180 (4.57)
8	-.279 (7.09)	-.135 (3.43)	41	+.045 (1.14)	-.270 (6.86)
9	-.279 (7.09)	-.225 (5.72)	42	+.045 (1.14)	-.360 (9.14)
10	-.201 (5.11)	+.270 (6.86)	43	+.123 (3.12)	+.315 (8.00)
11	-.201 (5.11)	+.180 (4.57)	44	+.123 (3.12)	+.225 (5.72)
12	-.201 (5.11)	+.090 (2.29)	45	+.123 (3.12)	+.135 (3.43)
13	-.201 (5.11)	+.000 (0.00)	46	+.123 (3.12)	+.045 (1.14)
14	-.201 (5.11)	-.090 (2.29)	47	+.123 (3.12)	-.045 (1.14)
15	-.201 (5.11)	-.180 (4.57)	48	+.123 (3.12)	-.135 (3.43)
16	-.201 (5.11)	-.270 (6.86)	49	+.123 (3.12)	-.225 (5.72)
17	-.123 (3.12)	+.315 (8.00)	50	+.123 (3.12)	-.315 (8.00)

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

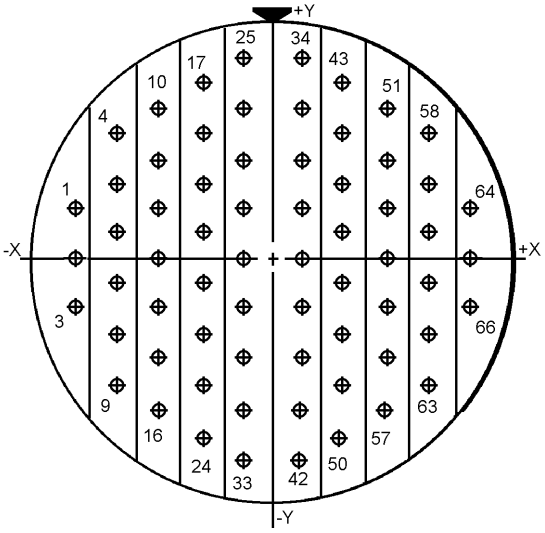
Contacts (Insert arrangement 19-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.123 (3.12)	+.225 (5.72)	51	+.201 (5.11)	+.270 (6.86)
19	-.123 (3.12)	+.135 (3.43)	52	+.201 (5.11)	+.180 (4.57)
20	-.123 (3.12)	+.045 (1.14)	53	+.201 (5.11)	+.090 (2.29)
21	-.123 (3.12)	-.045 (1.14)	54	+.201 (5.11)	+.000 (0.00)
22	-.123 (3.12)	-.135 (3.43)	55	+.201 (5.11)	-.090 (2.29)
23	-.123 (3.12)	-.225 (5.72)	56	+.201 (5.11)	-.180 (4.57)
24	-.123 (3.12)	-.315 (8.00)	57	+.201 (5.11)	-.270 (6.86)
25	-.045 (1.14)	+.360 (9.14)	58	+.279 (7.09)	+.225 (5.72)
26	-.045 (1.14)	+.270 (6.86)	59	+.279 (7.09)	+.135 (3.43)
27	-.045 (1.14)	+.180 (4.57)	60	+.279 (7.09)	+.045 (1.14)
28	-.045 (1.14)	+.090 (2.29)	61	+.279 (7.09)	-.045 (1.14)
29	-.045 (1.14)	+.000 (0.00)	62	+.279 (7.09)	-.135 (3.43)
30	-.045 (1.14)	-.090 (2.29)	63	+.279 (7.09)	-.225 (5.72)
31	-.045 (1.14)	-.180 (4.57)	64	+.357 (9.07)	+.090 (2.29)
32	-.045 (1.14)	-.270 (6.86)	65	+.357 (9.07)	+.000 (0.00)
33	-.045 (1.14)	-.360 (9.14)	66	+.357 (9.07)	-.090 (2.29)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-35	66	22D	M	All	MS27530-35

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 19-66)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.090 (2.29)	34	+.045 (1.14)	+.360 (9.14)
2	-.357 (9.07)	+.000 (0.00)	35	+.045 (1.14)	+.270 (6.86)
3	-.357 (9.07)	-.090 (2.29)	36	+.045 (1.14)	+.180 (4.57)
4	-.279 (7.09)	+.225 (5.72)	37	+.045 (1.14)	+.090 (2.29)
5	-.279 (7.09)	+.135 (3.43)	38	+.045 (1.14)	+.000 (0.00)
6	-.279 (7.09)	+.045 (1.14)	39	+.045 (1.14)	-.090 (2.29)
7	-.279 (7.09)	-.045 (1.14)	40	+.045 (1.14)	-.180 (4.57)
8	-.279 (7.09)	-.135 (3.43)	41	+.045 (1.14)	-.270 (6.86)
9	-.279 (7.09)	-.225 (5.72)	42	+.045 (1.14)	-.360 (9.14)
10	-.201 (5.11)	+.270 (6.86)	43	+.123 (3.12)	+.315 (8.00)
11	-.201 (5.11)	+.180 (4.57)	44	+.123 (3.12)	+.225 (5.72)
12	-.201 (5.11)	+.090 (2.29)	45	+.123 (3.12)	+.135 (3.43)
13	-.201 (5.11)	+.000 (0.00)	46	+.123 (3.12)	+.045 (1.14)
14	-.201 (5.11)	-.090 (2.29)	47	+.123 (3.12)	-.045 (1.14)
15	-.201 (5.11)	-.180 (4.57)	48	+.123 (3.12)	-.135 (3.43)
16	-.201 (5.11)	-.270 (6.86)	49	+.123 (3.12)	-.225 (5.72)
17	-.123 (3.12)	+.315 (8.00)	50	+.123 (3.12)	-.315 (8.00)

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

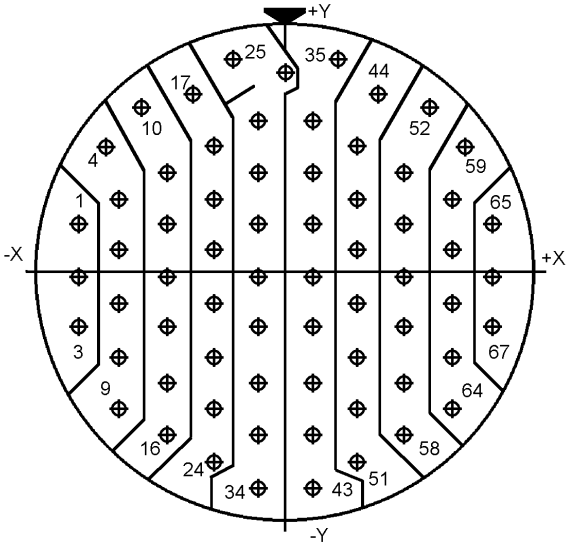
Contacts (Insert arrangement 19-66)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.123 (3.12)	+.225 (5.72)	51	+.201 (5.11)	+.270 (6.86)
19	-.123 (3.12)	+.135 (3.43)	52	+.201 (5.11)	+.180 (4.57)
20	-.123 (3.12)	+.045 (1.14)	53	+.201 (5.11)	+.090 (2.29)
21	-.123 (3.12)	-.045 (1.14)	54	+.201 (5.11)	+.000 (0.00)
22	-.123 (3.12)	-.135 (3.43)	55	+.201 (5.11)	-.090 (2.29)
23	-.123 (3.12)	-.225 (5.72)	56	+.201 (5.11)	-.180 (4.57)
24	-.123 (3.12)	-.315 (8.00)	57	+.201 (5.11)	-.270 (6.86)
25	-.045 (1.14)	+.360 (9.14)	58	+.279 (7.09)	+.225 (5.72)
26	-.045 (1.14)	+.270 (6.86)	59	+.279 (7.09)	+.135 (3.43)
27	-.045 (1.14)	+.180 (4.57)	60	+.279 (7.09)	+.045 (1.14)
28	-.045 (1.14)	+.090 (2.29)	61	+.279 (7.09)	-.045 (1.14)
29	-.045 (1.14)	+.000 (0.00)	62	+.279 (7.09)	-.135 (3.43)
30	-.045 (1.14)	-.090 (2.29)	63	+.279 (7.09)	-.225 (5.72)
31	-.045 (1.14)	-.180 (4.57)	64	+.357 (9.07)	+.090 (2.29)
32	-.045 (1.14)	-.270 (6.86)	65	+.357 (9.07)	+.000 (0.00)
33	-.045 (1.14)	-.360 (9.14)	66	+.357 (9.07)	-.090 (2.29)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-66	66	22M	M	All	MS20054-66

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 19-45)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.081 (2.06)	35	+.088 (2.24)	+.360 (9.14)
2	-.357 (9.07)	-.009 (0.23)	36	+.045 (1.14)	+.261 (6.63)
3	-.357 (9.07)	-.099 (2.51)	37	+.045 (1.14)	+.171 (4.34)
4	-.304 (7.72)	+.213 (5.41)	38	+.045 (1.14)	+.081 (2.06)
5	-.279 (7.09)	+.126 (3.20)	39	+.045 (1.14)	-.009 (.23)
6	-.279 (7.09)	+.036 (0.91)	40	+.045 (1.14)	-.099 (2.51)
7	-.279 (7.09)	-.054 (1.37)	41	+.045 (1.14)	-.189 (4.80)
8	-.279 (7.09)	-.144 (3.66)	42	+.045 (1.14)	-.279 (7.09)
9	-.279 (7.09)	-.234 (5.94)	43	+.045 (1.14)	-.369 (9.37)
10	-.244 (6.20)	+.280 (7.11)	44	+.156 (3.96)	+.301 (7.65)
11	-.201 (5.11)	+.171 (4.34)	45	+.123 (3.12)	+.216 (5.49)
12	-.201 (5.11)	+.081 (2.06)	46	+.123 (3.12)	+.126 (3.20)
13	-.201 (5.11)	-.009 (0.23)	47	+.123 (3.12)	+.036 (0.91)
14	-.201 (5.11)	-.099 (2.51)	48	+.123 (3.12)	-.054 (1.37)
15	-.201 (5.11)	-.189 (4.80)	49	+.123 (3.12)	-.144 (3.66)
16	-.201 (5.11)	-.279 (7.09)	50	+.123 (3.12)	-.234 (5.94)
17	-.156 (3.96)	+.301 (7.65)	51	+.123 (3.12)	-.324 (8.23)

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

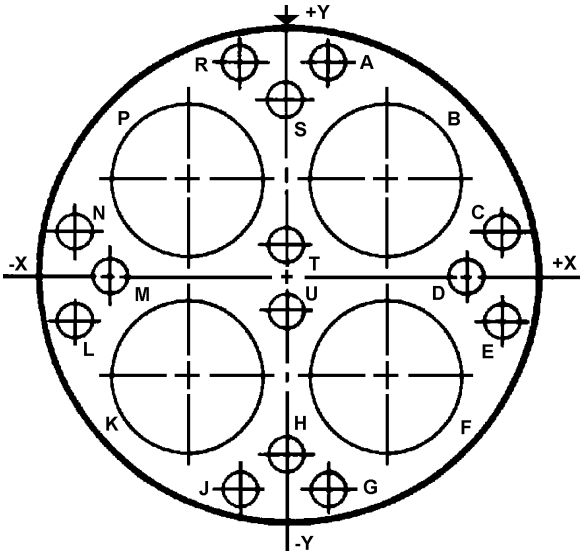
Contacts (Insert arrangement 19-45)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.123 (3.12)	+.216 (5.49)	52	+.244 (6.20)	+.280 (7.11)
19	-.123 (3.12)	+.126 (3.20)	53	+.201 (5.11)	+.171 (4.34)
20	-.123 (3.12)	+.036 (0.91)	54	+.201 (5.11)	+.081 (2.06)
21	-.123 (3.12)	-.054 (1.37)	55	+.201 (5.11)	-.009 (0.23)
22	-.123 (3.12)	-.144 (3.66)	56	+.201 (5.11)	-.099 (2.51)
23	-.123 (3.12)	-.234 (5.94)	57	+.201 (5.11)	-.189 (4.80)
24	-.123 (3.12)	-.324 (8.23)	58	+.201 (5.11)	-.279 (7.09)
25	-.088 (2.24)	+.360 (9.14)	59	+.304 (7.72)	+.213 (5.41)
26	+.000 (0.00)	+.339 (8.61)	60	+.279 (7.09)	+.126 (3.20)
27	-.045 (1.14)	+.261 (6.63)	61	+.279 (7.09)	+.036 (0.91)
28	-.045 (1.14)	+.171 (4.34)	62	+.279 (7.09)	-.054 (1.37)
29	-.045 (1.14)	+.081 (2.06)	63	+.279 (7.09)	-.144 (3.66)
30	-.045 (1.14)	-.009 (0.23)	64	+.279 (7.09)	-.234 (5.94)
31	-.045 (1.14)	-.099 (2.51)	65	+.357 (9.07)	+.081 (2.06)
32	-.045 (1.14)	-.189 (4.80)	66	+.357 (9.07)	-.009 (0.23)
33	-.045 (1.14)	-.279 (7.09)	67	+.357 (9.07)	-.099 (2.51)
34	-.045 (1.14)	-.369 (9.37)			

(Applicable to MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
19	-45	67	22D	M	All	MS27530-45

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



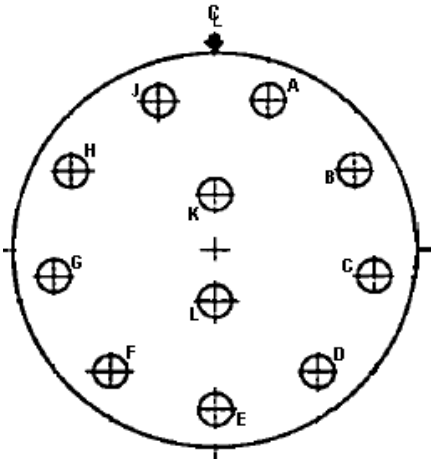
Contacts (Insert arrangement 19-18)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.068 (1.73)	+.380 (9.65)	K	-.180 (4.57)	-.180 (4.57)
B	+.180 (4.57)	+.180 (4.57)	L	-.377 (9.58)	-.065 (1.65)
C	+.377 (9.58)	+.065 (1.65)	M	-.317 (8.05)	+.000 (0.00)
D	+.317 (8.05)	+.000 (0.00)	N	-.377 (9.58)	+.065 (1.65)
E	+.377 (9.58)	-.065 (1.65)	P	-.180 (4.57)	+.180 (4.57)
F	+.180 (4.57)	-.180 (4.57)	R	-.068 (1.73)	+.380 (9.65)
G	+.068 (1.73)	-.380 (9.65)	S	+.000 (0.00)	+.319 (8.10)
H	+.000 (0.00)	-.319 (8.10)	T	+.000 (0.00)	+.044 (1.12)
J	-.068 (1.73)	-.380 (9.65)	U	+.000 (0.00)	-.044 (1.12)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
19	18	4	8 (See note)	TWINAX	B,F,K,P
		14	22D	M	All others

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 ºC (347 ºF) maximum.

FIGURE 6. Insert arrangement, electrical connector, shell size 19 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

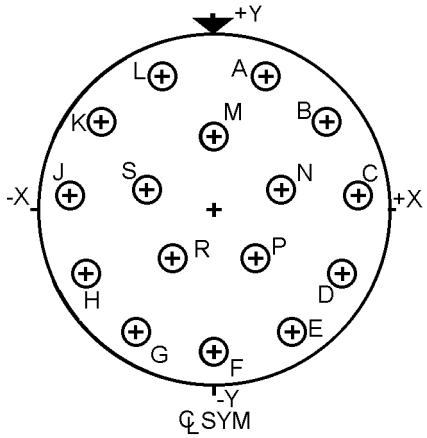


Contacts (Insert arrangement 21-11)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.121 (3.07)	+.332 (8.43)	G	-.348 (8.84)	-.061 (1.55)
B	+.306 (7.77)	+.177 (4.50)	H	-.306 (7.77)	+.177 (4.50)
C	+.348 (8.84)	-.061 (1.55)	J	-.121 (3.07)	+.332 (8.43)
D	+.227 (5.77)	-.270 (6.86)	K	+.000 (0.00)	+.123 (3.12)
E	+.000 (0.00)	-.353 (8.97)	L	+.000 (0.00)	-.115 (2.92)
F	-.227 (5.77)	-.270 (6.86)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-11	11	12	I	All	MS20055-11

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

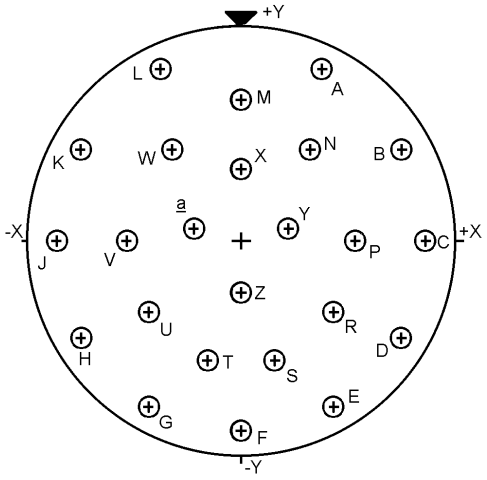


Contacts (Insert arrangement 21-16)					
Contact position ID	Location		Contact position ID	Location	
	X-axis	Y-axis		X-axis	Y-axis
A	+.118 (3.00)	+.322 (8.18)	J	-.341 (8.66)	+.036 (0.91)
B	+.271 (6.88)	+.211 (5.36)	K	-.271 (6.88)	+.211 (5.36)
C	+.341 (8.66)	+.036 (0.91)	L	-.118 (3.00)	+.322 (8.18)
D	+.308 (7.82)	-.150 (3.81)	M	+.000 (0.00)	+.175 (4.45)
E	+.182 (4.62)	-.290 (7.37)	N	+.154 (3.91)	+.062 (1.57)
F	+.000 (0.00)	-.343 (8.71)	P	+.094 (2.39)	-.122 (3.10)
G	-.182 (4.62)	-.290 (7.37)	R	-.094 (2.39)	-.122 (3.10)
H	-.308 (7.82)	-.150 (3.81)	S	-.154 (3.91)	+.062 (1.57)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-16	16	16	II	All	MS20055-16

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

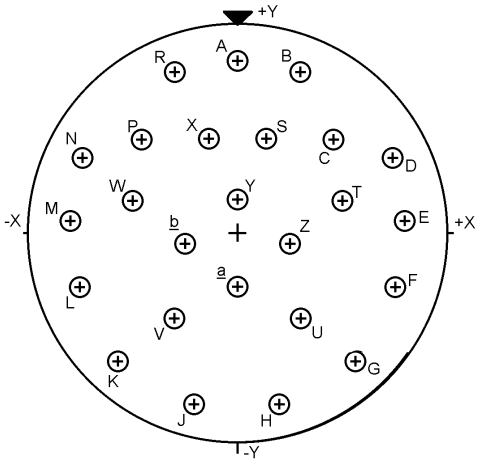


Contacts (Insert arrangement 21-24)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.175 (4.45)	+.375 (9.53)	N	+.150 (3.81)	+.200 (5.08)
B	+.350 (8.89)	+.200 (5.08)	P	+.250 (6.35)	+.000 (0.00)
C	+.400 (10.16)	+.000 (0.00)	R	+.200 (5.08)	-.150 (3.81)
D	+.350 (8.89)	-.200 (5.08)	S	+.075 (1.91)	-.250 (6.35)
E	+.200 (5.08)	-.350 (8.89)	T	-.075 (1.91)	-.250 (6.35)
F	+.000 (0.00)	-.400 (10.16)	U	-.200 (5.08)	-.150 (3.81)
G	-.200 (5.08)	-.350 (8.89)	V	-.250 (6.35)	+.000 (0.00)
H	-.350 (8.89)	-.200 (5.08)	W	-.150 (3.81)	+.200 (5.08)
J	-.400 (10.16)	+.000 (0.00)	X	+.000 (0.00)	+.150 (3.81)
K	-.350 (8.89)	+.200 (5.08)	Y	+.100 (2.54)	+.025 (0.64)
L	-.175 (4.45)	+.375 (9.53)	Z	+.000 (0.00)	-.100 (2.54)
M	+.000 (0.00)	+.300 (7.62)	a	-.100 (2.54)	+.025 (0.64)

size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-24	24	20	I	All	MS20055-24

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

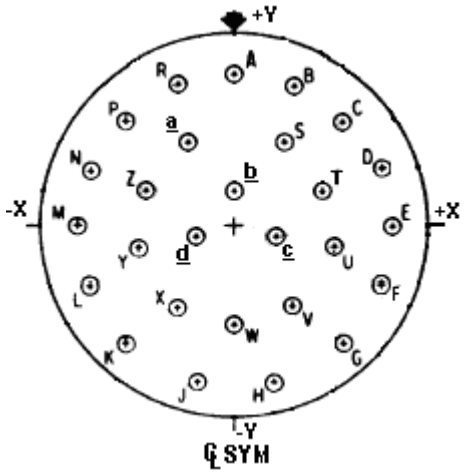


Contacts (Insert arrangement 21-25)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.400 (10.16)	P	-.225 (5.72)	+.225 (5.72)
B	+.150 (3.81)	+.375 (9.53)	R	-.150 (3.81)	+.375 (9.53)
C	+.225 (5.72)	+.225 (5.72)	S	+.075 (1.91)	+.225 (5.72)
D	+.375 (9.53)	+.175 (4.45)	T	+.250 (6.35)	+.075 (1.91)
E	+.400 (10.16)	+.025 (0.64)	U	+.150 (3.81)	-.200 (5.08)
F	+.375 (9.53)	-.125 (3.18)	V	-.150 (3.81)	-.200 (5.08)
G	+.275 (6.99)	-.300 (7.62)	W	-.250 (6.35)	+.075 (1.91)
H	+.100 (2.54)	-.400 (10.16)	X	-.075 (1.91)	+.225 (5.72)
J	-.100 (2.54)	-.400 (10.16)	Y	+.000 (0.00)	+.075 (1.91)
K	-.275 (6.99)	-.300 (7.62)	Z	+.125 (3.18)	-.025 (0.64)
L	-.375 (9.53)	-.125 (3.18)	a	+.000 (0.00)	-.125 (3.18)
M	-.400 (10.16)	+.025 (0.64)	b	-.125 (3.18)	-.025 (0.64)
N	-.375 (9.53)	+.175 (4.45)			

Shell size	Arrangement No.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-25	25	20	I	All	MS20055-25

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

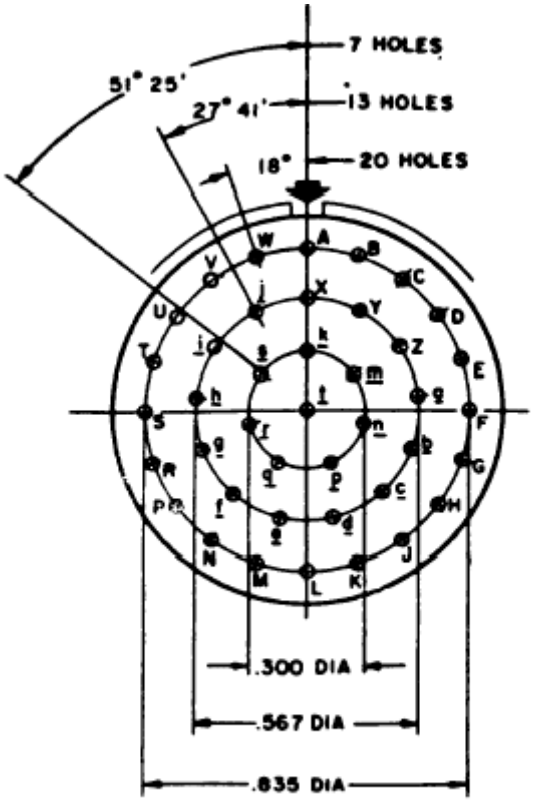


Contacts (Insert arrangement 21-27)					
Contact position ID	Contact location		Contact position ID	Contact location	
	X-axis	Y-axis		X-axis	Y-axis
A	+.000 (0.00)	+.400 (10.16)	R	-.150 (3.81)	+.375 (9.53)
B	+.150 (3.81)	+.375 (9.53)	S	+.125 (3.18)	+.225 (5.72)
C	+.275 (6.99)	+.275 (6.99)	T	+.225 (5.72)	+.100 (2.54)
D	+.375 (9.53)	+.150 (3.81)	U	+.250 (6.35)	-.050 (1.27)
E	+.400 (10.16)	+.000 (0.00)	V	+.150 (3.81)	-.200 (5.08)
F	+.375 (9.53)	-.150 (3.81)	W	+.000 (0.00)	-.250 (6.35)
G	+.275 (6.99)	-.300 (7.62)	X	-.150 (3.81)	-.200 (5.08)
H	+.100 (2.54)	-.400 (10.16)	Y	-.250 (6.35)	-.050 (1.27)
J	-.100 (2.54)	-.400 (10.16)	Z	-.225 (5.72)	+.100 (2.54)
K	-.275 (6.99)	-.300 (7.62)	a	-.125 (3.18)	+.225 (5.72)
L	-.375 (9.53)	-.150 (3.81)	b	+.000 (0.00)	+.100 (2.54)
M	-.400 (10.16)	+.000 (0.00)	c	+.100 (2.54)	-.025 (0.64)
N	-.375 (9.53)	+.150 (3.81)	d	-.100 (2.54)	-.025 (0.64)
P	-.275 (6.99)	+.275 (6.99)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-27	27	20	I	All	MS20055-27

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

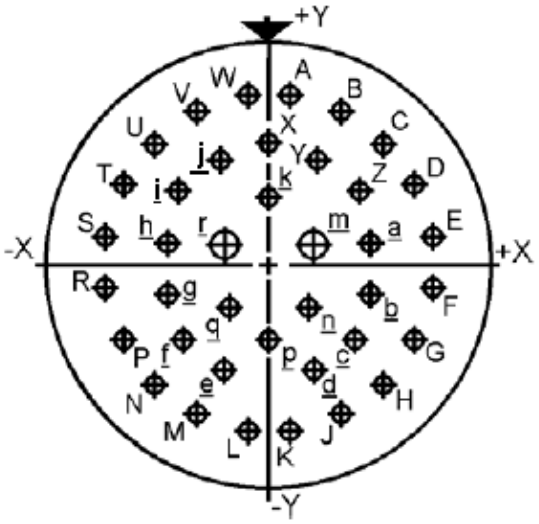


Arrangement (21-41)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-41	41	20	I	All	MS20055-41

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 21-39)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.411 (10.44)	X	+.000 (0.00)	+.295 (7.49)
B	+.189 (4.80)	+.371 (9.42)	Y	+.126 (3.20)	+.256 (6.50)
C	+.294 (7.47)	+.294 (7.47)	Z	+.232 (5.89)	+.179 (4.55)
D	+.371 (9.42)	+.189 (4.80)	a	+.280 (7.11)	+.057 (1.45)
E	+.411 (10.44)	+.065 (1.65)	b	+.280 (7.11)	-.074 (1.88)
F	+.411 (10.44)	-.065 (1.65)	c	+.217 (5.51)	-.189 (4.80)
G	+.371 (9.42)	-.189 (4.80)	d	+.112 (2.84)	-.265 (6.73)
H	+.294 (7.47)	-.294 (7.47)	e	-.112 (2.84)	-.265 (6.73)
J	+.189 (4.80)	-.371 (9.42)	f	-.217 (5.51)	-.189 (4.80)
K	+.065 (1.65)	-.411 (10.44)	g	-.280 (7.11)	-.074 (1.88)
L	-.065 (1.65)	-.411 (10.44)	h	-.280 (7.11)	+.057 (1.45)
M	-.189 (4.80)	-.371 (9.42)	i	-.232 (5.89)	+.179 (4.55)
N	-.294 (7.47)	-.294 (7.47)	j	-.126 (3.20)	+.256 (6.50)
P	-.371 (9.42)	-.189 (4.80)	k	+.000 (0.00)	+.164 (4.17)
R	-.411 (10.44)	-.065 (1.65)	m	+.114 (2.90)	+.048 (1.22)

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

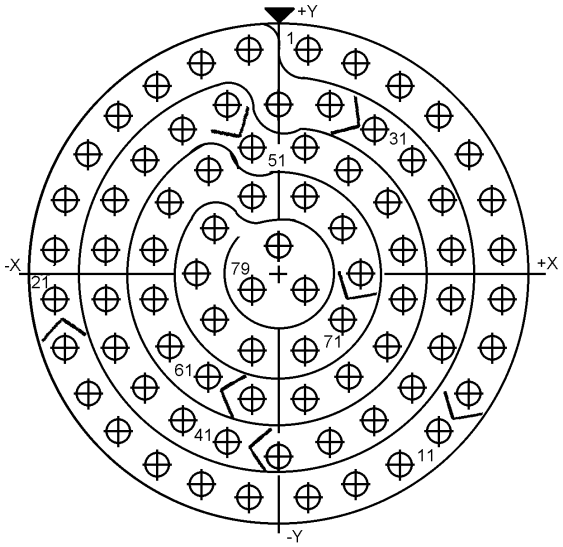
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 21-39)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
S	-.411 (10.44)	+.065 (1.65)	n	+.106 (2.69)	-.107 (2.72)
T	-.371 (9.42)	+.189 (4.80)	p	+.000 (0.00)	-.189 (4.80)
U	-.294 (7.47)	+.294 (7.47)	g	-.106 (2.69)	-.107 (2.72)
V	-.189 (4.80)	+.371 (9.42)	r	-.114 (2.90)	+.048 (1.09)
W	-.065 (1.65)	+.411 (10.44)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-39	2	16	I	r,m	MS20055-39
		37	20		All others	

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 21-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.053 (1.35)	+.426 (10.82)	41	-.098 (2.49)	-.322 (8.18)
2	+.146 (3.71)	+.404 (10.26)	42	-.184 (4.67)	-.280 (7.11)
3	+.232 (5.89)	+.362 (9.19)	43	-.258 (6.55)	-.220 (5.59)
4	+.306 (7.77)	+.302 (7.67)	44	-.311 (7.90)	-.141 (3.58)
5	+.365 (9.27)	+.227 (5.77)	45	-.332 (8.43)	-.048 (1.22)
6	+.406 (10.31)	+.141 (3.58)	46	-.332 (8.43)	+.048 (1.22)
7	+.427 (10.85)	+.048 (1.22)	47	-.311 (7.90)	+.141 (3.58)
8	+.427 (10.85)	-.048 (1.22)	48	-.258 (6.55)	+.220 (5.59)
9	+.406 (10.31)	-.141 (3.58)	49	-.184 (4.67)	+.280 (7.11)
10	+.365 (9.27)	-.227 (5.77)	50	-.098 (2.49)	+.322 (8.18)
11	+.306 (7.77)	-.302 (7.67)	51	-.048 (1.22)	+.241 (6.12)
12	+.232 (5.89)	-.362 (9.19)	52	+.048 (1.22)	+.241 (6.12)
13	+.146 (3.71)	-.404 (10.26)	53	+.134 (3.40)	+.199 (5.05)
14	+.053 (1.35)	-.426 (10.82)	54	+.208 (5.28)	+.139 (3.53)
15	-.053 (1.35)	-.426 (10.82)	55	+.237 (6.02)	+.048 (1.22)
16	-.146 (3.71)	-.404 (10.26)	56	+.237 (6.02)	-.048 (1.22)
17	-.232 (5.89)	-.362 (9.19)	57	+.208 (5.28)	-.139 (3.53)
18	-.306 (7.77)	-.302 (7.67)	58	+.134 (3.40)	-.199 (5.05)

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

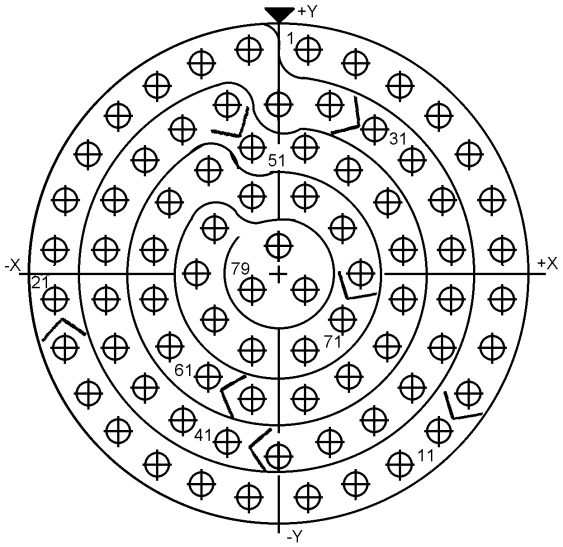
Contacts (Insert arrangement 21-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
19	-.365 (9.27)	-.227 (5.77)	59	+.048 (1.22)	-.241 (6.12)
20	-.406 (10.31)	-.141 (3.58)	60	-.048 (1.22)	-.241 (6.12)
21	-.427 (10.85)	-.048 (1.22)	61	-.134 (3.40)	-.199 (5.05)
22	-.427 (10.85)	+.048 (1.22)	62	-.208 (5.28)	-.139 (3.53)
23	-.406 (10.31)	+.141 (3.58)	63	-.237 (6.02)	-.048 (1.22)
24	-.365 (9.27)	+.227 (5.77)	64	-.237 (6.02)	+.048 (1.22)
25	-.306 (7.77)	+.302 (7.67)	65	-.208 (5.28)	+.139 (3.53)
26	-.232 (5.89)	+.362 (9.19)	66	-.134 (3.40)	+.199 (5.05)
27	-.146 (3.71)	+.404 (10.26)	67	-.048 (1.22)	+.146 (3.71)
28	-.053 (1.35)	+.426 (10.82)	68	+.048 (1.22)	+.146 (3.71)
29	+.000 (0.00)	+.323 (8.20)	69	+.125 (3.18)	+.090 (2.29)
30	+.098 (2.49)	+.322 (8.18)	70	+.155 (3.94)	+.000 (0.00)
31	+.184 (4.67)	+.280 (7.11)	71	+.125 (3.18)	-.090 (2.29)
32	+.258 (6.55)	+.220 (5.59)	72	+.048 (1.22)	-.146 (3.71)
33	+.311 (7.90)	+.141 (3.58)	73	-.048 (1.22)	-.146 (3.71)
34	+.332 (8.43)	+.048 (1.22)	74	-.125 (3.18)	-.090 (2.29)
35	+.332 (8.43)	-.048 (1.22)	75	-.155 (3.94)	+.000 (0.00)
36	+.311 (7.90)	-.141 (3.58)	76	-.125 (3.18)	+.090 (2.29)
37	+.258 (6.55)	-.220 (5.59)	77	+.000 (0.00)	+.053 (1.35)
38	+.184 (4.67)	-.280 (7.11)	78	+.048 (1.22)	-.029 (0.74)
39	+.098 (2.49)	-.322 (8.18)	79	-.048 (1.22)	-.029 (0.74)
40	+.000 (0.00)	-.347 (8.81)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-1	79	22M	M	79	MS20055-1

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 21-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.053 (1.35)	+.426 (10.82)	41	-.098 (2.49)	-.322 (8.18)
2	+.146 (3.71)	+.404 (10.26)	42	-.184 (4.67)	-.280 (7.11)
3	+.232 (5.89)	+.362 (9.19)	43	-.258 (6.55)	-.220 (5.59)
4	+.306 (7.77)	+.302 (7.67)	44	-.311 (7.90)	-.141 (3.58)
5	+.365 (9.27)	+.227 (5.77)	45	-.332 (8.43)	-.048 (1.22)
6	+.406 (10.31)	+.141 (3.58)	46	-.332 (8.43)	+.048 (1.22)
7	+.427 (10.85)	+.048 (1.22)	47	-.311 (7.90)	+.141 (3.58)
8	+.427 (10.85)	-.048 (1.22)	48	-.258 (6.55)	+.220 (5.59)
9	+.406 (10.31)	-.141 (3.58)	49	-.184 (4.67)	+.280 (7.11)
10	+.365 (9.27)	-.227 (5.77)	50	-.098 (2.49)	+.322 (8.18)
11	+.306 (7.77)	-.302 (7.67)	51	-.048 (1.22)	+.241 (6.12)
12	+.232 (5.89)	-.362 (9.19)	52	+.048 (1.22)	+.241 (6.12)
13	+.146 (3.71)	-.404 (10.26)	53	+.134 (3.40)	+.199 (5.05)
14	+.053 (1.35)	-.426 (10.82)	54	+.208 (5.28)	+.139 (3.53)
15	-.053 (1.35)	-.426 (10.82)	55	+.237 (6.02)	+.048 (1.22)
16	-.146 (3.71)	-.404 (10.26)	56	+.237 (6.02)	-.048 (1.22)
17	-.232 (5.89)	-.362 (9.19)	57	+.208 (5.28)	-.139 (3.53)

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

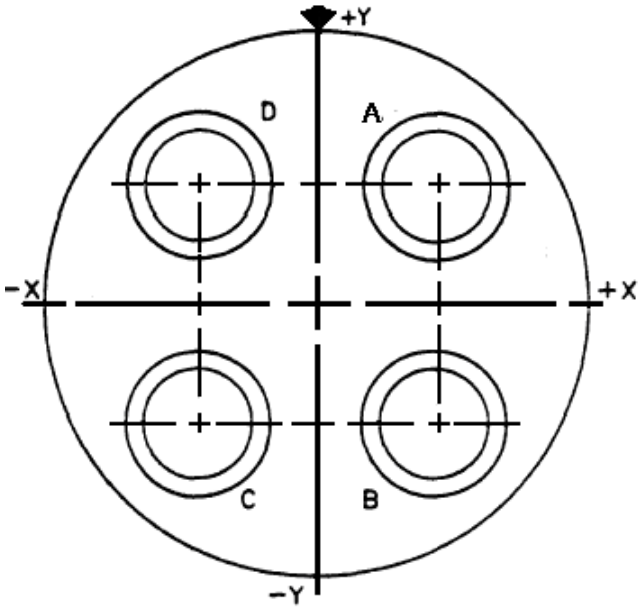
Contacts (Insert arrangement 21-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.306 (7.77)	-.302 (7.67)	58	+.134 (3.40)	-.199 (5.05)
19	-.365 (9.27)	-.227 (5.77)	59	+.048 (1.22)	-.241 (6.12)
20	-.406 (10.31)	-.141 (3.58)	60	-.048 (1.22)	-.241 (6.12)
21	-.427 (10.85)	-.048 (1.22)	61	-.134 (3.40)	-.199 (5.05)
22	-.427 (10.85)	+.048 (1.22)	62	-.208 (5.28)	-.139 (3.53)
23	-.406 (10.31)	+.141 (3.58)	63	-.237 (6.02)	-.048 (1.22)
24	-.365 (9.27)	+.227 (5.77)	64	-.237 (6.02)	+.048 (1.22)
25	-.306 (7.77)	+.302 (7.67)	65	-.208 (5.28)	+.139 (3.53)
26	-.232 (5.89)	+.362 (9.19)	66	-.134 (3.40)	+.199 (5.05)
27	-.146 (3.71)	+.404 (10.26)	67	-.048 (1.22)	+.146 (3.71)
28	-.053 (1.35)	+.426 (10.82)	68	+.048 (1.22)	+.146 (3.71)
29	+.000 (.00)	+.323 (8.20)	69	+.125 (3.18)	+.090 (2.29)
30	+.098 (2.49)	+.322 (8.18)	70	+.155 (3.94)	+.000 (0.00)
31	+.184 (4.67)	+.280 (7.11)	71	+.125 (3.18)	-.090 (2.29)
32	+.258 (6.55)	+.220 (5.59)	72	+.048 (1.22)	-.146 (3.71)
33	+.311 (7.90)	+.141 (3.58)	73	-.048 (1.22)	-.146 (3.71)
34	+.332 (8.43)	+.048 (1.22)	74	-.125 (3.18)	-.090 (2.29)
35	+.332 (8.43)	-.048 (1.22)	75	-.155 (3.94)	+.000 (0.00)
36	+.311 (7.90)	-.141 (3.58)	76	-.125 (3.18)	+.090 (2.29)
37	+.258 (6.55)	-.220 (5.59)	77	+.000 (0.00)	+.053 (1.35)
38	+.184 (4.67)	-.280 (7.11)	78	+.048 (1.22)	-.029 (0.74)
39	+.098 (2.49)	-.322 (8.18)	79	-.048 (1.22)	-.029 (0.74)
40	+.000 (0.00)	-.347 (8.81)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
21	-35	79	22D	M	ALL	MS27531-35

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



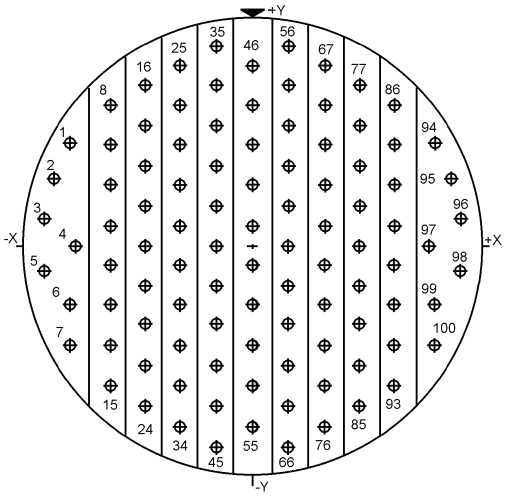
Contacts (Insert arrangement 21-75)		
Contact position ID	Location	
	X axis	Y axis
A	+.180 (4.57)	+.180 (4.57)
B	+.180 (4.57)	-.180 (4.57)
C	-.180 (4.57)	-.180 (4.57)
D	-.180 (4.57)	+.180 (4.57)

Shell size	Arrangement No.	Number of contacts	Size contacts	Service rating	Contact location	Standard contact	
						Pin	Socket
21	-75	4	8 (see note)	TWINAX	All	M39029/90-529	M39029/91-530

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 ºC (347 ºF) maximum.

FIGURE 7. Insert arrangement, electrical connector, shell size 21 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 23-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.428 (10.87)	+.241 (6.12)	51	+.000 (0.00)	-.047 (1.19)
2	-.467 (11.86)	+.154 (3.91)	52	+.000 (0.00)	-.142 (3.61)
3	-.488 (12.40)	+.061 (1.55)	53	+.000 (0.00)	-.237 (6.02)
4	-.415 (10.54)	+.000 (0.00)	54	+.000 (0.00)	-.332 (8.43)
5	-.488 (12.40)	-.061 (1.55)	55	+.000 (0.00)	-.427 (10.85)
6	-.428 (10.87)	-.142 (3.61)	56	+.083 (2.11)	+.475 (12.07)
7	-.428 (10.87)	-.237 (6.02)	57	+.083 (2.11)	+.380 (9.65)
8	-.332 (8.43)	+.333 (8.46)	58	+.083 (2.11)	+.285 (7.24)
9	-.332 (8.43)	+.238 (6.05)	59	+.083 (2.11)	+.190 (4.83)
10	-.332 (8.43)	+.143 (3.63)	60	+.083 (2.11)	+.095 (2.41)
11	-.332 (8.43)	+.048 (1.22)	61	+.083 (2.11)	+.000 (0.00)
12	-.332 (8.43)	-.047 (1.19)	62	+.083 (2.11)	-.095 (2.41)
13	-.332 (8.43)	-.142 (3.61)	63	+.083 (2.11)	-.190 (4.83)
14	-.332 (8.43)	-.237 (6.02)	64	+.083 (2.11)	-.285 (7.24)
15	-.332 (8.43)	-.332 (8.43)	65	+.083 (2.11)	-.380 (9.65)
16	-.249 (6.32)	+.380 (9.65)	66	+.083 (2.11)	-.475 (12.07)
17	-.249 (6.32)	+.285 (7.24)	67	+.166 (4.22)	+.428 (10.87)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 23-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.249 (6.32)	+.190 (4.83)	68	+.166 (4.22)	+.333 (8.46)
19	-.249 (6.32)	+.095 (2.41)	69	+.166 (4.22)	+.238 (6.05)
20	-.249 (6.32)	+.000 (0.00)	70	+.166 (4.22)	+.143 (3.63)
21	-.249 (6.32)	-.095 (2.41)	71	+.166 (4.22)	+.048 (1.22)
22	-.249 (6.32)	-.190 (4.83)	72	+.166 (4.22)	-.047 (1.19)
23	-.249 (6.32)	-.285 (7.24)	73	+.166 (4.22)	-.142 (3.61)
24	-.249 (6.32)	-.380 (9.65)	74	+.166 (4.22)	-.237 (6.02)
25	-.166 (4.22)	+.428 (10.87)	75	+.166 (4.22)	-.332 (8.43)
26	-.166 (4.22)	+.333 (8.46)	76	+.166 (4.22)	-.427 (10.85)
27	-.166 (4.22)	+.238 (6.05)	77	+.249 (6.32)	+.380 (9.65)
28	-.166 (4.22)	+.143 (3.63)	78	+.249 (6.32)	+.285 (7.24)
29	-.166 (4.22)	+.048 (1.22)	79	+.249 (6.32)	+.190 (4.83)
30	-.166 (4.22)	-.047 (1.19)	80	+.249 (6.32)	+.095 (2.41)
31	-.166 (4.22)	-.142 (3.61)	81	+.249 (6.32)	+.000 (0.00)
32	-.166 (4.22)	-.237 (6.02)	82	+.249 (6.32)	-.095 (2.41)
33	-.166 (4.22)	-.332 (8.43)	83	+.249 (6.32)	-.190 (4.83)
34	-.166 (4.22)	-.427 (10.85)	84	+.249 (6.32)	-.285 (7.24)
35	-.083 (2.11)	+.475 (12.07)	85	+.249 (6.32)	-.380 (9.65)
36	-.083 (2.11)	+.380 (9.65)	86	+.332 (8.43)	+.333 (8.46)
37	-.083 (2.11)	+.285 (7.24)	87	+.332 (8.43)	+.238 (6.05)
38	-.083 (2.11)	+.190 (4.83)	88	+.332 (8.43)	+.143 (3.63)
39	-.083 (2.11)	+.095 (2.41)	89	+.332 (8.43)	+.048 (1.22)
40	-.083 (2.11)	+.000 (0.00)	90	+.332 (8.43)	-.047 (1.19)
41	-.083 (2.11)	-.095 (2.41)	91	+.332 (8.43)	-.142 (3.61)
42	-.083 (2.11)	-.190 (4.83)	92	+.332 (8.43)	-.237 (6.02)
43	-.083 (2.11)	-.285 (7.24)	93	+.332 (8.43)	-.332 (8.43)
44	-.083 (2.11)	-.380 (9.65)	94	+.428 (10.87)	+.241 (6.12)
45	-.083 (2.11)	-.475 (12.07)	95	+.467 (11.86)	+.154 (3.91)
46	+.000 (0.00)	+.428 (10.87)	96	+.488 (12.40)	+.061 (1.55)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

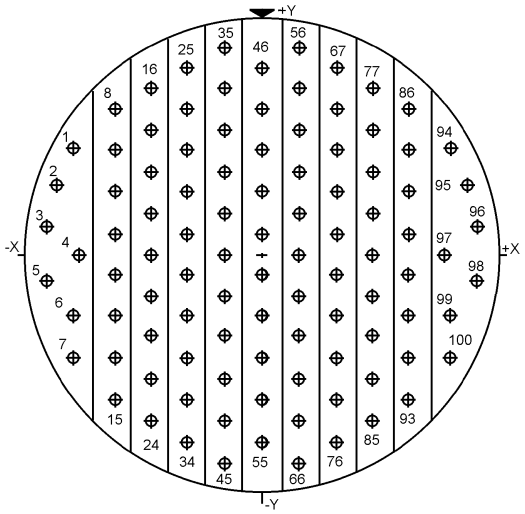
Contacts (Insert arrangement 23-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
47	+.000 (0.00)	+.333 (8.46)	97	+.415 (10.54)	+.000 (0.00)
48	+.000 (0.00)	+.238 (6.05)	98	+.488 (12.40)	-.061 (1.55)
49	+.000 (0.00)	+.143 (3.63)	99	+.428 (10.87)	-.142 (3.61)
50	+.000 (0.00)	+.048 (1.22)	100	+.428 (10.87)	-.237 (6.02)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-1	100	22M	M	All	MS20056-1

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 23-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.428 (10.87)	+.241 (6.12)	51	+.000 (0.00)	-.047 (1.19)
2	-.467 (11.86)	+.154 (3.91)	52	+.000 (0.00)	-.142 (3.61)
3	-.488 (12.40)	+.061 (1.55)	53	+.000 (0.00)	-.237 (6.02)
4	-.415 (10.54)	+.000 (0.00)	54	+.000 (0.00)	-.332 (8.43)
5	-.488 (12.40)	-.061 (1.55)	55	+.000 (0.00)	-.427 (10.85)
6	-.428 (10.87)	-.142 (3.61)	56	+.083 (2.11)	+.475 (12.07)
7	-.428 (10.87)	-.237 (6.02)	57	+.083 (2.11)	+.380 (9.65)
8	-.332 (8.43)	+.333 (8.46)	58	+.083 (2.11)	+.285 (7.24)
9	-.332 (8.43)	+.238 (6.05)	59	+.083 (2.11)	+.190 (4.83)
10	-.332 (8.43)	+.143 (3.63)	60	+.083 (2.11)	+.095 (2.41)
11	-.332 (8.43)	+.048 (1.22)	61	+.083 (2.11)	+.000 (0.00)
12	-.332 (8.43)	-.047 (1.19)	62	+.083 (2.11)	-.095 (2.41)
13	-.332 (8.43)	-.142 (3.61)	63	+.083 (2.11)	-.190 (4.83)
14	-.332 (8.43)	-.237 (6.02)	64	+.083 (2.11)	-.285 (7.24)
15	-.332 (8.43)	-.332 (8.43)	65	+.083 (2.11)	-.380 (9.65)
16	-.249 (6.32)	+.380 (9.65)	66	+.083 (2.11)	-.475 (12.07)
17	-.249 (6.32)	+.285 (7.24)	67	+.166 (4.22)	+.428 (10.87)
18	-.249 (6.32)	+.190 (4.83)	68	+.166 (4.22)	+.333 (8.46)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 23-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
19	-.249 (6.32)	+.095 (2.41)	69	+.166 (4.22)	+.238 (6.05)
20	-.249 (6.32)	+.000 (0.00)	70	+.166 (4.22)	+.143 (3.63)
21	-.249 (6.32)	-.095 (2.41)	71	+.166 (4.22)	+.048 (1.22)
22	-.249 (6.32)	-.190 (4.83)	72	+.166 (4.22)	-.047 (1.19)
23	-.249 (6.32)	-.285 (7.24)	73	+.166 (4.22)	-.142 (3.61)
24	-.249 (6.32)	-.380 (9.65)	74	+.166 (4.22)	-.237 (6.02)
25	-.166 (4.22)	+.428 (10.87)	75	+.166 (4.22)	-.332 (8.43)
26	-.166 (4.22)	+.333 (8.46)	76	+.166 (4.22)	-.427 (10.85)
27	-.166 (4.22)	+.238 (6.05)	77	+.249 (6.32)	+.380 (9.65)
28	-.166 (4.22)	+.143 (3.63)	78	+.249 (6.32)	+.285 (7.24)
29	-.166 (4.22)	+.048 (1.22)	79	+.249 (6.32)	+.190 (4.83)
30	-.166 (4.22)	-.047 (1.19)	80	+.249 (6.32)	+.095 (2.41)
31	-.166 (4.22)	-.142 (3.61)	81	+.249 (6.32)	+.000 (0.00)
32	-.166 (4.22)	-.237 (6.02)	82	+.249 (6.32)	-.095 (2.41)
33	-.166 (4.22)	-.332 (8.43)	83	+.249 (6.32)	-.190 (4.83)
34	-.166 (4.22)	-.427 (10.85)	84	+.249 (6.32)	-.285 (7.24)
35	-.083 (2.11)	+.475 (12.07)	85	+.249 (6.32)	-.380 (9.65)
36	-.083 (2.11)	+.380 (9.65)	86	+.332 (8.43)	+.333 (8.46)
37	-.083 (2.11)	+.285 (7.24)	87	+.332 (8.43)	+.238 (6.05)
38	-.083 (2.11)	+.190 (4.83)	88	+.332 (8.43)	+.143 (3.63)
39	-.083 (2.11)	+.095 (2.41)	89	+.332 (8.43)	+.048 (1.22)
40	-.083 (2.11)	+.000 (0.00)	90	+.332 (8.43)	-.047 (1.19)
41	-.083 (2.11)	-.095 (2.41)	91	+.332 (8.43)	-.142 (3.61)
42	-.083 (2.11)	-.190 (4.83)	92	+.332 (8.43)	-.237 (6.02)
43	-.083 (2.11)	-.285 (7.24)	93	+.332 (8.43)	-.332 (8.43)
44	-.083 (2.11)	-.380 (9.65)	94	+.428 (10.87)	+.241 (6.12)
45	-.083 (2.11)	-.475 (12.07)	95	+.467 (11.86)	+.154 (3.91)
46	+.000 (0.00)	+.428 (10.87)	96	+.488 (12.40)	+.061 (1.55)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

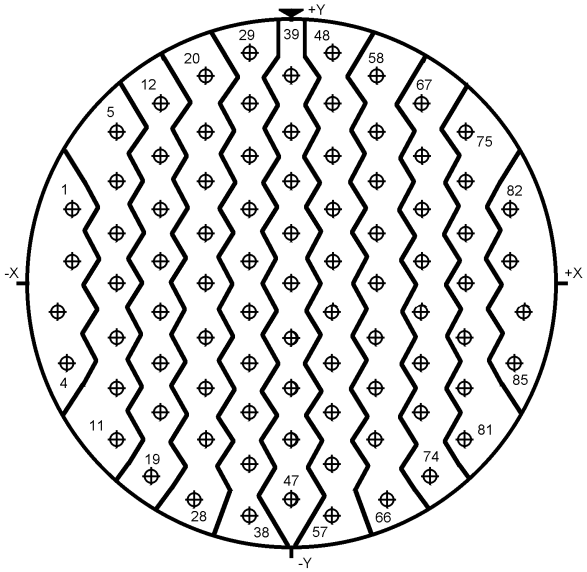
Contacts (Insert arrangement 23-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
47	+.000 (0.00)	+.333 (8.46)	97	+.415 (10.54)	+.000 (0.00)
48	+.000 (0.00)	+.238 (6.05)	98	+.488 (12.40)	-.061 (1.55)
49	+.000 (0.00)	+.143 (3.63)	99	+.428 (10.87)	-.142 (3.61)
50	+.000 (0.00)	+.048 (1.22)	100	+.428 (10.87)	-.237 (6.02)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-35	100	22D	M	All	MS27532-35

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 23-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.455 (11.56)	+.159 (4.04)	44	+.000 (0.00)	-.106 (2.69)
2	-.455 (11.56)	+.053 (1.35)	45	+.000 (0.00)	-.212 (5.38)
3	-.482 (12.24)	-.053 (1.35)	46	+.000 (0.00)	-.318 (8.08)
4	-.455 (11.56)	-.159 (4.04)	47	+.000 (0.00)	-.424 (10.77)
5	-.364 (9.25)	+.318 (8.08)	48	+.091 (2.31)	+.477 (12.12)
6	-.364 (9.25)	+.212 (5.38)	49	+.091 (2.31)	+.371 (9.42)
7	-.364 (9.25)	+.106 (2.69)	50	+.091 (2.31)	+.265 (6.73)
8	-.364 (9.25)	+.000 (0.00)	51	+.091 (2.31)	+.159 (4.04)
9	-.364 (9.25)	-.106 (2.69)	52	+.091 (2.31)	+.053 (1.35)
10	-.364 (9.25)	-.212 (5.38)	53	+.091 (2.31)	-.053 (1.35)
11	-.364 (9.25)	-.318 (8.08)	54	+.091 (2.31)	-.159 (4.04)
12	-.273 (6.93)	+.371 (9.42)	55	+.091 (2.31)	-.265 (6.73)
13	-.273 (6.93)	+.265 (6.73)	56	+.091 (2.31)	-.371 (9.42)
14	-.273 (6.93)	+.159 (4.04)	57	+.091 (2.31)	-.477 (12.12)
15	-.273 (6.93)	+.053 (1.35)	58	+.182 (4.62)	+.424 (10.77)
16	-.273 (6.93)	-.053 (1.35)	59	+.182 (4.62)	+.318 (8.08)
17	-.273 (6.93)	-.159 (4.04)	60	+.182 (4.62)	+.212 (5.38)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

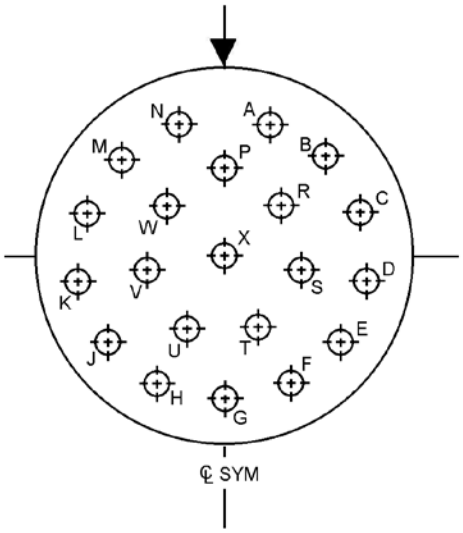
Contacts (Insert arrangement 23-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.273 (6.93)	-.265 (6.73)	61	+.182 (4.62)	+.106 (2.69)
19	-.287 (7.29)	-.390 (9.91)	62	+.182 (4.62)	+.000 (0.00)
20	-.182 (4.62)	+.424 (10.77)	63	+.182 (4.62)	-.106 (2.69)
21	-.182 (4.62)	+.318 (8.08)	64	+.182 (4.62)	-.212 (5.38)
22	-.182 (4.62)	+.212 (5.38)	65	+.182 (4.62)	-.318 (8.08)
23	-.182 (4.62)	+.106 (2.69)	66	+.192 (4.88)	-.446 (11.33)
24	-.182 (4.62)	+.000 (0.00)	67	+.273 (6.93)	+.371 (9.42)
25	-.182 (4.62)	-.106 (2.69)	68	+.273 (6.93)	+.265 (6.73)
26	-.182 (4.62)	-.212 (5.38)	69	+.273 (6.93)	+.159 (4.04)
27	-.182 (4.62)	-.318 (8.08)	70	+.273 (6.93)	+.053 (1.35)
28	-.192 (4.88)	-.446 (11.33)	71	+.273 (6.93)	-.053 (1.35)
29	-.091 (2.31)	+.477 (12.12)	72	+.273 (6.93)	-.159 (4.04)
30	-.091 (2.31)	+.371 (9.42)	73	+.273 (6.93)	-.265 (6.73)
31	-.091 (2.31)	+.265 (6.73)	74	+.287 (7.29)	-.390 (9.91)
32	-.091 (2.31)	+.159 (4.04)	75	+.364 (9.25)	+.318 (8.08)
33	-.091 (2.31)	+.053 (1.35)	76	+.364 (9.25)	+.212 (5.38)
34	-.091 (2.31)	-.053 (1.35)	77	+.364 (9.25)	+.106 (2.69)
35	-.091 (2.31)	-.159 (4.04)	78	+.364 (9.25)	+.000 (0.00)
36	-.091 (2.31)	-.265 (6.73)	79	+.364 (9.25)	-.106 (2.69)
37	-.091 (2.31)	-.371 (9.42)	80	+.364 (9.25)	-.212 (5.38)
38	-.091 (2.31)	-.477 (12.12)	81	+.364 (9.25)	-.318 (8.08)
39	+.000 (0.00)	+.424 (10.77)	82	+.455 (11.56)	+.159 (4.04)
40	+.000 (0.00)	+.318 (8.08)	83	+.455 (11.56)	+.053 (1.35)
41	+.000 (0.00)	+.212 (5.38)	84	+.482 (12.24)	-.053 (1.35)
42	+.000 (0.00)	+.106 (2.69)	85	+.455 (11.56)	-.159 (4.04)
43	+.000 (0.00)	+.000 (0.00)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-2	85	22	M	All	MS20056-2

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

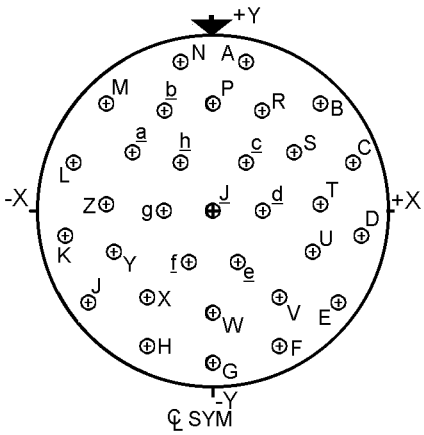


Contacts (Insert arrangement 23-21)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.128 (3.25)	+.385 (9.78)	M	-.289 (7.34)	+.285 (7.24)
B	+.289 (7.34)	+.285 (7.24)	N	-.128 (3.25)	+.385 (9.78)
C	+.386 (9.80)	+.123 (3.12)	P	+.000 (0.00)	+.245 (6.22)
D	+.400 (10.16)	-.065 (1.65)	R	+.160 (4.06)	+.146 (3.71)
E	+.328 (8.33)	-.239 (6.07)	S	+.214 (5.44)	-.035 (0.89)
F	+.183 (4.65)	-.362 (9.19)	T	+.094 (2.39)	-.194 (4.93)
G	+.000 (0.00)	-.406 (10.31)	U	-.094 (2.39)	-.194 (4.93)
H	-.183 (4.65)	-.362 (9.19)	V	-.214 (5.44)	-.035 (0.89)
J	-.328 (8.33)	-.239 (6.07)	W	-.160 (4.06)	+.146 (3.71)
K	-.400 (10.16)	-.065 (1.65)	X	+.000 (0.00)	+.000 (0.00)
L	-.386 (9.80)	+.123 (3.12)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-21	21	16	II	All	MS20056-21

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

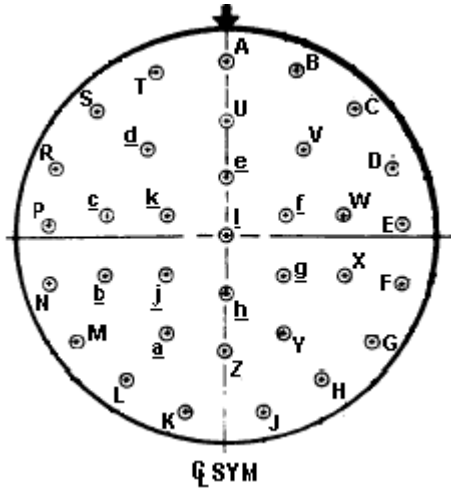


Contacts (Insert arrangement 23-32)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.450 (11.43)	T	+.325 (8.26)	+.025 (0.64)
B	+.325 (8.26)	+.325 (8.26)	U	+.300 (7.62)	-.125 (3.18)
C	+.425 (10.80)	+.150 (3.81)	V	+.200 (5.08)	-.250 (6.35)
D	+.450 (11.43)	-.075 (1.91)	W	+.000 (0.00)	-.300 (7.62)
E	+.375 (9.53)	-.275 (6.99)	X	-.200 (5.08)	-.250 (6.35)
F	+.200 (5.08)	-.400 (10.16)	Y	-.300 (7.62)	-.125 (3.18)
G	+.000 (0.00)	-.450 (11.43)	Z	-.325 (8.26)	+.025 (0.64)
H	-.200 (5.08)	-.400 (10.16)	a	-.250 (6.35)	+.175 (4.45)
J	-.375 (9.53)	-.275 (6.99)	b	-.150 (3.81)	+.300 (7.62)
K	-.450 (11.43)	-.075 (1.91)	c	+.100 (2.54)	+.150 (3.81)
L	-.425 (10.80)	+.150 (3.81)	d	+.150 (3.81)	+.000 (0.00)
M	-.325 (8.26)	+.325 (8.26)	e	+.075 (1.91)	-.150 (3.81)
N	-.100 (2.54)	+.450 (11.43)	f	-.075 (1.91)	-.150 (3.81)
P	+.000 (0.00)	+.325 (8.26)	g	-.150 (3.81)	+.000 (0.00)
R	+.150 (3.81)	+.300 (7.62)	h	-.100 (2.54)	+.150 (3.81)
S	+.250 (6.35)	+.175 (4.45)	J	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-32	32	20	I	All	MS20056-32

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

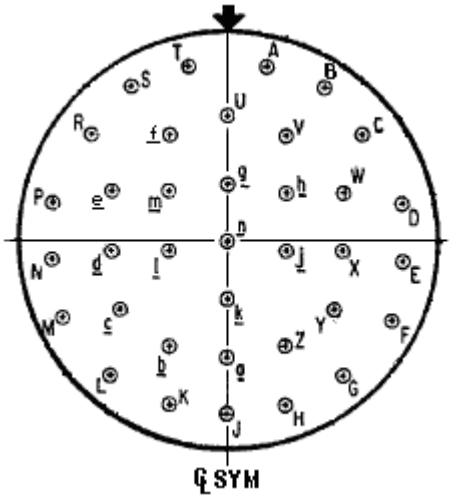


Contacts (Insert arrangement 23-34)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.450 (11.43)	U	+.000 (.00)	+.300 (7.62)
B	+.175 (4.45)	+.425 (10.80)	V	+.200 (5.08)	+.225 (5.72)
C	+.325 (8.26)	+.325 (8.26)	W	+.300 (7.62)	+.050 (1.27)
D	+.425 (10.80)	+.175 (4.45)	X	+.300 (7.62)	-.100 (2.54)
E	+.450 (11.43)	+.025 (0.64)	Y	+.150 (3.81)	-.250 (6.35)
F	+.450 (11.43)	-.125 (3.18)	Z	+.000 (0.00)	-.300 (7.62)
G	+.375 (9.53)	-.275 (6.99)	a	-.150 (3.81)	-.250 (6.35)
H	+.250 (6.35)	-.375 (9.53)	b	-.300 (7.62)	-.100 (2.54)
J	+.100 (2.54)	-.450 (11.43)	c	-.300 (7.62)	+.050 (1.27)
K	-.100 (2.54)	-.450 (11.43)	d	-.200 (5.08)	+.225 (5.72)
L	-.250 (6.35)	-.375 (9.53)	e	+.000 (0.00)	+.150 (3.81)
M	-.375 (9.53)	-.275 (6.99)	f	+.150 (3.81)	+.050 (1.27)
N	-.450 (11.43)	-.125 (3.18)	g	+.150 (3.81)	-.100 (2.54)
P	-.450 (11.43)	+.025 (0.64)	h	+.000 (0.00)	-.150 (3.81)
R	-.425 (10.80)	+.175 (4.45)	j	-.150 (3.81)	-.100 (2.54)
S	-.325 (8.26)	+.325 (8.26)	k	-.150 (3.81)	+.050 (1.27)
T	-.175 (4.45)	+.425 (10.80)	l	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-34	34	20	I	All	MS20056-34

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 23-36)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.450 (11.43)	V	+.150 (3.81)	+.275 (6.99)
B	+.250 (6.35)	+.400 (10.16)	W	+.300 (7.62)	+.125 (3.18)
C	+.350 (8.89)	+.275 (6.99)	X	+.300 (7.62)	-.025 (0.64)
D	+.450 (11.43)	+.100 (2.54)	Y	+.275 (6.99)	-.175 (4.45)
E	+.450 (11.43)	-.050 (1.27)	Z	+.150 (3.81)	-.275 (6.99)
F	+.425 (10.80)	-.200 (5.08)	a	+.000 (0.00)	-.300 (7.62)
G	+.300 (7.62)	-.350 (8.89)	b	-.150 (3.81)	-.275 (6.99)
H	+.150 (3.81)	-.425 (10.80)	c	-.275 (6.99)	-.175 (4.45)
J	+.000 (0.00)	-.450 (11.43)	d	-.300 (7.62)	-.025 (0.64)
K	-.150 (3.81)	-.425 (10.80)	e	-.300 (7.62)	+.125 (3.18)
L	-.300 (7.62)	-.350 (8.89)	f	-.150 (3.81)	+.275 (6.99)
M	-.425 (10.80)	-.200 (5.08)	g	+.000 (.00)	+.150 (3.81)
N	-.450 (11.43)	-.050 (1.27)	h	+.150 (3.81)	+.125 (3.18)
P	-.450 (11.43)	+.100 (2.54)	j	+.150 (3.81)	-.025 (0.64)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

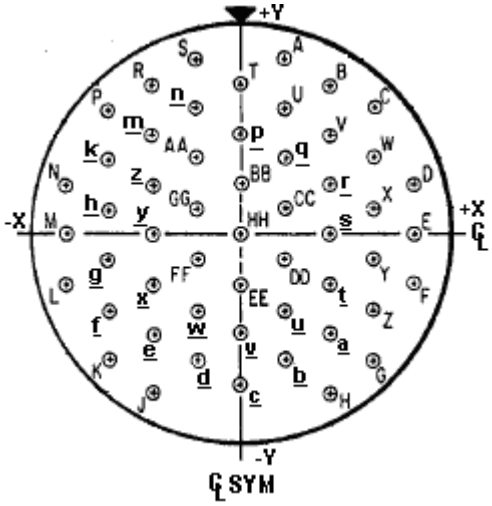
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 23-36)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
R	-.350 (8.89)	+.275 (6.99)	k	+.000 (0.00)	-.150 (3.81)
S	-.250 (6.35)	+.400 (10.16)	l	-.150 (3.81)	-.025 (0.64)
T	-.100 (2.54)	+.450 (11.43)	m	-.150 (3.81)	+.125 (3.18)
U	+.000 (0.00)	+.325 (8.26)	n	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-36	36	20	I	All	MS20056-36

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 23-53)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.112 (2.84)	+.455 (11.56)	e	-.225 (5.72)	-.260 (6.60)
B	+.225 (5.72)	+.390 (9.91)	f	-.336 (8.53)	-.195 (4.95)
C	+.336 (8.53)	+.325 (8.26)	g	-.336 (8.53)	-.065 (1.65)
D	+.450 (11.43)	+.130 (3.30)	h	-.336 (8.53)	+.065 (1.65)
E	+.450 (11.43)	+.000 (0.00)	k	-.336 (8.53)	+.195 (4.95)
F	+.450 (11.43)	-.130 (3.30)	m	-.225 (5.72)	+.260 (6.60)
G	+.336 (8.53)	-.325 (8.26)	n	-.112 (2.84)	+.325 (8.26)
H	+.225 (5.72)	-.410 (10.41)	p	+.000 (0.00)	+.260 (6.60)
J	-.225 (5.72)	-.410 (10.41)	q	+.112 (2.84)	+.195 (4.95)
K	-.336 (8.53)	-.325 (8.26)	r	+.225 (5.72)	+.130 (3.30)
L	-.450 (11.43)	-.130 (3.30)	s	+.225 (5.72)	+.000 (0.00)
M	-.450 (11.43)	+.000 (0.00)	t	+.225 (5.72)	-.130 (3.30)
N	-.450 (11.43)	+.130 (3.30)	u	+.112 (2.84)	-.195 (4.95)
P	-.336 (8.53)	+.325 (8.26)	v	+.000 (0.00)	-.260 (6.60)
R	-.225 (5.72)	+.390 (9.91)	w	-.112 (2.84)	-.195 (4.95)
S	-.112 (2.84)	+.455 (11.56)	x	-.225 (5.72)	-.130 (3.30)
T	+.000 (0.00)	+.390 (9.91)	y	-.225 (5.72)	+.000 (0.00)
U	+.112 (2.84)	+.325 (8.26)	z	-.225 (5.72)	+.130 (3.30)
V	+.225 (5.72)	+.260 (6.60)	AA	-.112 (2.84)	+.195 (4.95)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

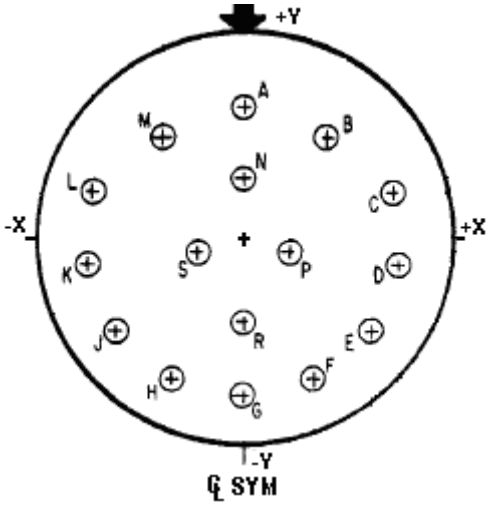
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 23-53)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
W	+.336 (8.53)	+.195 (4.95)	BB	+.000 (0.00)	+.130 (3.30)
X	+.336 (8.53)	+.065 (1.65)	CC	+.112 (2.84)	+.065 (1.65)
Y	+.336 (8.53)	-.065 (1.65)	DD	+.112 (2.84)	-.065 (1.65)
Z	+.336 (8.53)	-.195 (4.95)	EE	+.000 (0.00)	-.130 (3.30)
a	+.225 (5.72)	-.260 (6.60)	FF	-.112 (2.84)	-.065 (1.65)
b	+.112 (2.84)	-.325 (8.26)	GG	-.112 (2.84)	+.065 (1.65)
c	+.000 (0.00)	-.390 (9.91)	HH	+.000 (0.00)	+.000 (0.00)
d	-.112 (2.84)	-.325 (8.26)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-53	53	20	I	All	MS20056-53

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

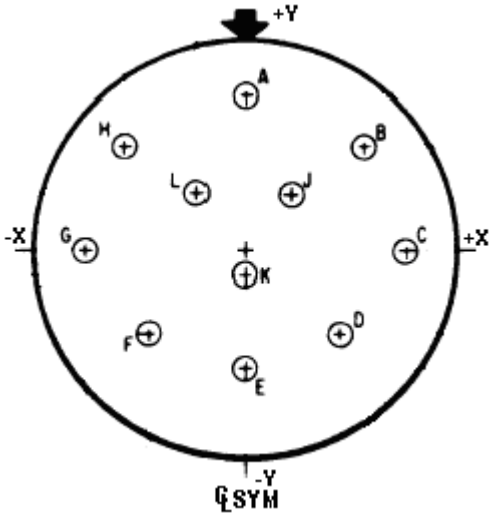


Contacts (Insert arrangement 23-97)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.344 (8.74)	J	-.328 (8.33)	-.239 (6.07)
B	+.210 (5.33)	+.270 (6.86)	K	-.400 (10.16)	-.065 (1.65)
C	+.386 (9.80)	+.123 (3.12)	L	-.386 (9.80)	+.123 (3.12)
D	+.400 (10.16)	-.065 (1.65)	M	-.210 (5.33)	+.270 (6.86)
E	+.328 (8.33)	-.239 (6.07)	N	+.000 (0.00)	+.156 (3.96)
F	+.183 (4.65)	-.362 (9.19)	P	+.120 (3.05)	-.030 (0.76)
G	+.000 (0.00)	-.406 (10.31)	R	+.000 (0.00)	-.218 (5.54)
H	-.183 (4.65)	-.362 (9.19)	S	-.120 (3.05)	-.030 (0.76)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-97	16	16	I	All	MS20056-97

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

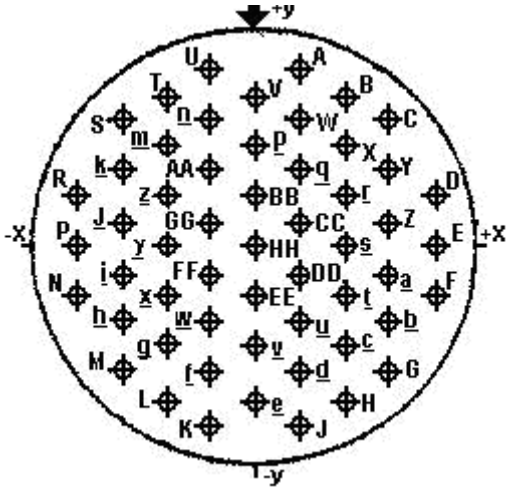


Contacts (Insert arrangement 23-99)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.404 (10.26)	G	-.404 (10.26)	+.000 (0.00)
B	+.300 (7.62)	+.270 (6.86)	H	-.300 (7.62)	+.270 (6.86)
C	+.404 (10.26)	+.000 (0.00)	J	+.120 (3.05)	+.150 (3.81)
D	+.240 (6.10)	-.210 (5.33)	K	+.000 (0.00)	-.060 (1.52)
E	+.000 (0.00)	-.300 (7.62)	L	-.120 (3.05)	+.150 (3.81)
F	-.240 (6.10)	-.210 (5.33)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-99	11	16	II	All	MS20056-99

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 23-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.112 (2.84)	+.455 (11.56)	f	-.112 (2.84)	-.325 (8.26)
B	+.225 (5.72)	+.390 (9.91)	g	-.225 (5.72)	-.260 (6.60)
C	+.336 (8.53)	+.325 (8.26)	h	-.336 (8.53)	-.195 (4.95)
D	+.450 (11.43)	+.130 (3.30)	i	-.336 (8.53)	-.065 (1.65)
E	+.450 (11.43)	+.000 (0.00)	J	-.336 (8.53)	+.065 (1.65)
F	+.450 (11.43)	-.130 (3.30)	k	-.336 (8.53)	+.195 (4.95)
G	+.336 (8.53)	-.325 (8.26)	m	-.225 (5.72)	+.260 (6.60)
H	+.225 (5.72)	-.390 (9.91)	n	-.112 (2.84)	+.325 (8.26)
J	+.112 (2.84)	-.455 (11.56)	p	+.000 (0.00)	+.260 (6.60)
K	-.112 (2.84)	-.455 (11.56)	q	+.112 (2.84)	+.195 (4.95)
L	-.225 (5.72)	-.390 (9.91)	r	+.225 (5.72)	+.130 (3.30)
M	-.336 (8.53)	-.325 (8.26)	s	+.225 (5.72)	+.000 (0.00)
N	-.450 (11.43)	-.130 (3.30)	t	+.225 (5.72)	-.130 (3.30)
P	-.450 (11.43)	+.000 (0.00)	u	+.112 (2.84)	-.195 (4.95)
R	-.450 (11.43)	+.130 (3.30)	v	+.000 (0.00)	-.260 (6.60)
S	-.336 (8.53)	+.325 (8.26)	w	-.112 (2.84)	-.195 (4.95)
T	-.225 (5.72)	+.390 (9.91)	x	-.225 (5.72)	-.130 (3.30)
U	-.112 (2.84)	+.455 (11.56)	y	-.225 (5.72)	+.000 (0.00)
V	+.000 (0.00)	+.390 (9.91)	z	-.225 (5.72)	+.130 (3.30)

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

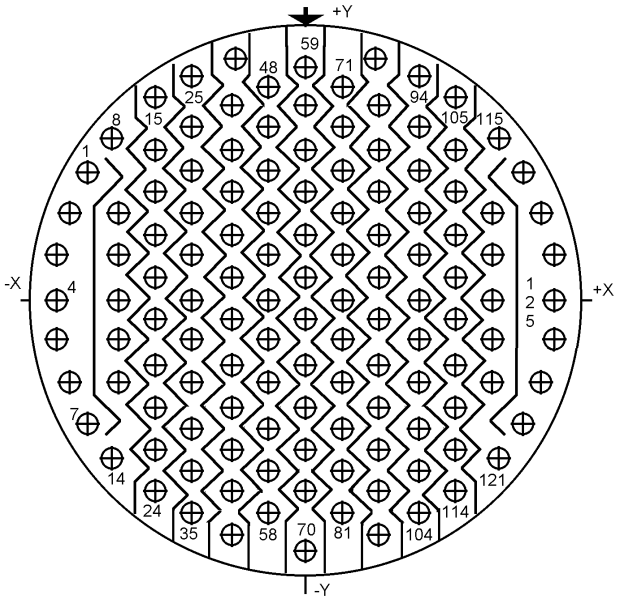
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 23-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
W	+.112 (2.84)	+.325 (8.26)	AA	-.112 (2.84)	+.195 (4.95)
X	+.225 (5.72)	+.260 (6.60)	BB	+.000 (0.00)	+.130 (3.30)
Y	+.336 (8.53)	+.195 (4.95)	CC	+.112 (2.84)	+.065 (1.65)
Z	+.336 (8.53)	+.065 (1.65)	DD	+.112 (2.84)	-.065 (1.65)
a	+.336 (8.53)	-.065 (1.65)	EE	+.000 (0.00)	-.130 (3.30)
b	+.336 (8.53)	-.195 (4.95)	FF	-.112 (2.84)	-.065 (1.65)
c	+.225 (5.72)	-.260 (6.60)	GG	-.112 (2.84)	+.065 (1.65)
d	+.112 (2.84)	-.325 (8.26)	HH	+.000 (0.00)	+.000 (0.00)
e	+.000 (0.00)	-.390 (9.91)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
23	-55	55	20	I	All	----

FIGURE 8. Insert arrangement, electrical connector, shell size 23 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.479 (12.17)	+.279 (7.09)	65	+.000 (0.00)	-.047 (1.19)
2	-.520 (13.21)	+.190 (4.83)	66	+.000 (0.00)	-.142 (3.61)
3	-.546 (13.87)	+.095 (2.41)	67	+.000 (0.00)	-.237 (6.02)
4	-.555 (14.10)	+.000 (0.00)	68	+.000 (0.00)	-.332 (8.43)
5	-.546 (13.87)	-.095 (2.41)	69	+.000 (0.00)	-.427 (10.85)
6	-.520 (13.21)	-.190 (4.83)	70	+.000 (0.00)	-.555 (14.10)
7	-.479 (12.17)	-.279 (7.09)	71	+.083 (2.11)	+.475 (12.07)
8	-.424 (10.77)	+.357 (9.07)	72	+.083 (2.11)	+.380 (9.65)
9	-.415 (10.54)	+.190 (4.83)	73	+.083 (2.11)	+.285 (7.24)
10	-.415 (10.54)	+.095 (2.41)	74	+.083 (2.11)	+.190 (4.83)
11	-.415 (10.54)	+.000 (0.00)	75	+.083 (2.11)	+.095 (2.41)
12	-.415 (10.54)	-.095 (2.41)	76	+.083 (2.11)	+.000 (0.00)
13	-.415 (10.54)	-.190 (4.83)	77	+.083 (2.11)	-.095 (2.41)
14	-.424 (10.77)	-.357 (9.07)	78	+.083 (2.11)	-.190 (4.83)
15	-.332 (8.43)	+.444 (11.28)	79	+.083 (2.11)	-.285 (7.24)
16	-.332 (8.43)	+.332 (8.43)	80	+.083 (2.11)	-.380 (9.65)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.332 (8.43)	+.237 (6.02)	81	+.083 (2.11)	-.475 (12.07)
18	-.332 (8.43)	+.142 (3.61)	82	+.160 (4.06)	+.531 (13.49)
19	-.332 (8.43)	+.047 (1.19)	83	+.166 (4.22)	+.427 (10.85)
20	-.332 (8.43)	-.047 (1.19)	84	+.166 (4.22)	+.332 (8.43)
21	-.332 (8.43)	-.142 (3.61)	85	+.166 (4.22)	+.237 (6.02)
22	-.332 (8.43)	-.237 (6.02)	86	+.166 (4.22)	+.142 (3.61)
23	-.332 (8.43)	-.332 (8.43)	87	+.166 (4.22)	+.047 (1.19)
24	-.332 (8.43)	-.427 (10.85)	88	+.166 (4.22)	-.047 (1.19)
25	-.249 (6.32)	+.496 (12.60)	89	+.166 (4.22)	-.142 (3.61)
26	-.249 (6.32)	+.380 (9.65)	90	+.166 (4.22)	-.237 (6.02)
27	-.249 (6.32)	+.285 (7.24)	91	+.166 (4.22)	-.332 (8.43)
28	-.249 (6.32)	+.190 (4.83)	92	+.166 (4.22)	-.427 (10.85)
29	-.249 (6.32)	+.095 (2.41)	93	+.166 (4.22)	-.522 (13.26)
30	-.249 (6.32)	+.000 (0.00)	94	+.249 (6.32)	+.496 (12.60)
31	-.249 (6.32)	-.095 (2.41)	95	+.249 (6.32)	+.380 (9.65)
32	-.249 (6.32)	-.190 (4.83)	96	+.249 (6.32)	+.285 (7.24)
33	-.249 (6.32)	-.285 (7.24)	97	+.249 (6.32)	+.190 (4.83)
34	-.249 (6.32)	-.380 (9.65)	98	+.249 (6.32)	+.095 (2.41)
35	-.249 (6.32)	-.475 (12.07)	99	+.249 (6.32)	+.000 (0.00)
36	-.160 (4.06)	+.531 (13.49)	100	+.249 (6.32)	-.095 (2.41)
37	-.166 (4.22)	+.427 (10.85)	101	+.249 (6.32)	-.190 (4.83)
38	-.166 (4.22)	+.332 (8.43)	102	+.249 (6.32)	-.285 (7.24)
39	-.166 (4.22)	+.237 (6.02)	103	+.249 (6.32)	-.380 (9.65)
40	-.166 (4.22)	+.142 (3.61)	104	+.249 (6.32)	-.475 (12.07)
41	-.166 (4.22)	+.047 (1.19)	105	+.332 (8.43)	+.444 (11.28)
42	-.166 (4.22)	-.047 (1.19)	106	+.332 (8.43)	+.332 (8.43)
43	-.166 (4.22)	-.142 (3.61)	107	+.332 (8.43)	+.237 (6.02)
44	-.166 (4.22)	-.237 (6.02)	108	+.332 (8.43)	+.142 (3.61)
45	-.166 (4.22)	-.332 (8.43)	109	+.332 (8.43)	+.047 (1.19)
46	-.166 (4.22)	-.427 (10.85)	110	+.332 (8.43)	-.047 (1.19)
47	-.166 (4.22)	-.522 (13.26)	111	+.332 (8.43)	-.142 (3.61)
48	-.083 (2.11)	+.475 (12.07)	112	+.332 (8.43)	-.237 (6.02)
49	-.083 (2.11)	+.380 (9.65)	113	+.332 (8.43)	-.332 (8.43)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

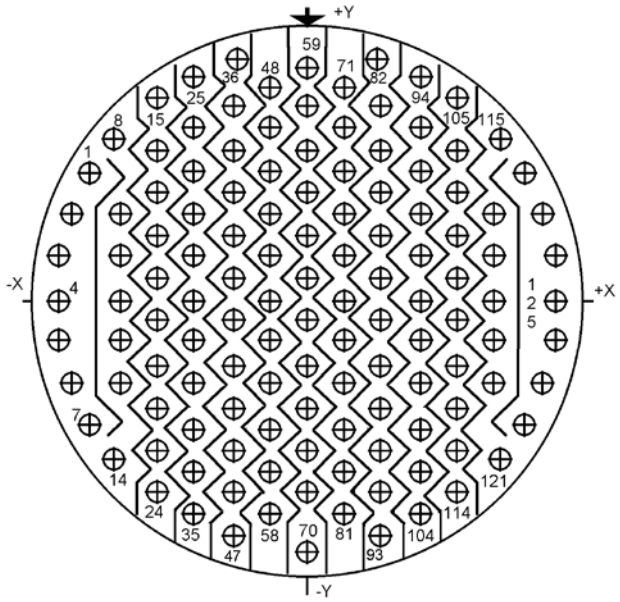
Contacts (Insert arrangement 25-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
50	-.083 (2.11)	+.285 (7.24)	114	+.332 (8.43)	-.427 (10.85)
51	-.083 (2.11)	+.190 (4.83)	115	+.424 (10.77)	+.357 (9.07)
52	-.083 (2.11)	+.095 (2.41)	116	+.415 (10.54)	+.190 (4.83)
53	-.083 (2.11)	+.000 (0.00)	117	+.415 (10.54)	+.095 (2.41)
54	-.083 (2.11)	-.095 (2.41)	118	+.415 (10.54)	+.000 (0.00)
55	-.083 (2.11)	-.190 (4.83)	119	+.415 (10.54)	-.095 (2.41)
56	-.083 (2.11)	-.285 (7.24)	120	+.415 (10.54)	-.190 (4.83)
57	-.083 (2.11)	-.380 (9.65)	121	+.424 (10.77)	-.357 (9.07)
58	-.083 (2.11)	-.475 (12.07)	122	+.479 (12.17)	+.279 (7.09)
59	+.000 (0.00)	+.522 (13.26)	123	+.520 (13.21)	+.190 (4.83)
60	+.000 (0.00)	+.427 (10.85)	124	+.546 (13.87)	+.095 (2.41)
61	+.000 (0.00)	+.332 (8.43)	125	+.555 (14.10)	+.000 (0.00)
62	+.000 (0.00)	+.237 (6.02)	126	+.546 (13.87)	-.095 (2.41)
63	+.000 (0.00)	+.142 (3.61)	127	+.520 (13.21)	-.190 (4.83)
64	+.000 (0.00)	+.047 (1.19)	128	+.479 (12.17)	-.279 (7.09)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-1	128	22M	M	All	MS20057-1

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.479 (12.17)	+.279 (7.09)	65	+.000 (0.00)	-.047 (1.19)
2	-.520 (13.21)	+.190 (4.83)	66	+.000 (0.00)	-.142 (3.61)
3	-.546 (13.87)	+.095 (2.41)	67	+.000 (0.00)	-.237 (6.02)
4	-.555 (14.10)	+.000 (0.00)	68	+.000 (0.00)	-.332 (8.43)
5	-.546 (13.87)	-.095 (2.41)	69	+.000 (0.00)	-.427 (10.85)
6	-.520 (13.21)	-.190 (4.83)	70	+.000 (0.00)	-.555 (14.10)
7	-.479 (12.17)	-.279 (7.09)	71	+.083 (2.11)	+.475 (12.07)
8	-.424 (10.77)	+.357 (9.07)	72	+.083 (2.11)	+.380 (9.65)
9	-.415 (10.54)	+.190 (4.83)	73	+.083 (2.11)	+.285 (7.24)
10	-.415 (10.54)	+.095 (2.41)	74	+.083 (2.11)	+.190 (4.83)
11	-.415 (10.54)	.000 (0.00)	75	+.083 (2.11)	+.095 (2.41)
12	-.415 (10.54)	-.095 (2.41)	76	+.083 (2.11)	+.000 (0.00)
13	-.415 (10.54)	-.190 (4.83)	77	+.083 (2.11)	-.095 (2.41)
14	-.424 (10.77)	-.357 (9.07)	78	+.083 (2.11)	-.190 (4.83)
15	-.332 (8.43)	+.444 (11.28)	79	+.083 (2.11)	-.285 (7.24)
16	-.332 (8.43)	+.332 (8.43)	80	+.083 (2.11)	-.380 (9.65)
17	-.332 (8.43)	+.237 (6.02)	81	+.083 (2.11)	-.475 (12.07)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.332 (8.43)	+.142 (3.61)	82	+.160 (4.06)	+.531 (13.49)
19	-.332 (8.43)	+.047 (1.19)	83	+.166 (4.22)	+.427 (10.85)
20	-.332 (8.43)	-.047 (1.19)	84	+.166 (4.22)	+.332 (8.43)
21	-.332 (8.43)	-.142 (3.61)	85	+.166 (4.22)	+.237 (6.02)
22	-.332 (8.43)	-.237 (6.02)	86	+.166 (4.22)	+.142 (3.61)
23	-.332 (8.43)	-.332 (8.43)	87	+.166 (4.22)	+.047 (1.19)
24	-.332 (8.43)	-.427 (10.85)	88	+.166 (4.22)	-.047 (1.19)
25	-.249 (6.32)	+.496 (12.60)	89	+.166 (4.22)	-.142 (3.61)
26	-.249 (6.32)	+.380 (9.65)	90	+.166 (4.22)	-.237 (6.02)
27	-.249 (6.32)	+.285 (7.24)	91	+.166 (4.22)	-.332 (8.43)
28	-.249 (6.32)	+.190 (4.83)	92	+.166 (4.22)	-.427 (10.85)
29	-.249 (6.32)	+.095 (2.41)	93	+.166 (4.22)	-.522 (13.26)
30	-.249 (6.32)	+.000 (0.00)	94	+.249 (6.32)	+.496 (12.60)
31	-.249 (6.32)	-.095 (2.41)	95	+.249 (6.32)	+.380 (9.65)
32	-.249 (6.32)	-.190 (4.83)	96	+.249 (6.32)	+.285 (7.24)
33	-.249 (6.32)	-.285 (7.24)	97	+.249 (6.32)	+.190 (4.83)
34	-.249 (6.32)	-.380 (9.65)	98	+.249 (6.32)	+.095 (2.41)
35	-.249 (6.32)	-.475 (12.07)	99	+.249 (6.32)	+.000 (0.00)
36	-.160 (4.06)	+.531 (13.49)	100	+.249 (6.32)	-.095 (2.41)
37	-.166 (4.22)	+.427 (10.85)	101	+.249 (6.32)	-.190 (4.83)
38	-.166 (4.22)	+.332 (8.43)	102	+.249 (6.32)	-.285 (7.24)
39	-.166 (4.22)	+.237 (6.02)	103	+.249 (6.32)	-.380 (9.65)
40	-.166 (4.22)	+.142 (3.61)	104	+.249 (6.32)	-.475 (12.07)
41	-.166 (4.22)	+.047 (1.19)	105	+.332 (8.43)	+.444 (11.28)
42	-.166 (4.22)	-.047 (1.19)	106	+.332 (8.43)	+.332 (8.43)
43	-.166 (4.22)	-.142 (3.61)	107	+.332 (8.43)	+.237 (6.02)
44	-.166 (4.22)	-.237 (6.02)	108	+.332 (8.43)	+.142 (3.61)
45	-.166 (4.22)	-.332 (8.43)	109	+.332 (8.43)	+.047 (1.19)
46	-.166 (4.22)	-.427 (10.85)	110	+.332 (8.43)	-.047 (1.19)
47	-.166 (4.22)	-.522 (13.26)	111	+.332 (8.43)	-.142 (3.61)
48	-.083 (2.11)	+.475 (12.07)	112	+.332 (8.43)	-.237 (6.02)
49	-.083 (2.11)	+.380 (9.65)	113	+.332 (8.43)	-.332 (8.43)
50	-.083 (2.11)	+.285 (7.24)	114	+.332 (8.43)	-.427 (10.85)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

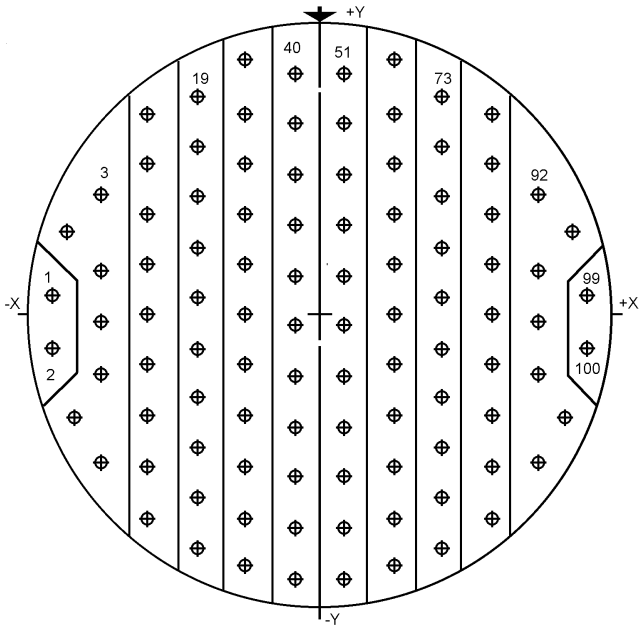
Contacts (Insert arrangement 25-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
51	-.083 (2.11)	+.190 (4.83)	115	+.424 (10.77)	+.357 (9.07)
52	-.083 (2.11)	+.095 (2.41)	116	+.415 (10.54)	+.190 (4.83)
53	-.083 (2.11)	+.000 (0.00)	117	+.415 (10.54)	+.095 (2.41)
54	-.083 (2.11)	-.095 (2.41)	118	+.415 (10.54)	+.000 (0.00)
55	-.083 (2.11)	-.190 (4.83)	119	+.415 (10.54)	-.095 (2.41)
56	-.083 (2.11)	-.285 (7.24)	120	+.415 (10.54)	-.190 (4.83)
57	-.083 (2.11)	-.380 (9.65)	121	+.424 (10.77)	-.357 (9.07)
58	-.083 (2.11)	-.475 (12.07)	122	+.479 (12.17)	+.279 (7.09)
59	+.000 (0.00)	+.522 (13.26)	123	+.520 (13.21)	+.190 (4.83)
60	+.000 (0.00)	+.427 (10.85)	124	+.546 (13.87)	+.095 (2.41)
61	+.000 (0.00)	+.332 (8.43)	125	+.555 (14.10)	+.000 (0.00)
62	+.000 (0.00)	+.237 (6.02)	126	+.546 (13.87)	-.095 (2.41)
63	+.000 (0.00)	+.142 (3.61)	127	+.520 (13.21)	-.190 (4.83)
64	+.000 (0.00)	+.047 (1.19)	128	+.479 (12.17)	-.279 (7.09)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-35	128	22D	M	All	MS27533-35

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.550 (13.97)	+.039 (.99)	51	+.053 (1.35)	+.502 (12.75)
2	-.550 (13.97)	-.068 (1.73)	52	+.053 (1.35)	+.397 (10.08)
3	-.459 (11.66)	+.253 (6.43)	53	+.053 (1.35)	+.292 (7.42)
4	-.523 (13.28)	+.175 (4.45)	54	+.053 (1.35)	+.187 (4.75)
5	-.459 (11.66)	+.092 (2.34)	55	+.053 (1.35)	+.082 (2.08)
6	-.459 (11.66)	-.014 (0.36)	56	+.053 (1.35)	-.023 (0.58)
7	-.459 (11.66)	-.122 (3.10)	57	+.053 (1.35)	-.128 (3.25)
8	-.509 (12.93)	-.215 (5.46)	58	+.053 (1.35)	-.233 (5.92)
9	-.459 (11.66)	-.307 (7.80)	59	+.053 (1.35)	-.338 (8.59)
10	-.354 (8.99)	+.420 (10.67)	60	+.053 (1.35)	-.443 (11.25)
11	-.354 (8.99)	+.315 (8.00)	61	+.053 (1.35)	-.548 (13.92)
12	-.354 (8.99)	+.210 (5.33)	62	+.154 (3.91)	+.525 (13.34)
13	-.354 (8.99)	+.105 (2.67)	63	+.154 (3.91)	+.420 (10.67)
14	-.354 (8.99)	+.000 (0.00)	64	+.154 (3.91)	+.315 (8.00)
15	-.354 (8.99)	-.105 (2.67)	65	+.154 (3.91)	+.210 (5.33)
16	-.354 (8.99)	-.210 (5.33)	66	+.154 (3.91)	+.105 (2.67)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.354 (8.99)	-.315 (8.00)	67	+.154 (3.91)	+.000 (0.00)
18	-.354 (8.99)	-.420 (10.67)	68	+.154 (3.91)	-.105 (2.67)
19	-.255 (6.48)	+.457 (11.61)	69	+.154 (3.91)	-.210 (5.33)
20	-.255 (6.48)	+.352 (8.94)	70	+.154 (3.91)	-.315 (8.00)
21	-.255 (6.48)	+.247 (6.27)	71	+.154 (3.91)	-.420 (10.67)
22	-.255 (6.48)	+.142 (3.61)	72	+.154 (3.91)	-.525 (13.34)
23	-.255 (6.48)	+.037 (0.94)	73	+.255 (6.48)	+.457 (11.61)
24	-.255 (6.48)	-.068 (1.73)	74	+.255 (6.48)	+.352 (8.94)
25	-.255 (6.48)	-.173 (4.39)	75	+.255 (6.48)	+.247 (6.27)
26	-.255 (6.48)	-.278 (7.06)	76	+.255 (6.48)	+.142 (3.61)
27	-.255 (6.48)	-.383 (9.73)	77	+.255 (6.48)	+.037 (0.94)
28	-.255 (6.48)	-.488 (12.40)	78	+.255 (6.48)	-.068 (1.73)
29	-.154 (3.91)	+.525 (13.34)	79	+.255 (6.48)	-.173 (4.39)
30	-.154 (3.91)	+.420 (10.67)	80	+.255 (6.48)	-.278 (7.06)
31	-.154 (3.91)	+.315 (8.00)	81	+.255 (6.48)	-.383 (9.73)
32	-.154 (3.91)	+.210 (5.33)	82	+.255 (6.48)	-.488 (12.40)
33	-.154 (3.91)	+.105 (2.67)	83	+.354 (8.99)	+.420 (10.67)
34	-.154 (3.91)	+.000 (0.00)	84	+.354 (8.99)	+.315 (8.00)
35	-.154 (3.91)	-.105 (2.67)	85	+.354 (8.99)	+.210 (5.33)
36	-.154 (3.91)	-.210 (5.33)	86	+.354 (8.99)	+.105 (2.67)
37	-.154 (3.91)	-.315 (8.00)	87	+.354 (8.99)	+.000 (0.00)
38	-.154 (3.91)	-.420 (10.67)	88	+.354 (8.99)	-.105 (2.67)
39	-.154 (3.91)	-.525 (13.34)	89	+.354 (8.99)	-.210 (5.33)
40	-.053 (1.35)	+.502 (12.75)	90	+.354 (8.99)	-.315 (8.00)
41	-.053 (1.35)	+.397 (10.08)	91	+.354 (8.99)	-.420 (10.67)
42	-.053 (1.35)	+.292 (7.42)	92	+.459 (11.66)	+.253 (6.43)
43	-.053 (1.35)	+.187 (4.75)	93	+.523 (13.28)	+.175 (4.45)
44	-.053 (1.35)	+.082 (2.08)	94	+.459 (11.66)	+.092 (2.34)
45	-.053 (1.35)	-.023 (0.58)	95	+.459 (11.66)	-.014 (0.36)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

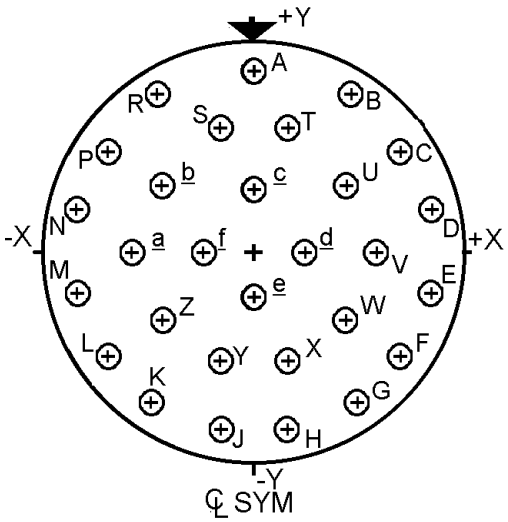
Contacts (Insert arrangement 25-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
46	-.053 (1.35)	-.128 (3.25)	96	+.459 (11.66)	-.122 (3.10)
47	-.053 (1.35)	-.233 (5.92)	97	+.509 (12.93)	-.215 (5.46)
48	-.053 (1.35)	-.338 (8.59)	98	+.459 (11.66)	-.307 (7.80)
49	-.053 (1.35)	-.443 (11.25)	99	+.550 (13.97)	+.039 (0.99)
50	-.053 (1.35)	-.548 (13.92)	100	+.550 (13.97)	-.068 (1.73)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-2	100	22	M	All	MS20057-2

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

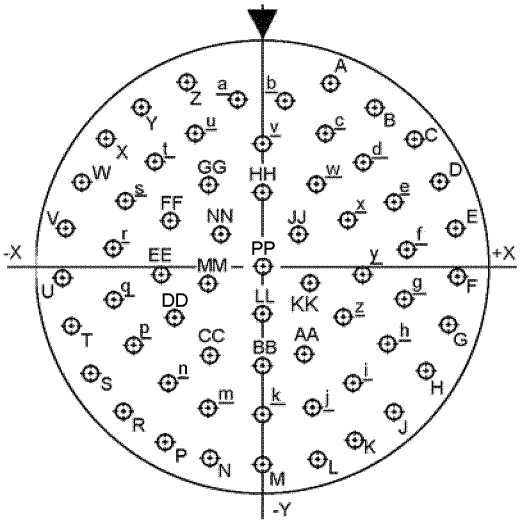


Contacts (Insert arrangement 25-29)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.481 (12.22)	S	-.091 (2.31)	+.321 (8.15)
B	+.258 (6.55)	+.406 (10.31)	T	+.091 (2.31)	+.321 (8.15)
C	+.395 (10.03)	+.277 (7.04)	U	+.228 (5.79)	+.194 (4.93)
D	+.469 (11.91)	+.109 (2.77)	V	+.319 (8.10)	+.000 (0.00)
E	+.469 (11.91)	-.109 (2.77)	W	+.240 (6.10)	-.181 (4.60)
F	+.395 (10.03)	-.277 (7.04)	X	+.091 (2.31)	-.290 (7.37)
G	+.263 (6.68)	-.406 (10.31)	Y	-.091 (2.31)	-.290 (7.37)
H	+.091 (2.31)	-.472 (11.99)	Z	-.240 (6.10)	-.181 (4.60)
J	-.091 (2.31)	-.472 (11.99)	a	-.319 (8.10)	+.000 (0.00)
K	-.263 (6.68)	-.406 (10.31)	b	-.228 (5.79)	+.194 (4.93)
L	-.395 (10.03)	-.277 (7.04)	c	+.000 (0.00)	+.161 (4.09)
M	-.469 (11.91)	-.109 (2.77)	d	+.134 (3.40)	+.000 (0.00)
N	-.469 (11.91)	+.109 (2.77)	e	+.000 (0.00)	-.130 (3.30)
P	-.395 (10.03)	+.277 (7.04)	f	-.134 (3.40)	+.000 (0.00)
R	-.258 (6.55)	+.406 (10.31)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-29	29	16	I	All	MS20057-29

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-61)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.196 (4.98)	+.500 (12.70)	i	+.251 (6.38)	-.314 (7.98)
B	+.314 (7.98)	+.435 (11.05)	j	+.133 (3.38)	-.379 (9.63)
C	+.413 (10.49)	+.343 (8.71)	k	+.000 (0.00)	-.402 (10.21)
D	+.485 (12.32)	+.230 (5.84)	m	-.133 (3.38)	-.379 (9.63)
E	+.527 (13.39)	+.101 (2.57)	n	-.251 (6.38)	-.314 (7.98)
F	+.536 (13.61)	-.030 (0.76)	p	-.341 (8.66)	-.213 (5.41)
G	+.511 (12.98)	-.164 (4.17)	q	-.392 (9.96)	-.088 (2.24)
H	+.454 (11.53)	-.287 (7.29)	r	-.399 (10.13)	+.046 (1.17)
J	+.368 (9.35)	-.391 (9.93)	s	-.362 (9.19)	+.175 (4.45)
K	+.259 (6.58)	-.470 (11.94)	t	-.285 (7.24)	+.283 (7.19)
L	+.134 (3.40)	-.519 (13.18)	u	-.173 (4.39)	+.363 (9.22)
M	+.000 (0.00)	-.537 (13.64)	v	+.000 (0.00)	+.338 (8.59)
N	-.134 (3.40)	-.519 (13.18)	w	+.147 (3.73)	+.223 (5.66)
P	-.259 (6.58)	-.470 (11.94)	x	+.237 (6.02)	+.122 (3.10)
R	-.368 (9.35)	-.391 (9.93)	y	+.267 (6.78)	-.010 (0.25)
S	-.454 (11.53)	-.287 (7.29)	z	+.228 (5.79)	-.139 (3.53)
T	-.511 (12.98)	-.164 (4.17)	AA	+.131 (3.33)	-.233 (5.92)
U	-.536 (13.61)	-.030 (0.76)	BB	+.000 (0.00)	-.267 (6.78)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

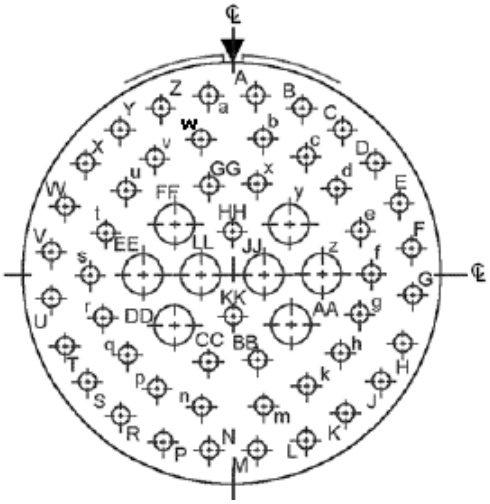
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-61)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
V	-.527 (13.39)	+.101 (2.57)	CC	-.131 (3.33)	-.233 (5.92)
W	-.485 (12.32)	+.230 (5.84)	DD	-.228 (5.79)	-.139 (3.53)
X	-.413 (10.49)	+.343 (8.71)	EE	-.267 (6.78)	-.010 (0.25)
Y	-.314 (7.98)	+.435 (11.05)	FF	-.237 (6.02)	+.122 (3.10)
Z	-.196 (4.98)	+.500 (12.70)	GG	-.147 (3.73)	+.223 (5.66)
a	-.068 (1.73)	+.454 (11.53)	HH	+.000 (0.00)	+.200 (5.08)
b	+.068 (1.73)	+.454 (11.53)	JJ	+.105 (2.67)	+.094 (2.39)
c	+.173 (4.39)	+.363 (9.22)	KK	+.135 (3.43)	-.041 (1.04)
d	+.285 (7.24)	+.283 (7.19)	LL	+.000 (0.00)	-.132 (3.35)
e	+.362 (9.19)	+.175 (4.45)	MM	-.135 (3.43)	-.041 (1.04)
f	+.399 (10.13)	+.046 (1.17)	NN	-.105 (2.67)	+.094 (2.39)
g	+.392 (9.96)	-.088 (2.24)	PP	+.000 (0.00)	+.000 (0.00)
h	+.341 (8.66)	-.213 (5.41)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-61	61	20	I	All	MS20057-61

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-4)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.069 (1.75)	+.531 (13.49)	f	+.412 (10.46)	+.000 (0.00)
B	+.203 (5.16)	+.495 (12.57)	g	+.377 (9.58)	-.132 (3.35)
C	+.324 (8.23)	+.425 (10.80)	h	+.311 (7.90)	-.251 (6.38)
D	+.424 (10.77)	+.326 (8.28)	k	+.212 (5.38)	-.344 (8.74)
E	+.493 (12.52)	+.205 (5.21)	m	+.086 (2.18)	-.397 (10.08)
F	+.531 (13.49)	+.069 (1.75)	n	-.086 (2.18)	-.397 (10.08)
G	+.531 (13.49)	-.069 (1.75)	p	-.212 (5.38)	-.344 (8.74)
H	+.493 (12.52)	-.205 (5.21)	q	-.311 (7.90)	-.251 (6.38)
J	+.424 (10.77)	-.326 (8.28)	r	-.377 (9.58)	-.132 (3.35)
K	+.324 (8.23)	-.425 (10.80)	s	-.412 (10.46)	+.000 (0.00)
L	+.203 (5.16)	-.495 (12.57)	t	-.377 (9.58)	+.132 (3.35)
M	+.069 (1.75)	-.531 (13.49)	u	-.311 (7.90)	+.251 (6.38)
N	-.069 (1.75)	-.531 (13.49)	v	-.212 (5.38)	+.344 (8.74)
P	-.203 (5.16)	-.495 (12.57)	w	-.086 (2.18)	+.397 (10.08)
R	-.324 (8.23)	-.425 (10.80)	x	+.069 (1.75)	+.263 (6.68)
S	-.424 (10.77)	-.326 (8.28)	y	+.172 (4.37)	+.149 (3.78)
T	-.493 (12.52)	-.205 (5.21)	z	+.258 (6.55)	+.000 (0.00)
U	-.531 (13.49)	-.069 (1.75)	AA	+.172 (4.37)	-.149 (3.78)
V	-.531 (13.49)	+.069 (1.75)	BB	+.069 (1.75)	-.263 (6.68)
W	-.493 (12.52)	+.205 (5.21)	CC	-.069 (1.75)	-.263 (6.68)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

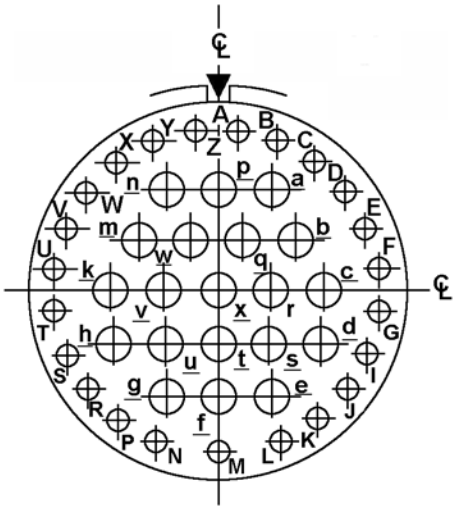
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-4)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
X	-.424 (10.77)	+.326 (8.28)	DD	-.172 (4.37)	-.149 (3.78)
Y	-.324 (8.23)	+.425 (10.80)	EE	-.258 (6.55)	+.000 (0.00)
Z	-.203 (5.16)	+.495 (12.57)	FF	-.172 (4.37)	+.149 (3.78)
a	-.069 (1.75)	+.531 (13.49)	GG	-.069 (1.75)	+.263 (6.68)
b	+.086 (2.18)	+.397 (10.08)	HH	+.000 (0.00)	+.132 (3.35)
c	+.212 (5.38)	+.344 (8.74)	JJ	+.086 (2.18)	+.000 (0.00)
d	+.311 (7.90)	+.251 (6.38)	KK	+.000 (0.00)	-.132 (3.35)
e	+.377 (9.58)	+.132 (3.35)	LL	-.086 (2.18)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-4	8	16	I	y, z, AA, DD, EE, FF, LL, JJ	MS20057-4
		48	20		All others	

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-43)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.069 (1.75)	+.531 (13.49)	Z	-.069 (1.75)	+.531 (13.49)
B	+.203 (5.16)	+.495 (12.57)	a	+.172 (4.37)	+.344 (8.74)
C	+.324 (8.23)	+.425 (10.80)	b	+.258 (6.55)	+.172 (4.37)
D	+.424 (10.77)	+.326 (8.28)	c	+.344 (8.74)	+.000 (0.00)
E	+.493 (12.52)	+.205 (5.21)	d	+.344 (8.74)	-.172 (4.37)
F	+.531 (13.49)	+.069 (1.75)	e	+.172 (4.37)	-.344 (8.74)
G	+.531 (13.49)	-.069 (1.75)	f	+.000 (0.00)	-.344 (8.74)
H	+.493 (12.52)	-.205 (5.21)	g	-.172 (4.37)	-.344 (8.74)
J	+.424 (10.77)	-.326 (8.28)	h	-.344 (8.74)	-.172 (4.37)
K	+.324 (8.23)	-.425 (10.80)	k	-.344 (8.74)	+.000 (0.00)
L	+.203 (5.16)	-.495 (12.57)	m	-.258 (6.55)	+.172 (4.37)
M	+.000 (0.00)	-.531 (13.49)	n	-.172 (4.37)	+.344 (8.74)
N	-.203 (5.16)	-.495 (12.57)	p	+.000 (0.00)	+.344 (8.74)
P	-.324 (8.23)	-.425 (10.80)	q	+.086 (2.18)	+.172 (4.37)
R	-.424 (10.77)	-.326 (8.28)	r	+.172 (4.37)	+.000 (0.00)
S	-.493 (12.52)	-.205 (5.21)	s	+.172 (4.37)	-.172 (4.37)
T	-.531 (13.49)	-.069 (1.75)	t	+.000 (0.00)	-.172 (4.37)
U	-.531 (13.49)	+.069 (1.75)	u	-.172 (4.37)	-.172 (4.37)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

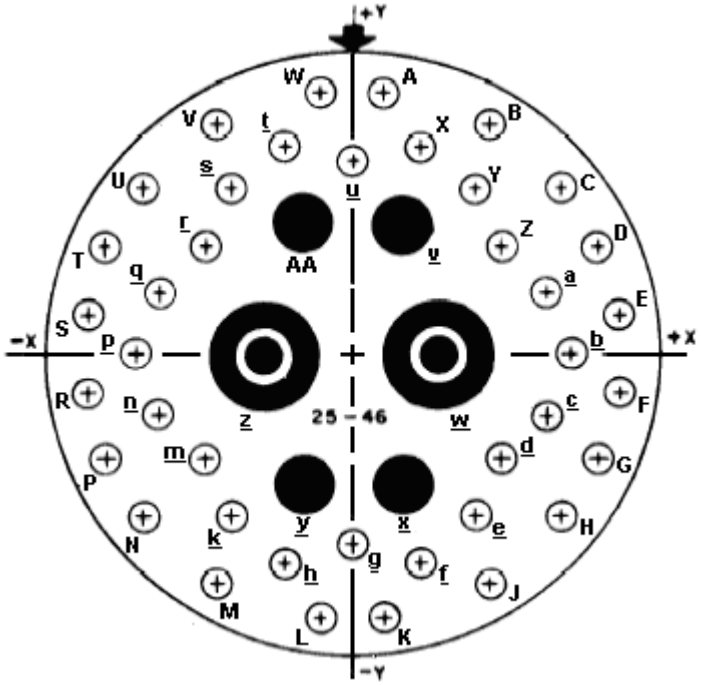
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-43)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
V	-.493 (12.52)	+.205 (5.21)	v	-.172 (4.37)	+.000 (0.00)
W	-.424 (10.77)	+.326 (8.28)	w	-.086 (2.18)	+.172 (4.37)
X	-.324 (8.23)	+.425 (10.80)	x	+.000 (0.00)	+.000 (0.00)
Y	-.203 (5.16)	+.495 (12.57)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-43	20	16	I	a,b,c,d,e,f,g,h,k,m, n,p,q,r,s,t,u,v,w,x	MS20057-43
		23	20		All others	

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-46)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.533 (13.54)	a	+.404 (10.26)	+.125 (3.18)
B	+.275 (6.99)	+.466 (11.84)	b	+.437 (11.10)	+.000 (0.00)
C	+.420 (10.67)	+.337 (8.56)	c	+.404 (10.26)	-.125 (3.18)
D	+.490 (12.45)	+.227 (5.77)	d	+.314 (7.98)	-.221 (5.61)
E	+.531 (13.49)	+.093 (2.36)	e	+.245 (6.22)	-.337 (8.56)
F	+.531 (13.49)	-.093 (2.36)	f	+.136 (3.45)	-.424 (10.77)
G	+.490 (12.45)	-.227 (5.77)	g	+.000 (0.00)	-.395 (10.03)
H	+.420 (10.67)	-.337 (8.56)	h	-.136 (3.45)	-.424 (10.77)
J	+.275 (6.99)	-.466 (11.84)	k	-.245 (6.22)	-.337 (8.56)
K	+.065 (1.65)	-.533 (13.54)	m	-.314 (7.98)	-.221 (5.61)
L	-.065 (1.65)	-.533 (13.54)	n	-.404 (10.26)	-.125 (3.18)
M	-.275 (6.99)	-.466 (11.84)	p	-.437 (11.10)	+.000 (0.00)
N	-.420 (10.67)	-.337 (8.56)	q	-.404 (10.26)	+.125 (3.18)
P	-.490 (12.45)	-.227 (5.77)	r	-.314 (7.98)	+.221 (5.61)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

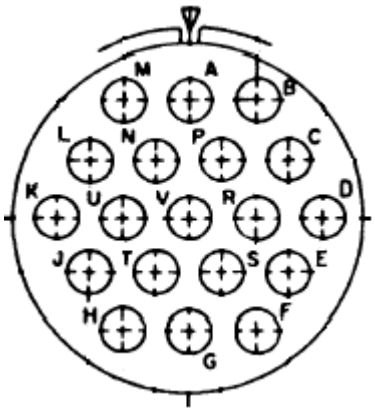
Contacts (Insert arrangement 25-46)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
R	-.531 (13.49)	-.093 (2.36)	s	-.245 (6.22)	+.337 (8.56)
S	-.531 (13.49)	+.093 (2.36)	t	-.136 (3.45)	+.424 (10.77)
T	-.490 (12.45)	+.227 (5.77)	u	+.000 (0.00)	+.395 (10.03)
U	-.420 (10.67)	+.337 (8.56)	v	+.097 (2.46)	+.265 (6.73)
V	-.275 (6.99)	+.466 (11.84)	w	+.180 (4.57)	+.000 (0.00)
W	-.065 (1.65)	+.533 (13.54)	x	+.097 (2.46)	-.265 (6.73)
X	+.136 (3.45)	+.424 (10.77)	y	-.097 (2.46)	-.265 (6.73)
Y	+.245 (6.22)	+.337 (8.56)	z	-.180 (4.57)	+.000 (0.00)
Z	+.314 (7.98)	+.221 (5.61)	AA	-.097 (2.46)	+.265 (6.73)

Shell size	Arrange-ment no.	Number of contacts	Size contacts	Service rating	Contact location	Standard contact	
						Pin	Socket
25	-46	2	No. 8 (see note)	COAX	z,w	M39029/60-367	M39029/59-366
		4	16	I	AA, v,x,y	M39029/58-364	M39029/56-352
		40	20		All others	M39029/58-363	M39029/56-351

NOTE: The SAE-AS39029/59 and /60 size 8 coaxial data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Use insert arrangement 25-90 if an identical insert which uses SAE-AS39029/90 and /91 twinax contacts is required.

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

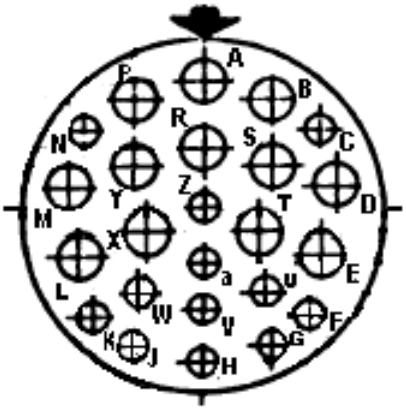


Contacts (Insert arrangement 25-19)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.409 (10.39)	L	-.354 (8.99)	+.205 (5.21)
B	+.236 (5.99)	+.409 (10.39)	M	-.236 (5.99)	+.409 (10.39)
C	+.354 (8.99)	+.205 (5.21)	N	-.118 (3.00)	+.205 (5.21)
D	+.472 (11.99)	+.000 (0.00)	P	+.118 (3.00)	+.205 (5.21)
E	+.354 (8.99)	-.205 (5.21)	R	+.236 (5.99)	+.000 (0.00)
F	+.236 (5.99)	-.409 (10.39)	S	+.118 (3.00)	-.205 (5.21)
G	+.000 (.00)	-.409 (10.39)	T	-.118 (3.00)	-.205 (5.21)
H	-.236 (5.99)	-.409 (10.39)	U	-.236 (5.99)	+.000 (0.00)
J	-.354 (8.99)	-.205 (5.21)	V	+.000 (0.00)	+.000 (0.00)
K	-.472 (11.99)	+.000 (0.00)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-19	19	12	I	All	MS20057-19

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

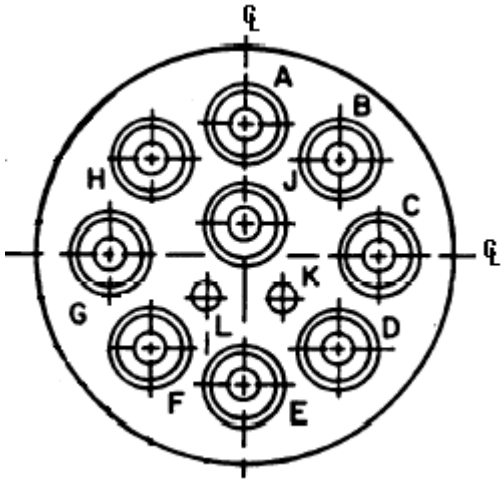


Contacts (Insert arrangement 25-24)					
Contact position ID	Location		Contact position ID	Location	
	X-axis	Y-axis		X-axis	Y-axis
A	+.000 (0.00)	+.472 (11.99)	N	-.403 (10.24)	+.298 (7.57)
B	+.230 (5.84)	+.410 (10.41)	P	-.230 (5.84)	+.410 (10.41)
C	+.403 (10.24)	+.298 (7.57)	R	+.000 (0.00)	+.234 (5.94)
D	+.461 (11.71)	+.100 (2.54)	S	+.230 (5.84)	+.172 (4.37)
E	+.413 (10.49)	-.134 (3.40)	T	+.186 (4.72)	-.062 (1.57)
F	+.370 (9.40)	-.336 (8.53)	U	+.211 (5.36)	-.267 (6.78)
G	+.230 (5.84)	-.441 (11.20)	V	+.000 (0.00)	-.323 (8.20)
H	+.000 (0.00)	-.495 (12.57)	W	-.211 (5.36)	-.267 (6.78)
J	-.230 (5.84)	-.441 (11.20)	X	-.186 (4.72)	-.062 (1.57)
K	-.370 (9.40)	-.336 (8.53)	Y	-.230 (5.84)	+.172 (4.37)
L	-.413 (10.49)	-.134 (3.40)	Z	+.000 (0.00)	+.028 (0.71)
M	-.461 (11.71)	-.100 (2.54)	a	+.000 (0.00)	-.151 (3.84)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
25	-24	12	12	I	A,B,D,E,LM P,R,S,T,X,Y
		12	16		All others

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



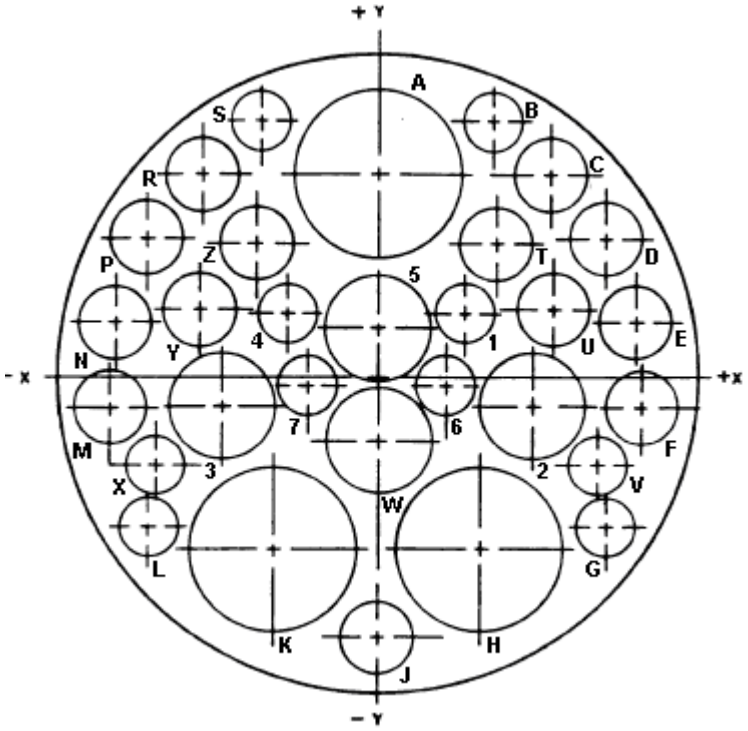
Contacts (Insert arrangement 25-11)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.431 (10.95)	G	-.431 (10.95)	+.000 (0.00)
B	+.304 (7.72)	+.304 (7.72)	H	-.304 (7.72)	+.304 (7.72)
C	+.431 (10.95)	+.000 (0.00)	J	+.000 (0.00)	+.102 (2.59)
D	+.304 (7.72)	-.304 (7.72)	K	+.092 (2.34)	-.160 (4.06)
E	+.000 (0.00)	-.431 (10.95)	L	-.092 (2.34)	-.160 (4.06)
F	-.304 (7.72)	-.304 (7.72)			

NOTES:
1. All dimensions are the true position.
2. Front face of pin insert shown.

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
25	-11	9	10	N	A,B,C,D,E,F, G,H,J
		2	20		L,K

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-20)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.407 (10.34)	S	-.234 (5.94)	+.511 (12.98)
B	+.234 (5.94)	+.511 (12.98)	T	+.243 (6.17)	+.270 (6.86)
C	+.352 (8.94)	+.411 (10.44)	U	+.357 (9.07)	+.136 (3.45)
D	+.466 (11.84)	+.275 (6.99)	V	+.450 (11.43)	-.183 (4.65)
E	+.530 (13.46)	+.111 (2.82)	w	+.000 (0.00)	-.131 (3.33)
F	+.537 (13.64)	-.060 (1.52)	X	-.450 (11.43)	-.183 (4.65)
G	+.467 (11.86)	-.312 (7.92)	Y	-.357 (9.07)	+.136 (3.45)
H	+.208 (5.28)	-.353 (8.97)	Z	-.243 (6.17)	+.270 (6.86)
J	+.000 (0.00)	-.541 (13.74)	1	+.195 (4.95)	+.115 (2.92)
K	-.208 (5.28)	-.353 (8.97)	2	+.317 (8.05)	-.061 (1.55)
L	-.467 (11.86)	-.312 (7.92)	3	-.317 (8.05)	-.061 (1.55)
M	-.537 (13.64)	-.060 (1.52)	4	-.195 (4.95)	+.115 (2.92)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

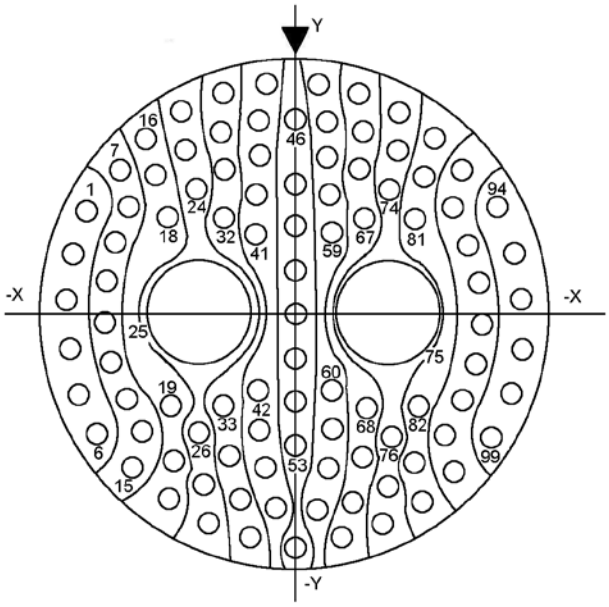
Contacts (Insert arrangement 25-20)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
N	-.530 (13.46)	+.111 (2.82)	5	+.000 (0.00)	+.099 (2.51)
P	-.466 (11.84)	+.275 (6.99)	6	+.141 (3.58)	-.018 (0.46)
R	-.352 (8.94)	+.411 (10.44)	7	-.141 (3.58)	-.018 (0.46)

Shell size	Arrange-ment no.	Number of contacts	Size contacts	Service rating	Contact location	Standard contact	
						Pin	Socket
25	-20	3	8	TWINAX	A,H,K	M39029/90-529	M39029/91-530
		4	12	COAX	2 ,3	M39029/28-211	M39029/75-416
					W, 5	M39029/102-558	M39029/103-559
		13	16	N	C,D,E,F, J,M,N,P, R,T,U,Y, Z	M39029/58-364	M39029/56-352
		10	20		B,G,L,S, V,X,1,4,6 ,7	M39029/58-363	M39029/56-351

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 ºC (347 ºF) maximum.

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-7)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.494 (12.55)	+.242 (6.15)	51	+.000 (0.00)	-.106 (2.69)
2	-.533 (13.54)	+.138 (3.51)	52	+.000 (0.00)	-.212 (5.38)
3	-.550 (13.97)	+.028 (0.71)	53	+.000 (0.00)	-.310 (7.87)
4	-.544 (13.82)	-.083 (2.11)	54	+.000 (0.00)	-.551 (14.00)
5	-.516 (13.11)	-.191 (4.85)	55	+.056 (1.42)	+.548 (13.92)
6	-.467 (11.86)	-.292 (7.42)	56	+.095 (2.41)	+.461 (11.71)
7	-.435 (11.05)	+.337 (8.56)	57	+.068 (1.73)	+.370 (9.40)
8	-.399 (10.13)	+.249 (6.32)	58	+.092 (2.34)	+.278 (7.06)
9	-.441 (11.20)	+.163 (4.14)	59	+.095 (2.41)	+.183 (4.65)
10	-.465 (11.81)	+.071 (1.80)	60	+.089 (2.26)	-.178 (4.52)
11	-.470 (11.94)	-.024 (0.61)	61	+.094 (2.39)	-.277 (7.04)
12	-.456 (11.58)	-.118 (3.00)	62	+.069 (1.75)	-.376 (9.55)
13	-.423 (10.74)	-.207 (5.26)	63	+.048 (1.22)	-.468 (11.89)
14	-.372 (9.45)	-.288 (7.32)	64	+.165 (4.19)	+.525 (13.34)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-7)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
15	-.399 (10.13)	-.379 (9.63)	65	+.186 (4.72)	+.433 (11.00)
16	-.359 (9.12)	+.418 (10.62)	66	+.164 (4.17)	+.340 (8.64)
17	-.341 (8.66)	+.324 (8.23)	67	+.181 (4.60)	+.225 (5.72)
18	-.308 (7.82)	+.222 (5.64)	68	+.172 (4.37)	-.223 (5.66)
19	-.303 (7.70)	-.223 (5.66)	69	+.159 (4.04)	-.347 (8.81)
20	-.307 (7.80)	-.357 (9.07)	70	+.141 (3.58)	-.449 (11.40)
21	-.314 (7.98)	-.452 (11.48)	71	+.111 (2.82)	-.539 (13.69)
22	-.267 (6.78)	+.481 (12.22)	72	+.267 (6.78)	+.481 (12.22)
23	-.269 (6.83)	+.386 (9.80)	73	+.269 (6.83)	+.386 (9.80)
24	-.247 (6.27)	+.294 (7.47)	74	+.247 (6.27)	+.294 (7.47)
25	-.238 (6.05)	+.000 (0.00)	75	+.238 (6.05)	+.000 (0.00)
26	-.237 (6.02)	-.292 (7.42)	76	+.237 (6.02)	-.292 (7.42)
27	-.228 (5.79)	-.412 (10.46)	77	+.228 (5.79)	-.412 (10.46)
28	-.217 (5.51)	-.506 (12.85)	78	+.217 (5.51)	-.506 (12.85)
29	-.165 (4.19)	+.525 (13.34)	79	+.359 (9.12)	+.418 (10.62)
30	-.186 (4.72)	+.433 (11.00)	80	+.341 (8.66)	+.324 (8.23)
31	-.164 (4.17)	+.340 (8.64)	81	+.308 (7.82)	+.222 (5.64)
32	-.181 (4.60)	+.225 (5.72)	82	+.303 (7.70)	-.223 (5.66)
33	-.172 (4.37)	-.223 (5.66)	83	+.307 (7.80)	-.357 (9.07)
34	-.159 (4.04)	-.347 (8.81)	84	+.314 (7.98)	-.452 (11.48)
35	-.141 (3.58)	-.449 (11.40)	85	+.435 (11.05)	+.337 (8.56)
36	-.111 (2.82)	-.539 (13.69)	86	+.399 (10.13)	+.249 (6.32)
37	-.056 (1.42)	+.548 (13.92)	87	+.441 (11.20)	+.163 (4.14)
38	-.095 (2.41)	+.461 (11.71)	88	+.465 (11.81)	+.071 (1.80)
39	-.068 (1.73)	+.370 (9.40)	89	+.470 (11.94)	-.024 (.61)
40	-.092 (2.34)	+.278 (7.06)	90	+.456 (11.58)	-.118 (3.00)
41	-.095 (2.41)	+.183 (4.65)	91	+.423 (10.74)	-.207 (5.26)
42	-.089 (2.26)	-.178 (4.52)	92	+.372 (9.45)	-.288 (7.32)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

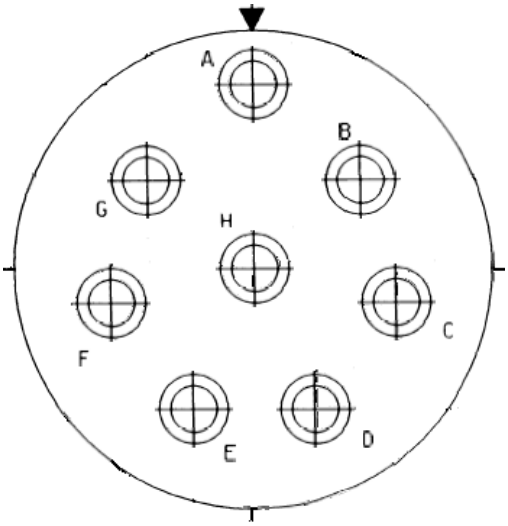
Contacts (Insert arrangement 25-7)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
43	-.094 (2.39)	-.277 (7.04)	93	+.399 (10.13)	-.379 (9.63)
44	-.069 (1.75)	-.376 (9.55)	94	+.494 (12.55)	+.242 (6.15)
45	-.048 (1.22)	-.468 (11.89)	95	+.533 (13.54)	+.138 (3.51)
46	+.000 (0.00)	+.471 (11.96)	96	+.550 (13.97)	+.028 (0.71)
47	+.000 (0.00)	+.303 (7.70)	97	+.544 (13.82)	-.083 (2.11)
48	+.000 (0.00)	+.208 (5.28)	98	+.516 (13.11)	-.191 (4.85)
49	+.000 (0.00)	+.104 (2.64)	99	+.467 (11.86)	-.292 (7.42)
50	+.000 (0.00)	+.000 (0.00)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Standard contact	
						Pin	Socket
25	-7	2	8 (See note)	Twinax	25, 75	M39029/90-529	M39029/91-530
		97	22D	M	All others	M39029/58-360	M39029/56-348

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 °C (347 °F) maximum.

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



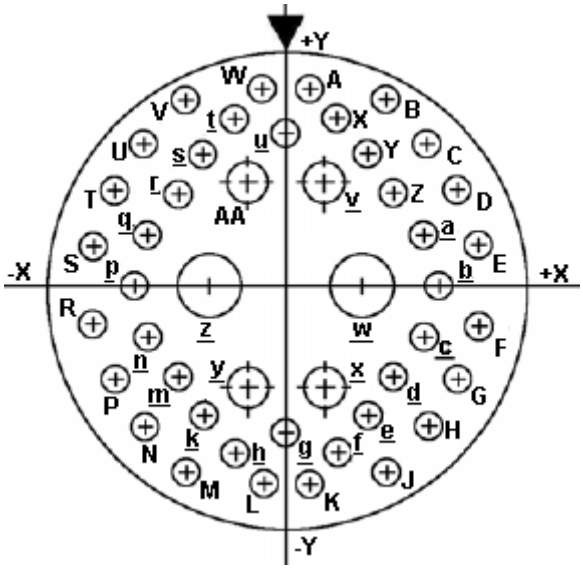
Contacts (Insert arrangement 25-8)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.426 (10.82)
B	+.333 (8.46)	+.266 (6.76)
C	+.415 (10.54)	-.095 (2.41)
D	+.185 (4.70)	-.384 (9.75)
E	-.185 (4.70)	-.384 (9.75)
F	-.415 (10.54)	-.095 (2.41)
G	-.333 (8.46)	+.266 (6.76)
H	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Standard contact	
						Pin	Socket
25	-8	8	8	Twinax	All	M39029/90-529	M39029/91-530

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 °C (347 °F) maximum.

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-90)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.533 (13.54)	a	+.404 (10.26)	+.125 (3.18)
B	+.275 (6.99)	+.466 (11.84)	b	+.437 (11.10)	+.000 (0.00)
C	+.420 (10.67)	+.337 (8.56)	c	+.404 (10.26)	-.125 (3.18)
D	+.490 (12.45)	+.227 (5.77)	d	+.314 (7.98)	-.221 (5.61)
E	+.531 (13.49)	+.093 (2.36)	e	+.245 (6.22)	-.337 (8.56)
F	+.531 (13.49)	-.093 (2.36)	f	+.136 (3.45)	-.424 (10.77)
G	+.490 (12.45)	-.227 (5.77)	g	+.000 (0.00)	-.395 (10.03)
H	+.420 (10.67)	-.337 (8.56)	h	-.136 (3.45)	-.424 (10.77)
J	+.275 (6.99)	-.466 (11.84)	k	-.245 (6.22)	-.337 (8.56)
K	+.065 (1.65)	-.533 (13.54)	m	-.314 (7.98)	-.221 (5.61)
L	-.065 (1.65)	-.533 (13.54)	n	-.404 (10.26)	-.125 (3.18)
M	-.275 (6.99)	-.466 (11.84)	p	-.437 (11.10)	+.000 (0.00)
N	-.420 (10.67)	-.337 (8.56)	q	-.404 (10.26)	+.125 (3.18)
P	-.490 (12.45)	-.227 (5.77)	r	-.314 (7.98)	+.221 (5.61)
R	-.531 (13.49)	-.093 (2.36)	s	-.245 (6.22)	+.337 (8.56)
S	-.531 (13.49)	+.093 (2.36)	t	-.136 (3.45)	+.424 (10.77)
T	-.490 (12.45)	+.227 (5.77)	u	+.000 (0.00)	+.395 (10.03)
U	-.420 (10.67)	+.337 (8.56)	v	+.097 (2.46)	+.265 (6.73)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

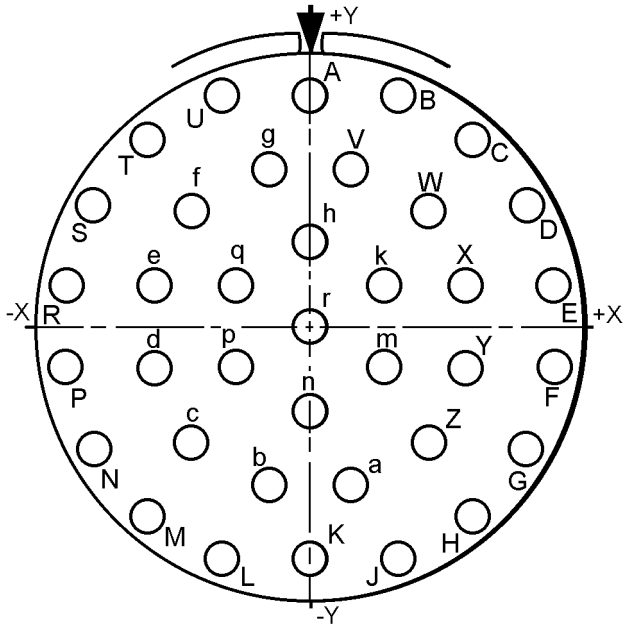
Contacts (Insert arrangement 25-90)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
V	-.275 (6.99)	+.466 (11.84)	w	+.180 (4.57)	+.000 (.00)
W	-.065 (1.65)	+.533 (13.54)	x	+.097 (2.46)	-.265 (6.73)
X	+.136 (3.45)	+.424 (10.77)	y	-.097 (2.46)	-.265 (6.73)
Y	+.245 (6.22)	+.337 (8.56)	z	-.180 (4.57)	+.000 (0.00)
Z	+.314 (7.98)	+.221 (5.61)	AA	-.097 (2.46)	+.265 (6.73)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Standard contact	
						Pin	Socket
25	-90	2	8 (see note)	Twinax	w, z	M39029/90-529	M39029/91-530
		4	16	I	v , x , y , AA	M39029/58-364	M39029/56-352
		40	20		All others	M39029/58-363	M39029/56-351

NOTE: Size 8 SAE-AS39029/90 and /91 twinax data bus contacts are not fully compatible with this insert arrangement and will require the use of a separate boot to seal properly and reduce the splaying of the contact. Heat shrink sleeves used with size 8 twinax contacts are rated to 175 °C (347 °F) maximum.

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 25-37)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (.00)	+.472 (11.99)	W	+.242 (6.15)	+.236 (5.99)
B	+.185 (4.70)	+.472 (11.99)	X	+.326 (8.28)	+.086 (2.18)
C	+.333 (8.46)	+.382 (9.70)	Y	+.326 (8.28)	-.086 (2.18)
D	+.441 (11.20)	+.249 (6.32)	Z	+.242 (6.15)	-.236 (5.99)
E	+.500 (12.70)	+.086 (2.18)	a	+.086 (2.18)	-.320 (8.13)
F	+.500 (12.70)	-.086 (2.18)	b	-.086 (2.18)	-.320 (8.13)
G	+.441 (11.20)	-.249 (6.32)	c	-.242 (6.15)	-.236 (5.99)
H	+.333 (8.46)	+.382 (9.70)	d	-.326 (8.28)	-.086 (2.18)
J	+.186 (4.72)	-.472 (11.99)	e	-.326 (8.28)	+.086 (2.18)
K	+.000 (.00)	-.472 (11.99)	f	-.242 (6.15)	+.236 (5.99)
L	-.186 (4.72)	-.472 (11.99)	g	-.086 (2.18)	+.320 (8.13)
M	-.333 (8.46)	-.382 (9.70)	h	+.000 (.00)	+.172 (4.37)
N	-.441 (11.20)	-.249 (6.32)	k	+.154 (3.91)	+.086 (2.18)
P	-.500 (12.70)	-.086 (2.18)	m	+.154 (3.91)	-.086 (2.18)
R	-.500 (12.70)	+.086 (2.18)	n	+.000 (.00)	-.172 (4.37)

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

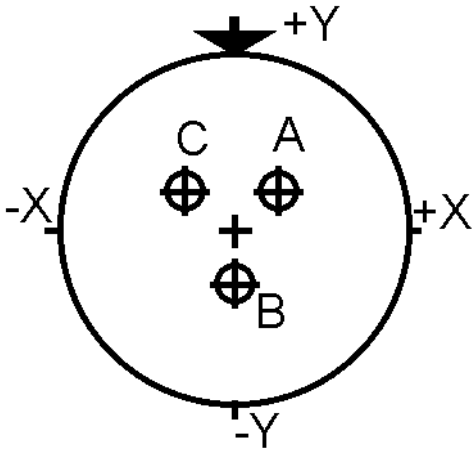
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 25-37)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
S	-.441 (11.20)	+.249 (6.32)	p	-.154 (3.91)	-.086 (2.18)
T	-.333 (8.46)	+.382 (9.70)	q	-.154 (3.91)	+.086 (2.18)
U	-.186 (4.72)	+.472 (11.99)	r	+.000 (.00)	+.000 (.00)
V	+.086 (2.18)	+.320 (8.13)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
25	-37	37	16	II	All	-----

FIGURE 9. Insert arrangement, electrical connector, shell size 25 (series I, III, IV of MIL-DTL-38999, series I of MIL-DTL-27599, and series A of SAE-AS29600) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



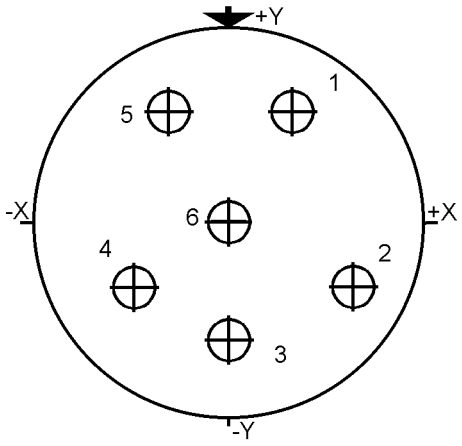
Contacts (Insert arrangement 8-3)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.056 (1.42)	+.018 (0.46)
B	+.000 (0.00)	-.079 (2.01)
C	-.056 (1.42)	+.018 (0.46)

(Applicable to MIL-DTL-27599 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
8	-3	3	20	M	All	MS27343-3

FIGURE 10. Insert arrangement, electrical connector, shell size 8 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



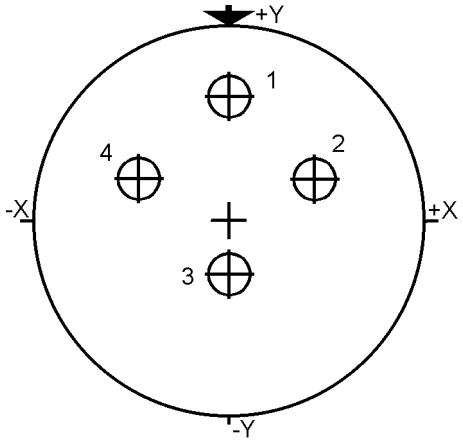
Contacts (Insert arrangement 8-6)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.078 (1.98)
2	+.078 (1.98)	-.045 (1.14)
3	+.000 (0.00)	-.090 (2.29)
4	-.078 (1.98)	-.045 (1.14)
5	-.045 (1.14)	+.078 (1.98)
6	+.000 (0.00)	+.000 (0.00)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
8	-6	6	22M	M	All	MS27343-6

FIGURE 10. Insert arrangement, electrical connector, shell size 8 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



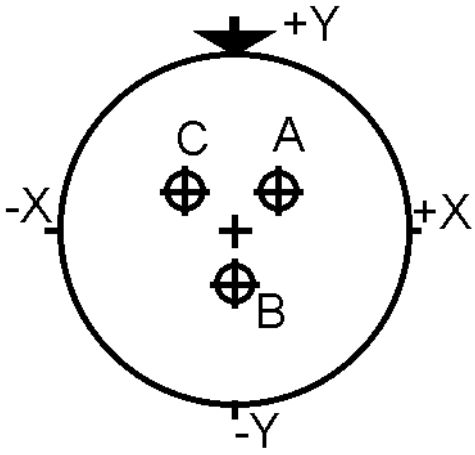
Contacts (Insert arrangement 8-44)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
1	+.000 (0.00)	+.081 (2.06)
2	+.080 (2.03)	+.021 (0.53)
3	+.000 (0.00)	-.050 (1.27)
4	-.080 (2.03)	+.021 (0.53)

(Applicable to MIL-DTL-27599 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
8	-44	4	22	M	All	MS27343-44

FIGURE 10. Insert arrangement, electrical connector, shell size 8 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

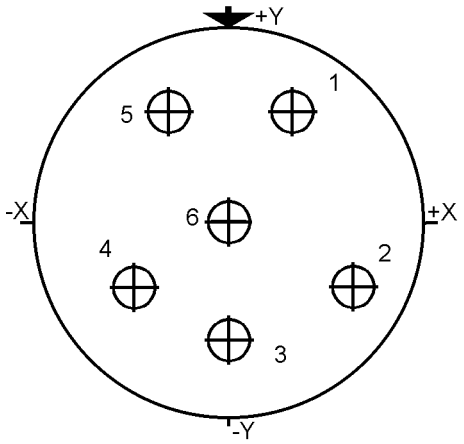


Contacts (Insert arrangement 8-98)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.038 (0.97)
B	+.000 (0.00)	-.075 (1.91)
C	-.065 (1.65)	+.038 (0.97)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
8	-98	3	20	I	All	MS27343-98

FIGURE 10. Insert arrangement, electrical connector, shell size 8 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



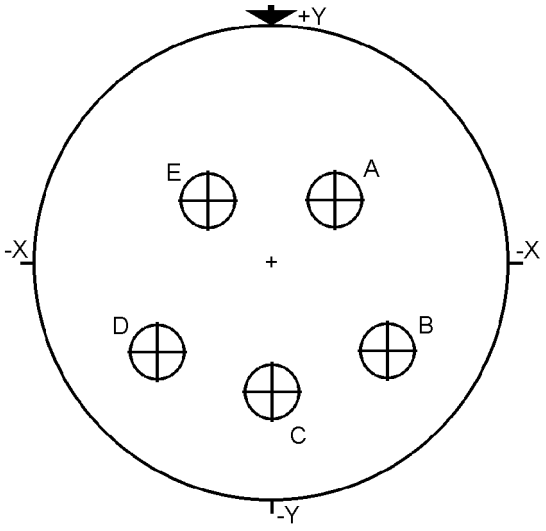
Contacts (Insert arrangement 8-35)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.078 (1.98)
2	+.078 (1.98)	-.045 (1.14)
3	+.000 (0.00)	-.090 (2.29)
4	-.078 (1.98)	-.045 (1.14)
5	-.045 (1.14)	+.078 (1.98)
6	+.000 (0.00)	+.000 (0.00)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
8	-35	6	22D	M	All	MS27516-35

FIGURE 10. Insert arrangement, electrical connector, shell size 8 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

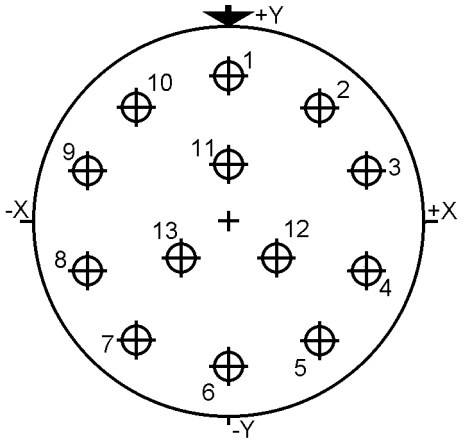


Contacts (Insert arrangement 10-5)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.056 (1.42)
B	+.113 (2.87)	-.065 (1.65)
C	+.000 (0.00)	-.130 (3.30)
D	-.113 (2.87)	-.065 (1.65)
E	-.065 (1.65)	+.056 (1.42)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
10	-5	5	20	I	All	MS27344-5

FIGURE 11. Insert arrangement, electrical connector, shell size 10 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



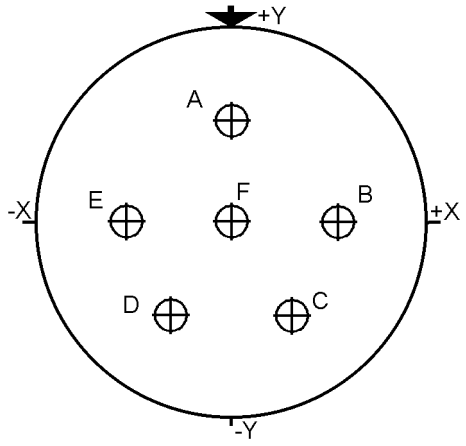
Contacts (Insert arrangement 10-13)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.000 (0.00)	+.146 (3.71)	8	-.138 (3.51)	-.045 (1.14)
2	+.085 (2.16)	+.118 (3.00)	9	-.138 (3.51)	+.045 (1.14)
3	+.138 (3.51)	+.045 (1.14)	10	-.085 (2.16)	+.118 (3.00)
4	+.138 (3.51)	-.045 (1.14)	11	+.000 (0.00)	+.056 (1.42)
5	+.085 (2.16)	-.118 (3.00)	12	+.049 (1.24)	-.035 (0.89)
6	+.000 (0.00)	-.146 (3.71)	13	-.049 (1.24)	-.035 (0.89)
7	-.085 (2.16)	-.118 (3.00)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
10	-13	13	22M	M	All	MS27344-13

FIGURE 11. Insert arrangement, electrical connector, shell size 10 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

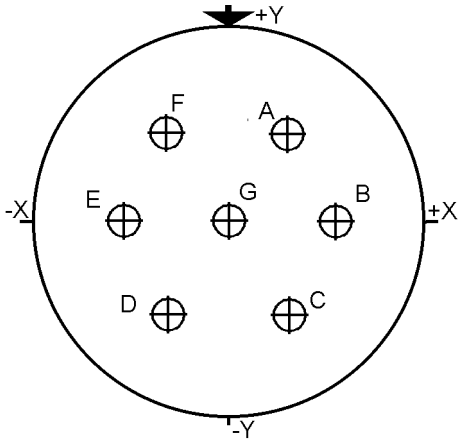


Contacts (Insert arrangement 10-98)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.130 (3.30)
B	+.130 (3.30)	+.000 (0.00)
C	+.065 (1.65)	-.113 (2.87)
D	-.065 (1.65)	-.113 (2.87)
E	-.130 (3.30)	+.000 (0.00)
F	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
10	-98	6	20	I	All	MS27344-98

FIGURE 11. Insert arrangement, electrical connector, shell size 10 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

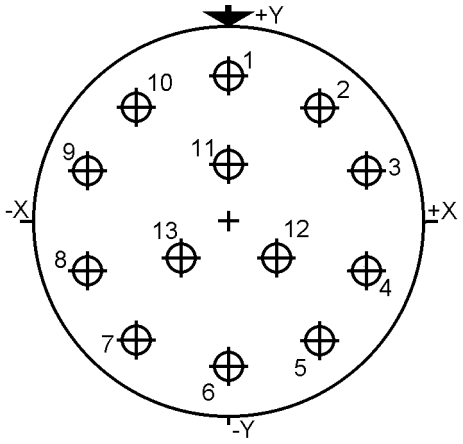


Contacts (Insert arrangement 10-99)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.113 (2.87)
B	+.130 (3.30)	+.000 (0.00)
C	+.065 (1.65)	-.113 (2.87)
D	-.065 (1.65)	-.113 (2.87)
E	-.130 (3.30)	+.000 (0.00)
F	-.065 (1.65)	+.113 (2.87)
G	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
10	-99	7	20	I	All	MS27344-99

FIGURE 11. Insert arrangement, electrical connector, shell size 10 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



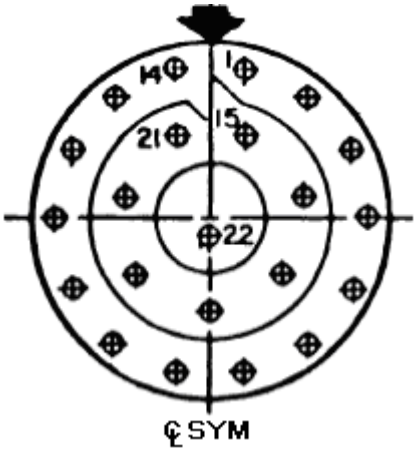
Contacts (Insert arrangement 10-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.000 (0.00)	+.146 (3.71)	8	-.138 (3.51)	-.045 (1.14)
2	+.085 (2.16)	+.118 (3.00)	9	-.138 (3.51)	+.045 (1.14)
3	+.138 (3.51)	+.045 (1.14)	10	-.085 (2.16)	+.118 (3.00)
4	+.138 (3.51)	-.045 (1.14)	11	+.000 (0.00)	+.056 (1.42)
5	+.085 (2.16)	-.118 (3.00)	12	+.049 (1.24)	-.035 (0.89)
6	+.000 (0.00)	-.146 (3.71)	13	-.049 (1.24)	-.035 (0.89)
7	-.085 (2.16)	-.118 (3.00)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
10	-35	13	22D	M	All	MS27517-35

FIGURE 11. Insert arrangement, electrical connector, shell size 10 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



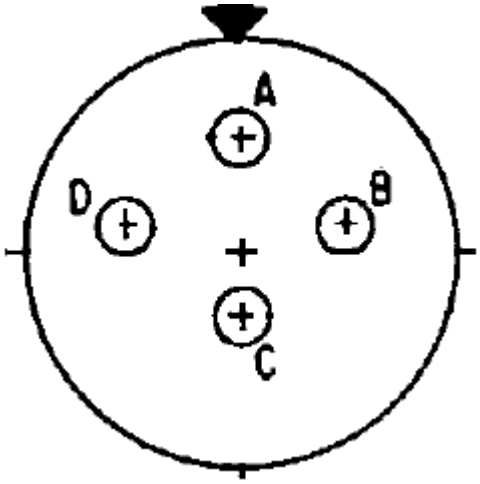
Contacts (Insert arrangement 12-22)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.197 (5.00)	12	-.182 (4.62)	+.088 (2.24)
2	+.126 (3.20)	+.158 (4.01)	13	-.126 (3.20)	+.158 (4.01)
3	+.182 (4.62)	+.088 (2.24)	14	-.045 (1.14)	+.197 (5.00)
4	+.203 (5.16)	+.000 (0.00)	15	+.045 (1.14)	+.107 (2.72)
5	+.182 (4.62)	-.088 (2.24)	16	+.117 (2.97)	+.026 (0.66)
6	+.126 (3.20)	-.158 (4.01)	17	+.093 (2.36)	-.075 (1.91)
7	+.045 (1.14)	-.197 (5.00)	18	+.000 (0.00)	-.120 (3.05)
8	-.045 (1.14)	-.197 (5.00)	19	-.093 (2.36)	-.075 (1.91)
9	-.126 (3.20)	-.158 (4.01)	20	-.117 (2.97)	+.026 (0.66)
10	-.182 (4.62)	-.088 (2.24)	21	-.045 (1.14)	+.107 (2.72)
11	-.203 (5.16)	+.000 (0.00)	22	+.000 (0.00)	-.030 (0.76)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
12	-22	22	22M	M	All	MS27345-22

FIGURE 12. Insert arrangement, electrical connector, shell size 12 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

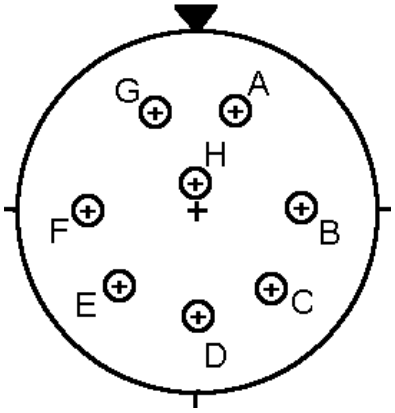


Contacts (Insert arrangement 12-4)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.150 (3.81)
B	+.146 (3.71)	+.035 (0.89)
C	+.000 (0.00)	-.083 (2.11)
D	-.146 (3.71)	+.035 (0.89)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
12	-4	4	16	I	All	MS27345-4

FIGURE 12. Insert arrangement, electrical connector, shell size 12 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

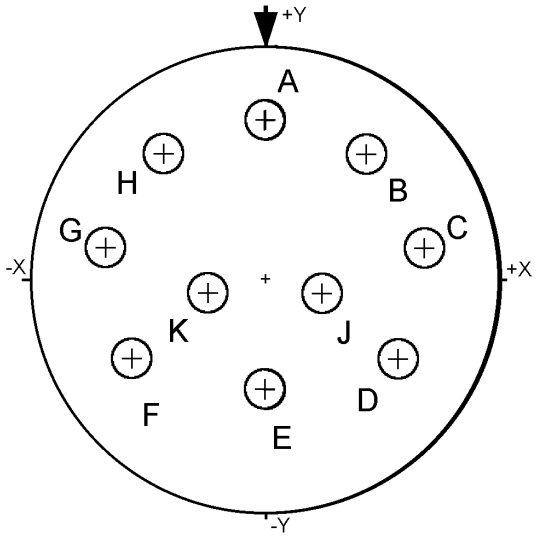


Contacts (Insert arrangement 12-8)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.157 (3.99)
B	+.170 (4.32)	+.000 (0.00)
C	+.120 (3.05)	-.120 (3.05)
D	+.000 (0.00)	-.170 (4.32)
E	-.120 (3.05)	-.120 (3.05)
F	-.170 (4.32)	+.000 (0.00)
G	-.065 (1.65)	+.157 (3.99)
H	+.000 (0.00)	+.044 (1.12)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
12	-8	8	20	I	All	MS27345-8

FIGURE 12. Insert arrangement, electrical connector, shell size 12 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

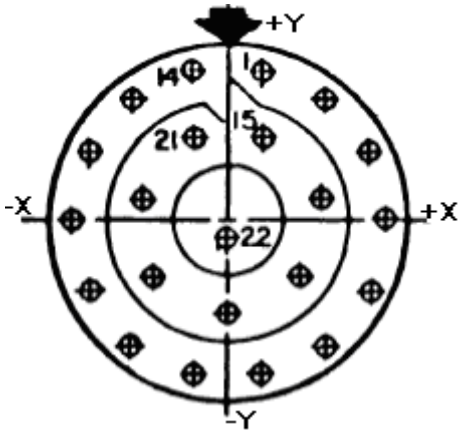


Contacts (Insert arrangement 12-98)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.195 (4.95)	F	-.164 (4.17)	-.105 (2.67)
B	+.125 (3.18)	+.150 (3.81)	G	-.193 (4.90)	+.030 (0.76)
C	+.193 (4.90)	+.030 (0.76)	H	-.125 (3.18)	+.150 (3.81)
D	+.164 (4.17)	-.105 (2.67)	J	+.065 (1.65)	-.015 (0.38)
E	+.000 (0.00)	-.135 (3.43)	K	-.065 (1.65)	-.015 (0.38)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
12	-98	10	20	I	All	MS27345-98

FIGURE 12. Insert arrangement, electrical connector, shell size 12 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



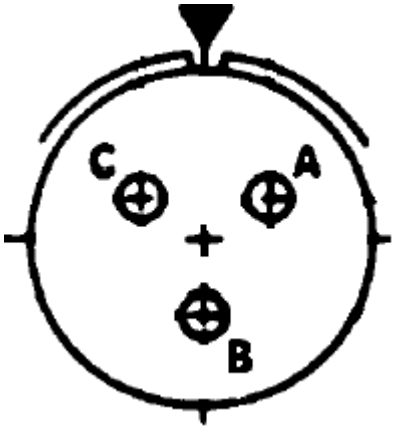
Contacts (Insert arrangement 12-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.197 (5.00)	12	-.182 (4.62)	+.088 (2.24)
2	+.126 (3.20)	+.158 (4.01)	13	-.126 (3.20)	+.158 (4.01)
3	+.182 (4.62)	+.088 (2.24)	14	-.045 (1.14)	+.197 (5.00)
4	+.203 (5.16)	+.000 (0.00)	15	+.045 (1.14)	+.107 (2.72)
5	+.182 (4.62)	-.088 (2.24)	16	+.117 (2.97)	+.026 (0.66)
6	+.126 (3.20)	-.158 (4.01)	17	+.093 (2.36)	-.075 (1.91)
7	+.045 (1.14)	-.197 (5.00)	18	+.000 (0.00)	-.120 (3.05)
8	-.045 (1.14)	-.197 (5.00)	19	-.093 (2.36)	-.075 (1.91)
9	-.126 (3.20)	-.158 (4.01)	20	-.117 (2.97)	+.026 (0.66)
10	-.182 (4.62)	-.088 (2.24)	21	-.045 (1.24)	+.107 (2.72)
11	-.203 (5.16)	+.000 (0.00)	22	+.000 (0.00)	-.030 (0.76)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
12	-35	22	22D	M	All	MS27518-35

FIGURE 12. Insert arrangement, electrical connector, shell size 12 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

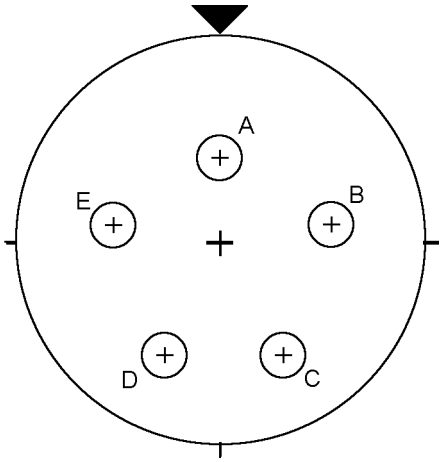


Contacts (Insert arrangement 12-3)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.094 (2.39)	+.058 (1.47)
B	+.000 (0.00)	-.111 (2.82)
C	-.094 (2.39)	+.058 (1.47)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
12	-3	3	16	II	All	MS27345-3

FIGURE 12. Insert arrangement, electrical connector, shell size 12 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

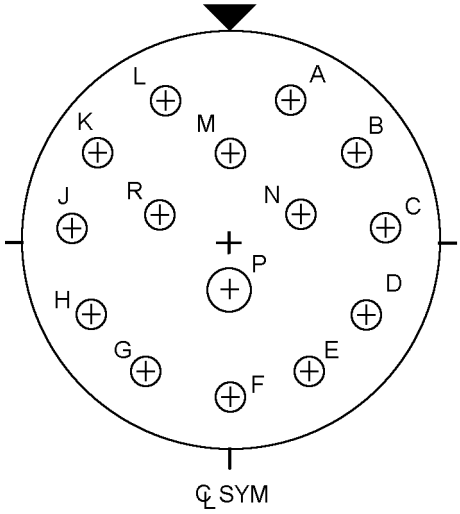


Contacts (Insert arrangement 14-5)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.100 (2.54)
B	+.174 (4.42)	+.024 (0.61)
C	+.094 (2.39)	-.148 (3.76)
D	-.094 (2.39)	-.148 (3.76)
E	-.174 (4.42)	+.024 (0.61)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
14	-5	5	16	II	All	MS27346-5

FIGURE 13. Insert arrangement, electrical connector, shell size 14 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

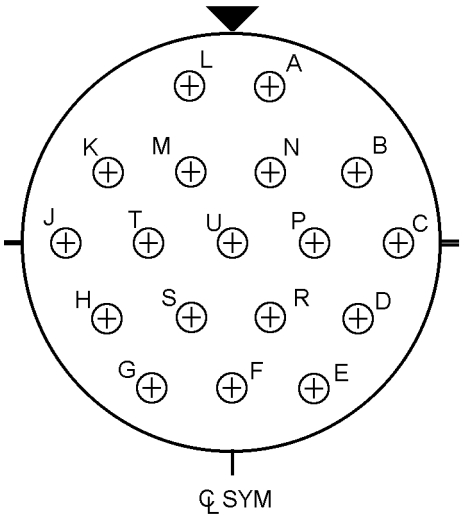


Contacts (Insert arrangement 14-15)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.225 (5.72)	J	-.244 (6.20)	+.014 (0.36)
B	+.202 (5.13)	+.140 (3.56)	K	-.202 (5.13)	+.140 (3.56)
C	+.244 (6.20)	+.014 (0.36)	L	-.100 (2.54)	+.225 (5.72)
D	+.218 (5.54)	-.113 (2.87)	M	+.000 (0.00)	+.140 (3.56)
E	+.126 (3.20)	-.209 (5.31)	N	+.110 (2.79)	+.040 (1.02)
F	+.000 (0.00)	-.245 (6.22)	P	+.000 (0.00)	-.077 (1.96)
G	-.126 (3.20)	-.209 (5.18)	R	-.110 (2.79)	+.040 (1.02)
H	-.218 (5.54)	-.113 (2.87)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
14	-15	1	16	I	P	MS27346-15
		14	20		All others	

FIGURE 13. Insert arrangement, electrical connector, shell size 14 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

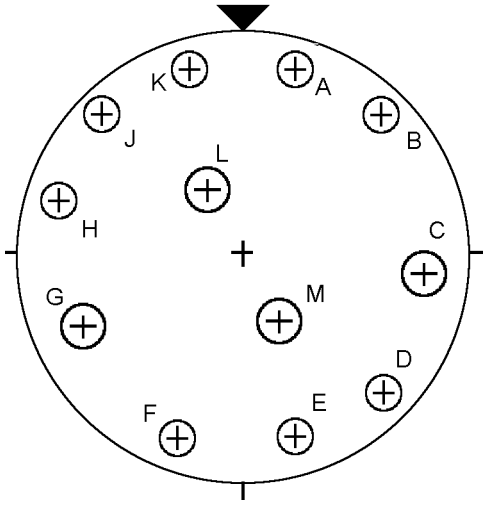


Contacts (Insert arrangement 14-18)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.252 (6.40)	K	-.195 (4.95)	+.113 (2.87)
B	+.195 (4.95)	+.113 (2.87)	L	-.065 (1.65)	+.252 (6.40)
C	+.260 (6.60)	+.000 (0.00)	M	-.065 (1.65)	+.113 (2.87)
D	+.195 (4.95)	-.113 (2.87)	N	+.065 (1.65)	+.113 (2.87)
E	+.130 (3.30)	-.225 (5.72)	P	+.130 (3.30)	+.000 (0.00)
F	+.000 (0.00)	-.225 (5.72)	R	+.065 (1.65)	-.113 (2.87)
G	-.130 (3.30)	-.225 (5.72)	S	-.065 (1.65)	-.113 (2.87)
H	-.195 (4.95)	-.113 (2.87)	T	-.130 (3.30)	+.000 (0.00)
J	-.260 (6.60)	+.000 (0.00)	U	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
14	-18	18	20	I	All	MS27346-18

FIGURE 13. Insert arrangement, electrical connector, shell size 14 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

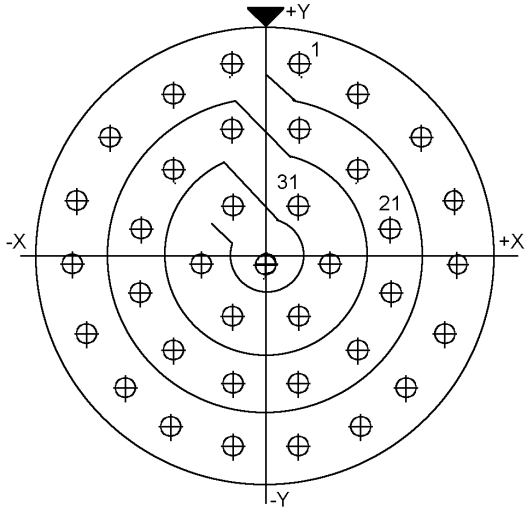


Contacts (Insert arrangement 14-97)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.234 (5.94)	G	-.207 (5.26)	-.095 (2.41)
B	+.178 (4.52)	+.178 (4.52)	H	-.234 (5.94)	+.065 (1.65)
C	+.230 (5.84)	-.023 (0.58)	J	-.178 (4.52)	+.178 (4.52)
D	+.178 (4.52)	-.178 (4.52)	K	-.065 (1.65)	+.234 (5.94)
E	+.065 (1.65)	-.234 (5.94)	L	-.047 (1.19)	+.081 (2.06)
F	-.089 (2.26)	-.235 (5.97)	M	+.047 (1.19)	-.081 (2.06)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
14	-97	4	16	I	C,M,G,L	MS27346-97
		8	20		All others	

FIGURE 13. Insert arrangement, electrical connector, shell size 14 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



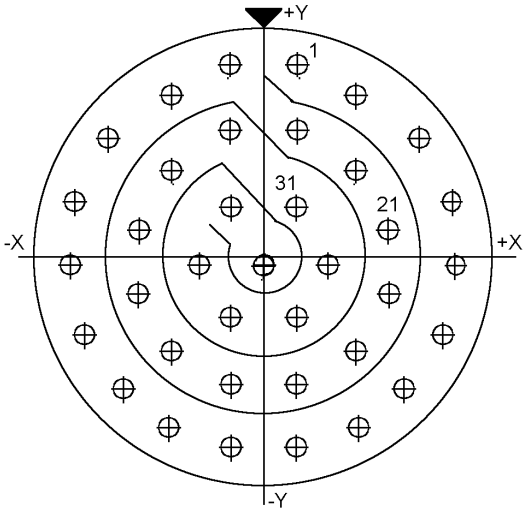
Contacts (Insert arrangement 14-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.262 (6.65)	20	+.123 (3.12)	+.119 (3.02)
2	+.123 (3.12)	+.217 (5.51)	21	+.170 (4.32)	+.040 (1.02)
3	+.211 (5.36)	+.160 (4.06)	22	+.170 (4.32)	-.050 (1.27)
4	+.254 (6.45)	+.080 (2.03)	23	+.123 (3.12)	-.127 (3.23)
5	+.266 (6.76)	-.010 (0.25)	24	+.045 (1.14)	-.172 (4.37)
6	+.247 (6.27)	-.098 (2.49)	25	-.045 (1.14)	-.172 (4.37)
7	+.200 (5.08)	-.175 (4.45)	26	-.123 (3.12)	-.127 (3.23)
8	+.130 (3.30)	-.232 (5.89)	27	-.170 (4.32)	-.050 (1.27)
9	+.045 (1.14)	-.262 (6.65)	28	-.170 (4.32)	+.040 (1.02)
10	-.045 (1.14)	-.262 (6.65)	29	-.123 (3.12)	+.119 (3.02)
11	-.130 (3.30)	-.232 (5.89)	30	-.045 (1.14)	+.172 (4.37)
12	-.200 (5.08)	-.175 (4.45)	31	+.045 (1.14)	+.074 (1.88)
13	-.247 (6.27)	-.098 (2.49)	32	+.090 (2.29)	-.004 (0.10)
14	-.266 (6.76)	-.010 (0.25)	33	+.045 (1.14)	-.082 (2.08)
15	-.254 (6.45)	+.080 (2.03)	34	-.045 (1.14)	-.082 (2.08)
16	-.211 (5.36)	+.160 (4.06)	35	-.090 (2.29)	-.004 (0.10)
17	-.123 (3.12)	+.217 (5.51)	36	-.045 (1.14)	+.074 (1.88)
18	-.045 (1.14)	+.262 (6.65)	37	+.000 (0.00)	-.004 (0.10)
19	+.045 (1.14)	+.172 (4.37)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
14	-35	37	22D	M	All	MS27519-35

FIGURE 13. Insert arrangement, electrical connector, shell size 14 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



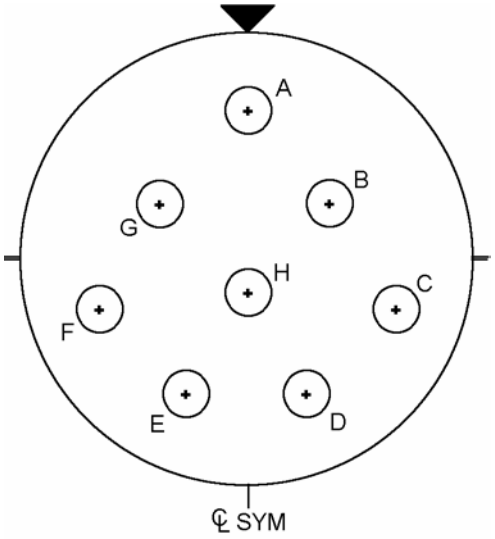
Contacts (Insert arrangement 14-37)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.045 (1.14)	+.262 (6.65)	20	+.123 (3.12)	+.119 (3.02)
2	+.123 (3.12)	+.217 (5.51)	21	+.170 (4.32)	+.040 (1.02)
3	+.211 (5.36)	+.160 (4.06)	22	+.170 (4.32)	-.050 (1.27)
4	+.254 (6.45)	+.080 (2.03)	23	+.123 (3.12)	-.127 (3.23)
5	+.266 (6.76)	-.010 (0.25)	24	+.045 (1.14)	-.172 (4.37)
6	+.247 (6.27)	-.098 (2.49)	25	-.045 (1.14)	-.172 (4.37)
7	+.200 (5.08)	-.175 (4.45)	26	-.123 (3.12)	-.127 (3.23)
8	+.130 (3.30)	-.232 (5.89)	27	-.170 (4.32)	-.050 (1.27)
9	+.045 (1.14)	-.262 (6.65)	28	-.170 (4.32)	+.040 (1.02)
10	-.045 (1.14)	-.262 (6.65)	29	-.123 (3.12)	+.119 (3.02)
11	-.130 (3.30)	-.232 (5.89)	30	-.045 (1.14)	+.172 (4.37)
12	-.200 (5.08)	-.175 (4.45)	31	+.045 (1.14)	+.074 (1.88)
13	-.247 (6.27)	-.098 (2.49)	32	+.090 (2.29)	-.004 (0.10)
14	-.266 (6.76)	-.010 (0.25)	33	+.045 (1.14)	-.082 (2.08)
15	-.254 (6.45)	+.080 (2.03)	34	-.045 (1.14)	-.082 (2.08)
16	-.211 (5.36)	+.160 (4.06)	35	-.090 (2.29)	-.004 (0.10)
17	-.123 (3.12)	+.217 (5.51)	36	-.045 (1.14)	+.074 (1.88)
18	-.045 (1.14)	+.262 (6.65)	37	+.000 (0.00)	-.004 (0.10)
19	+.045 (1.14)	+.172 (4.37)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
14	-37	37	22M	M	All	MS27346-37

FIGURE 13. Insert arrangement, electrical connector, shell size 14 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

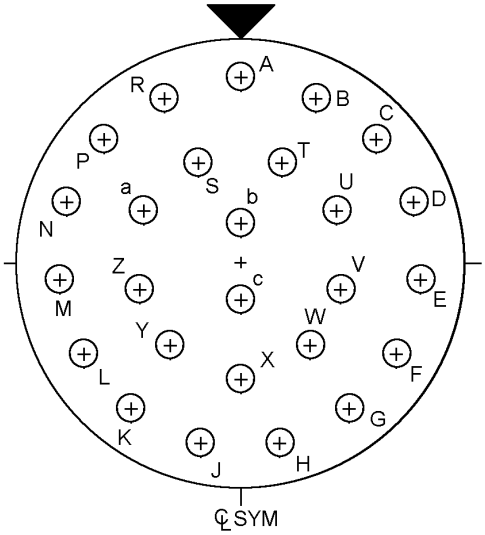


Contacts (Insert arrangement 16-8)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.236 (5.99)	E	-.094 (2.39)	-.216 (5.49)
B	+.128 (3.25)	+.086 (2.18)	F	-.230 (5.84)	-.078 (1.98)
C	+.230 (5.84)	-.078 (1.98)	G	-.128 (3.25)	+.086 (2.18)
D	+.094 (2.39)	-.216 (5.49)	H	+.000 (0.00)	-.052 (1.32)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-8	8	16	II	All	MS27347-8

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

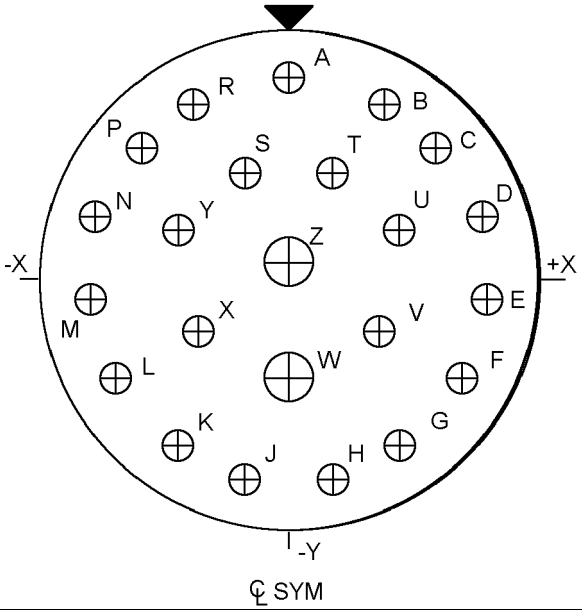


Contacts (Insert arrangement 16-26)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.321 (8.15)	P	-.239 (6.07)	+.214 (5.44)
B	+.131 (3.33)	+.293 (7.44)	R	-.131 (3.33)	+.293 (7.44)
C	+.239 (6.07)	+.214 (5.44)	S	-.070 (1.78)	+.177 (4.50)
D	+.305 (7.75)	+.099 (2.51)	T	+.070 (1.78)	+.177 (4.50)
E	+.319 (8.10)	-.034 (0.86)	U	+.175 (4.45)	+.094 (2.39)
F	+.278 (7.06)	-.161 (4.09)	V	+.178 (4.52)	-.036 (0.91)
G	+.189 (4.80)	-.260 (6.60)	W	+.119 (3.02)	-.151 (3.84)
H	+.067 (1.70)	-.314 (7.98)	X	+.000 (.00)	-.203 (5.16)
J	-.067 (1.70)	-.314 (7.98)	Y	-.119 (.005)	-.151 (3.84)
K	-.189 (4.80)	-.260 (6.60)	Z	-.178 (4.52)	-.036 (0.91)
L	-.278 (7.06)	-.161 (4.09)	a	-.175 (4.45)	+.094 (2.39)
M	-.319 (8.10)	-.034 (0.86)	b	+.000 (0.00)	+.065 (1.65)
N	-.305 (7.75)	+.099 (2.51)	c	+.000 (0.00)	-.065 (1.65)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-26	26	20	I	All	MS27347-26

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

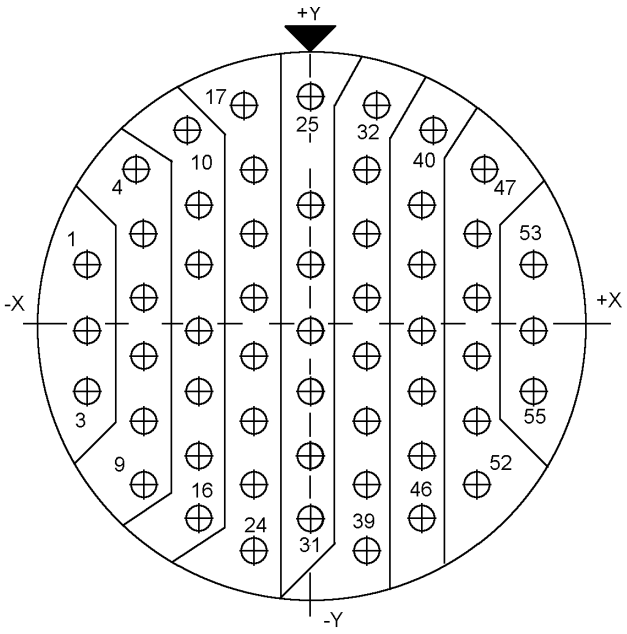


Contacts (Insert arrangement 16-99)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.321 (8.15)	N	-.305 (7.75)	+.099 (2.51)
B	+.131 (3.33)	+.293 (7.44)	P	-.239 (6.07)	+.214 (5.44)
C	+.239 (6.07)	+.214 (5.44)	R	-.131 (3.33)	+.293 (7.44)
D	+.305 (7.75)	+.099 (2.51)	S	-.070 (1.78)	+.177 (4.50)
E	+.319 (8.10)	-.034 (0.86)	T	+.070 (1.78)	+.177 (4.50)
F	+.278 (7.06)	-.161 (4.09)	U	+.175 (4.45)	+.094 (2.39)
G	+.189 (4.80)	-.260 (6.60)	V	+.150 (3.81)	-.075 (1.91)
H	+.067 (1.70)	-.314 (7.98)	W	+.000 (0.00)	-.161 (4.09)
J	-.067 (1.70)	-.314 (7.98)	X	-.150 (3.81)	-.075 (1.91)
K	-.189 (4.80)	-.260 (6.60)	Y	-.175 (4.45)	+.094 (2.39)
L	-.278 (7.06)	-.161 (4.09)	Z	+.000 (0.00)	+.025 (0.64)
M	-.319 (8.10)	-.034 (0.86)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-99	2	16	I	w,z	MS27347-99
		21	20		All others	

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 16-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.312 (7.92)	+.086 (2.18)	29	+.000 (0.00)	-.094 (2.39)
2	-.312 (7.92)	-.004 (0.10)	30	+.000 (0.00)	-.184 (4.67)
3	-.312 (7.92)	-.094 (2.39)	31	+.000 (0.00)	-.274 (6.96)
4	-.242 (6.15)	+.221 (5.61)	32	+.089 (2.26)	+.316 (8.03)
5	-.234 (5.94)	+.131 (3.33)	33	+.078 (1.98)	+.221 (5.61)
6	-.234 (5.94)	+.041 (1.04)	34	+.078 (1.98)	+.131 (3.33)
7	-.234 (5.94)	-.049 (1.24)	35	+.078 (1.98)	+.041 (1.04)
8	-.234 (5.94)	-.139 (3.53)	36	+.078 (1.98)	-.049 (1.24)
9	-.234 (5.94)	-.229 (5.82)	37	+.078 (1.98)	-.139 (3.53)
10	-.172 (4.37)	+.279 (7.09)	38	+.078 (1.98)	-.229 (5.82)
11	-.156 (3.96)	+.176 (4.47)	39	+.078 (1.98)	-.319 (8.10)

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

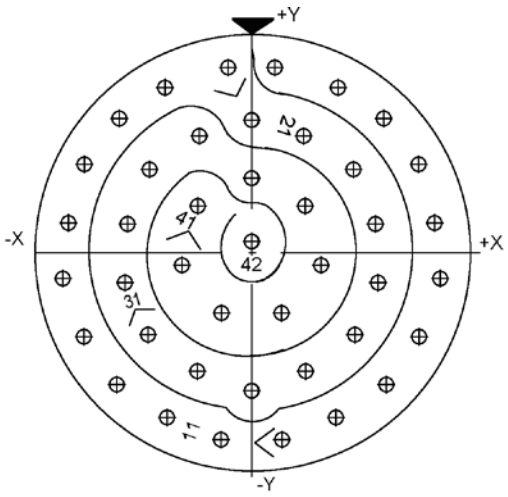
Contacts (Insert arrangement 16-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
12	-.156 (3.96)	+.086 (2.18)	40	+.172 (4.37)	+.279 (7.09)
13	-.156 (3.96)	-.004 (0.10)	41	+.156 (3.96)	+.176 (4.47)
14	-.156 (3.96)	-.094 (2.39)	42	+.156 (3.96)	+.086 (2.18)
15	-.156 (3.96)	-.184 (4.67)	43	+.156 (3.96)	-.004 (0.10)
16	-.156 (3.96)	-.274 (6.96)	44	+.156 (3.96)	-.094 (2.39)
17	-.089 (2.26)	+.316 (8.03)	45	+.156 (3.96)	-.184 (4.67)
18	-.078 (1.98)	+.221 (5.61)	46	+.156 (3.96)	-.274 (6.96)
19	-.078 (1.98)	+.131 (3.33)	47	+.242 (6.15)	+.221 (5.61)
20	-.078 (1.98)	+.041 (1.04)	48	+.234 (5.94)	+.131 (3.33)
21	-.078 (1.98)	-.049 (1.24)	49	+.234 (5.94)	+.041 (1.04)
22	-.078 (1.98)	-.139 (3.53)	50	+.234 (5.94)	-.049 (1.24)
23	-.078 (1.98)	-.229 (5.82)	51	+.234 (5.94)	-.139 (3.53)
24	-.078 (1.98)	-.319 (8.10)	52	+.234 (5.94)	-.229 (5.82)
25	+.000 (0.00)	+.329 (8.36)	53	+.312 (7.92)	+.086 (2.18)
26	+.000 (0.00)	+.176 (4.47)	54	+.312 (7.92)	-.004 (0.10)
27	+.000 (0.00)	+.086 (2.18)	55	+.312 (7.92)	-.094 (2.39)
28	+.000 (0.00)	-.004 (0.10)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-55	55	22M	M	All	MS27347-55

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 16-42)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.050 (1.27)	+.318 (8.08)	22	+.095 (2.41)	+.198 (5.03)
2	+.145 (3.68)	+.286 (7.26)	23	+.178 (4.52)	+.142 (3.61)
3	+.228 (5.79)	+.230 (5.84)	24	+.218 (5.54)	+.050 (1.27)
4	+.287 (7.29)	+.149 (3.78)	25	+.218 (5.54)	-.050 (1.27)
5	+.318 (8.08)	+.053 (1.35)	26	+.177 (4.50)	-.142 (3.61)
6	+.319 (8.10)	-.047 (1.19)	27	+.095 (2.41)	-.202 (5.13)
7	+.289 (7.34)	-.143 (3.63)	28	+.000 (0.00)	-.235 (5.97)
8	+.232 (5.89)	-.226 (5.74)	29	-.095 (2.41)	-.202 (5.13)
9	+.151 (3.84)	-.285 (7.24)	30	-.177 (4.50)	-.142 (3.61)
10	+.056 (1.42)	-.318 (8.08)	31	-.218 (5.54)	-.050 (1.27)
11	-.056 (1.42)	-.318 (8.08)	32	-.218 (5.54)	+.050 (1.27)
12	-.151 (3.84)	-.285 (7.24)	33	-.178 (4.52)	+.142 (3.61)
13	-.232 (5.89)	-.226 (5.74)	34	-.095 (2.41)	+.198 (5.03)
14	-.289 (7.34)	-.143 (3.63)	35	+.000 (0.00)	+.131 (3.33)
15	-.319 (8.10)	-.047 (1.19)	36	+.091 (2.31)	+.080 (2.03)
16	-.318 (8.08)	+.053 (1.35)	37	+.121 (3.07)	-.017 (0.43)
17	-.287 (7.29)	+.149 (3.78)	38	+.056 (1.42)	-.101 (2.57)
18	-.228 (5.79)	+.230 (5.84)	39	-.056 (1.42)	-.101 (2.57)

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

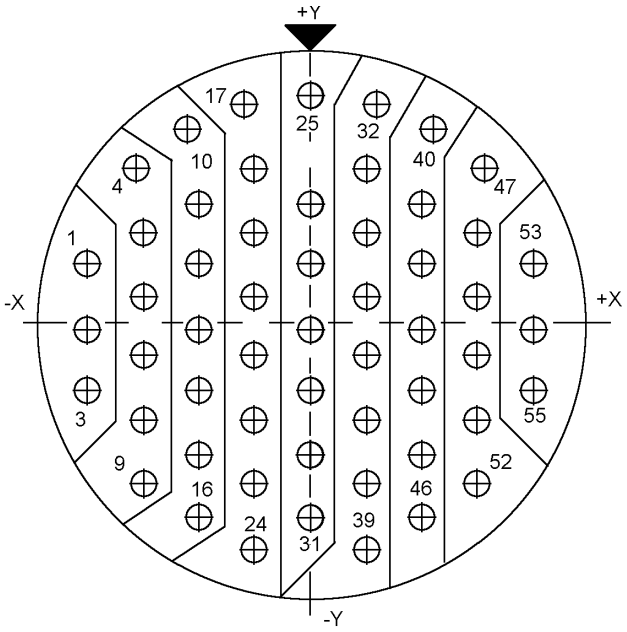
Contacts (Insert arrangement 16-42)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
19	-.149 (3.78)	+.286 (7.26)	40	-.121 (3.07)	-.017 (0.43)
20	-.050 (1.27)	+.318 (8.08)	41	-.091 (2.31)	+.080 (2.03)
21	+.000 (0.00)	+.231 (5.87)	42	+.000 (0.00)	+.031 (0.79)

(Applicable to MIL-DTL-27599 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-42	42	22	M	All	MS27347-42

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 16-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.312 (7.92)	+.086 (2.18)	29	+.000 (0.00)	-.094 (2.39)
2	-.312 (7.92)	-.004 (0.10)	30	+.000 (0.00)	-.184 (4.67)
3	-.312 (7.92)	-.094 (2.39)	31	+.000 (0.00)	-.274 (6.96)
4	-.242 (6.15)	+.221 (5.61)	32	+.089 (2.26)	+.316 (8.03)
5	-.234 (5.94)	+.131 (3.33)	33	+.078 (1.98)	+.221 (5.61)
6	-.234 (5.94)	+.041 (1.04)	34	+.078 (1.98)	+.131 (3.33)
7	-.234 (5.94)	-.049 (1.24)	35	+.078 (1.98)	+.041 (1.04)
8	-.234 (5.94)	-.139 (3.53)	36	+.078 (1.98)	-.049 (1.24)
9	-.234 (5.94)	-.229 (5.82)	37	+.078 (1.98)	-.139 (3.53)
10	-.172 (4.37)	+.279 (7.09)	38	+.078 (1.98)	-.229 (5.82)
11	-.156 (3.96)	+.176 (4.47)	39	+.078 (1.98)	-.319 (8.10)
12	-.156 (3.96)	+.086 (2.18)	40	+.172 (4.37)	+.279 (7.09)
13	-.156 (3.96)	-.004 (0.10)	41	+.156 (3.96)	+.176 (4.47)
14	-.156 (3.96)	-.094 (2.39)	42	+.156 (3.96)	+.086 (2.18)
15	-.156 (3.96)	-.184 (4.67)	43	+.156 (3.96)	-.004 (0.10)

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

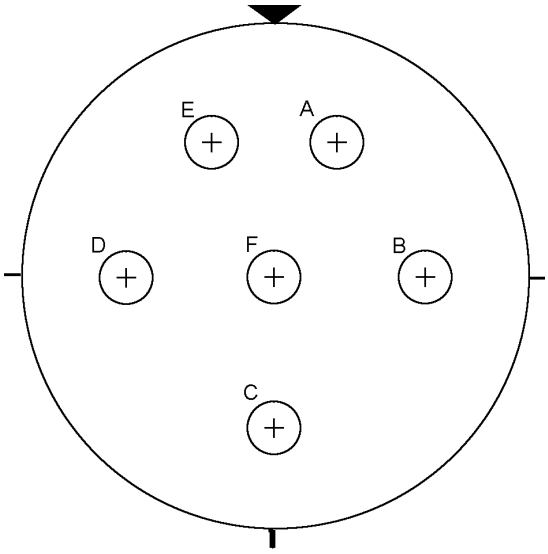
Contacts (Insert arrangement 16-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
16	-.156 (3.96)	-.274 (6.96)	44	+.156 (3.96)	-.094 (2.39)
17	-.089 (2.26)	+.316 (8.03)	45	+.156 (3.96)	-.184 (4.67)
18	-.078 (1.98)	+.221 (5.61)	46	+.156 (3.96)	-.274 (6.96)
19	-.078 (1.98)	+.131 (3.33)	47	+.242 (6.15)	+.221 (5.61)
20	-.078 (1.98)	+.041 (1.04)	48	+.234 (5.94)	+.131 (3.33)
21	-.078 (1.98)	-.049 (1.24)	49	+.234 (5.94)	+.041 (1.04)
22	-.078 (1.98)	-.139 (3.53)	50	+.234 (5.94)	-.049 (1.24)
23	-.078 (1.98)	-.229 (5.82)	51	+.234 (5.94)	-.139 (3.53)
24	-.078 (1.98)	-.319 (8.10)	52	+.234 (5.94)	-.229 (5.82)
25	+.000 (0.00)	+.329 (8.36)	53	-.312 (7.92)	+.086 (2.18)
26	+.000 (0.00)	+.176 (4.47)	54	+.312 (7.92)	-.004 (0.10)
27	+.000 (0.00)	+.086 (2.18)	55	+.312 (7.92)	-.094 (2.39)
28	+.000 (0.00)	-.004 (0.10)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-35	55	22D	M	All	MS27520-35

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

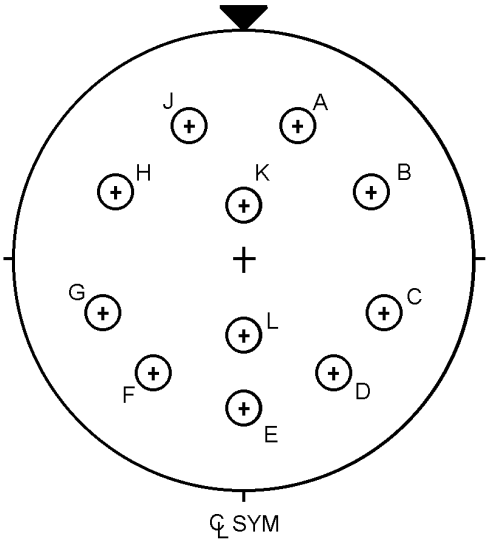


Contacts (Insert arrangement 16-6 or 16-7)		
Contact position ID	Contact location	
	X-axis (mm)	Y-axis (mm)
A	+.121 (3.07)	+.209 (5.31)
B	+.241 (6.12)	+.000 (0.00)
C	+.000 (0.00)	-.241 (6.12)
D	-.241 (6.12)	+.000 (0.00)
E	-.121 (3.07)	+.209 (5.31)
F	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
16	-6	6	12	I	All	MS27347-6
	-7	6	Shielded, Size 12A	II	All	-----

FIGURE 14. Insert arrangement, electrical connector, shell size 16 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

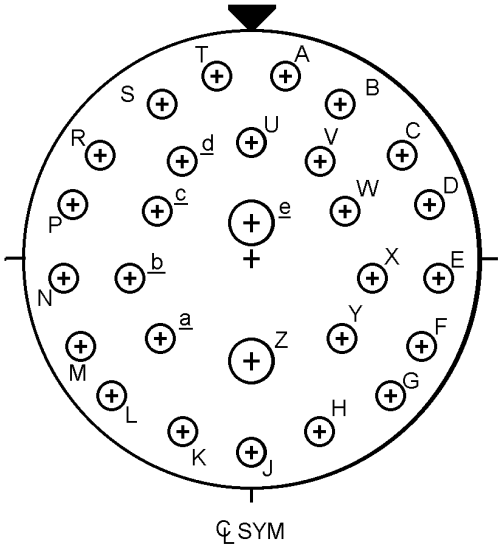


Contacts (Insert arrangement 18-11)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.105 (2.67)	+.260 (6.60)	G	-.275 (6.99)	-.053 (1.35)
B	+.250 (6.35)	+.132 (3.35)	H	-.250 (6.35)	+.132 (3.35)
C	+.275 (6.99)	-.053 (1.35)	J	-.105 (2.67)	+.260 (6.60)
D	+.179 (4.55)	-.215 (5.46)	K	+.000 (0.00)	+.105 (2.67)
E	+.000 (0.00)	-.281 (7.14)	L	+.000 (0.00)	-.092 (2.34)
F	-.179 (4.55)	-.215 (5.46)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-11	11	16	II	All	MS27348-11

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-28)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.066 (1.68)	+.353 (8.97)	R	-.286 (7.26)	+.217 (5.51)
B	+.189 (4.80)	+.305 (7.75)	S	-.189 (4.80)	+.305 (7.75)
C	+.286 (7.26)	+.217 (5.51)	T	-.066 (1.68)	+.353 (8.97)
D	+.345 (8.76)	+.098 (2.49)	U	+.000 (0.00)	+.230 (5.84)
E	+.357 (9.07)	-.033 (0.84)	V	+.124 (3.15)	+.193 (4.90)
F	+.321 (8.15)	-.160 (4.06)	W	+.209 (5.31)	+.095 (2.41)
G	+.242 (6.15)	-.265 (6.73)	X	+.228 (5.79)	-.033 (0.84)
H	+.130 (3.30)	-.335 (8.51)	Y	+.174 (4.42)	-.151 (3.84)
J	+.000 (0.00)	-.359 (9.12)	Z	+.000 (0.00)	-.191 (4.85)
K	-.130 (3.30)	-.335 (8.51)	a	-.174 (4.42)	-.151 (3.84)
L	-.242 (6.15)	-.265 (6.73)	b	-.228 (5.79)	-.033 (0.84)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

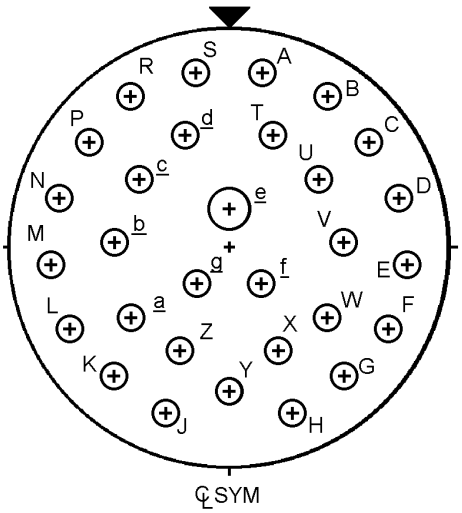
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 18-28)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
M	-.321 (8.15)	-.160 (4.06)	c	-.209 (5.31)	+.095 (2.41)
N	-.357 (9.07)	-.033 (0.84)	d	-.124 (3.15)	+.193 (4.90)
P	-.345 (8.76)	+.098 (2.49)	e	+.000 (0.00)	+.062 (1.57)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-28	2	16	I	e, Z	MS27348-28
		26	20		All others	

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-30)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.346 (8.79)	S	-.065 (1.65)	+.346 (8.79)
B	+.186 (4.72)	+.299 (7.59)	T	+.084 (2.13)	+.217 (5.51)
C	+.282 (7.16)	+.210 (5.33)	U	+.181 (4.60)	+.129 (3.28)
D	+.340 (8.64)	+.093 (2.36)	V	+.228 (5.79)	+.008 (0.20)
E	+.351 (8.92)	-.033 (0.84)	W	+.193 (4.90)	-.117 (2.97)
F	+.315 (8.00)	-.158 (4.01)	X	+.096 (2.44)	-.203 (5.16)
G	+.236 (5.99)	-.261 (6.63)	Y	+.000 (0.00)	-.290 (7.37)
H	+.124 (3.15)	-.330 (8.38)	Z	-.096 (2.44)	-.203 (5.16)
J	-.124 (3.15)	-.330 (8.38)	a	-.193 (4.90)	-.117 (2.97)
K	-.236 (5.99)	-.261 (6.63)	b	-.228 (5.79)	+.008 (0.20)
L	-.315 (8.00)	-.158 (4.01)	c	-.181 (4.60)	+.129 (3.28)
M	-.351 (8.92)	-.033 (0.84)	d	-.084 (2.13)	+.217 (5.51)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

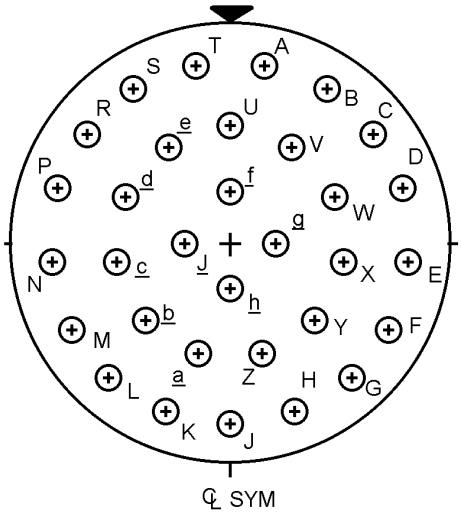
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 18-30)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
N	-.340 (8.64)	+.093 (2.36)	e	+.000 (0.00)	+.072 (1.83)
P	-.282 (7.16)	+.210 (5.33)	f	+.069 (1.75)	-.076 (1.93)
R	-.186 (4.72)	+.299 (7.59)	g	-.069 (1.75)	-.076 (1.93)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-30	1	16	I	e	MS27348-30
		29	20		All others	

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-32)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.066 (1.68)	+.353 (8.97)	T	-.066 (1.68)	+.353 (8.97)
B	+.189 (4.80)	+.305 (7.75)	U	+.000 (0.00)	+.230 (5.84)
C	+.286 (7.26)	+.217 (5.51)	V	+.124 (3.15)	+.193 (4.90)
D	+.345 (8.76)	+.098 (2.49)	W	+.209 (5.31)	+.095 (2.41)
E	+.357 (9.07)	-.033 (0.84)	X	+.228 (5.79)	-.033 (0.84)
F	+.321 (8.15)	-.160 (4.06)	Y	+.174 (4.42)	-.151 (3.84)
G	+.242 (6.15)	-.265 (6.73)	Z	+.065 (1.65)	-.221 (5.61)
H	+.130 (3.30)	-.335 (8.51)	a	-.065 (1.65)	-.221 (5.61)
J	+.000 (0.00)	-.359 (9.12)	b	-.174 (4.42)	-.151 (3.84)
K	-.130 (3.30)	-.335 (8.51)	c	-.228 (5.79)	-.033 (0.84)
L	-.242 (6.15)	-.265 (6.73)	d	-.209 (5.31)	+.095 (2.41)
M	-.321 (8.15)	-.160 (4.06)	e	-.124 (3.15)	+.193 (4.90)
N	-.357 (9.07)	-.033 (0.84)	f	+.000 (0.00)	+.096 (2.44)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

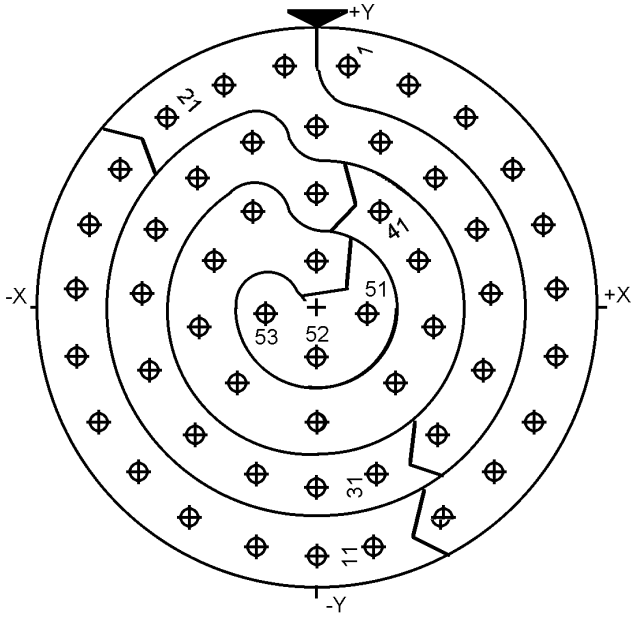
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 18-32)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
P	-.345 (8.76)	+.098 (2.49)	g	+.096 (2.44)	+.000 (0.00)
R	-.286 (7.26)	+.217 (5.51)	h	+.000 (0.00)	-.096 (2.44)
S	-.189 (4.80)	+.305 (7.75)	j	-.096 (2.44)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-32	32	20	I	All	MS27348-32

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-53)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.050 (1.27)	+.364 (9.25)	28	+.266 (6.76)	+.018 (0.46)
2	+.146 (3.71)	+.336 (8.53)	29	+.260 (6.60)	-.094 (2.39)
3	+.232 (5.89)	+.284 (7.21)	30	+.184 (4.67)	-.195 (4.95)
4	+.301 (7.65)	+.212 (5.38)	31	+.099 (2.51)	-.252 (6.40)
5	+.347 (8.81)	+.123 (3.12)	32	+.000 (0.00)	-.268 (6.81)
6	+.367 (9.32)	+.025 (0.64)	33	-.099 (2.51)	-.252 (6.40)
7	+.360 (9.14)	-.075 (1.91)	34	-.184 (4.67)	-.195 (4.95)
8	+.328 (8.33)	-.170 (4.32)	35	-.260 (6.60)	-.094 (2.39)
9	+.269 (6.83)	-.251 (6.38)	36	-.266 (6.76)	+.018 (0.46)
10	+.191 (4.85)	-.314 (7.98)	37	-.245 (6.22)	+.116 (2.95)
11	+.099 (2.51)	-.354 (8.99)	38	-.183 (4.65)	+.195 (4.95)
12	+.000 (0.00)	-.368 (9.35)	39	-.096 (2.44)	+.248 (6.30)
13	-.099 (2.51)	-.354 (8.99)	40	+.000 (0.00)	+.176 (4.47)
14	-.191 (4.85)	-.314 (7.98)	41	+.096 (2.44)	+.145 (3.68)
15	-.269 (6.83)	-.251 (6.38)	42	+.158 (4.01)	+.068 (1.73)
16	-.328 (8.33)	-.170 (4.32)	43	+.178 (4.52)	-.032 (0.81)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

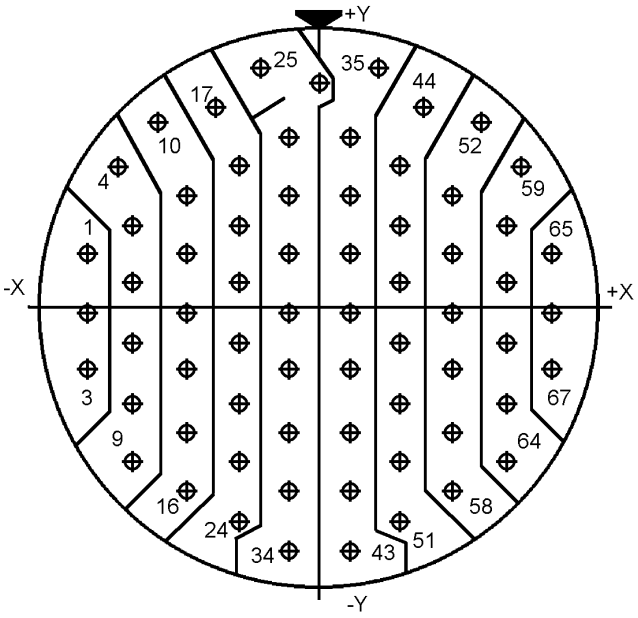
Contacts (Insert arrangement 18-53)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.360 (9.14)	-.075 (1.91)	44	+.120 (3.05)	-.114 (2.90)
18	-.367 (9.32)	+.025 (0.64)	45	+.000 (0.00)	-.168 (4.27)
19	-.347 (8.81)	+.123 (3.12)	46	-.120 (3.05)	-.114 (2.90)
20	-.301 (7.65)	+.212 (5.38)	47	-.178 (4.52)	-.032 (0.81)
21	-.232 (5.89)	+.284 (7.21)	48	-.158 (4.01)	+.068 (1.73)
22	-.146 (3.71)	+.336 (8.53)	49	-.096 (2.44)	+.145 (3.68)
23	-.050 (1.27)	+.364 (9.25)	50	+.000 (0.00)	+.076 (1.93)
24	+.000 (0.00)	+.276 (7.01)	51	+.080 (2.03)	-.002 (0.05)
25	+.096 (2.44)	+.248 (6.30)	52	+.000 (0.00)	-.068 (1.73)
26	+.183 (4.65)	+.195 (4.95)	53	-.080 (2.03)	-.002 (0.05)
27	+.245 (6.22)	+.116 (2.95)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-53	53	22	M	All	MS27348-53

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-67)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.081 (2.06)	35	+.088 (2.24)	+.360 (9.14)
2	-.357 (9.07)	-.009 (0.23)	36	+.045 (1.14)	+.261 (6.63)
3	-.357 (9.07)	-.099 (2.51)	37	+.045 (1.14)	+.171 (4.34)
4	-.304 (7.72)	+.213 (5.41)	38	+.045 (1.14)	+.081 (2.06)
5	-.279 (7.09)	+.126 (3.20)	39	+.045 (1.14)	-.009 (0.23)
6	-.279 (7.09)	+.036 (0.91)	40	+.045 (1.14)	-.099 (2.51)
7	-.279 (7.09)	-.054 (1.37)	41	+.045 (1.14)	-.189 (4.80)
8	-.279 (7.09)	-.144 (3.66)	42	+.045 (1.14)	-.279 (7.09)
9	-.279 (7.09)	-.234 (5.94)	43	+.045 (1.14)	-.369 (9.37)
10	-.244 (6.20)	+.280 (7.11)	44	+.156 (3.96)	+.301 (7.65)
11	-.201 (5.11)	+.171 (4.34)	45	+.123 (3.12)	+.216 (5.49)
12	-.201 (5.11)	+.081 (2.06)	46	+.123 (3.12)	+.126 (3.20)
13	-.201 (5.11)	-.009 (0.23)	47	+.123 (3.12)	+.036 (0.91)
14	-.201 (5.11)	-.099 (2.51)	48	+.123 (3.12)	-.054 (1.37)
15	-.201 (5.11)	-.189 (4.80)	49	+.123 (3.12)	-.144 (3.66)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

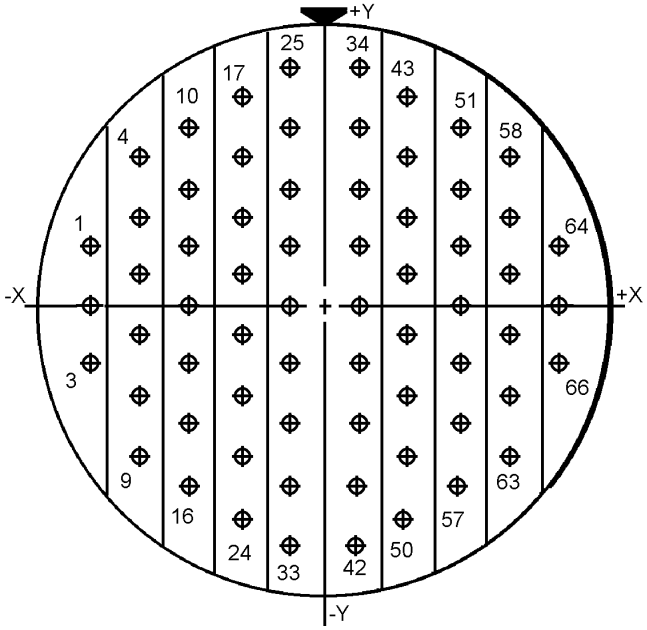
Contacts (Insert arrangement 18-67)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
16	-.201 (5.11)	-.279 (7.09)	50	+.123 (3.12)	-.234 (5.94)
17	-.156 (3.96)	+.301 (7.65)	51	+.123 (3.12)	-.324 (8.23)
18	-.123 (3.12)	+.216 (5.49)	52	+.244 (6.20)	+.280 (7.11)
19	-.123 (3.12)	+.126 (3.20)	53	+.201 (5.11)	+.171 (4.34)
20	-.123 (3.12)	+.036 (0.91)	54	+.201 (5.11)	+.081 (2.06)
21	-.123 (3.12)	-.054 (1.37)	55	+.201 (5.11)	-.009 (0.23)
22	-.123 (3.12)	-.144 (3.66)	56	+.201 (5.11)	-.099 (2.51)
23	-.123 (3.12)	-.234 (5.94)	57	+.201 (5.11)	-.189 (4.80)
24	-.123 (3.12)	-.324 (8.23)	58	+.201 (5.11)	-.279 (7.09)
25	-.088 (2.24)	+.360 (9.14)	59	+.304 (7.72)	+.213 (5.41)
26	-.000 (0.00)	+.339 (8.61)	60	+.279 (7.09)	+.126 (3.20)
27	-.045 (1.14)	+.261 (6.63)	61	+.279 (7.09)	+.036 (0.91)
28	-.045 (1.14)	+.171 (4.34)	62	+.279 (7.09)	-.054 (1.37)
29	-.045 (1.14)	+.081 (2.06)	63	+.279 (7.09)	-.144 (3.66)
30	-.045 (1.14)	-.009 (0.23)	64	+.279 (7.09)	-.234 (5.94)
31	-.045 (1.14)	-.099 (2.51)	65	+.357 (9.07)	+.081 (2.06)
32	-.045 (1.14)	-.189 (4.80)	66	+.357 (9.07)	-.009 (0.23)
33	-.045 (1.14)	-.279 (7.09)	67	+.357 (9.07)	-.099 (2.51)
34	-.045 (1.14)	-.369 (9.37)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-67	67	22M	M	All	MS27348-67

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.090 (2.29)	34	+.045 (1.14)	+.360 (9.14)
2	-.357 (9.07)	+.000 (0.00)	35	+.045 (1.14)	+.270 (6.86)
3	-.357 (9.07)	-.090 (2.29)	36	+.045 (1.14)	+.180 (4.57)
4	-.279 (7.09)	+.225 (5.72)	37	+.045 (1.14)	+.090 (2.29)
5	-.279 (7.09)	+.135 (3.43)	38	+.045 (1.14)	+.000 (0.00)
6	-.279 (7.09)	+.045 (1.14)	39	+.045 (1.14)	-.090 (2.29)
7	-.279 (7.09)	-.045 (1.14)	40	+.045 (1.14)	-.180 (4.57)
8	-.279 (7.09)	-.135 (3.43)	41	+.045 (1.14)	-.270 (6.86)
9	-.279 (7.09)	-.225 (5.72)	42	+.045 (1.14)	-.360 (9.14)
10	-.201 (5.11)	+.270 (6.86)	43	+.123 (3.12)	+.315 (8.00)
11	-.201 (5.11)	+.180 (4.57)	44	+.123 (3.12)	+.225 (5.72)
12	-.201 (5.11)	+.090 (2.29)	45	+.123 (3.12)	+.135 (3.43)
13	-.201 (5.11)	+.000 (0.00)	46	+.123 (3.12)	+.045 (1.14)
14	-.201 (5.11)	-.090 (2.29)	47	+.123 (3.12)	-.045 (1.14)
15	-.201 (5.11)	-.180 (4.57)	48	+.123 (3.12)	-.135 (3.43)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

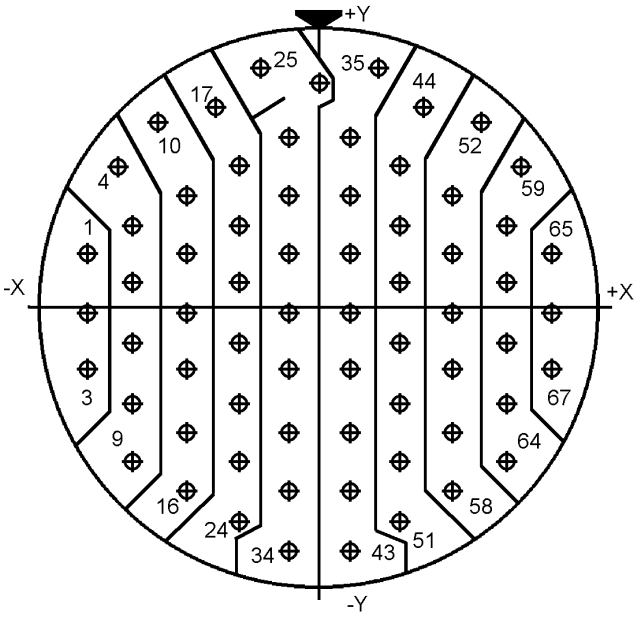
Contacts (Insert arrangement 18-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
16	-.201 (5.11)	-.270 (6.86)	49	+.123 (3.12)	-.225 (5.72)
17	-.123 (3.12)	+.315 (8.00)	50	+.123 (3.12)	-.315 (8.00)
18	-.123 (3.12)	+.225 (5.72)	51	+.201 (5.11)	+.270 (6.86)
19	-.123 (3.12)	+.135 (3.43)	52	+.201 (5.11)	+.180 (4.57)
20	-.123 (3.12)	+.045 (1.14)	53	+.201 (5.11)	+.090 (2.29)
21	-.123 (3.12)	-.045 (1.14)	54	+.201 (5.11)	+.000 (0.00)
22	-.123 (3.12)	-.135 (3.43)	55	+.201 (5.11)	-.090 (2.29)
23	-.123 (3.12)	-.225 (5.72)	56	+.201 (5.11)	-.180 (4.57)
24	-.123 (3.12)	-.315 (8.00)	57	+.201 (5.11)	-.270 (6.86)
25	-.045 (1.14)	+.360 (9.14)	58	+.279 (7.09)	+.225 (5.72)
26	-.045 (1.14)	+.270 (6.86)	59	+.279 (7.09)	+.135 (3.43)
27	-.045 (1.14)	+.180 (4.57)	60	+.279 (7.09)	+.045 (1.14)
28	-.045 (1.14)	+.090 (2.29)	61	+.279 (7.09)	-.045 (1.14)
29	-.045 (1.14)	+.000 (0.00)	62	+.279 (7.09)	-.135 (3.43)
30	-.045 (1.14)	-.090 (2.29)	63	+.279 (7.09)	-.225 (5.72)
31	-.045 (1.14)	-.180 (4.57)	64	+.357 (9.07)	+.090 (2.29)
32	-.045 (1.14)	-.270 (6.86)	65	+.357 (9.07)	+.000 (0.00)
33	-.045 (1.14)	-.360 (9.14)	66	+.357 (9.07)	-.090 (2.29)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-35	66	22D	M	All	MS27521-35

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-45)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.081 (2.06)	35	+.088 (2.24)	+.360 (9.14)
2	-.357 (9.07)	-.009 (0.23)	36	+.045 (1.14)	+.261 (6.63)
3	-.357 (9.07)	-.099 (2.51)	37	+.045 (1.14)	+.171 (4.34)
4	-.304 (7.72)	+.213 (5.41)	38	+.045 (1.14)	+.081 (2.06)
5	-.279 (7.09)	+.126 (3.20)	39	+.045 (1.14)	-.009 (0.23)
6	-.279 (7.09)	+.036 (0.91)	40	+.045 (1.14)	-.099 (2.51)
7	-.279 (7.09)	-.054 (1.37)	41	+.045 (1.14)	-.189 (4.80)
8	-.279 (7.09)	-.144 (3.66)	42	+.045 (1.14)	-.279 (7.09)
9	-.279 (7.09)	-.234 (5.94)	43	+.045 (1.14)	-.369 (9.37)
10	-.244 (6.20)	+.280 (7.11)	44	+.156 (3.96)	+.301 (7.65)
11	-.201 (5.11)	+.171 (4.34)	45	+.123 (3.12)	+.216 (5.49)
12	-.201 (5.11)	+.081 (2.06)	46	+.123 (3.12)	+.126 (3.20)
13	-.201 (5.11)	-.009 (0.23)	47	+.123 (3.12)	+.036 (0.91)
14	-.201 (5.11)	-.099 (2.51)	48	+.123 (3.12)	-.054 (1.37)
15	-.201 (5.11)	-.189 (4.80)	49	+.123 (3.12)	-.144 (3.66)
16	-.201 (5.11)	-.279 (7.09)	50	+.123 (3.12)	-.234 (5.94)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

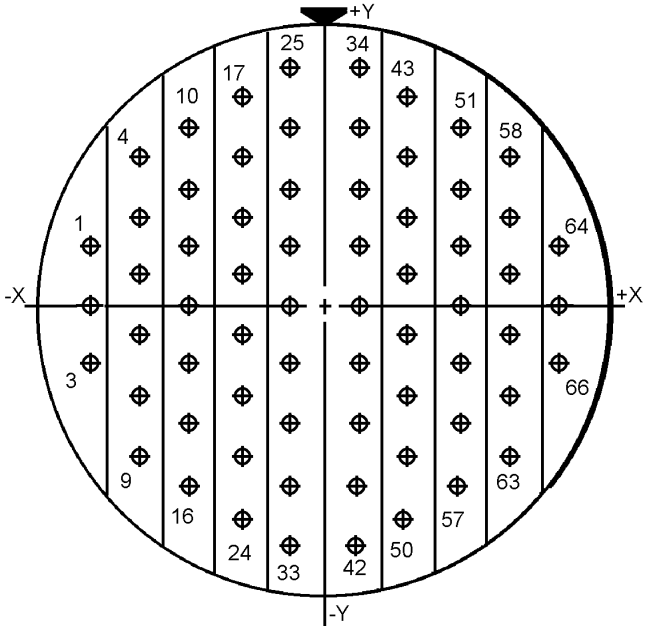
Contacts (Insert arrangement 18-45)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.156 (3.96)	+.301 (7.65)	51	+.123 (3.12)	-.324 (8.23)
18	-.123 (3.12)	+.216 (5.49)	52	+.244 (6.20)	+.280 (7.11)
19	-.123 (3.12)	+.126 (3.20)	53	+.201 (5.11)	+.171 (4.34)
20	-.123 (3.12)	+.036 (0.91)	54	+.201 (5.11)	+.081 (2.06)
21	-.123 (3.12)	-.054 (1.37)	55	+.201 (5.11)	-.009 (0.23)
22	-.123 (3.12)	-.144 (3.66)	56	+.201 (5.11)	-.099 (2.51)
23	-.123 (3.12)	-.234 (5.94)	57	+.201 (5.11)	-.189 (4.80)
24	-.123 (3.12)	-.324 (8.23)	58	+.201 (5.11)	-.279 (7.09)
25	-.088 (2.24)	+.360 (9.14)	59	+.304 (7.72)	+.213 (5.41)
26	-.000 (0.00)	+.339 (8.61)	60	+.279 (7.09)	+.126 (3.20)
27	-.045 (1.14)	+.261 (6.63)	61	+.279 (7.09)	+.036 (0.91)
28	-.045 (1.14)	+.171 (4.34)	62	+.279 (7.09)	-.054 (1.37)
29	-.045 (1.14)	+.081 (2.06)	63	+.279 (7.09)	-.144 (3.66)
30	-.045 (1.14)	-.009 (0.23)	64	+.279 (7.09)	-.234 (5.94)
31	-.045 (1.14)	-.099 (2.51)	65	+.357 (9.07)	+.081 (2.06)
32	-.045 (1.14)	-.189 (4.80)	66	+.357 (9.07)	-.009 (0.23)
33	-.045 (1.14)	-.279 (7.09)	67	+.357 (9.07)	-.099 (2.51)
34	-.045 (1.14)	-.369 (9.37)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-45	67	22D	M	All	MS27521-45

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 18-66)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.357 (9.07)	+.090 (2.29)	34	+.045 (1.14)	+.360 (9.14)
2	-.357 (9.07)	+.000 (0.00)	35	+.045 (1.14)	+.270 (6.86)
3	-.357 (9.07)	-.090 (2.29)	36	+.045 (1.14)	+.180 (4.57)
4	-.279 (7.09)	+.225 (5.72)	37	+.045 (1.14)	+.090 (2.29)
5	-.279 (7.09)	+.135 (3.43)	38	+.045 (1.14)	+.000 (0.00)
6	-.279 (7.09)	+.045 (1.14)	39	+.045 (1.14)	-.090 (2.29)
7	-.279 (7.09)	-.045 (1.14)	40	+.045 (1.14)	-.180 (4.57)
8	-.279 (7.09)	-.135 (3.43)	41	+.045 (1.14)	-.270 (6.86)
9	-.279 (7.09)	-.225 (5.72)	42	+.045 (1.14)	-.360 (9.14)
10	-.201 (5.11)	+.270 (6.86)	43	+.123 (3.12)	+.315 (8.00)
11	-.201 (5.11)	+.180 (4.57)	44	+.123 (3.12)	+.225 (5.72)
12	-.201 (5.11)	+.090 (2.29)	45	+.123 (3.12)	+.135 (3.43)
13	-.201 (5.11)	+.000 (0.00)	46	+.123 (3.12)	+.045 (1.14)
14	-.201 (5.11)	-.090 (2.29)	47	+.123 (3.12)	-.045 (1.14)
15	-.201 (5.11)	-.180 (4.57)	48	+.123 (3.12)	-.135 (3.43)
16	-.201 (5.11)	-.270 (6.86)	49	+.123 (3.12)	-.225 (5.72)

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

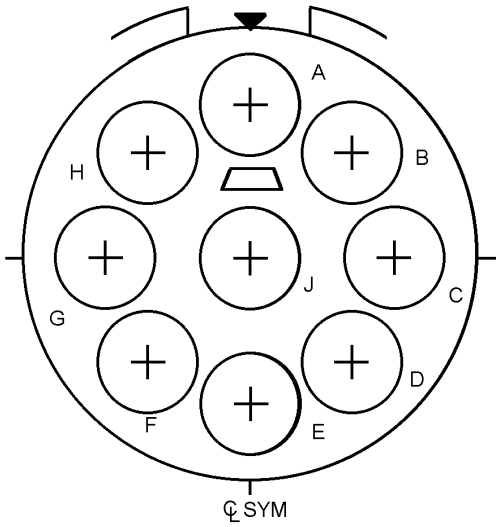
Contacts (Insert arrangement 18-66)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.123 (3.12)	+.315 (8.00)	50	+.123 (3.12)	-.315 (8.00)
18	-.123 (3.12)	+.225 (5.72)	51	+.201 (5.11)	+.270 (6.86)
19	-.123 (3.12)	+.135 (3.43)	52	+.201 (5.11)	+.180 (4.57)
20	-.123 (3.12)	+.045 (1.14)	53	+.201 (5.11)	+.090 (2.29)
21	-.123 (3.12)	-.045 (1.14)	54	+.201 (5.11)	+.000 (0.00)
22	-.123 (3.12)	-.135 (3.43)	55	+.201 (5.11)	-.090 (2.29)
23	-.123 (3.12)	-.225 (5.72)	56	+.201 (5.11)	-.180 (4.57)
24	-.123 (3.12)	-.315 (8.00)	57	+.201 (5.11)	-.270 (6.86)
25	-.045 (1.14)	+.360 (9.14)	58	+.279 (7.09)	+.225 (5.72)
26	-.045 (1.14)	+.270 (6.86)	59	+.279 (7.09)	+.135 (3.43)
27	-.045 (1.14)	+.180 (4.57)	60	+.279 (7.09)	+.045 (1.14)
28	-.045 (1.14)	+.090 (2.29)	61	+.279 (7.09)	-.045 (1.14)
29	-.045 (1.14)	+.000 (0.00)	62	+.279 (7.09)	-.135 (3.43)
30	-.045 (1.14)	-.090 (2.29)	63	+.279 (7.09)	-.225 (5.72)
31	-.045 (1.14)	-.180 (4.57)	64	+.357 (9.07)	+.090 (2.29)
32	-.045 (1.14)	-.270 (6.86)	65	+.357 (9.07)	+.000 (0.00)
33	-.045 (1.14)	-.360 (9.14)	66	+.357 (9.07)	-.090 (2.29)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
18	-66	66	22M	M	All	MS27348-66

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

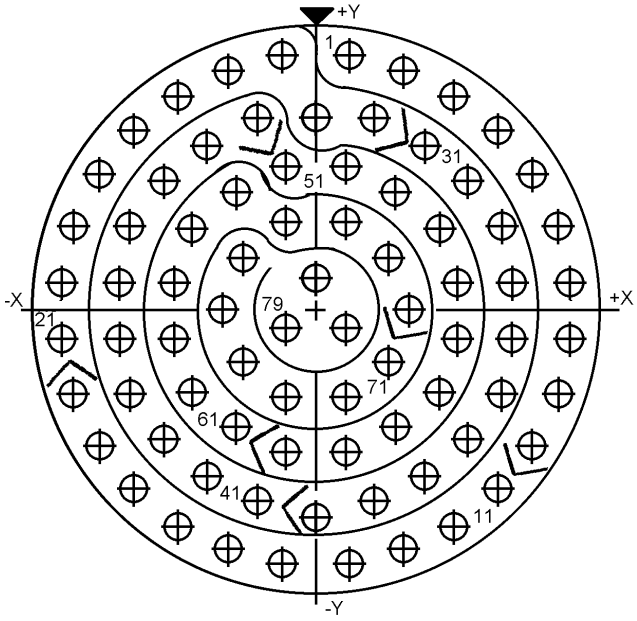


Contacts (Insert arrangement 18-96)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.301 (7.65)	F	-.213 (5.41)	-.213 (5.41)
B	+.213 (5.41)	+.213 (5.41)	G	-.301 (7.65)	+.000 (0.00)
C	+.301 (7.65)	+.000 (0.00)	H	-.213 (5.41)	+.213 (5.41)
D	+.213 (5.41)	-.213 (5.41)	J	+.000 (0.00)	+.000 (0.00)
E	+.000 (0.00)	-.301 (7.65)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
18	-96	9	12	I	All

FIGURE 15. Insert arrangement, electrical connector, shell size 18 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 20-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.053 (1.35)	+.426 (10.82)	41	-.098 (2.49)	-.322 (8.18)
2	+.146 (3.71)	+.404 (10.26)	42	-.184 (4.67)	-.280 (7.11)
3	+.232 (5.89)	+.362 (9.19)	43	-.258 (6.55)	-.220 (5.59)
4	+.306 (7.77)	+.302 (7.67)	44	-.311 (7.90)	-.141 (3.58)
5	+.365 (9.27)	+.227 (5.77)	45	-.332 (8.43)	-.048 (1.22)
6	+.406 (10.31)	+.141 (3.58)	46	-.332 (8.43)	+.048 (1.22)
7	+.427 (10.85)	+.048 (1.22)	47	-.311 (7.90)	+.141 (3.58)
8	+.427 (10.85)	-.048 (1.22)	48	-.258 (6.55)	+.220 (5.59)
9	+.406 (10.31)	-.141 (3.58)	49	-.184 (4.67)	+.280 (7.11)
10	+.365 (9.27)	-.227 (5.77)	50	-.098 (2.49)	+.322 (8.18)
11	+.306 (7.77)	-.302 (7.67)	51	-.048 (1.22)	+.241 (6.12)
12	+.232 (5.89)	-.362 (9.19)	52	+.048 (1.22)	+.241 (6.12)
13	+.146 (3.71)	-.404 (10.26)	53	+.134 (3.40)	+.199 (5.05)
14	+.053 (1.35)	-.426 (10.82)	54	+.208 (5.28)	+.139 (3.53)
15	-.053 (1.35)	-.426 (10.82)	55	+.237 (6.02)	+.048 (1.22)

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

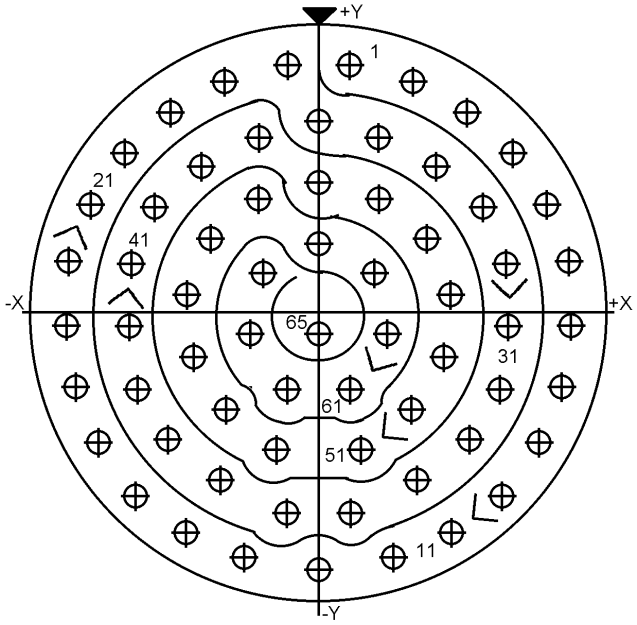
Contacts (Insert arrangement 20-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
16	-.146 (3.71)	-.404 (10.26)	56	+.237 (6.02)	-.048 (1.22)
17	-.232 (5.89)	-.362 (9.19)	57	+.208 (5.28)	-.139 (3.53)
18	-.306 (7.77)	-.302 (7.67)	58	+.134 (3.40)	-.199 (5.05)
19	-.365 (9.27)	-.227 (5.77)	59	+.048 (1.22)	-.241 (6.12)
20	-.406 (10.31)	-.141 (3.58)	60	-.048 (1.22)	-.241 (6.12)
21	-.427 (10.85)	-.048 (1.22)	61	-.134 (3.40)	-.199 (5.05)
22	-.427 (10.85)	+.048 (1.22)	62	-.208 (5.28)	-.139 (3.53)
23	-.406 (10.31)	+.141 (3.58)	63	-.237 (6.02)	-.048 (1.22)
24	-.365 (9.27)	+.227 (5.77)	64	-.237 (6.02)	+.048 (1.22)
25	-.306 (7.77)	+.302 (7.67)	65	-.208 (5.28)	+.139 (3.53)
26	-.232 (5.89)	+.362 (9.19)	66	-.134 (3.40)	+.199 (5.05)
27	-.146 (3.71)	+.404 (10.26)	67	-.048 (1.22)	+.146 (3.71)
28	-.053 (1.35)	+.426 (10.82)	68	+.048 (1.22)	+.146 (3.71)
29	+.000 (0.00)	+.323 (8.20)	69	+.125 (3.18)	+.090 (2.29)
30	+.098 (2.49)	+.322 (8.18)	70	+.155 (3.94)	+.000 (0.00)
31	+.184 (4.67)	+.280 (7.11)	71	+.125 (3.18)	-.090 (2.29)
32	+.258 (6.55)	+.220 (5.59)	72	+.048 (1.22)	-.146 (3.71)
33	+.311 (7.90)	+.141 (3.58)	73	-.048 (1.22)	-.146 (3.71)
34	+.332 (8.43)	+.048 (1.22)	74	-.125 (3.18)	-.090 (2.29)
35	+.332 (8.43)	-.048 (1.22)	75	-.155 (3.94)	+.000 (0.00)
36	+.311 (7.90)	-.141 (3.58)	76	-.125 (3.18)	+.090 (2.29)
37	+.258 (6.55)	-.220 (5.59)	77	+.000 (0.00)	+.053 (1.35)
38	+.184 (4.67)	-.280 (7.11)	78	+.048 (1.22)	-.029 (0.74)
39	+.098 (2.49)	-.322 (8.18)	79	-.048 (1.22)	-.029 (0.74)
40	+.000 (0.00)	-.347 (8.81)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-1	79	22M	M	All	MS27349-1

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 20-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.052 (1.32)	+.424 (10.77)	34	+.165 (4.19)	-.273 (6.93)
2	+.154 (3.91)	+.399 (10.13)	35	+.053 (1.35)	-.331 (8.41)
3	+.246 (6.25)	+.348 (8.84)	36	-.053 (1.35)	-.331 (8.41)
4	+.326 (8.28)	+.280 (7.11)	37	-.165 (4.19)	-.273 (6.93)
5	+.382 (9.70)	+.191 (4.85)	38	-.252 (6.40)	-.210 (5.33)
6	+.417 (10.59)	+.092 (2.34)	39	-.305 (7.75)	-.119 (3.02)
7	+.425 (10.80)	-.013 (.33)	40	-.320 (8.13)	-.015 (0.38)
8	+.410 (10.41)	-.117 (2.97)	41	-.312 (7.92)	+.090 (2.29)
9	+.370 (9.40)	-.214 (5.44)	42	-.274 (6.96)	+.188 (4.78)
10	+.308 (7.82)	-.299 (7.59)	43	-.194 (4.93)	+.256 (6.50)
11	+.224 (5.69)	-.362 (9.19)	44	-.102 (2.59)	+.307 (7.80)
12	+.128 (3.25)	-.405 (10.29)	45	+.000 (0.00)	+.227 (5.77)
13	+.000 (0.00)	-.422 (10.72)	46	+.102 (2.59)	+.202 (5.13)
14	-.128 (3.25)	-.405 (10.29)	47	+.183 (4.65)	+.135 (3.43)
15	-.224 (5.69)	-.362 (9.19)	48	+.220 (5.59)	+.037 (0.94)
16	-.308 (7.82)	-.299 (7.59)	49	+.213 (5.41)	-.068 (1.73)
17	-.370 (9.40)	-.214 (5.44)	50	+.158 (4.01)	-.158 (4.01)
18	-.410 (10.41)	-.117 (2.97)	51	+.072 (1.83)	-.224 (5.69)
19	-.425 (10.80)	-.013 (0.33)	52	-.072 (1.83)	-.224 (5.69)

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

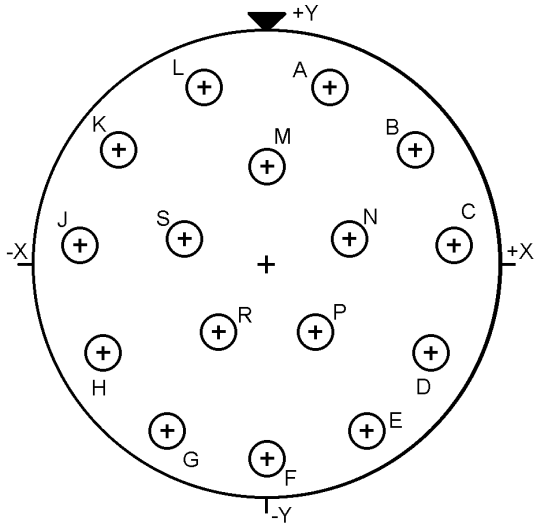
Contacts (Insert arrangement 20-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
20	-.417 (10.59)	+.092 (2.34)	53	-.158 (4.01)	-.158 (4.01)
21	-.382 (9.70)	+.191 (4.85)	54	-.213 (5.41)	-.068 (1.73)
22	-.326 (8.28)	+.280 (7.11)	55	-.220 (5.59)	+.037 (0.94)
23	-.246 (6.25)	+.348 (8.84)	56	-.183 (4.65)	+.135 (3.43)
24	-.154 (3.91)	+.399 (10.13)	57	-.102 (2.59)	+.202 (5.13)
25	-.052 (1.32)	+.424 (10.77)	58	+.000 (0.00)	+.122 (3.10)
26	+.000 (0.00)	+.332 (8.43)	59	+.095 (2.41)	+.075 (1.91)
27	+.102 (2.59)	+.307 (7.80)	60	+.115 (2.92)	-.029 (0.74)
28	+.194 (4.93)	+.256 (6.50)	61	+.056 (1.42)	-.120 (3.05)
29	+.274 (6.96)	+.188 (4.78)	62	-.056 (1.42)	-.120 (3.05)
30	+.312 (7.92)	+.090 (2.29)	63	-.115 (2.92)	-.029 (0.74)
31	+.320 (8.13)	-.015 (0.38)	64	-.095 (2.41)	+.075 (1.91)
32	+.305 (7.75)	-.119 (3.02)	65	+.000 (0.00)	-.029 (0.74)
33	+.252 (6.40)	-.210 (5.33)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-2	65	22	M	All	MS27349-2

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

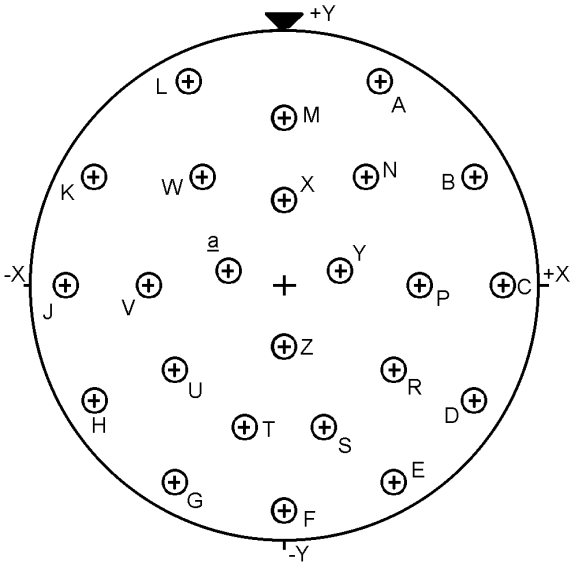


Contacts (Insert arrangement 20-16)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.118 (3.00)	+.322 (8.18)	J	-.341 (8.66)	+.036 (0.91)
B	+.271 (6.88)	+.211 (5.36)	K	-.271 (6.88)	+.211 (5.36)
C	+.341 (8.66)	+.036 (0.91)	L	-.118 (3.00)	+.322 (8.18)
D	+.308 (7.82)	-.150 (3.81)	M	+.000 (0.00)	+.175 (4.45)
E	+.182 (4.62)	-.290 (7.37)	N	+.154 (3.91)	+.062 (1.57)
F	+.000 (0.00)	-.343 (8.71)	P	+.094 (2.39)	-.122 (3.10)
G	-.182 (4.62)	-.290 (7.37)	R	-.094 (2.39)	-.122 (3.10)
H	-.308 (7.82)	-.150 (3.81)	S	-.154 (3.91)	+.062 (1.57)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-16	16	16	II	All	MS27349-16

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

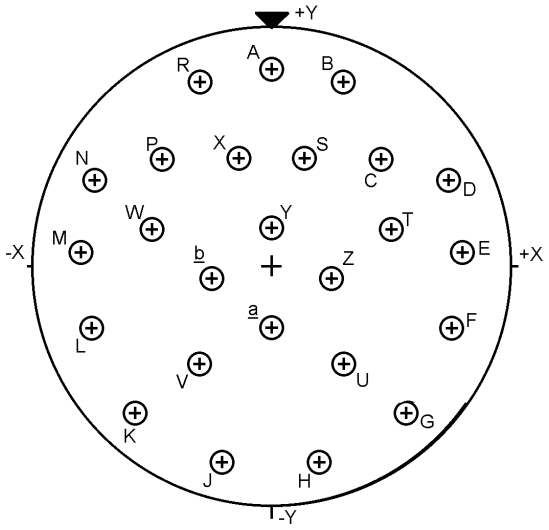


Contacts (Insert arrangement 20-24)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.175 (4.45)	+.375 (9.53)	N	+.150 (3.81)	+.200 (5.08)
B	+.350 (8.89)	+.200 (5.08)	P	+.250 (6.35)	+.000 (0.00)
C	+.400 (10.16)	+.000 (0.00)	R	+.200 (5.08)	-.150 (3.81)
D	+.350 (8.89)	-.200 (5.08)	S	+.075 (1.91)	-.250 (6.35)
E	+.200 (5.08)	-.350 (8.89)	T	-.075 (1.91)	-.250 (6.35)
F	+.000 (0.00)	-.400 (10.16)	U	-.200 (5.08)	-.150 (3.81)
G	-.200 (5.08)	-.350 (8.89)	V	-.250 (6.35)	+.000 (0.00)
H	-.350 (8.89)	-.200 (5.08)	W	-.150 (3.81)	+.200 (5.08)
J	-.400 (10.16)	+.000 (0.00)	X	+.000 (0.00)	+.150 (3.81)
K	-.350 (8.89)	+.200 (5.08)	Y	+.100 (2.54)	+.025 (0.64)
L	-.175 (4.45)	+.375 (9.53)	Z	+.000 (0.00)	-.100 (2.54)
M	+.000 (0.00)	+.300 (7.62)	a	-.100 (2.54)	+.025 (.64)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-24	24	20	I	All	MS27349-24

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

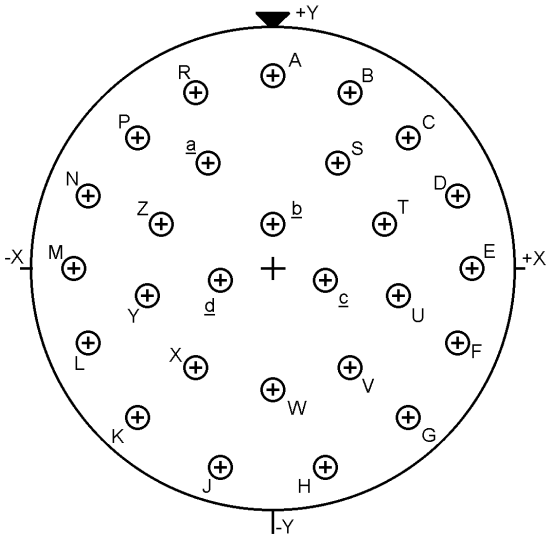


Contacts (Insert arrangement 20-25)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.400 (10.16)	P	-.225 (5.72)	+.225 (5.72)
B	+.150 (3.81)	+.375 (9.53)	R	-.150 (3.81)	+.375 (9.53)
C	+.225 (5.72)	+.225 (5.72)	S	+.075 (1.91)	+.225 (5.72)
D	+.375 (9.53)	+.175 (4.45)	T	+.250 (6.35)	+.075 (1.91)
E	+.400 (10.16)	+.025 (0.64)	U	+.150 (3.81)	-.200 (5.08)
F	+.375 (9.53)	-.125 (3.18)	V	-.150 (3.81)	-.200 (5.08)
G	+.275 (6.99)	-.300 (7.62)	W	-.250 (6.35)	+.075 (1.91)
H	+.100 (2.54)	-.400 (10.16)	X	-.075 (1.91)	+.225 (5.72)
J	-.100 (2.54)	-.400 (10.16)	Y	+.000 (0.00)	+.075 (1.91)
K	-.275 (6.99)	-.300 (7.62)	Z	+.125 (3.18)	-.025 (0.64)
L	-.375 (9.53)	-.125 (3.18)	a	+.000 (0.00)	-.125 (3.18)
M	-.400 (10.16)	+.025 (0.64)	b	-.125 (3.18)	-.025 (0.64)
N	-.375 (9.53)	+.175 (4.45)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-25	25	20	I	All	MS27349-25

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

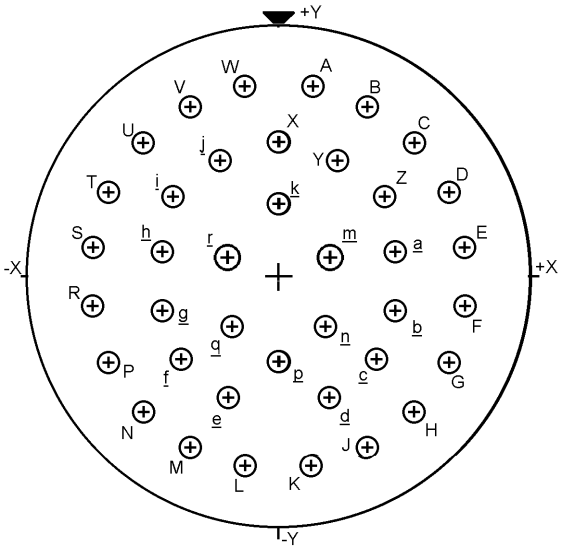


Contacts (Insert arrangement 20-27)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.400 (10.16)	R	-.150 (3.81)	+.375 (9.53)
B	+.150 (3.81)	+.375 (9.53)	S	+.125 (3.18)	+.225 (5.72)
C	+.275 (6.99)	+.275 (6.99)	T	+.225 (5.72)	+.100 (2.54)
D	+.375 (9.53)	+.150 (3.81)	U	+.250 (6.35)	-.050 (1.27)
E	+.400 (10.16)	+.000 (0.00)	V	+.150 (3.81)	-.200 (5.08)
F	+.375 (9.53)	-.150 (3.81)	W	+.000 (0.00)	-.250 (6.35)
G	+.275 (6.99)	-.300 (7.62)	X	-.150 (3.81)	-.200 (5.08)
H	+.100 (2.54)	-.400 (10.16)	Y	-.250 (6.35)	-.050 (1.27)
J	-.100 (2.54)	-.400 (10.16)	Z	-.225 (5.72)	+.100 (2.54)
K	-.275 (6.99)	-.300 (7.62)	a	-.125 (3.18)	+.225 (5.72)
L	-.375 (9.53)	-.150 (3.81)	b	+.000 (0.00)	+.100 (2.54)
M	-.400 (10.16)	+.000 (0.00)	c	+.100 (2.54)	-.025 (0.64)
N	-.375 (9.53)	+.150 (3.81)	d	-.100 (2.54)	-.025 (0.64)
P	-.275 (6.99)	+.275 (6.99)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-27	27	20	I	All	MS27349-27

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 20-39)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.065 (1.65)	+.411 (10.44)	X	+.000 (0.00)	+.295 (7.49)
B	+.189 (4.80)	+.371 (9.42)	Y	+.126 (3.20)	+.256 (6.50)
C	+.294 (7.47)	+.295 (7.49)	Z	+.232 (5.89)	+.179 (4.55)
D	+.371 (9.42)	+.189 (4.80)	a	+.280 (7.11)	+.057 (1.45)
E	+.411 (10.44)	+.065 (1.65)	b	+.280 (7.11)	-.074 (1.88)
F	+.411 (10.44)	-.065 (1.65)	c	+.217 (5.51)	-.189 (4.80)
G	+.371 (9.42)	-.189 (4.80)	d	+.112 (2.84)	-.265 (6.73)
H	+.294 (7.47)	-.294 (7.47)	e	-.112 (2.84)	-.265 (6.73)
J	+.189 (4.80)	-.371 (9.42)	f	-.217 (5.51)	-.189 (4.80)
K	+.065 (1.65)	-.411 (10.44)	g	-.280 (7.11)	-.074 (1.88)
L	-.065 (1.65)	-.411 (10.44)	h	-.280 (7.11)	+.057 (1.45)
M	-.189 (4.80)	-.371 (9.42)	i	-.232 (5.89)	+.179 (4.55)
N	-.294 (7.47)	-.294 (7.47)	j	-.126 (3.20)	+.256 (6.50)
P	-.371 (9.42)	-.189 (4.80)	k	+.000 (0.00)	+.164 (4.17)
R	-.411 (10.44)	-.065 (1.65)	m	+.114 (2.90)	+.048 (1.22)

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

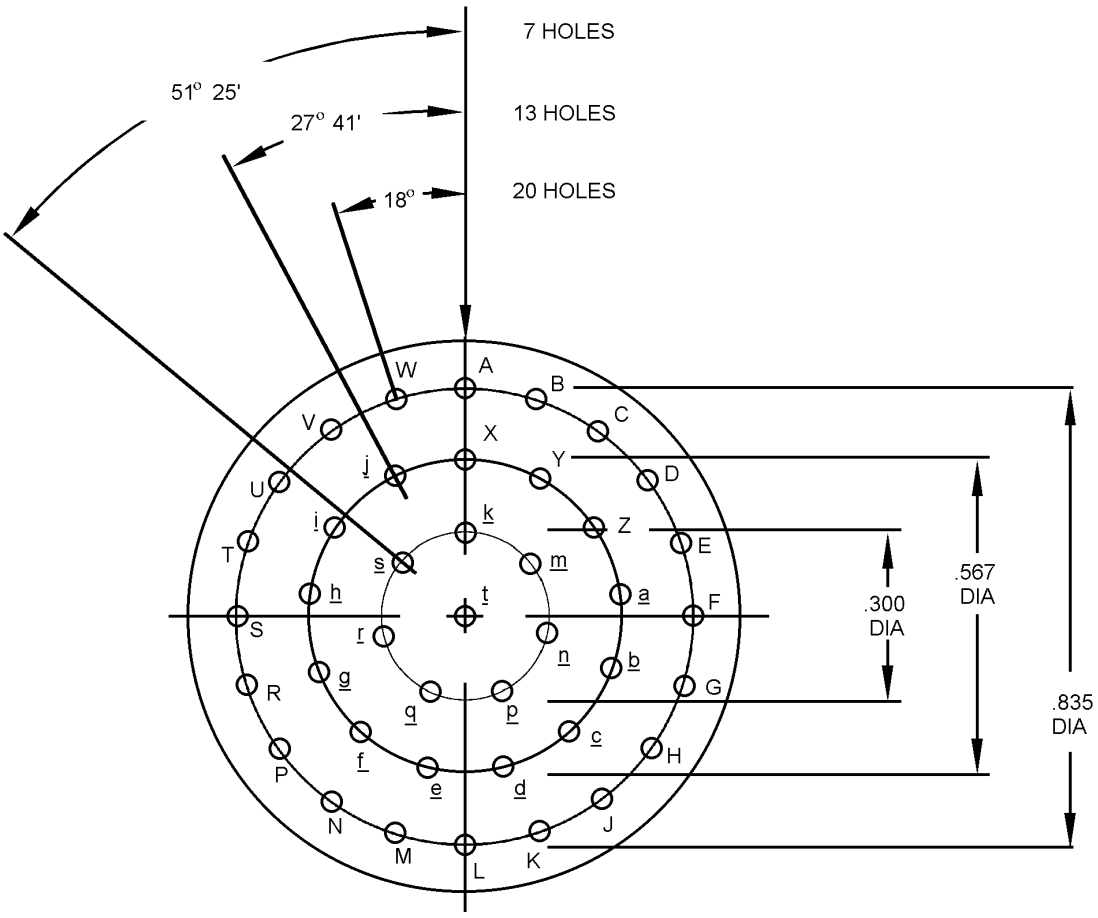
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 20-39)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
S	-.411 (10.44)	+.065 (1.65)	n	+.106 (2.69)	-.107 (2.72)
T	-.371 (9.42)	+.189 (4.80)	p	+.000 (0.00)	-.189 (4.80)
U	-.294 (7.47)	+.295 (7.49)	g	-.106 (2.69)	-.107 (2.72)
V	-.189 (4.80)	+.371 (9.42)	r	-.114 (2.90)	+.048 (1.22)
W	-.065 (1.65)	+.411 (10.44)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-39	2	16	I	m,r	MS27349-39
		37	20		All others	

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



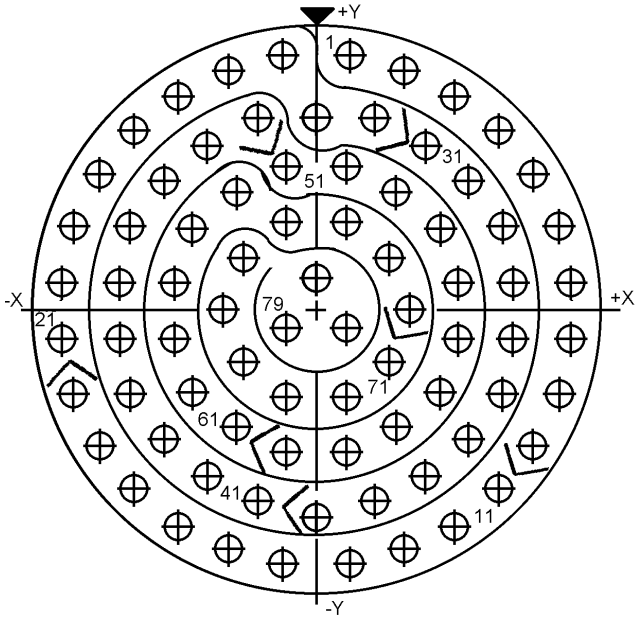
Inches	mm
.300	7.62
.567	14.40
.835	21.21

Insert arrangement (20-41)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-41	41	20	I	All	MS27349-41

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 20-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	+.053 (1.35)	+.426 (10.82)	41	-.098 (2.49)	-.322 (8.18)
2	+.146 (3.71)	+.404 (10.26)	42	-.184 (4.67)	-.280 (7.11)
3	+.232 (5.89)	+.362 (9.19)	43	-.258 (6.55)	-.220 (5.59)
4	+.306 (7.77)	+.302 (7.67)	44	-.311 (7.90)	-.141 (3.58)
5	+.365 (9.27)	+.227 (5.77)	45	-.332 (8.43)	-.048 (1.22)
6	+.406 (10.31)	+.141 (3.58)	46	-.332 (8.43)	+.048 (1.22)
7	+.427 (10.85)	+.048 (1.22)	47	-.311 (7.90)	+.141 (3.58)
8	+.427 (10.85)	-.048 (1.22)	48	-.258 (6.55)	+.220 (5.59)
9	+.406 (10.31)	-.141 (3.58)	49	-.184 (4.67)	+.280 (7.11)
10	+.365 (9.27)	-.227 (5.77)	50	-.098 (2.49)	+.322 (8.18)
11	+.306 (7.77)	-.302 (7.67)	51	-.048 (1.22)	+.241 (6.12)
12	+.232 (5.89)	-.362 (9.19)	52	+.048 (1.22)	+.241 (6.12)
13	+.146 (3.71)	-.404 (10.26)	53	+.134 (3.40)	+.199 (5.05)
14	+.053 (1.35)	-.426 (10.82)	54	+.208 (5.28)	+.139 (3.53)
15	-.053 (1.35)	-.426 (10.82)	55	+.237 (6.02)	+.048 (1.22)
16	-.146 (3.71)	-.404 (10.26)	56	+.237 (6.02)	-.048 (1.22)

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

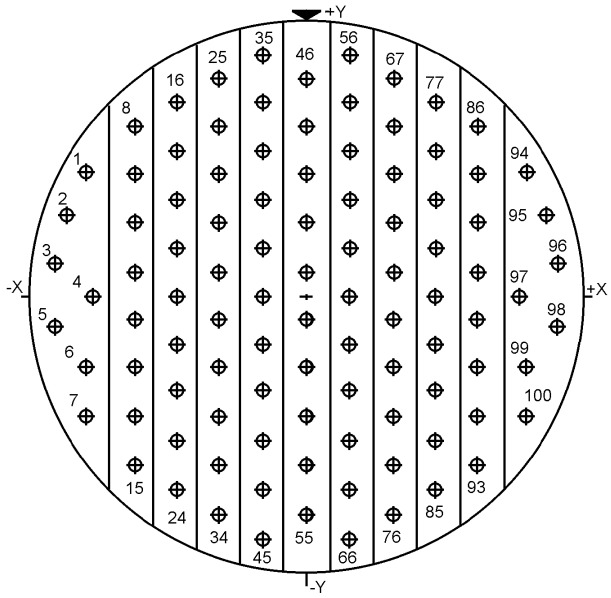
Contacts (Insert arrangement 20-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.232 (5.89)	-.362 (9.19)	57	+.208 (5.28)	-.139 (3.53)
18	-.306 (7.77)	-.302 (7.67)	58	+.134 (3.40)	-.199 (5.05)
19	-.365 (9.27)	-.227 (5.77)	59	+.048 (1.22)	-.241 (6.12)
20	-.406 (10.31)	-.141 (3.58)	60	-.048 (1.22)	-.241 (6.12)
21	-.427 (10.85)	-.048 (1.22)	61	-.134 (3.40)	-.199 (5.05)
22	-.427 (10.85)	+.048 (1.22)	62	-.208 (5.28)	-.139 (3.53)
23	-.406 (10.31)	+.141 (3.58)	63	-.237 (6.02)	-.048 (1.22)
24	-.365 (9.27)	+.227 (5.77)	64	-.237 (6.02)	+.048 (1.22)
25	-.306 (7.77)	+.302 (7.67)	65	-.208 (5.28)	+.139 (3.53)
26	-.232 (5.89)	+.362 (9.19)	66	-.134 (3.40)	+.199 (5.05)
27	-.146 (3.71)	+.404 (10.26)	67	-.048 (1.22)	+.146 (3.71)
28	-.053 (1.35)	+.426 (10.82)	68	+.048 (1.22)	+.146 (3.71)
29	+.000 (0.00)	+.323 (8.20)	69	+.125 (3.18)	+.090 (2.29)
30	+.098 (2.49)	+.322 (8.18)	70	+.155 (3.94)	+.000 (0.00)
31	+.184 (4.67)	+.280 (7.11)	71	+.125 (3.18)	-.090 (2.29)
32	+.258 (6.55)	+.220 (5.59)	72	+.048 (1.22)	-.146 (3.71)
33	+.311 (7.90)	+.141 (3.58)	73	-.048 (1.22)	-.146 (3.71)
34	+.332 (8.43)	+.048 (1.22)	74	-.125 (3.18)	-.090 (2.29)
35	+.332 (8.43)	-.048 (1.22)	75	-.155 (3.94)	+.000 (0.00)
36	+.311 (7.90)	-.141 (3.58)	76	-.125 (3.18)	+.090 (2.29)
37	+.258 (6.55)	-.220 (5.59)	77	+.000 (0.00)	+.053 (1.35)
38	+.184 (4.67)	-.280 (7.11)	78	+.048 (1.22)	-.029 (0.74)
39	+.098 (2.49)	-.322 (8.18)	79	-.048 (1.22)	-.029 (0.74)
40	+.000 (0.00)	-.347 (8.81)			

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
20	-35	79	22D	M	All	MS27522-35

FIGURE 16. Insert arrangement, electrical connector, shell size 20 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.428 (10.87)	+.241 (6.12)	51	+.000 (0.00)	-.047 (1.19)
2	-.467 (11.86)	+.154 (3.91)	52	+.000 (0.00)	-.142 (3.61)
3	-.488 (12.40)	+.061 (1.55)	53	+.000 (0.00)	-.237 (6.02)
4	-.415 (10.54)	+.000 (0.00)	54	+.000 (0.00)	-.332 (8.43)
5	-.488 (12.40)	-.061 (1.55)	55	+.000 (0.00)	-.427 (10.85)
6	-.428 (10.87)	-.142 (3.61)	56	+.083 (2.11)	+.475 (12.07)
7	-.428 (10.87)	-.237 (6.02)	57	+.083 (2.11)	+.380 (9.65)
8	-.332 (8.43)	+.333 (8.46)	58	+.083 (2.11)	+.285 (7.24)
9	-.332 (8.43)	+.238 (6.05)	59	+.083 (2.11)	+.190 (4.83)
10	-.332 (8.43)	+.143 (3.63)	60	+.083 (2.11)	+.095 (2.41)
11	-.332 (8.43)	+.048 (1.22)	61	+.083 (2.11)	+.000 (0.00)
12	-.332 (8.43)	-.047 (1.19)	62	+.083 (2.11)	-.095 (2.41)
13	-.332 (8.43)	-.142 (3.61)	63	+.083 (2.11)	-.190 (4.83)
14	-.332 (8.43)	-.237 (6.02)	64	+.083 (2.11)	-.285 (7.24)
15	-.332 (8.43)	-.332 (8.43)	65	+.083 (2.11)	-.380 (9.65)
16	-.249 (6.32)	+.380 (9.65)	66	+.083 (2.11)	-.475 (12.07)
17	-.249 (6.32)	+.285 (7.24)	67	+.166 (4.22)	+.428 (10.87)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 22-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.249 (6.32)	+.190 (4.83)	68	+.166 (4.22)	+.333 (8.46)
19	-.249 (6.32)	+.095 (2.41)	69	+.166 (4.22)	+.238 (6.05)
20	-.249 (6.32)	+.000 (0.00)	70	+.166 (4.22)	+.143 (3.63)
21	-.249 (6.32)	-.095 (2.41)	71	+.166 (4.22)	+.048 (1.22)
22	-.249 (6.32)	-.190 (4.83)	72	+.166 (4.22)	-.047 (1.19)
23	-.249 (6.32)	-.285 (7.24)	73	+.166 (4.22)	-.142 (3.61)
24	-.249 (6.32)	-.380 (9.65)	74	+.166 (4.22)	-.237 (6.02)
25	-.166 (4.22)	+.428 (10.87)	75	+.166 (4.22)	-.332 (8.43)
26	-.166 (4.22)	+.333 (8.46)	76	+.166 (4.22)	-.427 (10.85)
27	-.166 (4.22)	+.238 (6.05)	77	+.249 (6.32)	+.380 (9.65)
28	-.166 (4.22)	+.143 (3.63)	78	+.249 (6.32)	+.285 (7.24)
29	-.166 (4.22)	+.048 (1.22)	79	+.249 (6.32)	+.190 (4.83)
30	-.166 (4.22)	-.047 (1.19)	80	+.249 (6.32)	+.095 (2.41)
31	-.166 (4.22)	-.142 (3.61)	81	+.249 (6.32)	+.000 (0.00)
32	-.166 (4.22)	-.237 (6.02)	82	+.249 (6.32)	-.095 (2.41)
33	-.166 (4.22)	-.332 (8.43)	83	+.249 (6.32)	-.190 (4.83)
34	-.166 (4.22)	-.427 (10.85)	84	+.249 (6.32)	-.285 (7.24)
35	-.083 (2.11)	+.475 (12.07)	85	+.249 (6.32)	-.380 (9.65)
36	-.083 (2.11)	+.380 (9.65)	86	+.332 (8.43)	+.333 (8.46)
37	-.083 (2.11)	+.285 (7.24)	87	+.332 (8.43)	+.238 (6.05)
38	-.083 (2.11)	+.190 (4.83)	88	+.332 (8.43)	+.143 (3.63)
39	-.083 (2.11)	+.095 (2.41)	89	+.332 (8.43)	+.048 (1.22)
40	-.083 (2.11)	+.000 (0.00)	90	+.332 (8.43)	-.047 (1.19)
41	-.083 (2.11)	-.095 (2.41)	91	+.332 (8.43)	-.142 (3.61)
42	-.083 (2.11)	-.190 (4.83)	92	+.332 (8.43)	-.237 (6.02)
43	-.083 (2.11)	-.285 (7.24)	93	+.332 (8.43)	-.332 (8.43)
44	-.083 (2.11)	-.380 (9.65)	94	+.428 (10.87)	+.241 (6.12)
45	-.083 (2.11)	-.475 (12.07)	95	+.467 (11.86)	+.154 (3.91)
46	+.000 (0.00)	+.428 (10.87)	96	+.488 (12.40)	+.061 (1.55)
47	+.000 (0.00)	+.333 (8.46)	97	+.415 (10.54)	+.000 (0.00)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

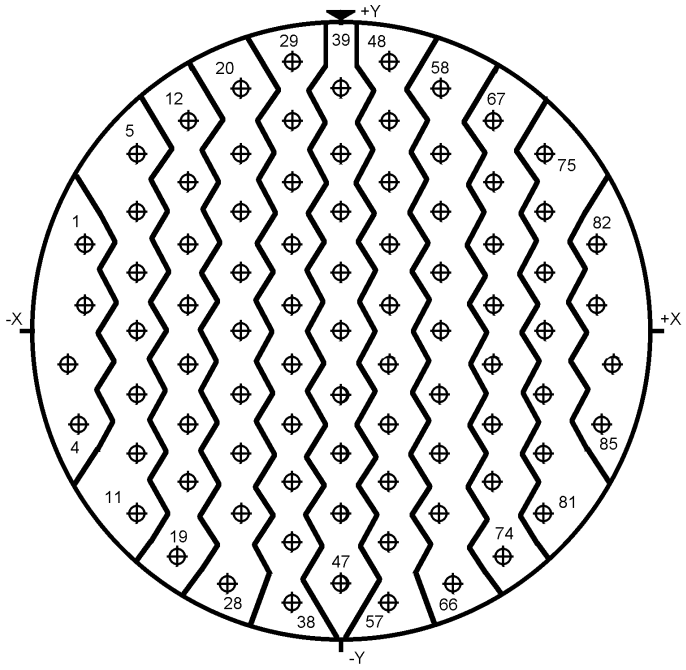
Contacts (Insert arrangement 22-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
48	+.000 (0.00)	+.238 (6.05)	98	+.488 (12.40)	-.061 (1.55)
49	+.000 (0.00)	+.143 (3.63)	99	+.428 (10.87)	-.142 (3.61)
50	+.000 (0.00)	+.048 (1.22)	100	+.428 (10.87)	-.237 (6.02)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-1	100	22M	M	All	MS27350-1

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.455 (11.56)	+.159 (4.04)	44	+.000 (0.00)	-.106 (2.69)
2	-.455 (11.56)	+.053 (1.35)	45	+.000 (0.00)	-.212 (5.38)
3	-.482 (12.24)	-.053 (1.35)	46	+.000 (0.00)	-.318 (8.08)
4	-.455 (11.56)	-.159 (4.04)	47	+.000 (0.00)	-.424 (10.77)
5	-.364 (9.25)	+.318 (8.08)	48	+.091 (2.31)	+.477 (12.12)
6	-.364 (9.25)	+.212 (5.38)	49	+.091 (2.31)	+.371 (9.42)
7	-.364 (9.25)	+.106 (2.69)	50	+.091 (2.31)	+.265 (6.73)
8	-.364 (9.25)	+.000 (0.00)	51	+.091 (2.31)	+.159 (4.04)
9	-.364 (9.25)	-.106 (2.69)	52	+.091 (2.31)	+.053 (1.35)
10	-.364 (9.25)	-.212 (5.38)	53	+.091 (2.31)	-.053 (1.35)
11	-.364 (9.25)	-.318 (8.08)	54	+.091 (2.31)	-.159 (4.04)
12	-.273 (6.93)	+.371 (9.42)	55	+.091 (2.31)	-.265 (6.73)
13	-.273 (6.93)	+.265 (6.73)	56	+.091 (2.31)	-.371 (9.42)
14	-.273 (6.93)	+.159 (4.04)	57	+.091 (2.31)	-.477 (12.12)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

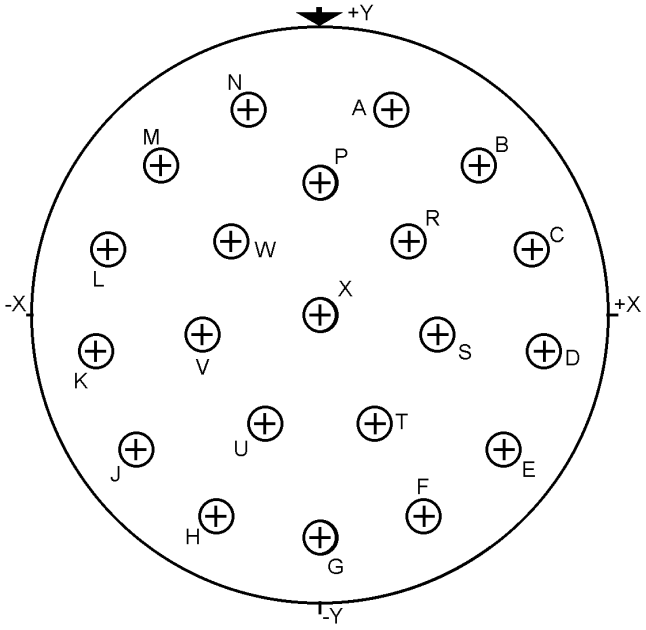
Contacts (Insert arrangement 22-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
15	-.273 (6.93)	+.053 (1.35)	58	+.182 (4.62)	+.424 (10.77)
16	-.273 (6.93)	-.053 (1.35)	59	+.182 (4.62)	+.318 (8.08)
17	-.273 (6.93)	-.159 (4.04)	60	+.182 (4.62)	+.212 (5.38)
18	-.273 (6.93)	-.265 (6.73)	61	+.182 (4.62)	+.106 (2.69)
19	-.287 (7.29)	-.390 (9.91)	62	+.182 (4.62)	+.000 (0.00)
20	-.182 (4.62)	+.424 (10.77)	63	+.182 (4.62)	-.106 (2.69)
21	-.182 (4.62)	+.318 (8.08)	64	+.182 (4.62)	-.212 (5.38)
22	-.182 (4.62)	+.212 (5.38)	65	+.182 (4.62)	-.318 (8.08)
23	-.182 (4.62)	+.106 (2.69)	66	+.192 (4.88)	-.446 (11.33)
24	-.182 (4.62)	+.000 (0.00)	67	+.273 (6.93)	+.371 (9.42)
25	-.182 (4.62)	-.106 (2.69)	68	+.273 (6.93)	+.265 (6.73)
26	-.182 (4.62)	-.212 (5.38)	69	+.273 (6.93)	+.159 (4.04)
27	-.182 (4.62)	-.318 (8.08)	70	+.273 (6.93)	+.053 (1.35)
28	-.192 (4.88)	-.446 (11.33)	71	+.273 (6.93)	-.053 (1.35)
29	-.091 (2.31)	+.477 (12.12)	72	+.273 (6.93)	-.159 (4.04)
30	-.091 (2.31)	+.371 (9.42)	73	+.273 (6.93)	-.265 (6.73)
31	-.091 (2.31)	+.265 (6.73)	74	+.287 (7.29)	-.390 (9.91)
32	-.091 (2.31)	+.159 (4.04)	75	+.364 (9.25)	+.318 (8.08)
33	-.091 (2.31)	+.053 (1.35)	76	+.364 (9.25)	+.212 (5.38)
34	-.091 (2.31)	-.053 (1.35)	77	+.364 (9.25)	+.106 (2.69)
35	-.091 (2.31)	-.159 (4.04)	78	+.364 (9.25)	+.000 (0.00)
36	-.091 (2.31)	-.265 (6.73)	79	+.364 (9.25)	-.106 (2.69)
37	-.091 (2.31)	-.371 (9.42)	80	+.364 (9.25)	-.212 (5.38)
38	-.091 (2.31)	-.477 (12.12)	81	+.364 (9.25)	-.318 (8.08)
39	+.000 (0.00)	+.424 (10.77)	82	+.455 (11.56)	+.159 (4.04)
40	+.000 (0.00)	+.318 (8.08)	83	+.455 (11.56)	+.053 (1.35)
41	+.000 (0.00)	+.212 (5.38)	84	+.482 (12.24)	-.053 (1.35)
42	+.000 (0.00)	+.106 (2.69)	85	+.455 (11.56)	-.159 (4.04)
43	+.000 (0.00)	+.000 (0.00)			

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-2	85	22	M	All	MS27350-2

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

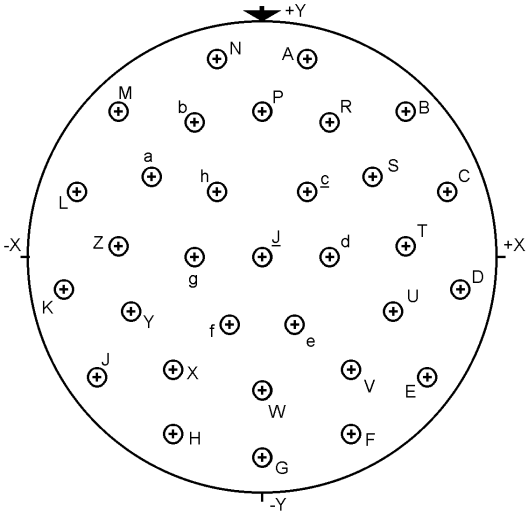


Contacts (Insert arrangement 22-21)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.128 (3.25)	+.385 (9.78)	M	-.289 (7.34)	+.285 (7.24)
B	+.289 (7.34)	+.285 (7.24)	N	-.128 (3.25)	+.385 (9.78)
C	+.386 (9.80)	+.123 (3.12)	P	+.000 (0.00)	+.245 (6.22)
D	+.400 (10.16)	-.065 (1.65)	R	+.160 (4.06)	+.146 (3.71)
E	+.328 (8.33)	-.239 (6.07)	S	+.214 (5.44)	-.035 (.89)
F	+.163 (4.14)	-.362 (9.19)	T	+.094 (2.39)	-.194 (4.93)
G	+.000 (0.00)	-.406 (10.31)	U	-.094 (2.39)	-.194 (4.93)
H	-.163 (4.14)	-.362 (9.19)	V	-.214 (5.44)	-.035 (0.89)
J	-.328 (8.33)	-.239 (6.07)	W	-.160 (4.06)	+.146 (3.71)
K	-.400 (10.16)	-.065 (1.65)	X	+.000 (0.00)	+.000 (0.00)
L	-.386 (9.80)	+.123 (3.12)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-21	21	16	II	All	MS27350-21

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

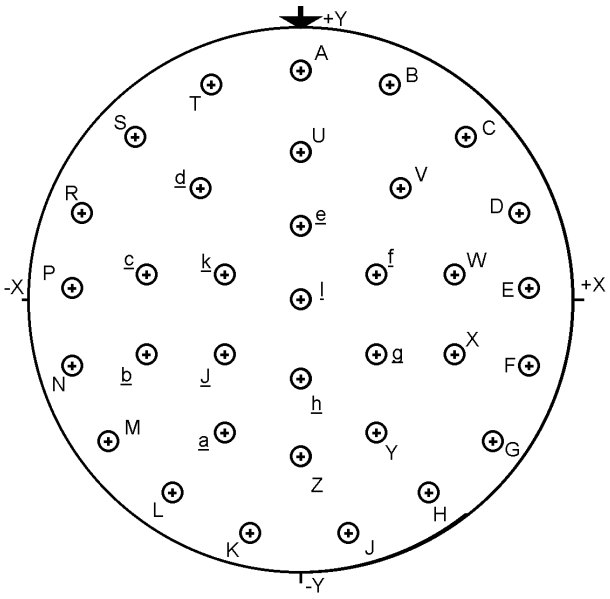


Contacts (Insert arrangement 22-32)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.450 (11.43)	T	+.325 (8.26)	+.025 (0.64)
B	+.325 (8.26)	+.325 (8.26)	U	+.300 (7.62)	-.125 (3.18)
C	+.425 (10.80)	+.150 (3.81)	V	+.200 (5.08)	-.250 (6.35)
D	+.450 (11.43)	-.075 (1.91)	W	+.000 (0.00)	-.300 (7.62)
E	+.375 (9.53)	-.275 (6.99)	X	-.200 (5.08)	-.250 (6.35)
F	+.200 (5.08)	-.400 (10.16)	Y	-.300 (7.62)	-.125 (3.18)
G	+.000 (0.00)	-.450 (11.43)	Z	-.325 (8.26)	+.025 (0.64)
H	-.200 (5.08)	-.400 (10.16)	a	-.250 (6.35)	+.175 (4.45)
J	-.375 (9.53)	-.275 (6.99)	b	-.150 (3.81)	+.300 (7.62)
K	-.450 (11.43)	-.075 (1.91)	c	+.100 (2.54)	+.150 (3.81)
L	-.425 (10.80)	+.150 (3.81)	d	+.150 (3.81)	+.000 (0.00)
M	-.325 (8.26)	+.325 (8.26)	e	+.075 (1.91)	-.150 (3.81)
N	-.100 (2.54)	+.450 (11.43)	f	-.075 (1.91)	-.150 (3.81)
P	+.000 (0.00)	+.325 (8.26)	g	-.150 (3.81)	+.000 (0.00)
R	+.150 (3.81)	+.300 (7.62)	h	-.100 (2.54)	+.150 (3.81)
S	+.250 (6.35)	+.175 (4.45)	J	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-32	32	20	I	All	MS27350-32

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-34)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.450 (11.43)	U	+.000 (0.00)	+.300 (7.62)
B	+.175 (4.45)	+.425 (10.80)	V	+.200 (5.08)	+.225 (5.72)
C	+.325 (8.26)	+.325 (8.26)	W	+.300 (7.62)	+.050 (1.27)
D	+.425 (10.80)	+.175 (4.45)	X	+.300 (7.62)	-.100 (2.54)
E	+.450 (11.43)	+.025 (0.64)	Y	+.150 (3.81)	-.250 (6.35)
F	+.450 (11.43)	-.125 (3.18)	Z	+.000 (0.00)	-.300 (7.62)
G	+.375 (9.53)	-.275 (6.99)	a	-.150 (3.81)	-.250 (6.35)
H	+.250 (6.35)	-.375 (9.53)	b	-.300 (7.62)	-.100 (2.54)
J	+.100 (2.54)	-.450 (11.43)	c	-.300 (7.62)	+.050 (1.27)
K	-.100 (2.54)	-.450 (11.43)	d	-.200 (5.08)	+.225 (5.72)
L	-.250 (6.35)	-.375 (9.53)	e	+.000 (0.00)	+.150 (3.81)
M	-.375 (9.53)	-.275 (6.99)	f	+.150 (3.81)	+.050 (1.27)
N	-.450 (11.43)	-.125 (3.18)	g	+.150 (3.81)	-.100 (2.54)
P	-.450 (11.43)	+.025 (0.64)	h	+.000 (0.00)	-.150 (3.81)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

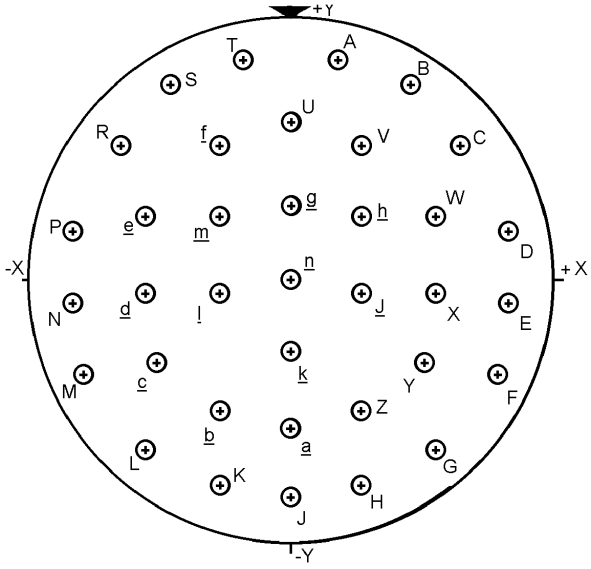
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 22-34)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
R	-.425 (10.80)	+.175 (4.45)	J	-.150 (3.81)	-.100 (2.54)
S	-.325 (8.26)	+.325 (8.26)	k	-.150 (3.81)	+.050 (1.27)
T	-.175 (4.45)	+.425 (10.80)	l	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-34	34	20	I	All	MS27350-34

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-36)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.100 (2.54)	+.450 (11.43)	V	+.150 (3.81)	+.275 (6.99)
B	+.250 (6.35)	+.400 (10.16)	W	+.300 (7.62)	+.125 (3.18)
C	+.350 (8.89)	+.275 (6.99)	X	+.300 (7.62)	-.025 (0.64)
D	+.450 (11.43)	+.100 (2.54)	Y	+.275 (6.99)	-.175 (4.45)
E	+.450 (11.43)	-.050 (1.27)	Z	+.150 (3.81)	-.275 (6.99)
F	+.425 (10.80)	-.200 (5.08)	a	+.000 (0.00)	-.300 (7.62)
G	+.300 (7.62)	-.350 (8.89)	b	-.150 (3.81)	-.275 (6.99)
H	+.150 (3.81)	-.425 (10.80)	c	-.275 (6.99)	-.175 (4.45)
J	+.000 (0.00)	-.450 (11.43)	d	-.300 (7.62)	-.025 (0.64)
K	-.150 (3.81)	-.425 (10.80)	e	-.300 (7.62)	+.125 (3.18)
L	-.300 (7.62)	-.350 (8.89)	f	-.150 (3.81)	+.275 (6.99)
M	-.425 (10.80)	-.200 (5.08)	g	+.000 (0.00)	+.150 (3.81)
N	-.450 (11.43)	-.050 (1.27)	h	+.150 (3.81)	+.125 (3.18)
P	-.450 (11.43)	+.100 (2.54)	J	+.150 (3.81)	-.025 (0.64)
R	-.350 (8.89)	+.275 (6.99)	k	+.000 (0.00)	-.150 (3.81)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

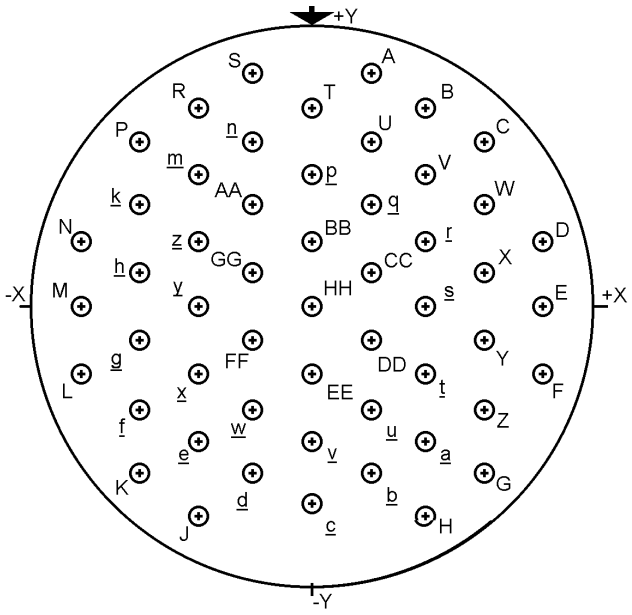
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 22-36)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
S	-.250 (6.35)	+.400 (10.16)	l	-.150 (3.81)	-.025 (0.64)
T	-.100 (2.54)	+.450 (11.43)	m	-.150 (3.81)	+.125 (3.18)
U	+.000 (0.00)	+.325 (8.26)	n	+.000 (0.00)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-36	36	20	I	All	MS27350-36

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-53)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.112 (2.84)	+.455 (11.56)	e	-.225 (5.72)	-.260 (6.60)
B	+.225 (5.72)	+.390 (9.91)	f	-.336 (8.53)	-.195 (4.95)
C	+.336 (8.53)	+.325 (8.26)	g	-.336 (8.53)	-.065 (1.65)
D	+.450 (11.43)	+.130 (3.30)	h	-.336 (8.53)	+.065 (1.65)
E	+.450 (11.43)	+.000 (0.00)	k	-.336 (8.53)	+.195 (4.95)
F	+.450 (11.43)	-.130 (3.30)	m	-.225 (5.72)	+.260 (6.60)
G	+.336 (8.53)	-.325 (8.26)	n	-.112 (2.84)	+.325 (8.26)
H	+.225 (5.72)	-.410 (10.41)	p	+.000 (0.00)	+.260 (6.60)
J	-.225 (5.72)	-.410 (10.41)	q	+.112 (2.84)	+.195 (4.95)
K	-.336 (8.53)	-.325 (8.26)	r	+.225 (5.72)	+.130 (3.30)
L	-.450 (11.43)	-.130 (3.30)	s	+.225 (5.72)	+.000 (0.00)
M	-.450 (11.43)	+.000 (0.00)	t	+.225 (5.72)	-.130 (3.30)
N	-.450 (11.43)	+.130 (3.30)	u	+.112 (2.84)	-.195 (4.95)
P	-.336 (8.53)	+.325 (8.26)	v	+.000 (0.00)	-.260 (6.60)
R	-.225 (5.72)	+.390 (9.91)	w	-.112 (2.84)	-.195 (4.95)
S	-.112 (2.84)	+.455 (11.56)	x	-.225 (5.72)	-.130 (3.30)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

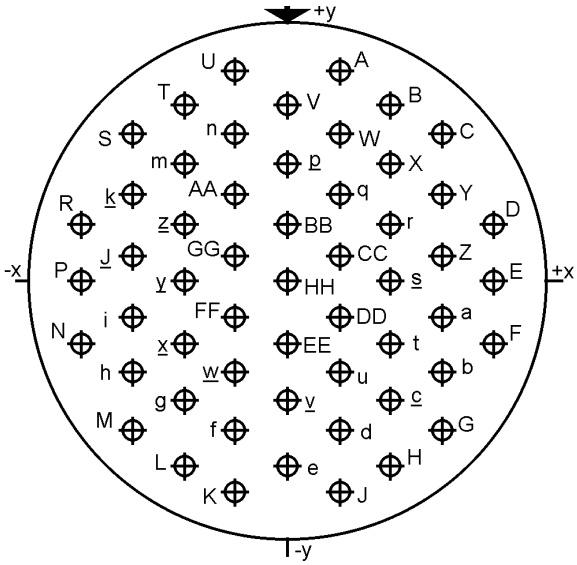
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 22-53)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
T	+.000 (0.00)	+.390 (9.91)	y	-.225 (5.72)	+.000 (0.00)
U	+.112 (2.84)	+.325 (8.26)	z	-.225 (5.72)	+.130 (3.30)
V	+.225 (5.72)	+.260 (6.60)	AA	-.112 (2.84)	+.195 (4.95)
W	+.336 (8.53)	+.195 (4.95)	BB	+.000 (0.00)	+.130 (3.30)
X	+.336 (8.53)	+.065 (1.65)	CC	+.112 (2.84)	+.065 (1.65)
Y	+.336 (8.53)	-.065 (1.65)	DD	+.112 (2.84)	-.065 (1.65)
Z	+.336 (8.53)	-.195 (4.95)	EE	+.000 (.00)	-.130 (3.30)
a	+.225 (5.72)	-.260 (6.60)	FF	-.112 (2.84)	-.065 (1.65)
b	+.112 (2.84)	-.325 (8.26)	GG	-.112 (2.84)	+.065 (1.65)
c	+.000 (0.00)	-.390 (9.91)	HH	+.000 (0.00)	+.000 (0.00)
d	-.112 (2.84)	-.325 (8.26)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-53	53	20	I	All	MS27350-53

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.112 (2.84)	+.455 (11.56)	f	-.112 (2.84)	-.325 (8.26)
B	+.225 (5.72)	+.390 (9.91)	g	-.225 (5.72)	-.260 (6.60)
C	+.336 (8.53)	+.325 (8.26)	h	-.336 (8.53)	-.195 (4.95)
D	+.450 (11.43)	+.130 (3.30)	i	-.336 (8.53)	-.065 (1.65)
E	+.450 (11.43)	+.000 (0.00)	J	-.336 (8.53)	+.065 (1.65)
F	+.450 (11.43)	-.130 (3.30)	k	-.336 (8.53)	+.195 (4.95)
G	+.336 (8.53)	-.325 (8.26)	m	-.225 (5.72)	+.260 (6.60)
H	+.225 (5.72)	-.390 (9.91)	n	-.112 (2.84)	+.325 (8.26)
J	+.112 (2.84)	-.455 (11.56)	p	+.000 (0.00)	+.260 (6.60)
K	-.112 (2.84)	-.455 (11.56)	q	+.112 (2.84)	+.195 (4.95)
L	-.225 (5.72)	-.390 (9.91)	r	+.225 (5.72)	+.130 (3.30)
M	-.336 (8.53)	-.325 (8.26)	s	+.225 (5.72)	+.000 (0.00)
N	-.450 (11.43)	-.130 (3.30)	t	+.225 (5.72)	-.130 (3.30)
P	-.450 (11.43)	+.000 (0.00)	u	+.112 (2.84)	-.195 (4.95)
R	-.450 (11.43)	+.130 (3.30)	v	+.000 (0.00)	-.260 (6.60)
S	-.336 (8.53)	+.325 (8.26)	w	-.112 (2.84)	-.195 (4.95)
T	-.225 (5.72)	+.390 (9.91)	x	-.225 (5.72)	-.130 (3.30)
U	-.112 (2.84)	+.455 (11.56)	y	-.225 (5.72)	+.000 (0.00)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

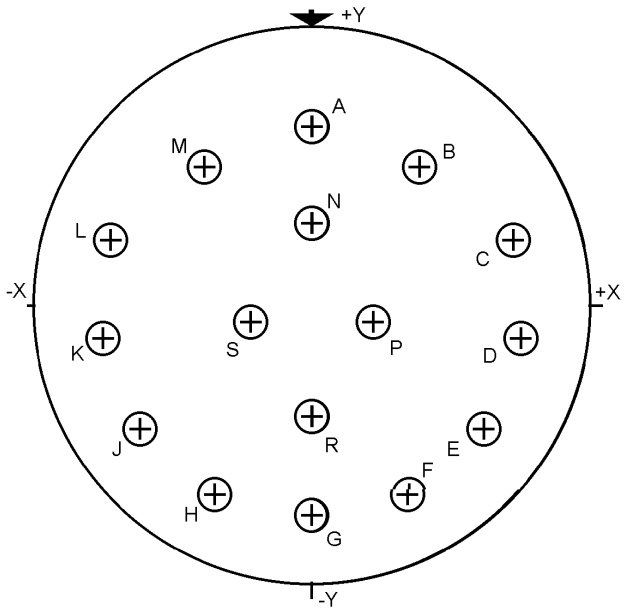
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 22-55)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
V	+.000 (0.00)	+.390 (9.91)	z	-.225 (5.72)	+.130 (3.30)
W	+.112 (2.84)	+.325 (8.26)	AA	-.112 (2.84)	+.195 (4.95)
X	+.225 (5.72)	+.260 (6.60)	BB	+.000 (0.00)	+.130 (3.30)
Y	+.336 (8.53)	+.195 (4.95)	CC	+.112 (2.84)	+.065 (1.65)
Z	+.336 (8.53)	+.065 (1.65)	DD	+.112 (2.84)	-.065 (1.65)
a	+.336 (8.53)	-.065 (1.65)	EE	+.000 (0.00)	-.130 (3.30)
b	+.336 (8.53)	-.195 (4.95)	FF	-.112 (2.84)	-.065 (1.65)
c	+.225 (5.72)	-.260 (6.60)	GG	-.112 (2.84)	+.065 (1.65)
d	+.112 (2.84)	-.325 (8.26)	HH	+.000 (0.00)	+.000 (0.00)
e	+.000 (0.00)	-.390 (9.91)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-55	55	20	I	All	MS27350-55

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

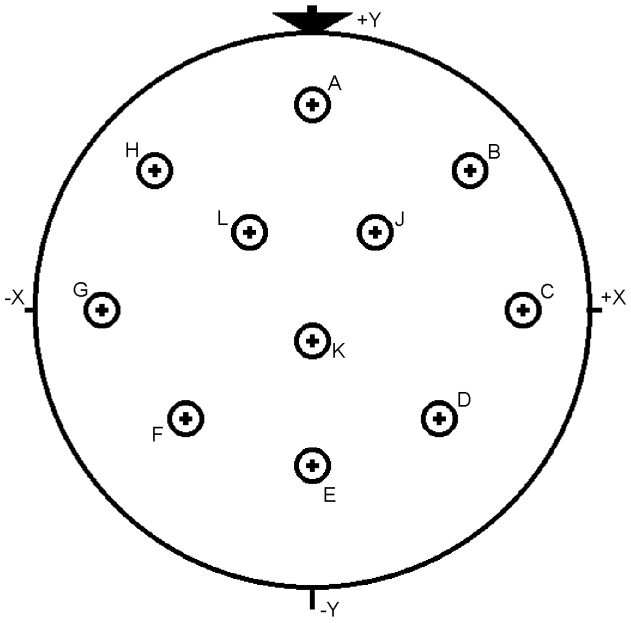


Contacts (Insert arrangement 22-97)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.344 (8.74)	J	-.328 (8.33)	-.239 (6.07)
B	+.210 (5.33)	+.270 (6.86)	K	-.400 (10.16)	-.065 (1.65)
C	+.386 (9.80)	+.123 (3.12)	L	-.386 (9.80)	+.123 (3.12)
D	+.400 (10.16)	-.065 (1.65)	M	-.210 (5.33)	+.270 (6.86)
E	+.328 (8.33)	-.239 (6.07)	N	+.000 (0.00)	+.156 (3.96)
F	+.183 (4.65)	-.362 (9.19)	P	+.120 (3.05)	-.030 (0.76)
G	+.000 (0.00)	-.406 (10.31)	R	+.000 (0.00)	-.218 (5.54)
H	-.183 (4.65)	-.362 (9.19)	S	-.120 (3.05)	-.030 (0.76)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-97	16	16	I	All	MS27350-97

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

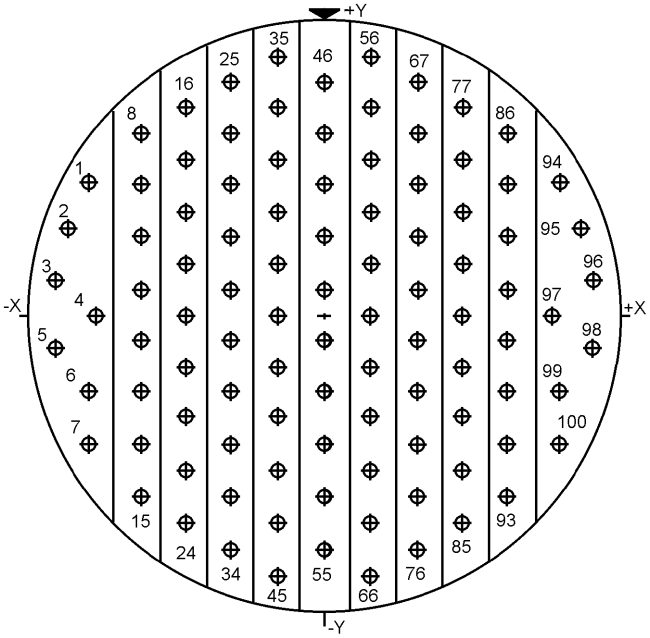


Contacts (Insert arrangement 22-99)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.404 (10.26)	G	-.404 (10.26)	+.000 (0.00)
B	+.300 (7.62)	+.270 (6.86)	H	-.300 (7.62)	+.270 (6.86)
C	+.404 (10.26)	+.000 (0.00)	J	+.120 (3.05)	+.150 (3.81)
D	+.240 (6.10)	-.210 (5.33)	K	+.000 (0.00)	-.060 (1.52)
E	+.000 (0.00)	-.300 (7.62)	L	-.120 (3.05)	+.150 (3.81)
F	-.240 (6.10)	-.210 (5.33)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-99	11	16	II	All	MS27350-99

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 22-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.428 (10.87)	+.241 (6.12)	51	+.000 (0.00)	-.047 (1.19)
2	-.467 (11.86)	+.154 (3.91)	52	+.000 (0.00)	-.142 (3.61)
3	-.488 (12.40)	+.061 (1.55)	53	+.000 (0.00)	-.237 (6.02)
4	-.415 (10.54)	+.000 (0.00)	54	+.000 (0.00)	-.332 (8.43)
5	-.488 (12.40)	-.061 (1.55)	55	+.000 (0.00)	-.427 (10.85)
6	-.428 (10.87)	-.142 (3.61)	56	+.083 (2.11)	+.475 (12.07)
7	-.428 (10.87)	-.237 (6.02)	57	+.083 (2.11)	+.380 (9.65)
8	-.332 (8.43)	+.333 (8.46)	58	+.083 (2.11)	+.285 (7.24)
9	-.332 (8.43)	+.238 (6.05)	59	+.083 (2.11)	+.190 (4.83)
10	-.332 (8.43)	+.143 (3.63)	60	+.083 (2.11)	+.095 (2.41)
11	-.332 (8.43)	+.048 (1.22)	61	+.083 (2.11)	+.000 (0.00)
12	-.332 (8.43)	-.047 (1.19)	62	+.083 (2.11)	-.095 (2.41)
13	-.332 (8.43)	-.142 (3.61)	63	+.083 (2.11)	-.190 (4.83)
14	-.332 (8.43)	-.237 (6.02)	64	+.083 (2.11)	-.285 (7.24)
15	-.332 (8.43)	-.332 (8.43)	65	+.083 (2.11)	-.380 (9.65)
16	-.249 (6.32)	+.380 (9.65)	66	+.083 (2.11)	-.475 (12.07)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 22-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.249 (6.32)	+.285 (7.24)	67	+.166 (4.22)	+.428 (10.87)
18	-.249 (6.32)	+.190 (4.83)	68	+.166 (4.22)	+.333 (8.46)
19	-.249 (6.32)	+.095 (2.41)	69	+.166 (4.22)	+.238 (6.05)
20	-.249 (6.32)	+.000 (0.00)	70	+.166 (4.22)	+.143 (3.63)
21	-.249 (6.32)	-.095 (2.41)	71	+.166 (4.22)	+.048 (1.22)
22	-.249 (6.32)	-.190 (4.83)	72	+.166 (4.22)	-.047 (1.19)
23	-.249 (6.32)	-.285 (7.24)	73	+.166 (4.22)	-.142 (3.61)
24	-.249 (6.32)	-.380 (9.65)	74	+.166 (4.22)	-.237 (6.02)
25	-.166 (4.22)	+.428 (10.87)	75	+.166 (4.22)	-.332 (8.43)
26	-.166 (4.22)	+.333 (8.46)	76	+.166 (4.22)	-.427 (10.85)
27	-.166 (4.22)	+.238 (6.05)	77	+.249 (6.32)	+.380 (9.65)
28	-.166 (4.22)	+.143 (3.63)	78	+.249 (6.32)	+.285 (7.24)
29	-.166 (4.22)	+.048 (1.22)	79	+.249 (6.32)	+.190 (4.83)
30	-.166 (4.22)	-.047 (1.19)	80	+.249 (6.32)	+.095 (2.41)
31	-.166 (4.22)	-.142 (3.61)	81	+.249 (6.32)	+.000 (0.00)
32	-.166 (4.22)	-.237 (6.02)	82	+.249 (6.32)	-.095 (2.41)
33	-.166 (4.22)	-.332 (8.43)	83	+.249 (6.32)	-.190 (4.83)
34	-.166 (4.22)	-.427 (10.85)	84	+.249 (6.32)	-.285 (7.24)
35	-.083 (2.11)	+.475 (12.07)	85	+.249 (6.32)	-.380 (9.65)
36	-.083 (2.11)	+.380 (9.65)	86	+.332 (8.43)	+.333 (8.46)
37	-.083 (2.11)	+.285 (7.24)	87	+.332 (8.43)	+.238 (6.05)
38	-.083 (2.11)	+.190 (4.83)	88	+.332 (8.43)	+.143 (3.63)
39	-.083 (2.11)	+.095 (2.41)	89	+.332 (8.43)	+.048 (1.22)
40	-.083 (2.11)	+.000 (0.00)	90	+.332 (8.43)	-.047 (1.19)
41	-.083 (2.11)	-.095 (2.41)	91	+.332 (8.43)	-.142 (3.61)
42	-.083 (2.11)	-.190 (4.83)	92	+.332 (8.43)	-.237 (6.02)
43	-.083 (2.11)	-.285 (7.24)	93	+.332 (8.43)	-.332 (8.43)
44	-.083 (2.11)	-.380 (9.65)	94	+.428 (10.87)	+.241 (6.12)
45	-.083 (2.11)	-.475 (12.07)	95	+.467 (11.86)	+.154 (3.91)
46	+.000 (0.00)	+.428 (10.87)	96	+.488 (12.40)	+.061 (1.55)
47	+.000 (0.00)	+.333 (8.46)	97	+.415 (10.54)	+.000 (0.00)

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

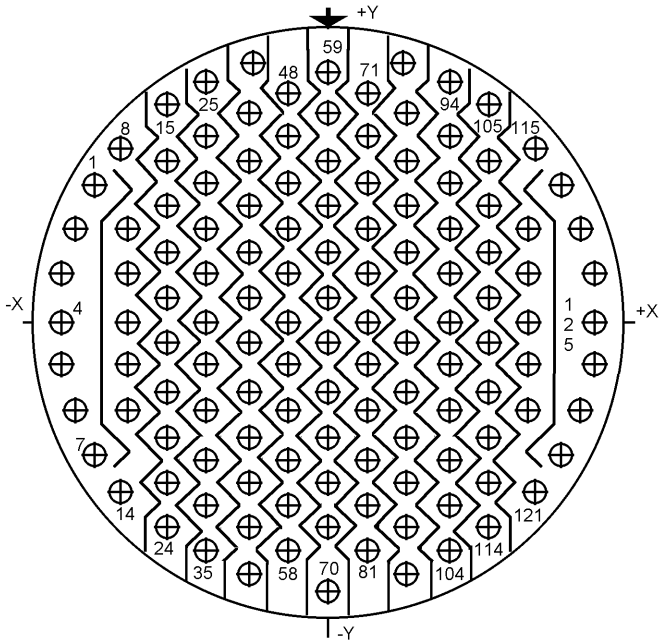
Contacts (Insert arrangement 22-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
48	+.000 (0.00)	+.238 (6.05)	98	+.488 (12.40)	-.061 (1.55)
49	+.000 (0.00)	+.143 (3.63)	99	+.428 (10.87)	-.142 (3.61)
50	+.000 (0.00)	+.048 (1.22)	100	+.428 (10.87)	-.237 (6.02)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
22	-35	100	22D	M	All	MS27523-35

FIGURE 17. Insert arrangement, electrical connector, shell size 22 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 24-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.479 (12.17)	+.279 (7.09)	65	+.000 (0.00)	-.047 (1.19)
2	-.520 (13.21)	+.190 (4.83)	66	+.000 (0.00)	-.142 (3.61)
3	-.546 (13.87)	+.095 (2.41)	67	+.000 (0.00)	-.237 (6.02)
4	-.555 (14.10)	+.000 (0.00)	68	+.000 (0.00)	-.332 (8.43)
5	-.546 (13.87)	-.095 (2.41)	69	+.000 (0.00)	-.427 (10.85)
6	-.520 (13.21)	-.190 (4.83)	70	+.000 (0.00)	-.555 (14.10)
7	-.479 (12.17)	-.279 (7.09)	71	+.083 (2.11)	+.475 (12.07)
8	-.424 (10.77)	+.357 (9.07)	72	+.083 (2.11)	+.380 (9.65)
9	-.415 (10.54)	+.190 (4.83)	73	+.083 (2.11)	+.285 (7.24)
10	-.415 (10.54)	+.095 (2.41)	74	+.083 (2.11)	+.190 (4.83)
11	-.415 (10.54)	+.000 (0.00)	75	+.083 (2.11)	+.095 (2.41)
12	-.415 (10.54)	-.095 (2.41)	76	+.083 (2.11)	+.000 (0.00)
13	-.415 (10.54)	-.190 (4.83)	77	+.083 (2.11)	-.095 (2.41)
14	-.424 (10.77)	-.357 (9.07)	78	+.083 (2.11)	-.190 (4.83)
15	-.332 (8.43)	+.444 (11.28)	79	+.083 (2.11)	-.285 (7.24)
16	-.332 (8.43)	+.332 (8.43)	80	+.083 (2.11)	-.380 (9.65)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 24-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
17	-.332 (8.43)	+.237 (6.02)	81	+.083 (2.11)	-.475 (12.07)
18	-.332 (8.43)	+.142 (3.61)	82	+.160 (4.06)	+.531 (13.49)
19	-.332 (8.43)	+.047 (1.19)	83	+.166 (4.22)	+.427 (10.85)
20	-.332 (8.43)	-.047 (1.19)	84	+.166 (4.22)	+.332 (8.43)
21	-.332 (8.43)	-.142 (3.61)	85	+.166 (4.22)	+.237 (6.02)
22	-.332 (8.43)	-.237 (6.02)	86	+.166 (4.22)	+.142 (3.61)
23	-.332 (8.43)	-.332 (8.43)	87	+.166 (4.22)	+.047 (1.19)
24	-.332 (8.43)	-.427 (10.85)	88	+.166 (4.22)	-.047 (1.19)
25	-.249 (6.32)	+.496 (12.60)	89	+.166 (4.22)	-.142 (3.61)
26	-.249 (6.32)	+.380 (9.65)	90	+.166 (4.22)	-.237 (6.02)
27	-.249 (6.32)	+.285 (7.24)	91	+.166 (4.22)	-.332 (8.43)
28	-.249 (6.32)	+.190 (4.83)	92	+.166 (4.22)	-.427 (10.85)
29	-.249 (6.32)	+.095 (2.41)	93	+.166 (4.22)	-.522 (13.26)
30	-.249 (6.32)	+.000 (0.00)	94	+.249 (6.32)	+.496 (12.60)
31	-.249 (6.32)	-.095 (2.41)	95	+.249 (6.32)	+.380 (9.65)
32	-.249 (6.32)	-.190 (4.83)	96	+.249 (6.32)	+.285 (7.24)
33	-.249 (6.32)	-.285 (7.24)	97	+.249 (6.32)	+.190 (4.83)
34	-.249 (6.32)	-.380 (9.65)	98	+.249 (6.32)	+.095 (2.41)
35	-.249 (6.32)	-.475 (12.07)	99	+.249 (6.32)	+.000 (0.00)
36	-.160 (4.06)	+.531 (13.49)	100	+.249 (6.32)	-.095 (2.41)
37	-.166 (4.22)	+.427 (10.85)	101	+.249 (6.32)	-.190 (4.83)
38	-.166 (4.22)	+.332 (8.43)	102	+.249 (6.32)	-.285 (7.24)
39	-.166 (4.22)	+.237 (6.02)	103	+.249 (6.32)	-.380 (9.65)
40	-.166 (4.22)	+.142 (3.61)	104	+.249 (6.32)	-.475 (12.07)
41	-.166 (4.22)	+.047 (1.19)	105	+.332 (8.43)	+.444 (11.28)
42	-.166 (4.22)	-.047 (1.19)	106	+.332 (8.43)	+.332 (8.43)
43	-.166 (4.22)	-.142 (3.61)	107	+.332 (8.43)	+.237 (6.02)
44	-.166 (4.22)	-.237 (6.02)	108	+.332 (8.43)	+.142 (3.61)
45	-.166 (4.22)	-.332 (8.43)	109	+.332 (8.43)	+.047 (1.19)
46	-.166 (4.22)	-.427 (10.85)	110	+.332 (8.43)	-.047 (1.19)
47	-.166 (4.22)	-.522 (13.26)	111	+.332 (8.43)	-.142 (3.61)
48	-.083 (2.11)	+.475 (12.07)	112	+.332 (8.43)	-.237 (6.02)
49	-.083 (2.11)	+.380 (9.65)	113	+.332 (8.43)	-.332 (8.43)
50	-.083 (2.11)	+.285 (7.24)	114	+.332 (8.43)	-.427 (10.85)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

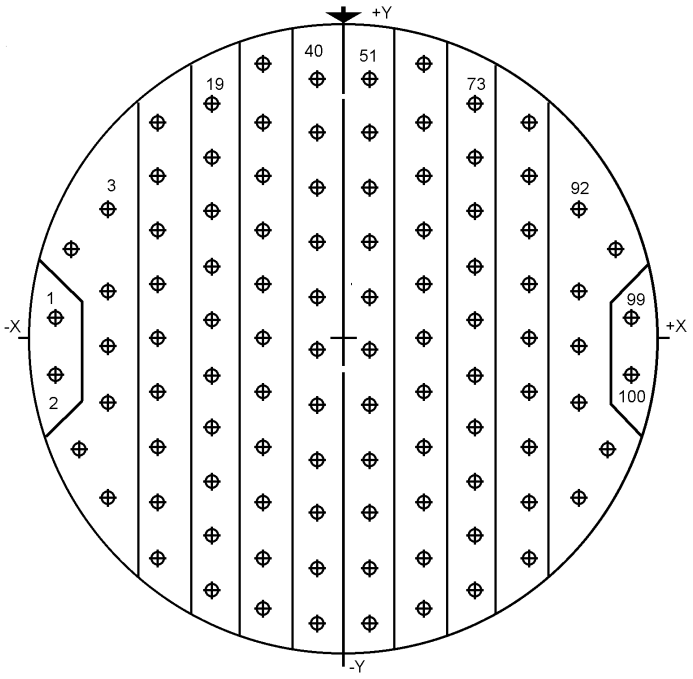
Contacts (Insert arrangement 24-1)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
51	-.083 (2.11)	+.190 (4.83)	115	+.424 (10.77)	+.357 (9.07)
52	-.083 (2.11)	+.095 (2.41)	116	+.415 (10.54)	+.190 (4.83)
53	-.083 (2.11)	+.000 (0.00)	117	+.415 (10.54)	+.095 (2.41)
54	-.083 (2.11)	-.095 (2.41)	118	+.415 (10.54)	+.000 (0.00)
55	-.083 (2.11)	-.190 (4.83)	119	+.415 (10.54)	-.095 (2.41)
56	-.083 (2.11)	-.285 (7.24)	120	+.415 (10.54)	-.190 (4.83)
57	-.083 (2.11)	-.380 (9.65)	121	+.424 (10.77)	-.357 (9.07)
58	-.083 (2.11)	-.475 (12.07)	122	+.479 (12.17)	+.279 (7.09)
59	+.000 (0.00)	+.522 (13.26)	123	+.520 (13.21)	+.190 (4.83)
60	+.000 (0.00)	+.427 (10.85)	124	+.546 (13.87)	+.095 (2.41)
61	+.000 (0.00)	+.332 (8.43)	125	+.555 (14.10)	+.000 (0.00)
62	+.000 (0.00)	+.237 (6.02)	126	+.546 (13.87)	-.095 (2.41)
63	+.000 (0.00)	+.142 (3.61)	127	+.520 (13.21)	-.190 (4.83)
64	+.000 (0.00)	+.047 (1.19)	128	+.479 (12.17)	-.279 (7.09)

(Inactive for new design for MIL-DTL-38999)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-1	128	22M	M	All	MS27351-1

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 24-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.550 (13.97)	+.039 (.99)	51	+.053 (1.35)	+.502 (12.75)
2	-.550 (13.97)	-.068 (1.73)	52	+.053 (1.35)	+.397 (10.08)
3	-.459 (11.66)	+.253 (6.43)	53	+.053 (1.35)	+.292 (7.42)
4	-.523 (13.28)	+.175 (4.45)	54	+.053 (1.35)	+.187 (4.75)
5	-.459 (11.66)	+.092 (2.34)	55	+.053 (1.35)	+.082 (2.08)
6	-.459 (11.66)	-.014 (0.36)	56	+.053 (1.35)	-.023 (0.58)
7	-.459 (11.66)	-.122 (3.10)	57	+.053 (1.35)	-.128 (3.25)
8	-.509 (12.93)	-.215 (5.46)	58	+.053 (1.35)	-.233 (5.92)
9	-.459 (11.66)	-.307 (7.80)	59	+.053 (1.35)	-.338 (8.59)
10	-.354 (8.99)	+.420 (10.67)	60	+.053 (1.35)	-.443 (11.25)
11	-.354 (8.99)	+.315 (8.00)	61	+.053 (1.35)	-.548 (13.92)
12	-.354 (8.99)	+.210 (5.33)	62	+.154 (3.91)	+.525 (13.34)
13	-.354 (8.99)	+.105 (2.67)	63	+.154 (3.91)	+.420 (10.67)
14	-.354 (8.99)	+.000 (0.00)	64	+.154 (3.91)	+.315 (8.00)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 24-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
15	-.354 (8.99)	-.105 (2.67)	65	+.154 (3.91)	+.210 (5.33)
16	-.354 (8.99)	-.210 (5.33)	66	+.154 (3.91)	+.105 (2.67)
17	-.354 (8.99)	-.315 (8.00)	67	+.154 (3.91)	+.000 (0.00)
18	-.354 (8.99)	-.420 (10.67)	68	+.154 (3.91)	-.105 (2.67)
19	-.255 (6.48)	+.457 (11.61)	69	+.154 (3.91)	-.210 (5.33)
20	-.255 (6.48)	+.352 (8.94)	70	+.154 (3.91)	-.315 (8.00)
21	-.255 (6.48)	+.247 (6.27)	71	+.154 (3.91)	-.420 (10.67)
22	-.255 (6.48)	+.142 (3.61)	72	+.154 (3.91)	-.525 (13.34)
23	-.255 (6.48)	+.037 (0.94)	73	+.255 (6.48)	+.457 (11.61)
24	-.255 (6.48)	-.068 (1.73)	74	+.255 (6.48)	+.352 (8.94)
25	-.255 (6.48)	-.173 (4.39)	75	+.255 (6.48)	+.247 (6.27)
26	-.255 (6.48)	-.278 (7.06)	76	+.255 (6.48)	+.142 (3.61)
27	-.255 (6.48)	-.383 (9.73)	77	+.255 (6.48)	+.037 (0.94)
28	-.255 (6.48)	-.488 (12.40)	78	+.255 (6.48)	-.068 (1.73)
29	-.154 (3.91)	+.525 (13.34)	79	+.255 (6.48)	-.173 (4.39)
30	-.154 (3.91)	+.420 (10.67)	80	+.255 (6.48)	-.278 (7.06)
31	-.154 (3.91)	+.315 (8.00)	81	+.255 (6.48)	-.383 (9.73)
32	-.154 (3.91)	+.210 (5.33)	82	+.255 (6.48)	-.488 (12.40)
33	-.154 (3.91)	+.105 (2.67)	83	+.354 (8.99)	+.420 (10.67)
34	-.154 (3.91)	+.000 (0.00)	84	+.354 (8.99)	+.315 (8.00)
35	-.154 (3.91)	-.105 (2.67)	85	+.354 (8.99)	+.210 (5.33)
36	-.154 (3.91)	-.210 (5.33)	86	+.354 (8.99)	+.105 (2.67)
37	-.154 (3.91)	-.315 (8.00)	87	+.354 (8.99)	+.000 (0.00)
38	-.154 (3.91)	-.420 (10.67)	88	+.354 (8.99)	-.105 (2.67)
39	-.154 (3.91)	-.525 (13.34)	89	+.354 (8.99)	-.210 (5.33)
40	-.053 (1.35)	+.502 (12.75)	90	+.354 (8.99)	-.315 (8.00)
41	-.053 (1.35)	+.397 (10.08)	91	+.354 (8.99)	-.420 (10.67)
42	-.053 (1.35)	+.292 (7.42)	92	+.459 (11.66)	+.253 (6.43)
43	-.053 (1.35)	+.187 (4.75)	93	+.523 (13.28)	+.175 (4.45)
44	-.053 (1.35)	+.082 (2.08)	94	+.459 (11.66)	+.092 (2.34)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

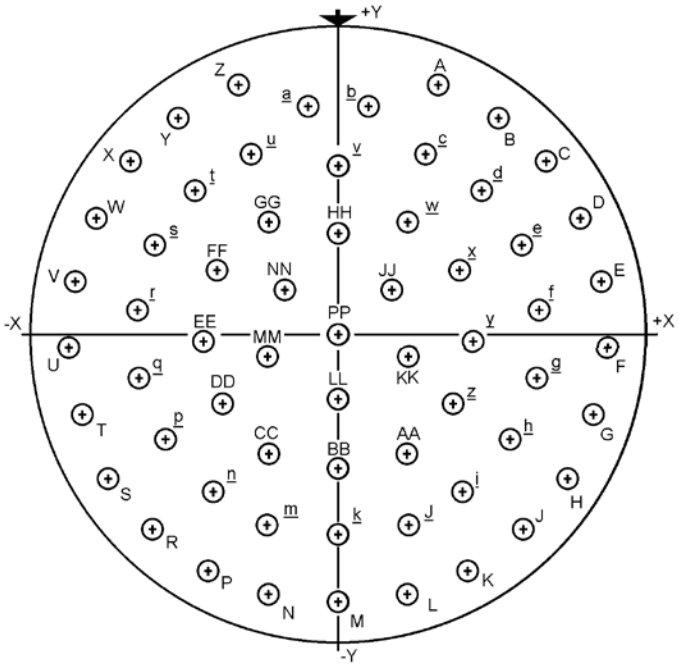
Contacts (Insert arrangement 24-2)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
45	-.053 (1.35)	-.023 (0.58)	95	+.459 (11.66)	-.014 (0.36)
46	-.053 (1.35)	-.128 (3.25)	96	+.459 (11.66)	-.122 (3.10)
47	-.053 (1.35)	-.233 (5.92)	97	+.509 (12.93)	-.215 (5.46)
48	-.053 (1.35)	-.338 (8.59)	98	+.459 (11.66)	-.307 (7.80)
49	-.053 (1.35)	-.443 (11.25)	99	+.550 (13.97)	+.039 (0.99)
50	-.053 (1.35)	-.548 (13.92)	100	+.550 (13.97)	-.068 (1.73)

(Inactive for new design for MIL-DTL-38999.)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-2	100	22	M	All	MS27351-2

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 24-61)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.196 (4.98)	+.500 (12.70)	i	+.251 (6.38)	-.314 (7.98)
B	+.314 (7.98)	+.435 (11.05)	J	+.133 (3.38)	-.379 (9.63)
C	+.413 (10.49)	+.343 (8.71)	k	+.000 (0.00)	-.402 (10.21)
D	+.485 (12.32)	+.230 (5.84)	m	-.133 (3.38)	-.379 (9.63)
E	+.527 (13.39)	+.101 (2.57)	n	-.251 (6.38)	-.314 (7.98)
F	+.536 (13.61)	-.030 (0.76)	p	-.341 (8.66)	-.213 (5.41)
G	+.511 (12.98)	-.164 (4.17)	q	-.392 (9.96)	-.088 (2.24)
H	+.454 (11.53)	-.287 (7.29)	r	-.399 (10.13)	+.046 (1.17)
J	+.368 (9.35)	-.391 (9.93)	s	-.362 (9.19)	+.175 (4.45)
K	+.259 (6.58)	-.470 (11.94)	t	-.285 (7.24)	+.283 (7.19)
L	+.134 (3.40)	-.519 (13.18)	u	-.173 (4.39)	+.363 (9.22)
M	+.000 (0.00)	-.537 (13.64)	v	+.000 (0.00)	+.338 (8.59)
N	-.134 (3.40)	-.519 (13.18)	w	+.147 (3.73)	+.223 (5.66)
P	-.259 (6.58)	-.470 (11.94)	x	+.237 (6.02)	+.122 (3.10)
R	-.368 (9.35)	-.391 (9.93)	y	+.267 (6.78)	-.010 (0.25)
S	-.454 (11.53)	-.287 (7.29)	z	+.228 (5.79)	-.139 (3.53)
T	-.511 (12.98)	-.164 (4.17)	AA	+.131 (3.33)	-.233 (5.92)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

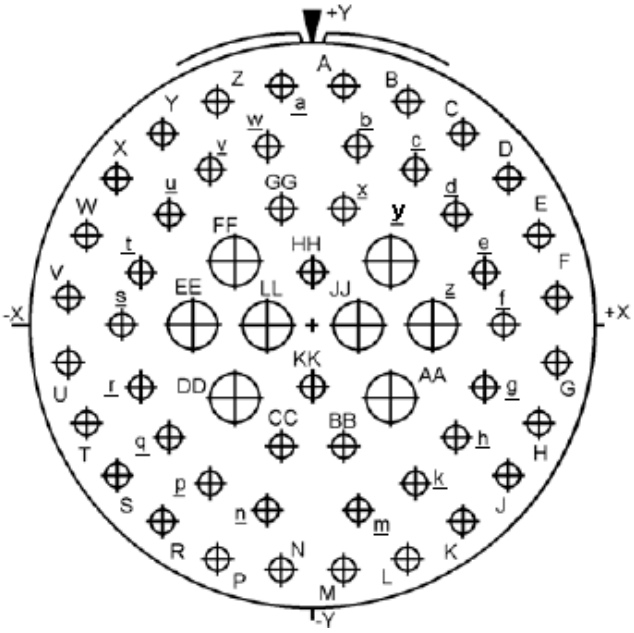
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 24-61)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
U	-.536 (13.61)	-.030 (.76)	BB	+.000 (.00)	-.267 (6.78)
V	-.527 (13.39)	+.101 (2.57)	CC	-.131 (3.33)	-.233 (5.92)
W	-.485 (12.32)	+.230 (5.84)	DD	-.228 (5.79)	-.139 (3.53)
X	-.413 (10.49)	+.343 (8.71)	EE	-.267 (6.78)	-.010 (0.25)
Y	-.314 (7.98)	+.435 (11.05)	FF	-.237 (6.02)	+.122 (3.10)
Z	-.196 (4.98)	+.500 (12.70)	GG	-.147 (3.73)	+.223 (5.66)
a	-.068 (1.73)	+.454 (11.53)	HH	+.000 (0.00)	+.200 (5.08)
b	+.068 (1.73)	+.454 (11.53)	JJ	+.105 (2.67)	+.094 (2.39)
c	+.173 (4.39)	+.363 (9.22)	KK	+.135 (3.43)	-.041 (1.04)
d	+.285 (7.24)	+.283 (7.19)	LL	+.000 (0.00)	-.132 (3.35)
e	+.362 (9.19)	+.175 (4.45)	MM	-.135 (3.43)	-.041 (1.04)
f	+.399 (10.13)	+.046 (1.17)	NN	-.105 (2.67)	+.094 (2.39)
g	+.392 (9.96)	-.088 (2.24)	PP	+.000 (0.00)	+.000 (0.00)
h	+.341 (8.66)	-.213 (5.41)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-61	61	20	I	All	MS27351-61

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 24-4)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.069 (1.75)	+.531 (13.49)	f	+.412 (10.46)	+.000 (0.00)
B	+.203 (5.16)	+.495 (12.57)	g	+.377 (9.58)	-.132 (3.35)
C	+.324 (8.23)	+.425 (10.80)	h	+.311 (7.90)	-.251 (6.38)
D	+.424 (10.77)	+.326 (8.28)	k	+.212 (5.38)	-.344 (8.74)
E	+.493 (12.52)	+.205 (5.21)	m	+.086 (2.18)	-.397 (10.08)
F	+.531 (13.49)	+.069 (1.75)	n	-.086 (2.18)	-.397 (10.08)
G	+.531 (13.49)	-.069 (1.75)	p	-.212 (5.38)	-.344 (8.74)
H	+.493 (12.52)	-.205 (5.21)	q	-.311 (7.90)	-.251 (6.38)
J	+.424 (10.77)	-.326 (8.28)	r	-.377 (9.58)	-.132 (3.35)
K	+.324 (8.23)	-.425 (10.80)	s	-.412 (10.46)	+.000 (0.00)
L	+.203 (5.16)	-.495 (12.57)	t	-.377 (9.58)	+.132 (3.35)
M	+.069 (1.75)	-.531 (13.49)	u	-.311 (7.90)	+.251 (6.38)
N	-.069 (1.75)	-.531 (13.49)	v	-.212 (5.38)	+.344 (8.74)
P	-.203 (5.16)	-.495 (12.57)	w	-.086 (2.18)	+.397 (10.08)
R	-.324 (8.23)	-.425 (10.80)	x	+.069 (1.75)	+.263 (6.68)
S	-.424 (10.77)	-.326 (8.28)	y	+.172 (4.37)	+.149 (3.78)
T	-.493 (12.52)	-.205 (5.21)	z	+.258 (6.55)	+.000 (0.00)
U	-.531 (13.49)	-.069 (1.75)	AA	+.172 (4.37)	-.149 (3.78)
V	-.531 (13.49)	+.069 (1.75)	BB	+.069 (1.75)	-.263 (6.68)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

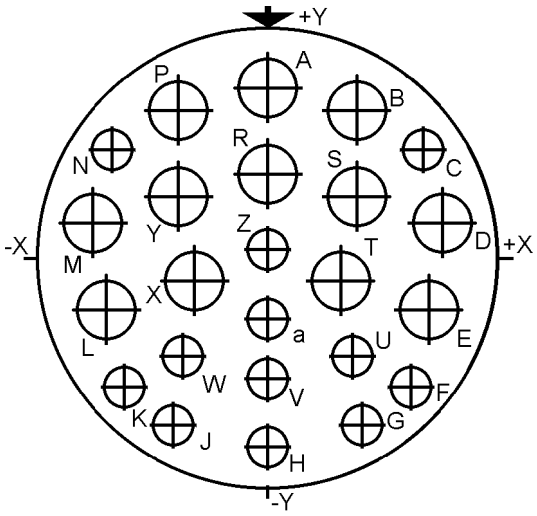
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 24-4)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
W	-.493 (12.52)	+.205 (5.21)	CC	-.069 (1.75)	-.263 (6.68)
X	-.424 (10.77)	+.326 (8.28)	DD	-.172 (4.37)	-.149 (3.78)
Y	-.324 (8.23)	+.425 (10.80)	EE	-.258 (6.55)	+.000 (0.00)
Z	-.203 (5.16)	+.495 (12.57)	FF	-.172 (4.37)	+.149 (3.78)
a	-.069 (1.75)	+.531 (13.49)	GG	-.069 (1.75)	+.263 (6.68)
b	+.086 (2.18)	+.397 (10.08)	HH	+.000 (0.00)	+.132 (3.35)
c	+.212 (5.38)	+.344 (8.74)	JJ	+.086 (2.18)	+.000 (0.00)
d	+.311 (7.90)	+.251 (6.38)	KK	+.000 (0.00)	-.132 (3.35)
e	+.377 (9.58)	+.132 (3.35)	LL	-.086 (2.18)	+.000 (0.00)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-4	8	16	I	AA,DD,EE, FF,JJ,LL,v,z	MS27351-4
		48	20		All others	

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

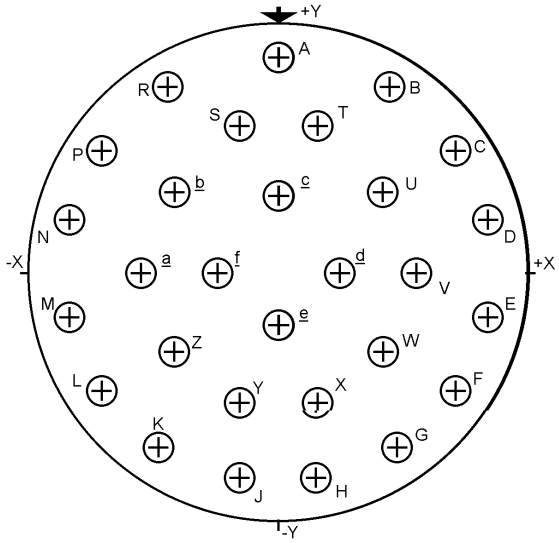


Contacts (Insert arrangement 24-24)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.472 (11.99)	N	-.403 (10.24)	+.298 (7.57)
B	+.230 (5.84)	+.410 (10.41)	P	-.230 (5.84)	+.410 (10.41)
C	+.403 (10.24)	+.298 (7.57)	R	+.000 (0.00)	+.234 (5.94)
D	+.461 (11.71)	+.100 (2.54)	S	+.230 (5.84)	+.172 (4.37)
E	+.413 (10.49)	-.134 (3.40)	T	+.186 (4.72)	-.062 (1.57)
F	+.370 (9.40)	-.336 (8.53)	U	+.211 (5.36)	-.267 (6.78)
G	+.230 (5.84)	-.441 (11.20)	V	+.000 (0.00)	-.323 (8.20)
H	+.000 (0.00)	-.495 (12.57)	W	-.211 (5.36)	-.267 (6.78)
J	-.230 (5.84)	-.441 (11.20)	X	-.186 (4.72)	-.062 (1.57)
K	-.370 (9.40)	-.336 (8.53)	Y	-.230 (5.84)	+.172 (4.37)
L	-.413 (10.49)	-.134 (3.40)	Z	+.000 (0.00)	+.028 (0.71)
M	-.461 (11.71)	+.100 (2.54)	a	+.000 (0.00)	-.151 (3.84)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-24	12	12	I	A,B,D,E,L,M, P,R,S,T,X,Y	MS27351-24
		12	16		All others	

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

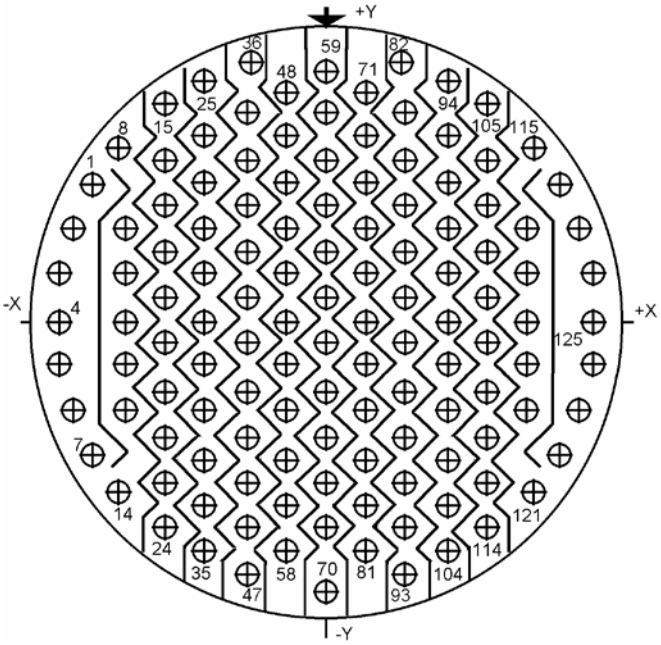


Contacts (Insert arrangement 24-29)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.481 (12.22)	S	-.091 (2.31)	+.321 (8.15)
B	+.258 (6.55)	+.406 (10.31)	T	+.091 (2.31)	+.321 (8.15)
C	+.395 (10.03)	+.277 (7.04)	U	+.228 (5.79)	+.194 (4.93)
D	+.469 (11.91)	+.109 (2.77)	V	+.319 (8.10)	+.000 (0.00)
E	+.469 (11.91)	-.109 (2.77)	W	+.240 (6.10)	-.181 (4.60)
F	+.395 (10.03)	-.277 (7.04)	X	+.091 (2.31)	-.290 (7.37)
G	+.263 (6.68)	-.406 (10.31)	Y	-.091 (2.31)	-.290 (7.37)
H	+.091 (2.31)	-.472 (11.99)	Z	-.240 (6.10)	-.181 (4.60)
J	-.091 (2.31)	-.472 (11.99)	a	-.319 (8.10)	+.000 (0.00)
K	-.263 (6.68)	-.406 (10.31)	b	-.228 (5.79)	+.194 (4.93)
L	-.395 (10.03)	-.277 (7.04)	c	+.000 (0.00)	+.161 (4.09)
M	-.469 (11.91)	-.109 (2.77)	d	+.134 (3.40)	+.000 (0.00)
N	-.469 (11.91)	+.109 (2.77)	e	+.000 (0.00)	-.130 (3.30)
P	-.395 (10.03)	+.277 (7.04)	f	-.134 (3.40)	+.000 (0.00)
R	-.258 (6.55)	+.406 (10.31)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-29	29	16	I	All	MS27351-29

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 24-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
1	-.479 (12.17)	+.279 (7.09)	65	+.000 (0.00)	-.047 (1.19)
2	-.520 (13.21)	+.190 (4.83)	66	+.000 (0.00)	-.142 (3.61)
3	-.546 (13.87)	+.095 (2.41)	67	+.000 (0.00)	-.237 (6.02)
4	-.555 (14.10)	+.000 (0.00)	68	+.000 (0.00)	-.332 (8.43)
5	-.546 (13.87)	-.095 (2.41)	69	+.000 (0.00)	-.427 (10.85)
6	-.520 (13.21)	-.190 (4.83)	70	+.000 (0.00)	-.555 (14.10)
7	-.479 (12.17)	-.279 (7.09)	71	+.083 (2.11)	+.475 (12.07)
8	-.424 (10.77)	+.357 (9.07)	72	+.083 (2.11)	+.380 (9.65)
9	-.415 (10.54)	+.190 (4.83)	73	+.083 (2.11)	+.285 (7.24)
10	-.415 (10.54)	+.095 (2.41)	74	+.083 (2.11)	+.190 (4.83)
11	-.415 (10.54)	+.000 (0.00)	75	+.083 (2.11)	+.095 (2.41)
12	-.415 (10.54)	-.095 (2.41)	76	+.083 (2.11)	+.000 (.00)
13	-.415 (10.54)	-.190 (4.83)	77	+.083 (2.11)	-.095 (2.41)
14	-.424 (10.77)	-.357 (9.07)	78	+.083 (2.11)	-.190 (4.83)
15	-.332 (8.43)	+.444 (11.28)	79	+.083 (2.11)	-.285 (7.24)
16	-.332 (8.43)	+.332 (8.43)	80	+.083 (2.11)	-.380 (9.65)
17	-.332 (8.43)	+.237 (6.02)	81	+.083 (2.11)	-.475 (12.07)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

Contacts (Insert arrangement 24-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
18	-.332 (8.43)	+.142 (3.61)	82	+.160 (4.06)	+.531 (13.49)
19	-.332 (8.43)	+.047 (1.19)	83	+.166 (4.22)	+.427 (10.85)
20	-.332 (8.43)	-.047 (1.19)	84	+.166 (4.22)	+.332 (8.43)
21	-.332 (8.43)	-.142 (3.61)	85	+.166 (4.22)	+.237 (6.02)
22	-.332 (8.43)	-.237 (6.02)	86	+.166 (4.22)	+.142 (3.61)
23	-.332 (8.43)	-.332 (8.43)	87	+.166 (4.22)	+.047 (1.19)
24	-.332 (8.43)	-.427 (10.85)	88	+.166 (4.22)	-.047 (1.19)
25	-.249 (6.32)	+.496 (12.60)	89	+.166 (4.22)	-.142 (3.61)
26	-.249 (6.32)	+.380 (9.65)	90	+.166 (4.22)	-.237 (6.02)
27	-.249 (6.32)	+.285 (7.24)	91	+.166 (4.22)	-.332 (8.43)
28	-.249 (6.32)	+.190 (4.83)	92	+.166 (4.22)	-.427 (10.85)
29	-.249 (6.32)	+.095 (2.41)	93	+.166 (4.22)	-.522 (13.26)
30	-.249 (6.32)	+.000 (0.00)	94	+.249 (6.32)	+.496 (12.60)
31	-.249 (6.32)	-.095 (2.41)	95	+.249 (6.32)	+.380 (9.65)
32	-.249 (6.32)	-.190 (4.83)	96	+.249 (6.32)	+.285 (7.24)
33	-.249 (6.32)	-.285 (7.24)	97	+.249 (6.32)	+.190 (4.83)
34	-.249 (6.32)	-.380 (9.65)	98	+.249 (6.32)	+.095 (2.41)
35	-.249 (6.32)	-.475 (12.07)	99	+.249 (6.32)	+.000 (0.00)
36	-.160 (4.06)	+.531 (13.49)	100	+.249 (6.32)	-.095 (2.41)
37	-.166 (4.22)	+.427 (10.85)	101	+.249 (6.32)	-.190 (4.83)
38	-.166 (4.22)	+.332 (8.43)	102	+.249 (6.32)	-.285 (7.24)
39	-.166 (4.22)	+.237 (6.02)	103	+.249 (6.32)	-.380 (9.65)
40	-.166 (4.22)	+.142 (3.61)	104	+.249 (6.32)	-.475 (12.07)
41	-.166 (4.22)	+.047 (1.19)	105	+.332 (8.43)	+.444 (11.28)
42	-.166 (4.22)	-.047 (1.19)	106	+.332 (8.43)	+.332 (8.43)
43	-.166 (4.22)	-.142 (3.61)	107	+.332 (8.43)	+.237 (6.02)
44	-.166 (4.22)	-.237 (6.02)	108	+.332 (8.43)	+.142 (3.61)
45	-.166 (4.22)	-.332 (8.43)	109	+.332 (8.43)	+.047 (1.19)
46	-.166 (4.22)	-.427 (10.85)	110	+.332 (8.43)	-.047 (1.19)
47	-.166 (4.22)	-.522 (13.26)	111	+.332 (8.43)	-.142 (3.61)
48	-.083 (2.11)	+.475 (12.07)	112	+.332 (8.43)	-.237 (6.02)
49	-.083 (2.11)	+.380 (9.65)	113	+.332 (8.43)	-.332 (8.43)
50	-.083 (2.11)	+.285 (7.24)	114	+.332 (8.43)	-.427 (10.85)
51	-.083 (2.11)	+.190 (4.83)	115	+.424 (10.77)	+.357 (9.07)
52	-.083 (2.11)	+.095 (2.41)	116	+.415 (10.54)	+.190 (4.83)
53	-.083 (2.11)	+.000 (0.00)	117	+.415 (10.54)	+.095 (2.41)

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

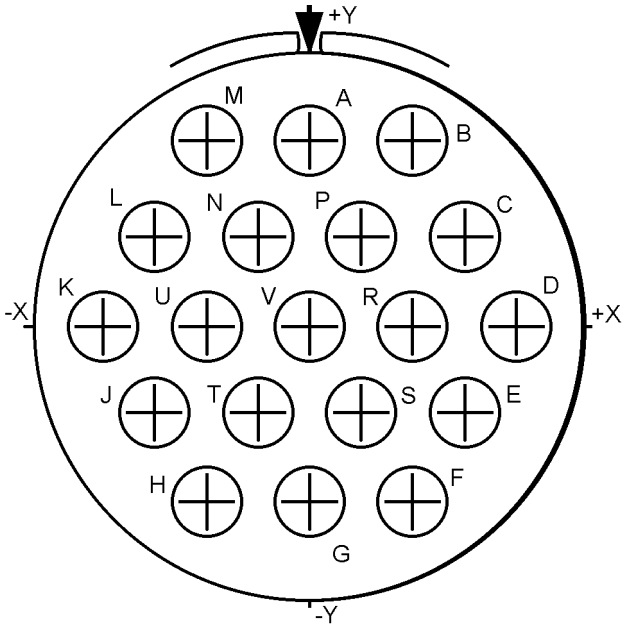
Contacts (Insert arrangement 24-35)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
54	-.083 (2.11)	-.095 (2.41)	118	+.415 (10.54)	+.000 (0.00)
55	-.083 (2.11)	-.190 (4.83)	119	+.415 (10.54)	-.095 (2.41)
56	-.083 (2.11)	-.285 (7.24)	120	+.415 (10.54)	-.190 (4.83)
57	-.083 (2.11)	-.380 (9.65)	121	+.424 (10.77)	-.357 (9.07)
58	-.083 (2.11)	-.475 (12.07)	122	+.479 (12.17)	+.279 (7.09)
59	+.000 (0.00)	+.522 (13.26)	123	+.520 (13.21)	+.190 (4.83)
60	+.000 (0.00)	+.427 (10.85)	124	+.546 (13.87)	+.095 (2.41)
61	+.000 (0.00)	+.332 (8.43)	125	+.555 (14.10)	+.000 (.00)
62	+.000 (0.00)	+.237 (6.02)	126	+.546 (13.87)	-.095 (2.41)
63	+.000 (0.00)	+.142 (3.61)	127	+.520 (13.21)	-.190 (4.83)
64	+.000 (0.00)	+.047 (1.19)	128	+.479 (12.17)	-.279 (7.09)

(Applicable to MIL-DTL-38999 only)

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Supersedes
24	-35	128	22D	M	All	MS27524-35

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.



Contacts (Insert arrangement 24-19)					
Contact position ID	Location		Contact position ID	Location	
	X-axis (mm)	Y-axis (mm)		X-axis (mm)	Y-axis (mm)
A	+.000 (0.00)	+.409 (10.39)	L	-.354 (8.99)	+.205 (5.21)
B	+.236 (5.99)	+.409 (10.39)	M	-.236 (5.99)	+.409 (10.39)
C	+.354 (8.99)	+.205 (5.21)	N	-.118 (3.00)	+.205 (5.21)
D	+.472 (11.99)	+.000 (0.00)	P	+.118 (3.00)	+.205 (5.21)
E	+.354 (8.99)	-.205 (5.21)	R	+.236 (5.99)	+.000 (0.00)
F	+.236 (5.99)	-.409 (10.39)	S	+.118 (3.00)	-.205 (5.21)
G	+.000 (0.00)	-.409 (10.39)	T	-.118 (3.00)	-.205 (5.21)
H	-.236 (5.99)	-.409 (10.39)	U	-.236 (5.99)	+.000 (0.00)
J	-.354 (8.99)	-.205 (5.21)	V	+.000 (.00)	+.000 (0.00)
K	-.472 (11.99)	+.000 (0.00)			

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location
24	-19	19	12	I	All

FIGURE 18. Insert arrangement, electrical connector, shell size 24 (series II of MIL-DTL-38999 and series II of MIL-DTL-27599) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935–2010-085)

Review activities:
Army – AR, MI
Navy – EC, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:49Z
Check the source to verify that this is the current version before use.